

HUAWEI NetEngine40E Universal Service Router V800R011C00

Feature Description - IP Routing

Issue 01
Date 2019-04-15



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1 About This Document

Purpose

This document describes the IP Routing feature in terms of its overview, principles, and applications.

Related Version

The following table lists the product version related to this document.

Product Name	Version
NE40E Series	V800R011C00
U2000	V200R018C60

Intended Audience

This document is intended for:

- Network planning engineers
- Commissioning engineers
- Data configuration engineers
- System maintenance engineers

Security Declaration

- Encryption algorithm declaration
The encryption algorithms DES/3DES/RSA (RSA-1024 or lower)/MD5 (in digital signature scenarios and password encryption)/SHA1 (in digital signature scenarios) have a low security, which may bring security risks. If protocols allowed, using more secure encryption algorithms, such as AES/RSA (RSA-2048 or higher)/SHA2/HMAC-SHA2 is recommended.
- Password configuration declaration
 - Do not set both the start and end characters of a password to "%^%#". This causes the password to be displayed directly in the configuration file.

- To further improve device security, periodically change the password.
- Personal data declaration

Your purchased products, services, or features may use users' some personal data during service operation or fault locating. You must define user privacy policies in compliance with local laws and take proper measures to fully protect personal data.
- Feature declaration
 - The NetStream feature may be used to analyze the communication information of terminal customers for network traffic statistics and management purposes. Before enabling the NetStream feature, ensure that it is performed within the boundaries permitted by applicable laws and regulations. Effective measures must be taken to ensure that information is securely protected.
 - The mirroring feature may be used to analyze the communication information of terminal customers for a maintenance purpose. Before enabling the mirroring function, ensure that it is performed within the boundaries permitted by applicable laws and regulations. Effective measures must be taken to ensure that information is securely protected.
 - The packet header obtaining feature may be used to collect or store some communication information about specific customers for transmission fault and error detection purposes. Huawei cannot offer services to collect or store this information unilaterally. Before enabling the function, ensure that it is performed within the boundaries permitted by applicable laws and regulations. Effective measures must be taken to ensure that information is securely protected.
- Reliability design declaration

Network planning and site design must comply with reliability design principles and provide device- and solution-level protection. Device-level protection includes planning principles of dual-network and inter-board dual-link to avoid single point or single link of failure. Solution-level protection refers to a fast convergence mechanism, such as FRR and VRRP. If solution-level protection is used, ensure that the primary and backup paths do not share links or transmission devices. Otherwise, solution-level protection may fail to take effect.

Special Declaration

- This document serves only as a guide. The content is written based on device information gathered under lab conditions. The content provided by this document is intended to be taken as general guidance, and does not cover all scenarios. The content provided by this document may be different from the information on user device interfaces due to factors such as version upgrades and differences in device models, board restrictions, and configuration files. The actual user device information takes precedence over the content provided by this document. The preceding differences are beyond the scope of this document.
- The maximum values provided in this document are obtained in specific lab environments (for example, only a certain type of board or protocol is configured on a tested device). The actually obtained maximum values may be different from the maximum values provided in this document due to factors such as differences in hardware configurations and carried services.
- Interface numbers used in this document are examples. Use the existing interface numbers on devices for configuration.
- The pictures of hardware in this document are for reference only.

- In this document, public IP addresses may be used in feature introduction and configuration examples and are for reference only unless otherwise specified.

Symbol Conventions

The symbols that may be found in this document are defined as follows.

Symbol	Description
	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
	Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
	Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.
	Indicates a potentially hazardous situation which, if not avoided, could result in equipment damage, data loss, performance deterioration, or unanticipated results. NOTICE is used to address practices not related to personal injury.
	Calls attention to important information, best practices and tips. NOTE is used to address information not related to personal injury, equipment damage, and environment deterioration.

Change History

Changes between document issues are cumulative. The latest document issue contains all the changes made in earlier issues.

- **Changes in Issue 01 (2019-03-30)**

This issue is the first official release. The software version of this issue is V800R011C00SPC200.

2 IP Routing Overview

About This Chapter

- [2.1 Overview of IP Routing](#)
- [2.2 Understanding IP Routing](#)
- [2.3 Application Scenarios for IP Routing](#)
- [2.4 Appendix List of Port Numbers of Common Protocols](#)
- [2.5 Terminology for IP Routing](#)

2.1 Overview of IP Routing

Definition

As a basic concept on data communication networks, routing is the process of packet relaying or forwarding, and the process provides route information for packet forwarding.

Purpose

During data forwarding, routers, routing tables, and routing protocols are indispensable. Routing protocols are used to discover routes and contribute to the generation of routing tables. Routing tables store the routes discovered by various routing protocols, and routers select routes and implement data forwarding.

2.2 Understanding IP Routing

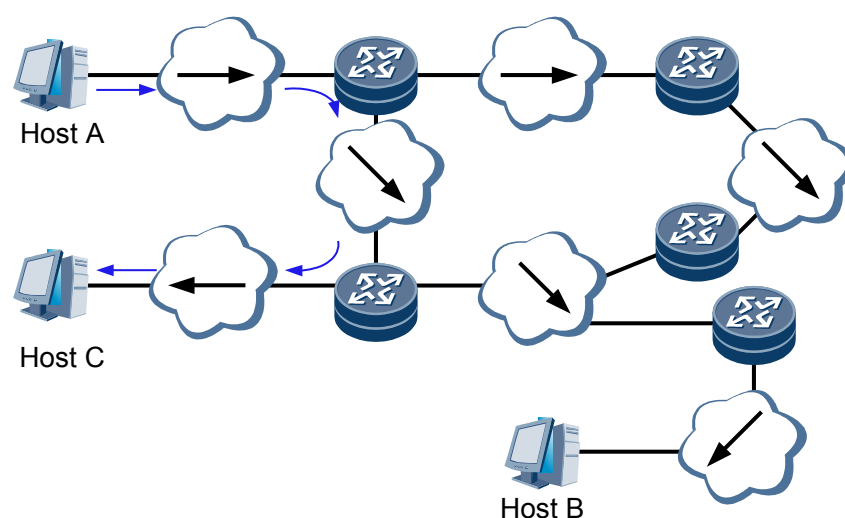
2.2.1 Routers

On the Internet, network connection devices control network traffic and ensure data transmission quality on networks. Common network connection devices include hubs, bridges, switches, and routers.

As a standard network connection device, a router is used to select routes and forward packets. Based on the destination address in the received packet, a router selects a path to send the packet to the next router. The last router is responsible for sending the packet to the destination host. In addition, a router can select an optimal path for data transmission.

For example, in **Figure 2-1**, traffic from Host A to Host C needs to pass through three networks and two routers. The hop count from a router to its directly connected network is zero. The hop count from a router to a network that the router can reach through another router is one. The rest can be deduced by analogy. If a router is connected to another router through a network segment, a network segment exists between the two routers, and they are considered adjacent on the Internet. In **Figure 2-1**, the bold arrows indicate network segments. The routers do not need to know about the physical link composition of each network segment.

Figure 2-1 Network segment and hop count



Network sizes may vary greatly, and the actual lengths of network segments vary as well. Therefore, you can set a weighted coefficient for the network segments of each network and then measure the cost of a route based on the number of network segments.

A route with the minimal network segments is not necessarily optimal. For example, a route passing through three high-speed Local Area Network (LAN) network segments may be a better choice than one passing through two low-speed Wide Area Network (WAN) network segments.

2.2.2 Routing Protocols

Routing protocols are rules used by routers to discover routes, add routes, and maintain routing tables for packet forwarding.

2.2.3 Routing Tables

A router searches a routing table for routes, and each router maintains at least one routing table.

Routing tables store the routes discovered by various routing protocols. Based on the generation method, routes in a routing table consist of the following types:

- Routes discovered by link layer protocols, which are also called interface routes or direct routes
- Static routes configured by the network administrator
- Dynamic routes that are discovered by dynamic routing protocols

Routing Table Types

Each router maintains a local core routing table, and each routing protocol maintains its own routing table.

- Protocol routing table

A protocol routing table stores routing information discovered by the protocol.

A routing protocol can import and advertise routes generated by other routing protocols. For example, if a router that runs Open Shortest Path First (OSPF) needs to use OSPF to advertise direct routes, static routes, or Intermediate System to Intermediate System (IS-IS) routes, the router needs to import these routes into the OSPF routing table.

- Local core routing table

A local core routing table stores protocol routes and optimal routes and selects routes based on the priorities of routing protocols and costs of routes. You can run the **display ip routing-table** command to view the local core routing table of a router.

NOTE

Each router that supports Layer 3 virtual private network (L3VPN) maintains a management routing table (local core routing table) for each VPN instance.

Contents in the Routing Table

On the NE40E, the **display ip routing-table** command displays brief information about the routing table.

```
<HUAWEI> display ip routing-table
Route Flags: R - relay, D - download
to fib, T - to vpn-instance, B - black hole route
-----
Routing Table: Public
Destinations : 8      Routes : 8

Destination/Mask  Proto  Pre  Cost   Flags NextHop         Interface
0.0.0.0/0         Static 60    0      D      1.1.4.2
GigabitEthernet1/0/0
1.1.4.0/30        OSPF   10    0      D      1.1.4.1
GigabitEthernet1/0/0
1.1.4.1/32        Direct 0      0      D      127.0.0.1        InLoopBack0
1.1.4.2/32        OSPF   10    0      D      1.1.4.2
GigabitEthernet1/0/0
127.0.0.0/8       Direct 0      0      D      127.0.0.1        InLoopBack0
127.0.0.1/32      Direct 0      0      D      127.0.0.1        InLoopBack0
127.255.255.255/32 Direct 0      0      D      127.0.0.1        InLoopBack0
255.255.255.255/32 Direct 0      0      D      127.0.0.1        InLoopBack0
```

A routing table contains the following key entries:

- Destination: indicates the destination IP address or the destination network address of an IP packet.
- Mask: indicates the network mask. The network mask and the destination address are used together to identify the address of the network segment where the destination host or router resides.

- The address of the network segment where the destination host or router resides can be calculated using after the AND operation on the destination address and network mask. For example, if the destination address is 1.1.1.1 and the mask is 255.255.255.0, the address of the network segment where the host or the router resides is 1.1.1.0.
- The mask, which consists of several consecutive 1s, can be expressed either in dotted decimal notation or by the number of consecutive 1s in the mask. For example, the length of the mask 255.255.255.0 is 24, and therefore, the mask can also be expressed as 24.
- Protocol: indicates the name of a routing protocol.
- Pre: indicates the priority of a route that is added to the IP routing table. If multiple routes have the same destination but different next hops or outbound interfaces or these routes are static routes or discovered by different routing protocols, the one with the highest priority (the smallest value) is selected as the optimal route. For the route priority of each routing protocol, see [Table 2-1](#).
- Cost: indicates the route cost. When multiple routes to the same destination have the same priority, the route with the smallest cost is selected as the optimal route.

 NOTE

The Preference is used during the selection of routes discovered by different routing protocols, whereas the Cost is used during the selection of routes discovered by the same routing protocol.

- Flags:
Route flag:
 - R: indicates an iterated route.
 - D: indicates a route that is downloaded to the FIB.
 - T: indicates a route whose next hop belongs to a VPN instance.
 - B: indicates a black-hole rout
- Next hop: indicates the IP address of the next router through which an IP packet passes.
- Interface: indicates the outbound interface that forwards an IP packet.

Based on the destination addresses, routes can be classified into the following types:

- Network segment route: The destination is a network segment.
- Host route: The destination is a host.

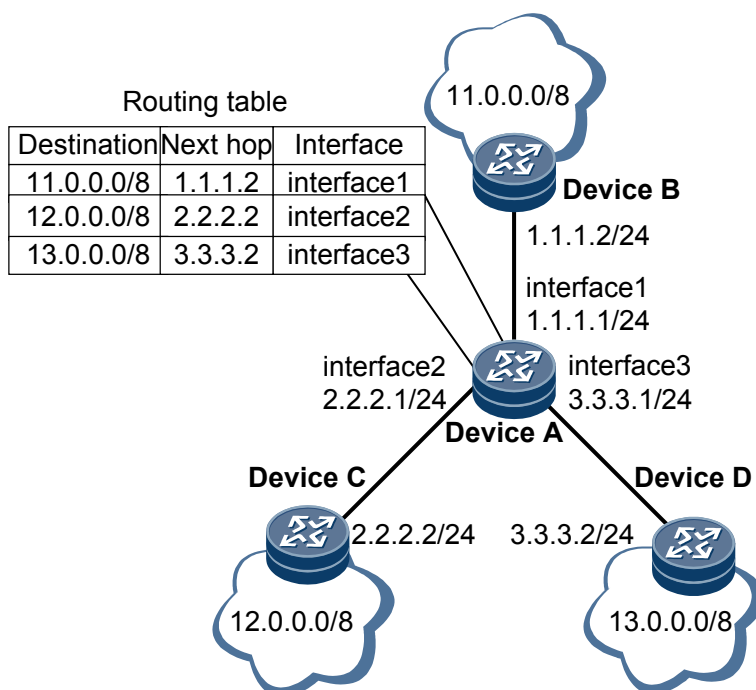
In addition, based on whether the destination is directly connected to the router, route types are as follows:

- Direct route: The router is directly connected to the destination network.
- Indirect route: The router is indirectly connected to the destination network.

Setting a default route can reduce the number of routing entries in the routing table. When a router cannot find a route in the routing table, the router uses the default route (destined for 0.0.0.0/0) to send packets.

In [Figure 2-2](#), Device A is connected to three networks, and therefore, it has three IP addresses and three outbound interfaces. [Figure 2-2](#) shows the routing table on Device A.

Figure 2-2 Routing table



2.2.4 Route Recursion

Routes can be used to forward traffic only when they have directly connected next hops. However, this condition may not be met when routes are generated. Therefore, the system needs to search for the directly connected next hops and corresponding outbound interfaces, and this process is called route recursion. In most cases, BGP routes, static routes, and UNRs do not have directly connected next hops, and route recursion is required.

For example, the next hop IP address of a BGP route is the IP address of a non-directly connected peer's loopback interface, and therefore, the BGP route needs to perform recursion. Specifically, the system searches the IP routing table for a direct route (IGP route in most cases) that is destined for the next hop IP address of the BGP route and then adds the next hop IP address and outbound interface of the IGP route to the IP routing table to generate a FIB entry.

The next hop IP address of a BGP VPN route is the IP address of a non-directly connected PE's loopback interface, and the BGP route needs to recurse to a tunnel. Specifically, the system searches the tunnel list for a tunnel that is destined for this loopback IP address and then adds the tunnel information to the routing table to generate a FIB entry.

2.2.5 Static and Dynamic Routes

Static routes can be easily configured and have low requirements on the system. They apply to simple, stable, and small-scale networks. However, they cannot automatically adapt to network topology changes. Therefore, static routes require subsequent maintenance.

Dynamic routing protocols have their routing algorithms and can automatically adapt to network topology changes. They apply to the network equipped with a number of Layer 3 devices. Dynamic route configurations are complex. Dynamic routes have higher requirements on a system than static ones do and consume network resources.

2.2.6 Classification of Dynamic Routing Protocols

Dynamic routing protocols can be classified based on the following criteria.

Based on the Application Scope

Based on the application scope, routing protocol types can be defined as follows:

- Interior Gateway Protocol (IGP): runs within an Autonomous System (AS), such as RIP, OSPF, and IS-IS.
- Exterior Gateway Protocol (EGP): runs between ASs. At present, BGP is the most widely used EGP.

Based on the Routing Algorithm

Based on the routing algorithm, routing protocol types can be defined as follows:

- Distance-vector routing protocol: includes RIP and BGP. BGP is also called a path-vector protocol.
- Link-state routing protocol: includes OSPF and IS-IS.

These routing algorithms differ in their methods of discovering and calculating routes.

Based on the Destination Address Type

Based on the destination address type, routing protocol types can be defined as follows:

- Unicast routing protocol: includes RIP, OSPF, BGP, and IS-IS.
- Multicast routing protocol: includes Protocol Independent Multicast-Sparse Mode (PIM-SM).

This chapter describes unicast routing protocols only. For details on multicast routing protocols, see the *HUAWEI NetEngine40E Universal Service Router Feature Description - IP Multicast*.

Routers manage both static and dynamic routes. These routes can be exchanged between different routing protocols to implement [readvertisement of routing information](#).

2.2.7 Routing Protocol and Route Priority

Route Priority

Dynamic routes of different routing protocols and static routes may have the same destination, but not all these routes are optimal. At a certain moment, only one routing protocol determines the current route to a certain destination. To select the optimal route, each routing protocol and static route are configured with priorities, and the route with the highest priority becomes the optimal route. [Table 2-1](#) lists routing protocols and their default priorities.

In [Table 2-1](#), 0 indicates a direct route, and 255 indicates any route learned from unreliable sources. The smaller the value, the higher the priority.

Table 2-1 Routing protocols and default route priorities

Routing Protocol or Route Type	Route Priority
Direct	0
OSPF	10
IS-IS	15
Static	60
RIP	100
OSPF ASE	150
OSPF NSSA	150
IBGP	255
EBGP	255

The priorities of routing protocols can be configured, except for direct routes. In addition, the priority of each static route can be different.

The NE40E defines external and internal priorities. The external priority refers to the priority set by users for each routing protocol. [Table 2-1](#) lists the default external priorities.

When different routing protocols are configured with the same priority, the system selects the optimal route based on the internal priority. For the internal priority of each routing protocol, see [Table 2-2](#).

Table 2-2 Internal priority of each routing protocol

Routing Protocol or Route Type	Route Priority
DIRECT	0
OSPF INTER AREA	10
OSPFv3 INTER AREA	10
IS-IS Level-1	15
IS-IS Level-2	18
EBGP	20
STATIC	60
UNR	65
RIP	100
RIPNG	100

Routing Protocol or Route Type	Route Priority
OSPF ASE	150
OSPFv3 ASE	150
OSPF NSSA	150
OSPFv3 NSSA	150
IBGP	200

For example, two routes, an OSPF route and a static route, can reach 10.1.1.0/24, and the priorities of the two routes are set to 5. In this case, the NE40E selects the optimal route based on the internal priorities listed in [Table 2-2](#). The internal priority of OSPF (10) is higher than that of the static route (60). Therefore, the system selects the route discovered by OSPF as the optimal route.

 NOTE

- If multiple OSPFv2 processes learn routes to the same destination and the external and internal priorities of the routes are the same, the system selects the route with the smallest link cost; if the link costs of the routes are the same, the routes participate in load balancing. If multiple OSPFv3 processes learn routes to the same destination and the external and internal priorities of the routes are the same, the system selects the route with the smallest process ID.
- If multiple IS-IS processes learn routes to the same destination and the external and internal priorities of the routes are the same, the system selects the route with the smallest link cost; if the link costs of the routes are the same, the routes participate in load balancing.
- If multiple RIP or RIPng processes learn routes to the same destination and the external and internal priorities of the routes are the same, the system selects the route with the smallest link cost; if the link costs of the routes are the same, the routes participate in load balancing.

2.2.8 Priority-based Route Convergence

Definition

Priority-based route convergence is an important technology that improves network reliability. It provides faster route convergence for key services. For example, to minimize the interruption of key services in case of network faults, real-time multicast services require that the routes to the multicast source quickly converge, and the Multiprotocol Label Switching (MPLS) VPN bearer network requires that routes between PEs also quickly converge.

Convergence priorities provide references for the system to converge routes for service forwarding. Different routes can be set with different convergence priorities, which can be identified as critical, high, medium, and low listed in descending order.

Purpose

With the integration of network services, requirements on service differentiation increase. Carriers require that the routes for key services, such as Voice over IP (VoIP) and video conferencing services converge faster than those for common services. Therefore, routes need to converge based on their convergence priorities to improve network reliability.

Route Convergence Priority

Table 2-3 lists the default convergence priorities of public network routes. You can set convergence priorities for routes based on the requirements of a live network.

Table 2-3 Default convergence priorities of public network routes

Routing Protocol or Route Type	Convergence Priority
Direct	Critical
Static	Medium
32-bit host routes of OSPF and IS-IS	Medium
OSPF route (except 32-bit host routes)	Low
IS-IS route (except 32-bit host routes)	Low
RIP	Low
BGP	Low

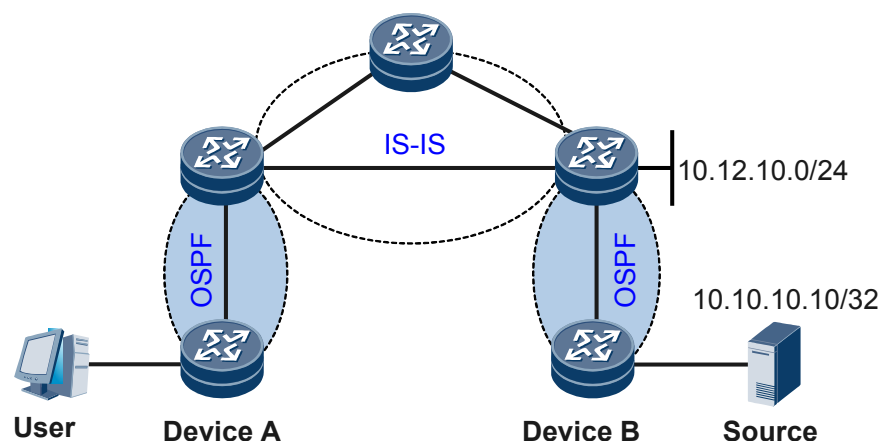
 NOTE

For VPN route priorities, only 32-bit host routes of OSPF and IS-IS are identified as medium, and the other routes are identified as low.

Applications

Figure 2-3 shows networking for multicast services. An IGP runs on the network; Device A is the receiver, and Device B is the multicast source server with IP address 10.10.10.10/32. The route to the multicast source server is required to converge faster than other routes, such as 10.12.10.0/24. In this case, you can set a higher convergence priority for 10.10.10.10/32 than that of 10.12.10.0/24. Then, when routes converge on the network, the route to the multicast source server 10.10.10.10/32 converges first, ensuring the transmission of multicast services.

Figure 2-3 Networking for priority-based route convergence



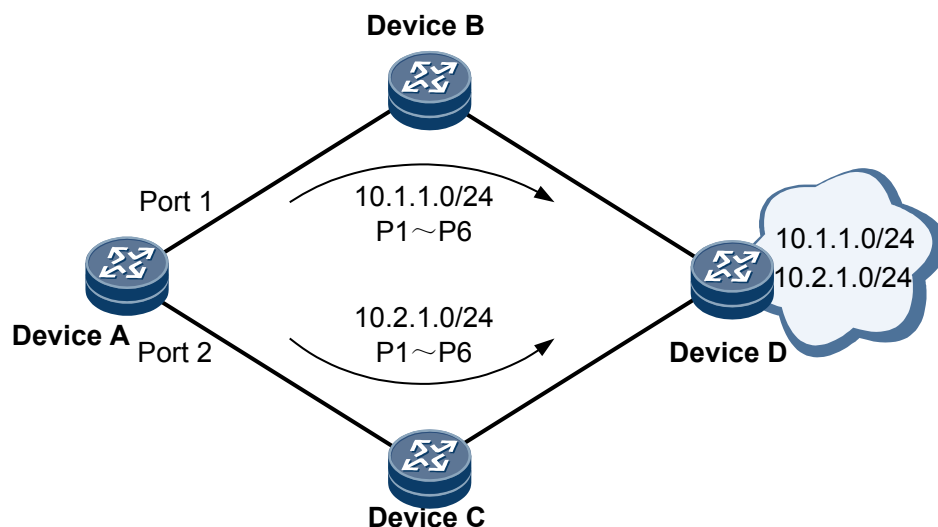
2.2.9 Load Balancing and Route Backup

Load Balancing

The NE40E supports the multi-route model (multiple routes with the same destination and priority). Routes discovered by one routing protocol with the same destination and cost can load-balance traffic. In each routing protocol view, you can run the **maximum load-balancing number** command to perform load balancing. Load balancing can work per-destination or per-packet.

- Per-packet load balancing
With per-packet load balancing, the router forwards packets destined for the same destination through equal-cost routes, and each time the next hop address is different from the last one.
- Per-destination load balancing
After per-destination load balancing is configured, the router forwards packets based on the quintuple (the source address, destination address, source port, destination port, and protocol in the packets). When the quintuple is the same, the router always chooses the next hop address that is the same as the last one to send packets. [Figure 2-4](#) per-destination load balancing.

Figure 2-4 Networking for per-destination load balancing



Device A needs to forward packets to 10.1.1.0/24 and 10.2.1.0/24. Based on per-destination load balancing, packets of the same flow are transmitted along the same path. The processes for forwarding packets on Device A are as follows:

- The first packet P1 to 10.1.1.0/24 is forwarded through Port 1, and all subsequent packets to 10.1.1.0/24 are forwarded through Port 1.
- The first packet P1 to 10.2.1.0/24 is forwarded through Port 2, and all subsequent packets to 10.2.1.0/24 are forwarded through Port 2.

Currently, RIP, OSPF, BGP, and IS-IS support load balancing, and static routes also support load balancing.

NOTE

The number of equal-cost routes for load balancing varies with products.

Route Backup

The NE40E supports route backup to improve network reliability. You can configure multiple routes to the same destination as required. The route with the highest priority functions as the primary route, and the other routes with lower priorities function as backup routes.

In most cases, the NE40E uses the primary route to forward packets. If the link fails, the primary route becomes inactive. The NE40E then selects a backup route with the highest priority to forward packets, and the primary route is switched to the backup route. When the original primary route recovers, the NE40E restores and reselects the optimal route. Because the original primary route has the highest priority, the NE40E selects this route to send packets. Therefore, the backup route is switched to the primary route.

2.2.10 Principles of IP FRR

Overview

Fast Reroute (FRR) functions when the lower layer (physical layer or data link layer) detects a fault. The lower layer reports the fault to the upper layer routing system and immediately forwards packets through a backup link.

If a link fails, FRR helps reduce the impact of the link failure on services transmitted on the link.

Background

On traditional IP networks, when a fault occurs at the lower layer of the forwarding link, the physical interface on the router goes Down. After the router detects the fault, it instructs the upper layer routing system to recalculate routes and then update routing information. The routing system takes several seconds to reselect an available route.

For services that are sensitive to packet loss and delay, a convergence time of several seconds is intolerable because it may lead to service interruptions. For example, the maximum convergence time tolerable for Voice over IP (VoIP) services is within milliseconds. IP FRR enables the forwarding system to detect a fault and then to take measures to restore services as soon as possible.

Classification and Implementation

IP FRR, which is designed for routes on IP networks, consists of public network IP FRR and VPN IP FRR.

- Public network IP FRR: protects routers on the public network.
- VPN IP FRR: protects Customer Edges (CEs).



NOTE

The static routes that are imported between public and private networks do not support IP FRR.

IP FRR is implemented as follows:

- IP FRR can be enabled or disabled using commands.
- When optimal routes are selected from the routes discovered by routing protocols, a backup link is selected for each preferred primary link based on the protocol priority, and then the forwarding information of primary and backup links is provided for the forwarding engine.

Implementation of IP FRR Between Different Protocols

When IP FRR between different protocols is enabled, and optimal routes are selected from protocol routes, a backup link is selected for each preferred primary link based on the protocol priority, and then the forwarding information of primary and backup links is provided for the forwarding engine.

If the forwarding engine detects that the primary link is unavailable after IP FRR between different protocols is enabled, the system can use the backup link to forward traffic before the routes converge on the control plane.

Comparison Between IP FRR and Load Balancing

Table 2-4 Comparison between IP FRR and load balancing

Feature	Description
IP FRR	Implements FRR through a backed up route. IP FRR is applicable to networks where a master link and a backup link exist and load balancing is not configured.
Load balancing	Implements fast route switching through equal-cost routes and applies to the multi-link networking with load balancing.

2.2.11 Readvertisement of Routing Information

Different routing protocols may discover different routes because they adopt different routing algorithms. When the scale of a network is large and multiple routing protocols run on the network, these protocols need to readvertise their discovered routes.

On the NE40E, the routes discovered by a routing protocol can be imported into the routing table of another routing protocol. Each protocol has its mechanism to import routes. For details, see "Routing Policy."

2.2.12 Indirect Next Hop

Definition

Indirect next hop is a technique used to speed up route convergence. This technique can change the direct association between route prefixes and next hop information into an indirect association. Indirect next hop allows next hop information to be refreshed independently of the prefixes of the same next hop, which speeds up route convergence.

Purpose

In the scenario requiring route recursion, when IGP routes or tunnels are switched, forwarding entries are rapidly refreshed, which implements fast route convergence and reduces the impact of route or tunnel switching on services.

Mapping Between the Route Prefix and the Next Hop

Mapping between route prefixes and next hops is the basis of indirect next hop. To meet the requirements of route recursion and tunnel recursion in different scenarios, next hop information includes the address family, original next hop address, and tunnel policy. The system assigns an index to each next hop, performs route recursion, communicates the recursion result to the routing protocol, and then delivers forwarding entries.

On-Demand Route Recursion

On the NE40E, the route to a reachable address is called a dependent route. The system forwards packets based on dependent routes. The process of finding a dependent route based on the next hop address is called route recursion.

On-demand route recursion indicates that when a dependent route changes, only the next hop associated with the dependent route performs recursion again. If the route destination address is the original next hop address or network segment address of next hop information, any route changes affect the recursion result of the next hop information. Otherwise, route changes do not affect next hop information. Therefore, when a route changes, you can perform recursion again only on the associated next hop by assessing the destination address of the route. For example, if the original next hop address of the route 2.2.2.2/32 is 1.1.1.1, the route that the original next hop 1.1.1.1 depends on may be 1.1.1.1/32 or 1.1.0.0/16. If the route 1.1.1.1/32 or 1.1.0.0/16 changes, the recursion result of the original next hop 1.1.1.1 is affected.

With respect to tunnel recursion, when a tunnel alternates between Up and Down, perform recursion again on the next hop whose original next hop address is the same as the destination address of the tunnel.

Recursion Policy

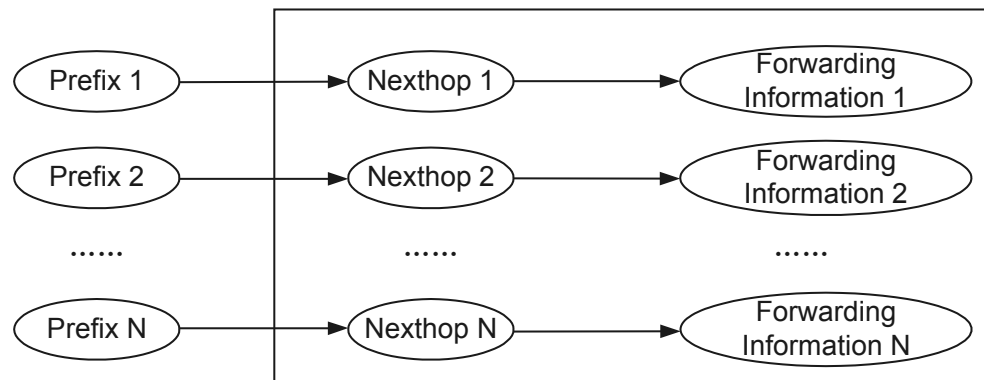
A recursion policy is used to control the recursion result of the next hop to meet requirements of different scenarios. In route recursion, behaviors do not need to be controlled by the recursion policy. Instead, recursion behaviors only need to comply with the longest match rule. In addition, the recursion policy needs to be applied only when VPN routes recurse to tunnels.

By default, the system selects Label Switched Paths (LSPs) for VPNs without performing load balancing. If load balancing or other types of tunnels are required, configure a tunnel policy and bind it to a tunnel. After the tunnel policy is applied, the system uses the tunnel bound to the tunnel policy or selects a tunnel based on the priorities specified in the tunnel policy during next hop recursion.

Mechanism for Indirect Next Hop

Without indirect next hop, the forwarding information corresponds to the prefix, and therefore, the route convergence time is decided by the number of route prefixes. With indirect next hop, multiple route prefixes correspond to one next hop. Forwarding information is added to the forwarding table using the next hop, and traffic with relevant route prefixes can be switched, which speeds up route convergence.

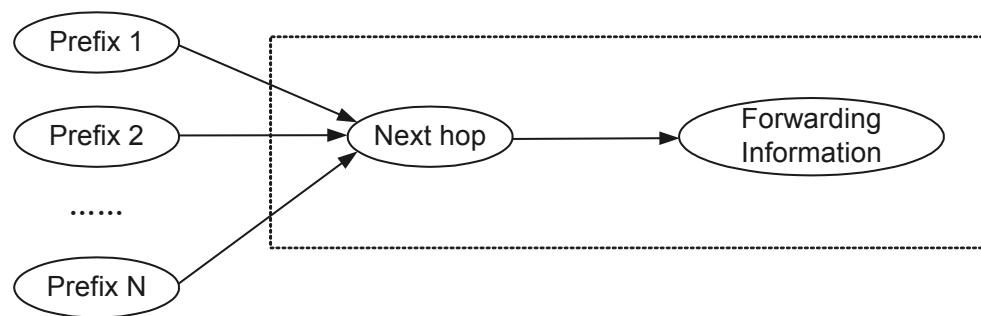
Figure 2-5 Implementation without indirect next hop



As shown in [Figure 2-5](#), without indirect next hop, prefixes are totally independent, each corresponding to its next hop and forwarding information. When a dependent route changes, the next hop corresponding to each prefix performs recursion and forwarding information is updated based on the prefix. In this case, the convergence time is decided by the number of prefixes.

Note that prefixes of a BGP peer have the same next hop, forwarding information, and refreshed forwarding information.

Figure 2-6 Implementation with indirect next hop



As shown in [Figure 2-6](#), with indirect next hop, prefixes of routes from the same BGP peer share the same next hop. When a dependent route changes, only the shared next hop performs recursion and forwarding information is updated based on the next hop. In this case, routes of all prefixes can converge at a time. Therefore, the convergence time is irrelevant to the number of prefixes.

Comparison Between Route Recursion and Tunnel Recursion

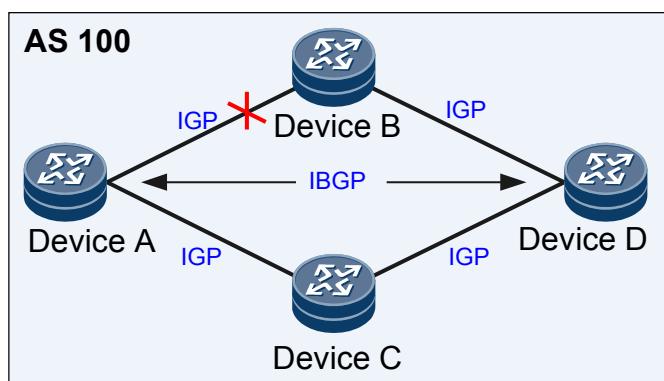
The following table lists differences between route recursion and tunnel recursion.

Table 2-5 Differences between route recursion and tunnel recursion

Recursion Type	Description
Route recursion	● Applies to BGP public network routes. ● Is triggered by route changes. ● Supports next hop recursion based on the specified routing policy.
Tunnel recursion	● Applies to BGP VPN routes. ● Is triggered by tunnel or tunnel policy changes. ● Recursion behaviors can be controlled using a tunnel policy to meet requirements of different scenarios.

IBGP Route Recursion to an IGP Route

Figure 2-7 Networking for IBGP route recursion



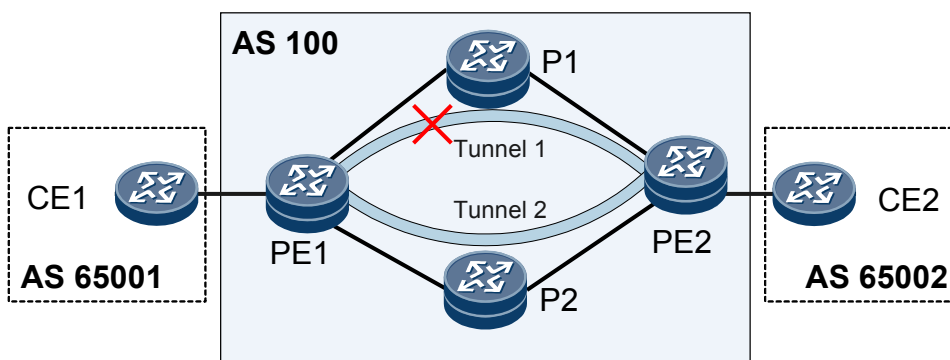
In [Figure 2-7](#), an IBGP peer relationship is established between Device A and Device D. The IBGP peer relationship is established between two loopback interfaces on the routers, but the next hop cannot be used to guide packet forwarding, because it is not directly reachable. Therefore, to refresh the forwarding table and guide packet forwarding, the system needs to search for the actual outbound interface and directly connected next hop based on the original IBGP next hop.

Device D receives 100,000 routes from Device A. These routes have the same original BGP next hop. After recursion, these routes eventually follow the same IGP path (A->B->D). If the IGP path (A->B->D) fails, these IBGP routes do not need to perform recursion separately, and the relevant forwarding entries do not need to be refreshed one by one. Note that only the shared next hop needs to perform recursion and be refreshed. Consequently, these IBGP routes converge to the path (A->C->D) on the forwarding plane. Therefore, the convergence time depends on only the number of next hops, not the number of prefixes.

If Device A and Device D establish a multi-hop EBGP peer relationship, the convergence procedure is the same as the preceding one. Indirect next hop also applies to the recursion of a multi-hop EBGP route.

VPN Routes Recursion to a Tunnel

Figure 2-8 Networking for VPN route recursion



In [Figure 2-8](#), a neighbor relationship is established between PE1 and PE2, and PE2 receives 100,000 VPN routes from PE1. These routes have the same original BGP next hop. After recursion, these VPN routes eventually follow the same public network tunnel (tunnel 1). If tunnel 1 fails, these routes do not need to perform recursion separately, and the relevant forwarding entries do not need to be refreshed one by one. Note that only the shared next hop needs to perform recursion, and the relevant forwarding entries need to be refreshed. Consequently, these VPN routes converge to tunnel 2 on the forwarding plane. In this manner, the convergence time depends on only the number of next hops, not the number of prefixes.

2.2.13 Default Route

Default routes are special routes. In most cases, they are configured by administrators. Default routes can also be generated by dynamic routing protocols, such as OSPF and IS-IS.

Default routes are used only when no matching routing entry is available for packet forwarding in the routing table. A default route in the routing table is the route to the network 0.0.0.0 (with mask 0.0.0.0). You can check whether the default route is configured using the **display ip routing-table** command.

If the destination address of a packet does not match any entry in the routing table, the packet is sent along a default route. If no default route exists and the destination address of the packet does not match any entry in the routing table, the packet is discarded. An Internet Control Message Protocol (ICMP) packet is then sent, informing the originating host that the destination host or network is unreachable.

2.2.14 Multi-Topology

Multi-Topology Overview

On a traditional IP network, only one unicast topology exists, and only one unicast forwarding table is available on the forwarding plane, which forces services transmitted from one router to the same destination address to share the same next hop, and various end-to-end services, such as voice and data services, to share the same physical links. As a result, some links may become heavily congested whereas others remain relatively idle. To address this problem, configure multi-topology to divide a physical network into different logical topologies for different services.

By default, the base topology is created on the public network. The class-specific topology can be added or deleted in the public network address family view. Each topology contains its own routing table. The class-specific topology supports the addition, deletion, and import of protocol routes.

The base topology cannot be deleted.

Direct Routes Supporting Multi-Topology

Direct routes can be added to or deleted from the routing table of any topology. The same routes can also be added to multiple topologies, independent of each other.

Direct routes associated with interfaces are added to the base topology by default. Direct routes in the base topology are not deleted, and the base topology contains all direct routes.

Static Routes Supporting Multi-Topology

Static routes can be added to or deleted from the routing table of any topology. The routes with the same prefix, outbound interface, and next hop can also be added to multiple topologies, independent of each other.

Static routes, by default, are configured in the base topology. However, they can be configured in a specified class-specific topology and can be changed or deleted.

Static routes have no outbound interfaces, and therefore, need to perform recursion based on the next hop. In this case, you cannot specify the topology in which the next hop resides.

Public network static route recursion to a VPN next hop or VPN static route recursion to a public network next hop can be configured only in the base topology. When configuring static routes, you cannot specify the name of the topology in which the destination resides.

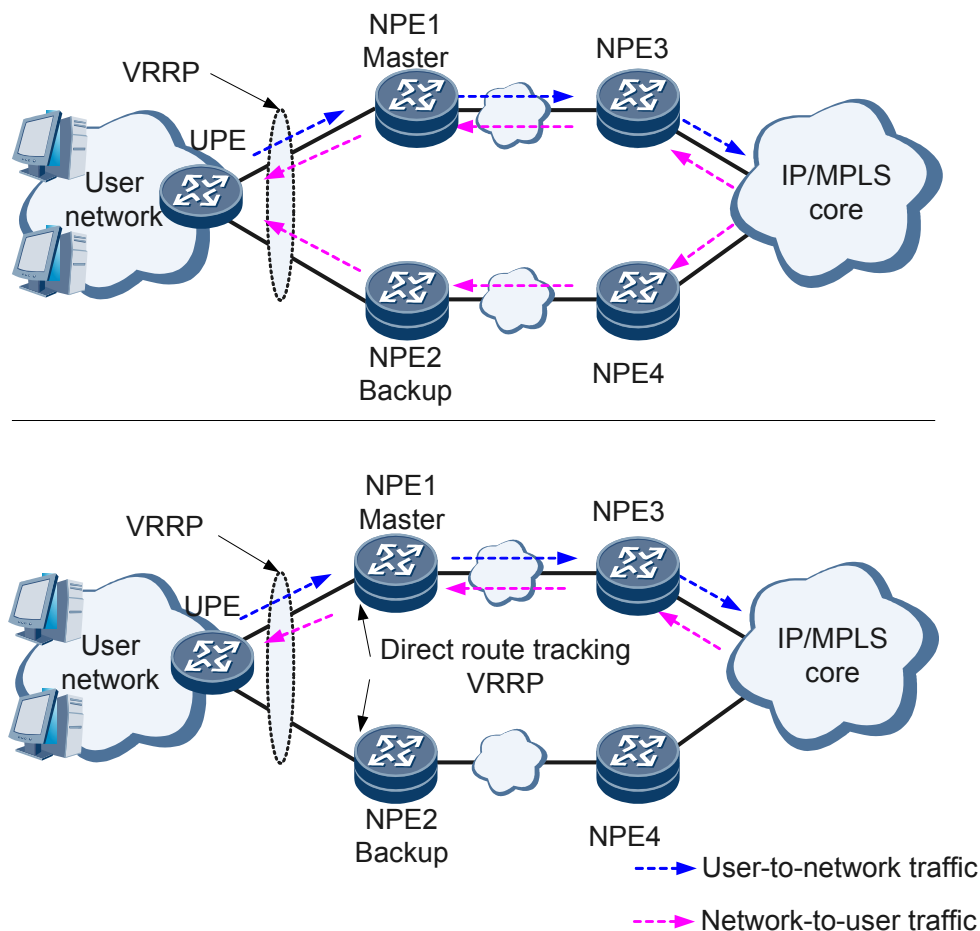
2.2.15 Association Between Direct Routes and a VRRP Backup Group

Background

A VRRP backup group is configured on Device1 and Device2 on the network shown in [Figure 2-9](#). Device1 is a master device, whereas Device2 is a backup device. The VRRP backup group serves as a gateway for users. User-to-network traffic travels through Device1. However, network-to-user traffic may travel through Device1, Device2, or both of them over a path determined by a dynamic routing protocol. Therefore, user-to-network traffic and network-to-user traffic may travel along different paths, which interrupts services if firewalls are attached to devices in the VRRP backup group, complicates traffic monitoring or statistics collection, and increases costs.

To address the preceding problems, the routing protocol is expected to select a route passing through the master device so that the user-to-network and network-to-user traffic travels along the same path. Association between direct routes and a VRRP backup group can meet expectations by allowing the dynamic routing protocol to select a route based on the VRRP status.

Figure 2-9 Association between direct routes and a VRRP backup group



Related Concepts

VRRP is a widely used fault-tolerant protocol that groups multiple routing devices into a backup group, improving network reliability. A VRRP backup group consists of a master device and one or more backup devices. If the master device fails, the VRRP backup group switches services to a backup device to ensure communication continuity and reliability.

A device in a VRRP backup group operates in one of three states:

- **Master:** If a network is working correctly, the master device transmits all services.
- **Backup:** If the master device fails, the VRRP backup group selects a backup device as the new master device to take over traffic and ensure uninterrupted service transmissions.
- **Initialize:** A device in the Initialize state is waiting for an interface Startup message to switch its status to Master or Backup.

NOTE

For details about VRRP, see *HUAWEI NetEngine40E Universal Service Router Feature Description - Network Reliability - VRRP*.

Implementation

Association between direct routes and a VRRP backup group allows VRRP interfaces to adjust the costs of direct network segment routes based on the VRRP status. The direct route

with the master device as the next hop has the lowest cost. A dynamic routing protocol imports the direct routes and selects the direct route with the lowest cost. For example, VRRP interfaces on Device1 and Device2 on the network shown in [Figure 2-9](#) are configured with association between direct routes and the VRRP backup group. The implementation is as follows:

- Device1 in the Master state sets the cost of its route to the directly connected virtual IP network segment to 0 (default value).
- Device2 in the Backup state increases the cost of its route to the directly connected virtual IP network segment.

A dynamic routing protocol selects the route with Device1 as the next hop because this route costs less than the other route. Therefore, both user-to-network and network-to-user traffic travels through Device1.

Usage Scenario

When a data center is used, firewalls are attached to devices in a VRRP backup group to improve network security. Network-to-user traffic cannot pass through a firewall if it travels over a path different than the one used by user-to-network traffic.

When an IP radio access network (RAN) is configured, VRRP is configured to set the master/backup status of aggregation site gateways (ASGs) and radio service gateways (RSGs). Network-to-user and user-to-network traffic may pass through different paths, complicating network operation and management.

Association between direct routes and a VRRP backup group can address the preceding problems by ensuring the user-to-network and network-to-user traffic travels along the same path.

2.2.16 Direct Routes Responding to L3VE Interface Status Changes After a Delay

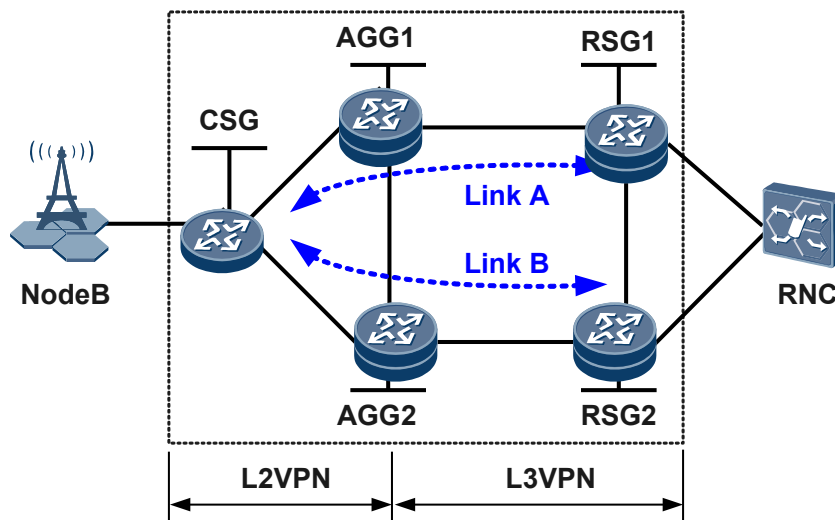
Background

In [Figure 2-10](#), a Layer 2 virtual private network (VPN) connection is set up between each AGG and the CSG through L2 virtual Ethernet (VE) interfaces, and BGP VPNv4 peer relationships are set up between the AGGs and RSGs on an L3VPN. L3VE interfaces are configured on the AGGs, and VPN instances are bound to the L3VE interfaces so that the CSG can access the L3VPN. BGP is configured on the AGGs to import direct routes between the CSG and AGGs. The AGGs convert these direct routes to BGP VPNv4 routes before advertising them to the RSGs.

AGG1 functions as the master device in [Figure 2-10](#). In most cases, the RSGs select routes advertised by AGG1, and traffic travels along Link A. If AGG1 or the CSG-AGG1 link fails, traffic switches over to Link B. After AGG1 or the CSG-AGG1 link recovers, the L3VE interface on AGG1 goes from Down to Up, and AGG1 immediately generates a direct route destined for the CSG and advertises the route to the RSGs. Downstream traffic then switches over to Link A. However, AGG1 has not learned the MAC address of the NodeB yet. As a result, downstream traffic is lost.

To address this problem, configure the direct route to respond to L3VE interface status changes after a delay. After you configure the delay, the RSG preferentially selects routes advertised by AGG1 only after AGG1 learns the MAC address of the NodeB.

Figure 2-10 Networking for the direct route responding to L3VE interface status changes after a delay



Implementation

After you configure the direct route to respond to L3VE interface status changes after a delay, the cost of the direct route between the CSG and AGG1 is modified to the configured cost (greater than 0) when the L3VE interface on AGG1 goes from Down to Up. After the configured delay expires, the cost of the direct route to the CSG restores to the default value 0. Because BGP has imported the direct route and has advertised it to RSGs, the cost value determines whether RSGs preferentially select the direct route.

RSGs preferentially transmit traffic over Link B before AGG1 has learned the MAC address of the NodeB, which reduces traffic loss.

Usage Scenario

This feature applies to IP radio access networks (RANs) on which an L2VPN accesses an L3VPN.

2.2.17 Association Between the Direct Route and PW Status

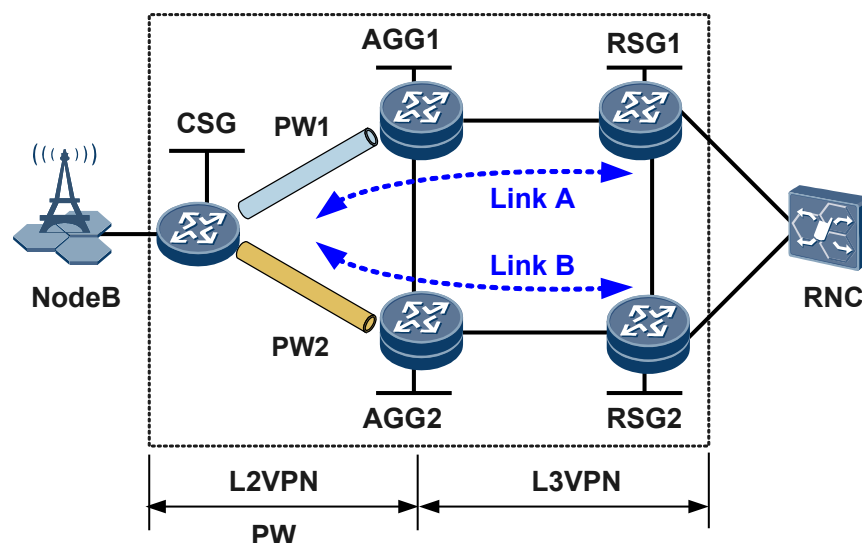
Background

In [Figure 2-11](#), PWs are set up between the AGGs and the CSG. BGP virtual private network version 4 (VPNv4) peer relationships are set up between the AGGs and RSGs. Layer 3 virtual Ethernet (L3VE) interfaces are configured on the AGGs, and VPN instances are bound to the L3VE interfaces so that the CSG can access the L3VPN. BGP is configured on the AGGs to import direct routes between the CSG and AGGs. The AGGs convert these direct routes to BGP VPNv4 routes before advertising them to the RSGs.

AGG1 functions as the master device in [Figure 2-11](#). In most cases, the RSGs select routes advertised by AGG1, and traffic travels along Link A. If AGG1 or the CSG-AGG1 link fails, traffic switches over to Link B. After AGG1 or the CSG-AGG1 link recovers, the L3VE interface on AGG1 goes from Down to Up, and AGG1 immediately generates a direct route destined for the CSG and advertises the route to the RSGs. Downstream traffic then switches over to Link A. However, PW1 is on standby. As a result, downstream traffic is lost.

To address this problem, associate the direct route and PW status. After the association is configured, the RSG preferentially selects the direct route only after PW1 becomes active.

Figure 2-11 Networking for the association between the direct route and PW status



Implementation

Configuring the association between the direct route and PW status allows a VE interface to adjust the cost value of the direct route based on PW status. The cost value determines whether the RSGs preferentially select the direct route because BGP has imported the direct route and has advertised it to RSGs. For example, if you associate the direct route and PW status on the network shown in [Figure 2-11](#), the implementation is as follow:

- When PW1 becomes active, the cost value of the direct route between the CSG and AGG1 restores to the default value 0. RSGs preferentially transmit traffic over Link A.
- When PW1 is on standby, the cost value of the direct route between the CSG and AGG1 is modified to a configured value (greater than 0). RSGs preferentially transmit traffic over Link B, which reduces traffic loss.

Usage Scenario

This feature applies to IP radio access networks (RANs) on which primary/secondary PWs are configured between the CSG and AGGs.

2.2.18 Vlink Direct Route Advertisement

Background

By default, IPv4 Address Resolution Protocol (ARP) Vlink direct routes or IPv6 Neighbor Discovery Protocol (NDP) Vlink direct routes are only used for packet forwarding in the same VLAN and cannot be imported to dynamic routing protocols. This is because importing Vlink direct routes to dynamic routing protocols will increase the number of routing entries and affect routing table stability. In some cases, some operations need to be performed based on Vlink direct routes of VLAN users. For example, different VLAN users use different route exporting policies to guide traffic from the remote device. In this scenario, ARP or NDP Vlink direct routes are needed to be imported by a dynamic routing protocol and advertised to

the remote device. After advertisement of ARP or NDP Vlink direct routes is enabled, these direct routes can be imported by a dynamic routing protocol (IGP or BGP) and advertised to the remote device.

Related Concepts

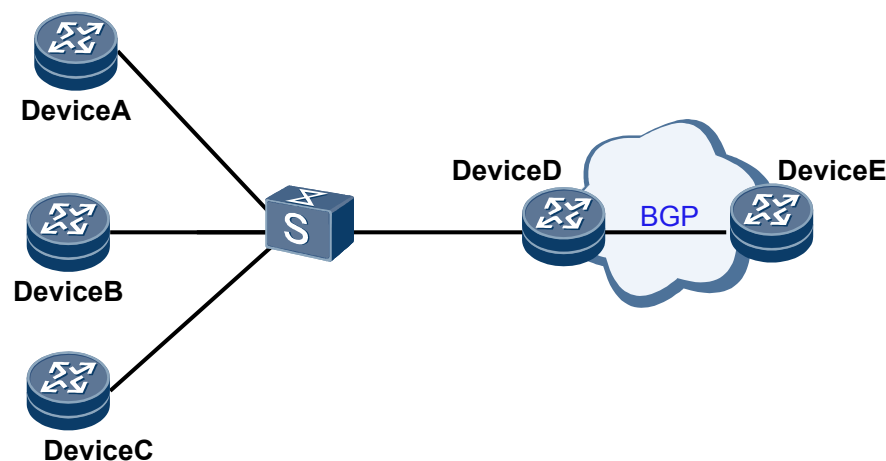
ARP Vlink direct routes: routing entries with physical interfaces of VLAN users and used to forward IP packets. These physical interfaces are learned using ARP. On networks with VLANs, IP packets can be forwarded only by physical interfaces rather than logical interfaces. After learning the ARP entry of a peer end, VLANIF interfaces, QinQ interfaces, or QinQ VLAN tag termination sub-interfaces generate a 32-bit ARP Vlink direct route, and the route is displayed in the routing table. Regular physical interfaces do not generate a 32-bit ARP Vlink direct route in this case.

NDP Vlink direct routes: routing entries carrying IPv6 addresses of VLAN users' physical interfaces. These IPv6 addresses are learned and resolved using NDP.

Implementation

On the network shown in [Figure 2-12](#), Device A, Device B, and Device C are connected to the logical interface of Device D which is a Border Gateway Protocol (BGP) peer of Device E. However, Device E needs to communicate only with Device B rather than Device A and Device C. In this scenario, Vlink direct route advertisement must be enabled on Device D. Then Device D obtains each physical interface of Device A, Device B, and Device C, uses a routing policy to filter out network segment routes and routes destined for Device A and Device C, and advertises the route destined for Device B to Device E.

Figure 2-12 Networking for Vlink direct route advertisement



Usage Scenario

Vlink direct route advertisement is applicable to networks in which a device needs to add Vlink direct routes with physical interfaces of VLAN users to the routing table of a dynamic routing protocol before advertising the routes to remote ends.

Advantages

With Vlink direct route advertisement, a device can add Vlink direct routes to the routing table of a dynamic routing protocol (such as an Interior Gateway Protocol or BGP) and then use different export policies to advertise routes required by remote ends.

2.2.19 Association Between the IPv4 Direct Routes and IPsec Instance Status

Figure 2-13 shows an IP radio access network (IPRAN) scenario, and some services require high security. To meet such requirements, cell site gateways (CSGs) encrypt data of these services using IPsec. After the data flows to RSGs (IPsec gateways) through an IPsec tunnel, the RSGs decrypt the data. In most cases, carriers deploy master and backup RSGs and configure the same IP address for the IPsec tunnel interfaces of the master and backup RSGs to improve network reliability.

Without the association between IPv4 direct routes and IPsec instance status, IPv4 direct routes with the same prefix generated on the IPsec tunnel interfaces of the master and backup RSGs share the same default cost (0). As a result, after receiving these routes from the master and backup RSGs, CSGs cannot select an optimal one based on the cost.

Association between IPv4 direct routes and IPsec instance status can address this problem. After the association is configured:

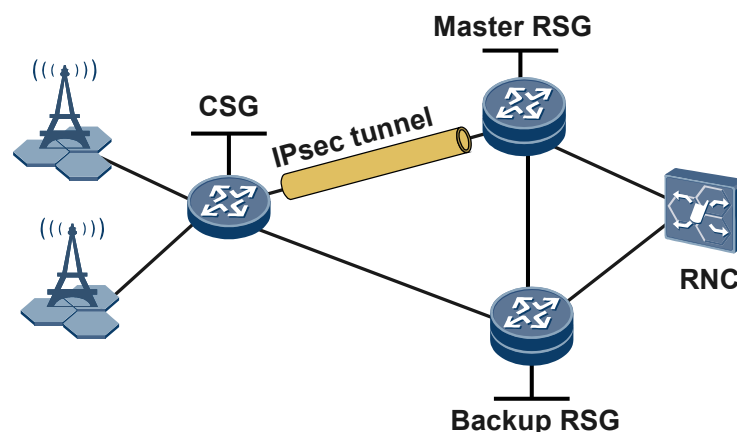
- If the IPsec instance status is master on an IPsec tunnel interface, the cost of the IPv4 direct routes generated on the interface is 0.
- If the IPsec instance status is backup on an IPsec tunnel interface or the system cannot detect the IPsec instance status, the cost of the IPv4 direct routes generated on the interface is the cost configured on the interface.

NOTE

In IPsec dual-device hot backup scenarios, if a tunnel interface borrows the IP address of another interface, the cost of direct routes on the tunnel interface cannot be associated with the IPsec instance status.

After receiving the IPv4 direct routes with the same prefix from the master and backup RSGs, CSGs can select an optimal one based on the cost. Therefore, the CSGs can transmit data encrypted using IPsec to the correct RSG.

Figure 2-13 Networking for the association between IPv4 direct routes and IPsec instance status

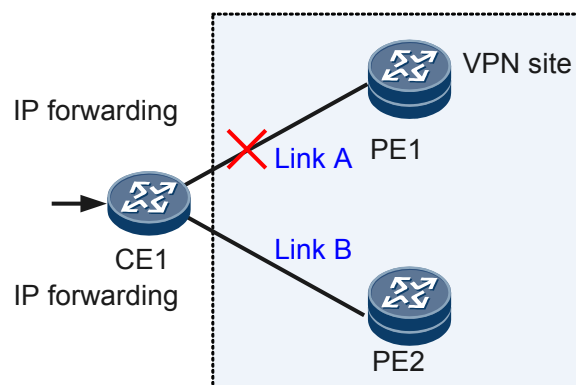


2.3 Application Scenarios for IP Routing

2.3.1 Typical Application of IP FRR

In [Figure 2-14](#), CE1 is dual-homed to PE1 and PE2. CE1 is configured with two outbound interfaces and two next hops. Link B functions as the backup of link A. If link A fails, traffic can be rapidly switched to link B.

Figure 2-14 Configuring IP FRR



2.3.2 Data Center Applications of Association Between Direct Routes and a VRRP Backup Group

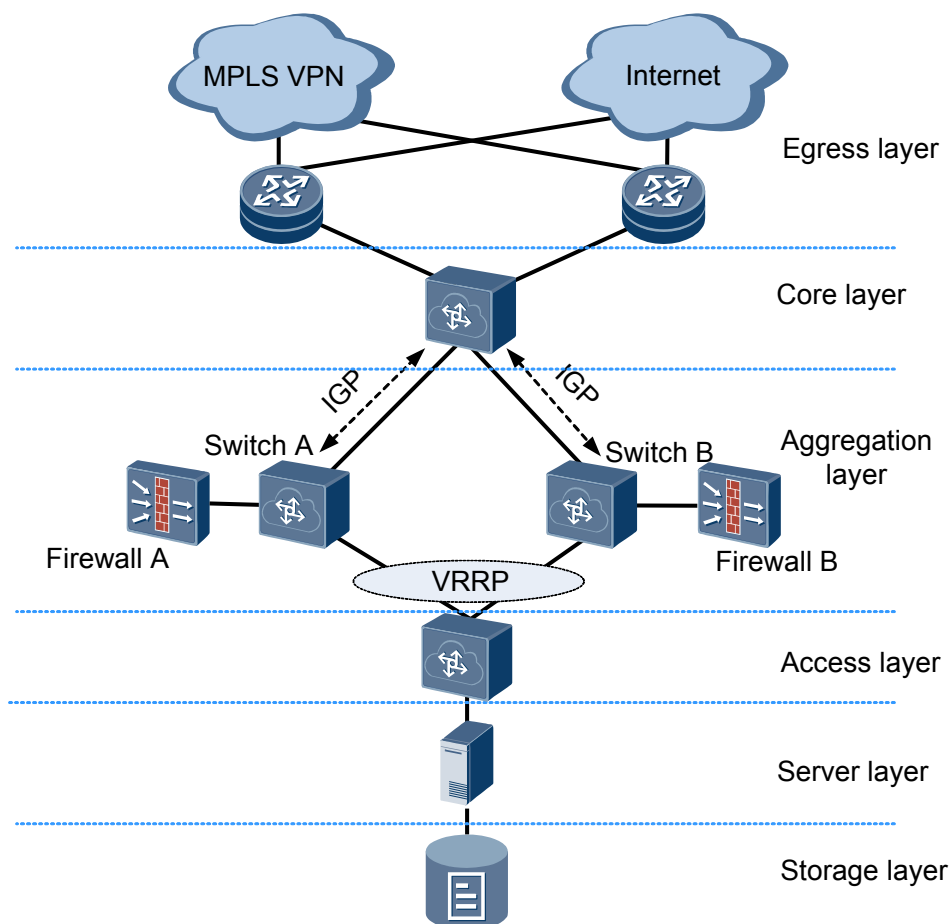
Service Overview

A data center, used for service access and transmission, consists of many servers, disk arrays, security devices, and network devices that store and process a great number of services and applications. Firewalls are used to improve data security, and VRRP backup groups are configured to improve communication reliability. VRRP may cause user-to-network traffic and network-to-user traffic to travel along different paths, and as a result, the firewall may discard the network-to-user traffic because of path inconsistency. To address this problem, association between direct routes and a VRRP backup group must be configured.

Networking Description

[Figure 2-15](#) shows a data center network. A server functions as a core service module in the data center. A VRRP backup group protects data exchanged between the server and core devices, improving service security. Firewalls are attached to devices in the VRRP backup group to improve network security.

Figure 2-15 Data center network



Feature Deployment

The master device transmits server traffic to a core device. When the core device attempts to send traffic to the server, the traffic can only pass through a firewall attached to the master device. On the network shown in [Figure 2-15](#), the server sends data destined for the core device through the master device, and the core device sends data destined for the server along a path that an Interior Gateway Protocol (IGP) selects. The association between the direct routes and a VRRP backup group can be configured on switch A and switch B so that the IGP selects a route based on VRRP status. The IGP forwards core-device-to-server traffic over the same path as the one over which server-to-core-device traffic is transmitted, which prevents the firewall from discarding traffic.

2.3.3 IPRAN Applications of Association Between Direct Routes and a VRRP Backup Group

Service Overview

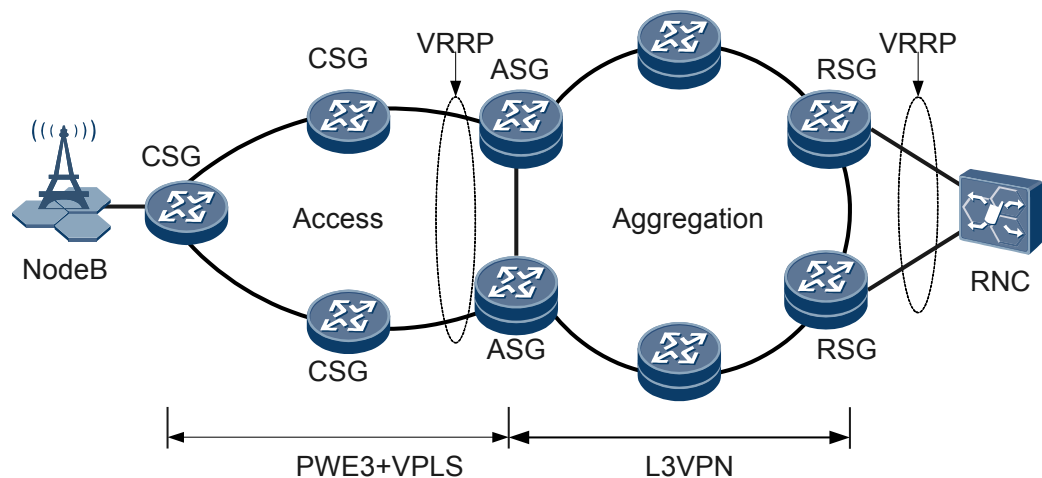
NodeBs and radio network controllers (RNCs) on an IP radio access network (RAN) do not have dynamic routing capabilities. Therefore, static routes must be configured to allow NodeBs to communicate with aggregation site gateways (ASGs) and allow RNCs to communicate with remote service gateways (RSGs) that are at the aggregation layer. VRRP is configured to provide ASG and RSG redundancy, improving device reliability and ensuring

non-stop transmission of value-added services, such as voice, video, and cloud computation services over mobile bearer networks.

Networking Description

Figure 2-16 shows VRRP-based gateway protection applications on an IPRAN. A NodeB is dual-homed to VRRP-enabled ASGs to communicate with the aggregation network. The NodeB sends traffic destined for the RNC through the master ASG, whereas the RNC sends traffic destined for the NodeB through either the master or backup ASG over a path selected by a dynamic routing protocol. As a result, traffic in opposite directions may travel along different paths. Similarly, the RNC is dual-homed to VRRP-enabled RSGs. Path inconsistency may also occur.

Figure 2-16 VRRP-based gateway protection on an IPRAN



Feature Deployment

On the IPRAN shown in **Figure 2-16**, both ASGs and RSGs may send and receive traffic over different paths. For example, user-to-network traffic enters the aggregation network through the master ASG, whereas network-to-user traffic flows out of the aggregation network from the backup ASG. Path inconsistency complicates traffic monitoring or statistics collection and increases the cost. In addition, when the master ASG is working properly, the backup ASG also transmits services, which is counterproductive to VRRP redundancy backup implementation. Association between direct routes and the VRRP backup group can be configured to ensure path consistency.

On the NodeB side, the direct network segment routes of ASG VRRP interfaces can be associated with VRRP status. The route with the master ASG as the next hop has a lower cost than the route with the backup ASG as the next hop. The dynamic routing protocol imports the direct routes and selects the route with a lower cost, ensuring path consistency. Implementation on the RNC side is similar to that on the NodeB side.

2.4 Appendix List of Port Numbers of Common Protocols

Table 2-6 Port numbers of routing protocols

Routing Protocol	UDP Port Number	TCP Port Number
RIP	520	-
RIPv2	520	-
RIPng	521	-
BGP	-	179
OSPF	-	-
IS-IS	-	-
Note that "-" indicates that the related transport layer protocol is not used.		

Table 2-7 Port numbers of application layer protocols

Application Layer Protocol	UDP Port Number	TCP Port Number
DHCP	67	-
DNS	53	53
FTP	-	20/21
HTTP	-	80
IMAP	-	993
NetBIOS	137/138	137/139
POP3	-	995
SMB	445	445
SMTP	25	25
SNMP	161	-
TELNET	-	23
TFTP	69	-
Note that "-" indicates that the related transport layer protocol is not used.		

2.5 Terminology for IP Routing

Terms

Term	Description
ARP Vlink direct routes	IP packets are forwarded through a specified physical interface. IP packets cannot be forwarded through a VLANIF interface, because a VLANIF interface is a logical interface with several physical interfaces as its member interfaces. If an IPv4 packet reaches a VLANIF interface, the device obtains information about the physical interface using ARP and generates the relevant routing entry. The route recorded in the routing entry is called an ARP Vlink direct route.
FRR	FRR is applicable to services that are very sensitive to packet loss and delay. When a fault is detected at the lower layer, the lower layer informs the upper layer routing system of the fault. Then, the routing system forwards packets through a backup link. In this manner, the impact of the link fault on services is minimized.
NDP Vlink direct routes	IP packets are forwarded through a specified physical interface. IP packets cannot be forwarded through a VLANIF interface, because a VLANIF interface is a logical interface with several physical interfaces as its member interfaces. If an IPv6 packet reaches a VLANIF interface, the device obtains information about the physical interface using the neighbor discovery protocol (NDP) and generates the relevant routing entry. The route recorded in the routing entry is called an NDP Vlink direct route.
UNR	When a user goes online through a Layer 2 device, such as a switch, but there is no available Layer 3 interface and the user is assigned an IP address, no dynamic routing protocol can be used. To enable devices to use IP routes to forward the traffic of this user, use the Huawei User Network Route (UNR) technology to assign a route to forward the traffic of the user.

Abbreviations

Abbreviation	Full Name
ARP	Address Resolution Protocol
BGP	Border Gateway Protocol
CE	Customer Edge
FIB	Forwarding Information Base
IBGP	Internal Border Gateway Protocol
IGP	Internal Gateway Protocol
IS-IS	Intermediate System-Intermediate System
NDP	Neighbor Discover Protocol

Abbreviation	Full Name
OSPF	Open Shortest Path First
PE	Provider Edge
RIP	Routing Information Protocol
RM	Route Management
Vlink	Virtual Link
VoIP	Voice Over IP
VPN	Virtual Private Network
VRP	Versatile Routing Platform

3 Static Routes

About This Chapter

[3.1 Overview of Static Routes](#)

[3.2 Understanding Static Routes](#)

3.1 Overview of Static Routes

Definition

Static routes are special routes that are configured by network administrators.

Purpose

On a simple network, only static routes can ensure that the network runs properly. If a router cannot run dynamic routing protocols or cannot generate routes to a destination network, you can configure static routes on the router.

Route selection can be controlled using static routes. Properly configuring and using static routes can improve network performance and guarantee the required bandwidth for important applications. When a network fault occurs or the network topology changes, however, static routes must be changed manually by the administrator.

3.2 Understanding Static Routes

3.2.1 Components

On the NE40E, you can run the **ip route-static** command to configure a static route, which consists of the following components:

- **Destination address and mask**
- **Outbound interface and next hop address**

Destination Address and Mask

In the **ip route-static** command, the IPv4 address is expressed in dotted decimal notation, and the mask is expressed in dotted decimal notation or represented by the mask length (the number of consecutive 1s in the mask).

Outbound Interface and Next Hop Address

When configuring a static route, you can specify an outbound interface through *interface-type* *interface-number*, a next-hop IP address through *nexthop-address*, or both an outbound interface and a next-hop IP address. In addition, you can configure the Next-Table function by specifying a VPN instance name for the next hop of the static route without an outbound interface or next-hop IP address. Set the parameters as required.

Every route requires a next-hop IP address. Before sending a packet, a device needs to search its routing table for the route matching the destination address in the packet using the longest match rule. The link layer can find the corresponding link-layer address and then forward the packet only when a next-hop IP address is available.

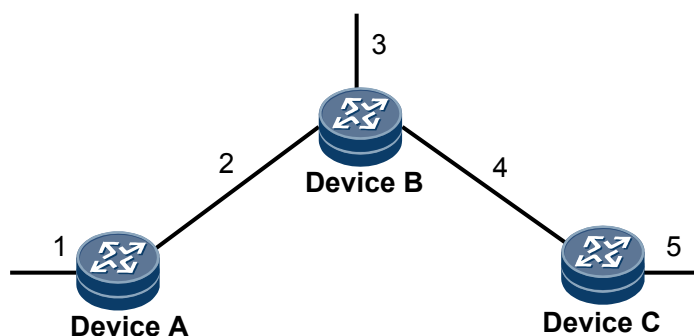
When specifying an outbound interface, note the following:

- For a Point-to-Point (P2P) interface, if the outbound interface is specified, the next hop address is the address of the remote interface connected to the outbound interface. For example, when a GE interface is encapsulated with Point-to-Point Protocol (PPP) and obtains the remote IP address through PPP negotiation, you need to specify only the outbound interface rather than the next hop address.
- Non-Broadcast Multiple-Access (NBMA) interfaces are applicable to Point-to-Multipoint networks. Therefore, IP routes and the mappings between IP addresses and link layer addresses are required. In this case, you need to configure next hop addresses.
- An Ethernet interface is a broadcast interface and a virtual-template (VT) interface can be associated with multiple virtual access (VA) interfaces. If the Ethernet interface or the VT interface is specified as the outbound interface of a static route, the next hop cannot be determined because multiple next hops exist. Therefore, do not specify an Ethernet interface or a VT interface as the outbound interface unless necessary. If you need to specify a broadcast interface (such as an Ethernet interface) or a VT interface as the outbound interface, specify the associated next hop address.

3.2.2 Application Scenarios for Static Routes

In [Figure 3-1](#), the network topology is simple, and network communication can be implemented through static routes. You need to specify an address for each physical network, identify indirectly connected physical networks for each router, and configure static routes for indirectly connected physical networks.

Figure 3-1 Networking for static routes



In [Figure 3-1](#), static routes to networks 3, 4, and 5 need to be configured on Device A; static routes to networks 1 and 5 need to be configured on Device B; static routes to networks 1, 2, and 3 need to be configured on Device C.

Default Static Route

Default routes are a special kind of routes and can be configured. The default route is used only when no matched entry is available in the routing table. In a routing table, both the destination address and the mask of the default route are 0.0.0.0.

If the destination address of a packet does not match any entry in the routing table, the router selects the default route to forward this packet. If no default route exists and the destination address of the packet does not match any entry in the routing table, the packet is discarded. An Internet Control Message Protocol (ICMP) packet is then sent, informing the originating host that the destination host or network is unreachable.

The static route with the destination address and mask 0s (0.0.0.0 0.0.0.0) configured using the **ip route-static** command is a default route intended to simplify network configuration.

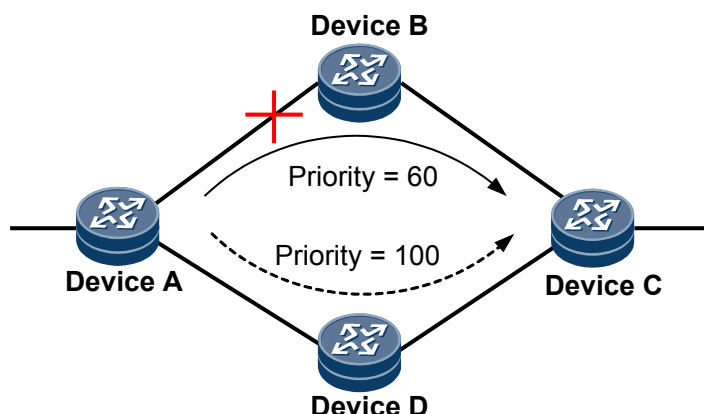
In [Figure 3-1](#), because the next hop of the packets from Device A to networks 3, 4, and 5 is Device B, a default route can be configured on Device A to replace the three static routes destined for networks 3, 4, and 5. Similarly, only a default route from Device C to Device B needs to be configured to replace the three static routes destined for networks 1, 2, and 3.

Floating Static Routes

Different static routes can be configured with different priorities so that routing management policies can be flexibly applied. Route backup can be implemented by specifying different priorities for multiple routes to the same destination.

In [Figure 3-2](#), there are two static routes from Device A to Device C. In most cases, the only Active route is the static route with Device B as the next hop in the routing table because it has a higher priority. The other static route with Device D as the next hop functions as a backup route. The backup route is only activated to forward traffic when the primary link fails. After the primary link recovers, the static route with Device B as the next hop becomes Active to take over the traffic. Therefore, the backup route is also called a floating static route. The floating static route becomes ineffective if a fault occurs on the link between Device B and Device C.

Figure 3-2 Networking for a floating static route

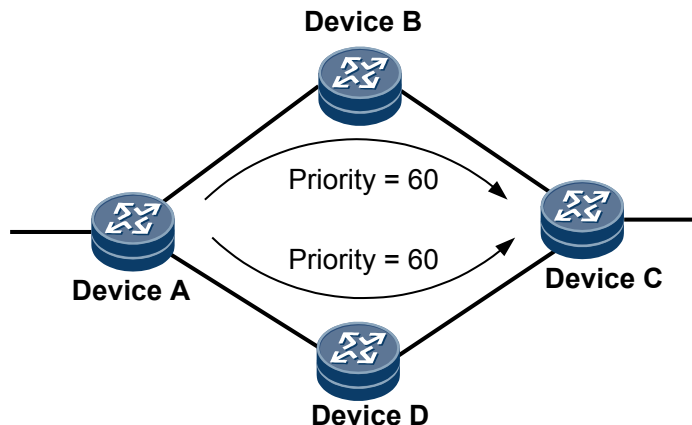


Load Balancing Among Static Routes

Routes to the same destination with the same priority can be used to load-balance traffic.

As shown in [Figure 3-3](#), there are two static routes with the same priority from Device A to Device C. The two routes both exist in the routing table and forward traffic at the same time.

Figure 3-3 Load balancing among static routes



FRR for Static Routes

When routes are delivered to the routing management (RM) module, the optimal route is delivered with a backup route. If the optimal route fails, traffic is immediately switched to the backup route, minimizing traffic loss.

You need to configure two routes with the same prefix but different priorities to implement FRR. The route with the higher priority is the primary route, and the route with the lower priority is the backup route. FRR is implemented only on static routes that are manually configured. That is, FRR is not implemented on recursive next hops.

3.2.3 Functions

IPv4 Static Routes

The NE40E supports common static routes and the static routes associated with VPN instances. The static routes associated with VPN instances are used to manage VPN routes. For details about VPN instances, see the *HUAWEI NetEngine40E Universal Service Router Feature Description - VPN*.

Attributes and Functions of IPv6 Static Routes

Similar to IPv4 static routes, IPv6 static routes are configured by the administrator and are applicable to simple IPv6 networks.

The major difference between IPv6 static routes and IPv4 static routes lies in their destination addresses and next hop addresses.

An IPv6 static route with destination address `::/0` (mask length 0) is a default IPv6 route. If the destination address of an IPv6 packet fails to match any entry in the routing table, a router selects the default IPv6 route to forward the IPv6 packet.

3.2.4 BFD for Static Routes

Different from dynamic routing protocols, static routes do not have a detection mechanism. If a fault occurs on a network, an administrator must manually address it. Bidirectional Forwarding Detection (BFD) for static routes is introduced to associate a static route with a BFD session so that the BFD session can detect the status of the link that the static route passes through.

After BFD for static routes is configured, each static route can be associated with a BFD session. In addition to route selection rules, whether a static route can be selected as the optimal route is subject to BFD session status.

- If a BFD session associated with a static route detects a link failure when the BFD session is Down, the BFD session reports the link failure to the system. The system then deletes the static route from the IP routing table.
- If a BFD session associated with a static route detects that a faulty link recovers when the BFD session is Up, the BFD session reports the fault recovery to the system. The system then adds the static route to the IP routing table again.
- By default, a static route can still be selected even though the BFD session associated with it is AdminDown (triggered by the **shutdown** command run either locally or remotely). If a device is restarted, the BFD session needs to be re-negotiated. In this case, whether the static route associated with the BFD session can be selected as the optimal route is subject to the re-negotiated BFD session status.

BFD for static routes has two detection modes:

- Single-hop detection
In single-hop detection mode, the configured outbound interface and next hop address are the information about the directly connected next hop. The outbound interface associated with the BFD session is the outbound interface of the static route, and the peer address is the next hop address of the static route.
- Multi-hop detection

In multi-hop detection mode, only the next hop address is configured. Therefore, the static route must recurse to the directly connected next hop and outbound interface. The peer address of the BFD session is the original next hop address of the static route, and the outbound interface is not specified. In most cases, the original next hop is an indirect next hop. Multi-hop detection is performed on the static routes that support route recursion.

 NOTE

For details about BFD, see the *HUAWEI NetEngine40E Universal Service Router Feature Description - Reliability*.

3.2.5 NQA for Static Routes

Background

Static routes do not have a dedicated detection mechanism. If a link fails, a network administrator must manually delete the corresponding static route from the IP routing table. This process delays link switchovers and can cause lengthy service interruptions.

Bidirectional Forwarding Detection (BFD) for static routes can use BFD sessions to monitor the link status of a static route. However, both ends of the link must support BFD. Network quality analysis (NQA) for static routes, however, can monitor the link status of a static route as long as only one end supports NQA.

Table 3-1 compares BFD and NQA for static routes.

Table 3-1 Comparison between BFD and NQA for static routes

Item	BFD for Static Routes	NQA for Static Routes
Detection mode	Bidirectional session	Unidirectional detection
Requirements for communicating devices	Two communicating devices; both must support BFD.	Two communicating devices; only one must support NQA.
Detection speed	Millisecond-level	Second-level

Related Concepts

NQA monitors network quality of service (QoS) in real time. If a network fails, NQA can be used to diagnose the fault.

NQA relies on a test instance to monitor link status. The two ends of an NQA test are called the NQA client and the NQA server. The NQA client initiates an NQA test that can return any of the following results:

- **success:** The test is successful. NQA instructs the routing management module to set the static route to active and add the static route to the routing table.
- **failed:** The test fails. NQA instructs the routing management module to set the static route to inactive and delete the static route from the routing table.
- **no result:** The test is running and no result has been obtained, which does not change the status of the static route.



For NQA details, see the chapter "System Monitor" in the *HUAWEI NetEngine40E Universal Service Router Feature Description*.

Implementation

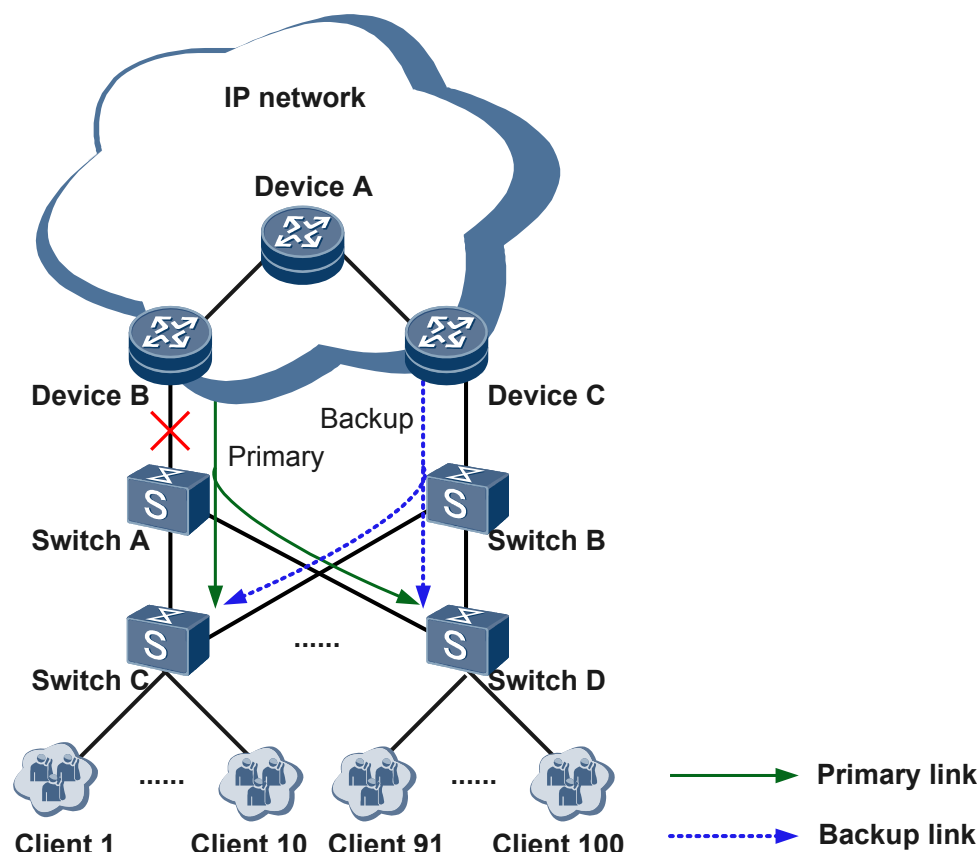
NQA for static routes associates an NQA test instance with a static route and uses the NQA test instance to monitor the link. The routing management module determines whether a static route is active or inactive based on the test result. If the static route is inactive, the routing management module deletes it from the IP routing table and selects a backup link for data forwarding, which prevents lengthy service interruptions.

In [Figure 3-4](#), each access switch is connected to 10 clients, and a total of 100 clients are connected. Because no dynamic routing protocol can be deployed between Device B and the clients, static routes to the clients must be configured on Device B, and backup static routes to the clients can be configured on Device C.

Device A, Device B, and Device C run a dynamic routing protocol and learn routes from one another. Device B and Device C are configured to import static routes to the routing table of the dynamic routing protocol, and different costs are set for the static routes. Device A can contact Device B and Device C using the dynamic routing protocol to learn routes to the clients. Device A selects one primary link and one backup link based on link costs.

NQA for static routes, configured on Device B, uses NQA test instances to monitor the status of the primary link. If the primary link fails, the corresponding static route is deleted and network-to-client traffic switches to the backup link. When both the primary and backup links are running properly, network-to-client traffic is preferentially transmitted along the primary link.

Figure 3-4 NQA for static routes



NOTE

NQA test instances support both IPv4 and IPv6 static routes. The mechanisms for monitoring IPv4 and IPv6 static routes are the same.

Each static route can be associated with only one NQA test instance.

Usage Scenario

NQA for static routes applies to a network on which BFD for static routes cannot be deployed due to device connectivity limitations. For example, switches, optical line terminals (OLTs), digital subscriber line access multiplexers (DSLAMs), multiservice access nodes (MSANs), or x digital subscriber lines (xDSLs) exist on the network.

Benefits

NQA for static routes can monitor the link status of static routes and implement rapid primary/backup link switchovers, preventing lengthy service interruptions.

3.2.6 Static Route Permanent Advertisement

Background

When the link over which a static route runs fails, the static route will be deleted from the IP routing table to trigger a route re-selection. After a new route is selected, traffic is switched to

the new route. Some carriers, however, may require that specific traffic always travel along a fixed link, regardless of the link status. Static route permanent advertisement is introduced to meet this service need.

Implementation

With static route permanent advertisement, a static route can still be advertised and added to the IP routing table for route selection even when the link over which the static route runs fails. After static route permanent advertisement is configured, the static route can be advertised and added to the IP routing table in both of the following scenarios:

- An outbound interface is configured for the static route, and the outbound interface has an IP address. Static route permanent advertisement is not affected no matter whether the outbound interface is Up.
- No outbound interface is configured for the static route. Static route permanent advertisement is not affected no matter whether the static route can obtain an outbound interface through route recursion.

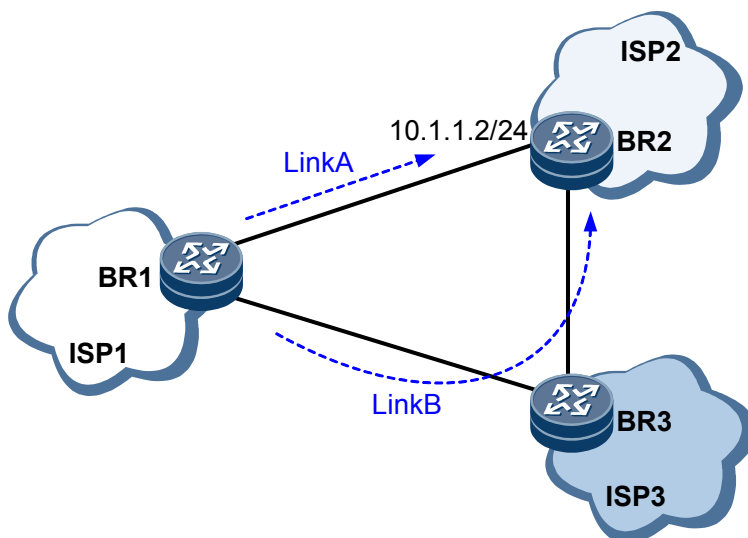
NOTE

After static route permanent advertisement is enabled, a static route always remains in the IP routing table regardless of route reachability. If the destination of the route becomes unreachable, traffic interruption occurs.

Typical Networking

On the network shown in [Figure 3-5](#), BR1, BR2, and BR3 belong to ISP1, ISP2, and ISP3 respectively. Two links (Link A and Link B) exist between BR1 and BR2, but ISP1 expects its service traffic destined for ISP2 to be always transmitted over Link A.

Figure 3-5 Networking with static route permanent advertisement



A direct EBGP peer relationship is established between BR1 and BR2. A static route is created on BR1, with 10.1.1.2/24 (IP address of BR2) as the destination address and the local interface connected to BR2 as the outbound interface.

Without static route permanent advertisement, Link A is used to transmit traffic. If Link A fails, BGP will switch the traffic to Link B.

With static route permanent advertisement, Link A is used to transmit traffic regardless of whether the destination is reachable through Link A. If Link A fails, no link switchover is performed, causing traffic interruption. To check whether the destination is reachable through the static route, ping the destination address of the static route to which static route permanent advertisement is applied.

3.2.7 Association between LDP and Static Routes

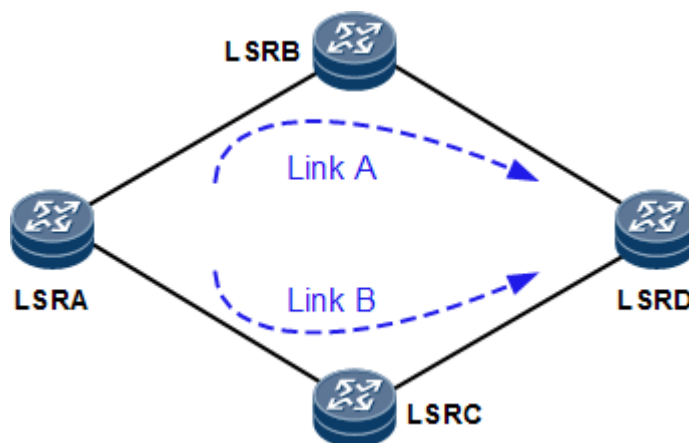
Background

If a backup path exists between LSRs, packet loss may occur due to status inconsistency between static routes and LDP sessions. To solve this problem, you can configure association between LDP and static routes.

Typical Networking

Figure 3-6 shows the typical networking of association between LDP and static routes. LSRA and LSRD interwork through static routes. Primary and secondary static routes are deployed on LSRA, with the next-hop devices being LSRB and LSRC, respectively. Primary and secondary LDP LSPs are established based on the static routes. The primary LSP uses Link A, and the secondary LSP uses Link B. In normal cases, Link A is preferred. Association between LDP and static routes in switchover and switchback scenarios is described as follows.

Figure 3-6 Networking of LSP switching scenario where association between LDP and static routes is configured



Switchover scenario

In the switchover scenario, traffic of the primary static route is not switched to the secondary link when the LDP session on the primary link fails (not because of a link fault). As a result, traffic on the LSP over the primary link is interrupted.

After an LDP session is established, LSP traffic travels along the primary link, Link A (LSRA → LSRB → LSRD). If the LDP session between LSRA and LSRB is interrupted, traffic of the primary LSP is switched immediately to the secondary link, Link B (LSRA → LSRC → LSRD). However, because the link between LSRA and LSRB is normal, traffic of the primary

static route is not switched to the secondary link. The asynchronous state between LDP and the primary static route causes an LSP traffic interruption.

If association between LDP and static routes is enabled, traffic is automatically switched to the secondary link when the LDP session goes Down, ensuring uninterrupted traffic forwarding.

Switchback scenario

In the switchback scenario, when the primary link recovers from a fault, the traffic of the primary static route is switched back to Link A earlier than the traffic of the primary LSP because the convergence of static routes is faster than that of LDP LSPs. As a result, the secondary LSP on Link B cannot be used, and the LSP on Link A has not been set up yet. As a result, LSP traffic is interrupted.

If the link between LSRA and LSRB fails, traffic is switched immediately to the secondary link, Link B (LSRA → LSRC → LSRD). After the link between LSRA and LSRB recovers, traffic of the primary static route is immediately switched back to Link A (LSRA → LSRB → LSRD). However, the secondary LSP cannot be used, and the LSP on Link A has not recovered yet. As a result, traffic is interrupted.

If association between LDP and static routes is enabled, the static route on Link A becomes active only when the LDP session on Link A goes Up. In this manner, the states of the primary static route and LSP are asynchronous during the switchback, which prevents traffic loss.

Usage Scenario

Association between LDP and static routes applies to scenarios where a static route backup path exists between LSRs.

Benefits

Association between LDP and static routes ensures state consistency between LDP and static routes, prevents traffic loss, and improves network reliability.

4 RIP

About This Chapter

[4.1 Overview of RIP](#)

[4.2 Understanding RIP](#)

4.1 Overview of RIP

Definition

Routing Information Protocol (RIP) is a simple Interior Gateway Protocol (IGP). RIP is used in small-scale networks, such as campus networks and simple regional networks.

As a distance-vector routing protocol, RIP exchanges routing information through User Datagram Protocol (UDP) packets with port number 520.

RIP employs the hop count as the metric to measure the distance to the destination. In RIP, by default, the number of hops from the router to its directly connected network is 0; the number of hops from the router to a network that is reachable through another router is 1, and so on. The hop count (the metric) equals the number of routers along the path from the local network to the destination network. To speed up route convergence, RIP defines the hop count as an integer that ranges from 0 to 15. A hop count that is equal to or greater than 16 is classified as infinite, indicating that the destination network or host is unreachable. Due to the hop limit, RIP is not applicable to large-scale networks.

RIP has two versions:

- RIP version 1 (RIP-1), a classful routing protocol
- RIP version 2 (RIP-2), a classless routing protocol

RIP supports split horizon, poison reverse, and triggered update, which improves the performance and prevents routing loops.

Purpose

As the earliest IGP, RIP is used in small and medium-sized networks. Its implementation is simple, and the configuration and maintenance of RIP are easier than those of Open Shortest

Path First (OSPF) and Intermediate System-to-Intermediate System (IS-IS). Therefore, RIP is widely used on live networks.

4.2 Understanding RIP

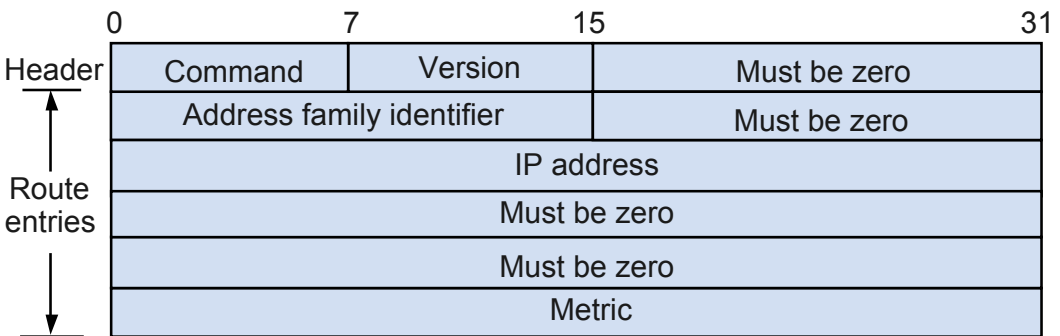
RIP is a distance-vector routing protocol. It forwards packets through UDP and uses timers to control the advertisement, update, and aging of routing information. However, design defects in RIP may cause routing loops. Therefore, split horizon, poison reverse, and triggered update were introduced into RIP to avoid routing loops.

In addition, RIP periodically advertises its routing table to neighbors, and route summarization was introduced to reduce the size of the routing table.

4.2.1 RIP-1

RIP version 1 (RIP-1) is a classful routing protocol, which supports only the broadcast of protocol packets. [Figure 4-1](#) shows the format of a RIP-1 packet. A RIP packet can carry a maximum of 25 routing entries. RIP is based on UDP, and a RIP-1 packet cannot be longer than 512 bytes. RIP-1 packets do not carry any mask information, and RIP-1 can identify only the routes to natural network segments, such as Class A, Class B, and Class C. Therefore, RIP-1 does not support route summarization or discontinuous subnets.

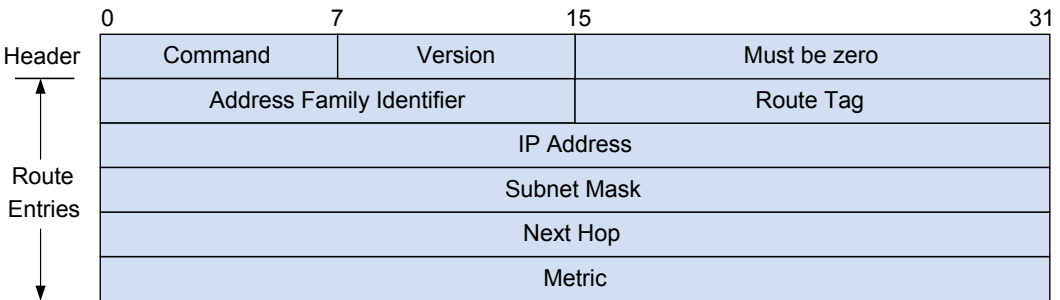
Figure 4-1 RIP-1 packet format



4.2.2 RIP-2

RIP version 2 (RIP-2) is a classless routing protocol. [Figure 4-2](#) shows the format of a RIP-2 packet.

Figure 4-2 RIP-2 packet format



Compared with RIP-1, RIP-2 has the following advantages:

- Supports external route tags and uses a routing policy to flexibly control routes based on the tag.
- Supports route summarization and classless inter-domain routing (CIDR) by adding mask information to RIP-2 packets.
- Supports next hop specification so that the optimal next hop address can be specified on the broadcast network.
- Supports Update packets transmission along multicast routes. Only the routers that support RIP-2 can receive RIP-2 packets, which reduces resource consumption.
- Provides two packet authentication modes: simple authentication and Message Digest 5 (MD5) authentication, which improves security.

4.2.3 Timers

RIP uses the following timers:

- Update timer: The Update timer periodically triggers Update packet transmission. By default, the interval at which Update packets are sent is 30s.
- Age timer: If a RIP device does not receive any packets from its neighbor to update a route before the route expires, the RIP device considers the route unreachable. By default, the age timer interval is 180s.
- Garbage-collect timer: If a route becomes invalid after the age timer expires or a route unreachable message is received, the route is placed into a garbage queue instead of being immediately deleted from the RIP routing table. The garbage-collect timer monitors the garbage queue and deletes expired routes. If an Update packet of a route is received before the garbage-collect timer expires, the route is placed back into the age queue. The garbage-collect timer is set to avoid route flapping. By default, the garbage collect timer interval is 120s.
- Hold-down timer: If a RIP device receives an updated route with cost 16 from a neighbor, the route enters the hold-down state, and the hold-down timer is started. To avoid route flapping, the RIP device does not accept any updated routes until the hold-down timer expires, even if the cost is less than 16 except in the following scenarios:
 - a. The cost carried in the Update packet is less than or equal to that carried in the last update packet.
 - b. The hold-down timer expires, and the corresponding route enters the Garbage state.

The relationship between RIP routes and the four timers is as follows:

- The advertisement of RIP routing updates is triggered by the update timer with a default value 30 seconds.
- Each routing entry is associated with two timers: the age timer and garbage-collect timer.
 - a. Each time a route is learned and added to the routing table, the age timer is started.
 - b. If no Update packet is received from the neighbor within 180 seconds after the age timer is started, the metric of the corresponding route is set to 16, and the garbage-collect timer is started.
- If no Update packet is received within 120 seconds after the garbage-collect timer is started, the corresponding routing entry is deleted from the routing table after the garbage-collect timer expires.

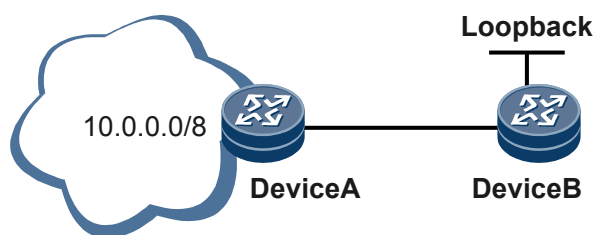
- By default, the hold-down timer is disabled. If you configure a hold-down timer, it starts after the system receives a route with a cost greater than 16 from its neighbor.

4.2.4 Split Horizon

Split Horizon on Broadcast, P2PMP, and P2P Networks

Split horizon prevents a RIP-enabled interface from sending back the routes it learns, which reduces bandwidth consumption and prevents routing loops.

Figure 4-3 Networking for interface-based split horizon



In [Figure 4-3](#), Device A sends Device B a route to 10.0.0.0/8. If split horizon is not configured, Device B will send this route back to Device A after learning it from Device A. As a result, Device A learns the following routes to 10.0.0.0/8:

- A direct route with zero hops
- A route with Device B as the next hop and total two hops

Only direct routes, however, are active in the RIP routing table of Device A.

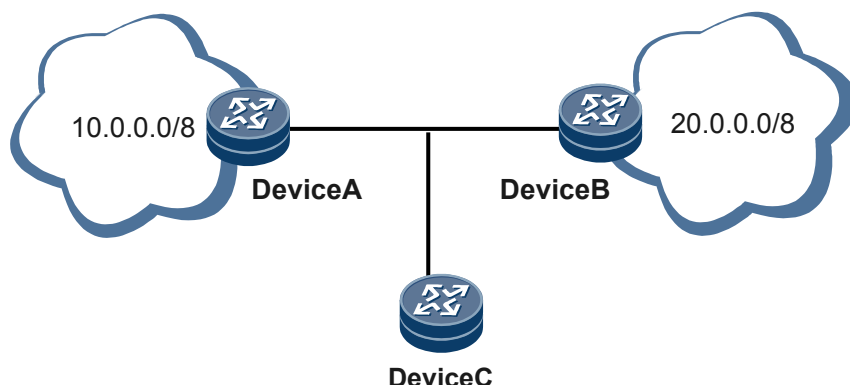
If the route from Device A to 10.0.0.0/8 becomes unreachable and Device B is not notified, Device B still considers the route to 10.0.0.0/8 reachable and continues sending this route to Device A. Then, Device A receives incorrect routing information and considers the route to 10.0.0.0/8 reachable through Device B; Device B considers the route to 10.0.0.0/8 reachable through Device A. As a result, a loop occurs on the network.

After split horizon is configured, Router B no longer sends the route back after learning the route, which prevents such a loop.

Split Horizon on NBMA Networks

On a Non-Broadcast Multi-Access (NBMA) network where an interface is connected to multiple neighbors, RIP supports neighbor-based split horizon. On NBMA networks, routes are sent in unicast mode, and an interface can differentiate which neighbor each route was learned from, and the interface will not send the routes back to the neighbor it learned them from.

Figure 4-4 Networking for neighbor-based split horizon on an NBMA network

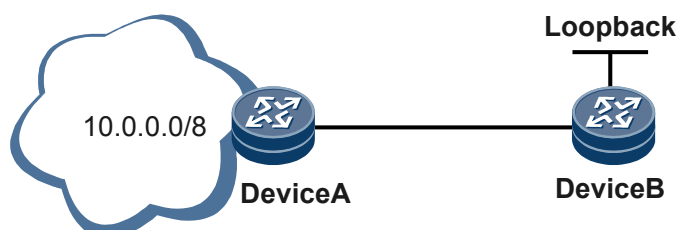


In [Figure 4-4](#), Device A sends the route to 10.0.0.0/8 that it learns from Device B only to Device C.

4.2.5 Poison Reverse

Poison reverse allows a RIP-enabled interface to set the cost of the route that it learns from a neighbor to 16 (indicating that the route is unreachable) and then send the route back. After receiving this route, the neighbor deletes the useless route from its routing table, which prevents loops.

Figure 4-5 Networking for poison reverse



In [Figure 4-5](#), Device A sends Device B a route to 10.0.0.0/8. If poison reverse is not configured, Device B will send this route back to Device A after learning it from Device A. As a result, Device A learns the following routes to 10.0.0.0/8:

- A direct route with zero hops
- A route with Device B as the next hop and total two hops

Only direct routes, however, are active in the RIP routing table of Device A.

If the route from Device A to 10.0.0.0 becomes unreachable and Device B is not notified, Device B still considers the route to 10.0.0.0/8 reachable and continues sending this route to Device A. Then, Device A receives incorrect routing information and considers the route to 10.0.0.0/8 reachable through Device B; Device B considers the route to 10.0.0.0/8 reachable through Device A. As a result, a loop occurs on the network.

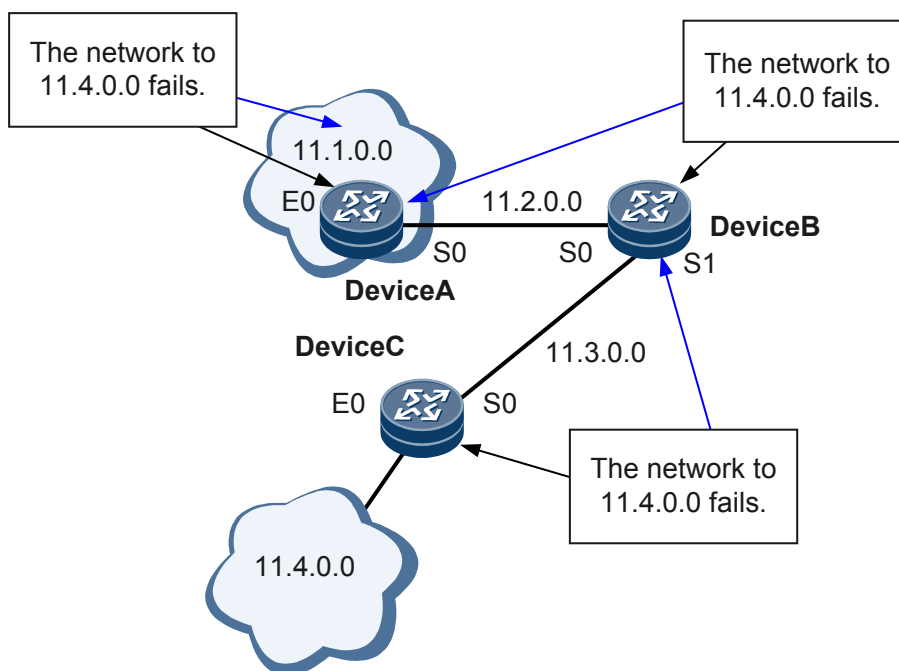
With poison reverse, after Device B receives the route from Device A, Device B sends a route unreachable message to Device A with cost 16. Device A then no longer learns the reachable route from Device B, which prevents routing loops.

If both split horizon and poison reverse are configured, only poison reverse takes effect.

4.2.6 Triggered Update

Triggered update allows a device to advertise routing information changes immediately, which speeds up network convergence.

Figure 4-6 Networking for triggered update



In **Figure 4-6**, if the route to 10.4.0.0 becomes unreachable, Device C learns the information first. By default, a RIP-enabled device sends routing updates to its neighbors every 30s. If Device C receives an Update packet from Device B within 30s while Device C is still waiting to send Update packets, Device C learns the incorrect route to 10.4.0.0. In this case, the next hops of the routes from Device B or Device C to network 10.4.0.0 are Device C and Device B respectively, which results in routing loops. If Device C sends an Update packet to Device B immediately after it detects a network, Device B can rapidly update its routing table, which prevents routing loops.

In addition, if the next hop of a route becomes unavailable due to a link failure, the local Device sets the cost of the route to 16 and then advertises the route immediately to its neighbors. This process is called route poisoning.

4.2.7 Route Summarization

Route summarization allows routes to the same natural network segment but different subnets to be summarized into a single route belonging to the same network segment before it is transmitted to other network segments. RIP-1 packets do not carry mask information, and therefore RIP-1 can advertise only routes with natural masks. RIP-2 supports route summarization because RIP-2 packets carry mask information. Therefore, RIP-2 supports subnetting.

In RIP-2, route summarization can reduce the size of the routing table and improve the extensibility and efficiency of a large-scale network.

Route summarization has two modes:

- Process-based classful summarization

Summary routes are advertised with natural masks. If split horizon or poison reverse is configured, classful summarization becomes invalid because split horizon or poison reverse suppresses some routes from being advertised. In addition, when classful summarization is configured, routes learned from different interfaces may be summarized into a single route. As a result, a conflict occurs in the advertisement of the summary route.

For example, a RIP process summarizes the route 10.1.1.0 /24 with metric 2 and route 10.2.2.0/24 with metric 3 into the route 10.0.0.0/8 with metric 2.

- Interface-based summarization

Users can specify a summary address.

For example, users can configure a RIP-enabled interface to summarize the route 10.1.1.0/24 with metric 2 and route 10.2.2.0/24 with metric 3 into the route 10.1.0.0/16 with metric 2.

4.2.8 Multi-Process and Multi-Instance

RIP supports multi-process and multi-instance to simplify network management and improve service control efficiency. Multi-process allows a set of interfaces to be associated with a specific RIP process, which ensures that the specific RIP process performs all the protocol operations only on this set of interfaces. Therefore, multiple RIP processes can run on one router, and each process manages a unique set of interfaces. In addition, the routing data of each RIP process is independent; however, processes can import routes from each other.

On routers that support VPN, each RIP process is associated with a specific VPN instance. Therefore, all the interfaces associated with the RIP process need to be associated with the RIP process-related VPN instance.

4.2.9 BFD for RIP

Background

Routing Information Protocol (RIP)-capable devices monitor the neighbor status by exchanging Update packets periodically. During the period local devices detect link failures, carriers or users may lose a large number of packets. Bidirectional forwarding detection (BFD) for RIP can speed up fault detection and route convergence, which improves network reliability.

After BFD for RIP is configured on the router, BFD can detect a fault (if any) within milliseconds and notify the RIP module of the fault. The router then deletes the route that passes through the faulty link and switches traffic to a backup link. This process speeds up RIP convergence.

Table 4-1 describes the differences before and after BFD for RIP is configured.

Table 4-1 Differences before and after BFD for RIP is configured

Item	Link Fault Detection Mechanism	Convergence Speed
BFD for RIP is not configured.	A RIP aging timer expires.	Second-level
BFD for RIP is configured.	A BFD session goes Down.	Millisecond-level

Related Concepts

The BFD mechanism bidirectionally monitors data protocol connectivity over the link between two routers. After BFD is associated with a routing protocol, BFD can rapidly detect a fault (if any) and notify the protocol module of the fault, which speeds up route convergence and minimizes traffic loss.

BFD is classified into the following modes:

- Static BFD

In static BFD mode, BFD session parameters (including local and remote discriminators) must be configured, and requests must be delivered manually to establish BFD sessions.

Static BFD is applicable to networks on which only a few links require high reliability.

- Dynamic BFD

In dynamic BFD mode, the establishment of BFD sessions is triggered by routing protocols, and the local discriminator is dynamically allocated, whereas the remote discriminator is obtained from BFD packets sent by the neighbor.

When a new neighbor relationship is set up, a BFD session is established based on the neighbor and detection parameters, including source and destination IP addresses. When a fault occurs on the link, the routing protocol associated with BFD can detect the BFD session Down event. Traffic is switched to the backup link immediately, which minimizes data loss.

Dynamic BFD is applicable to networks that require high reliability.

Implementation

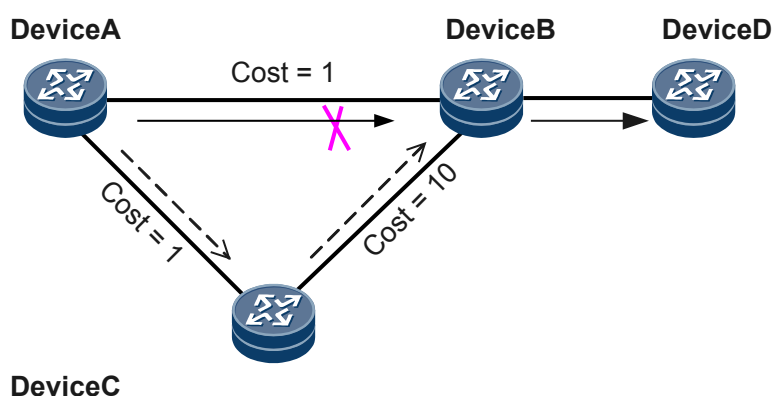
For details about BFD implementation, see "BFD" in *Universal Service Router Feature Description - Reliability*. [Figure 4-7](#) shows a typical network topology for BFD for RIP.

- Dynamic BFD for RIP implementation:

- RIP neighbor relationships are established among Device A, Device B, and Device C and between Device B and Device D.
- BFD for RIP is enabled on Device A and Device B.
- Device A calculates routes, and the next hop along the route from Device A to Device D is Device B.
- If a fault occurs on the link between Device A and Device B, BFD will rapidly detect the fault and report it to Device A. Device A then deletes the route whose next hop is Device B from the routing table.

- e. Device A recalculates routes and selects a new path Device C → Device B → Device D.
- f. After the link between Device A and Device B recovers, a new BFD session is established between the two routers. Device A then reselects an optimal link to forward packets.
- Static BFD for RIP implementation:
 - a. RIP neighbor relationships are established among Device A, Device B, and Device C and between Device B and Device D.
 - b. Static BFD is configured on the interface that connects Device A to Device B.
 - c. If a fault occurs on the link between Device A and Device B, BFD will rapidly detect the fault and report it to Device A. Device A then deletes the route whose next hop is Device B from the routing table.
 - d. After the link between Device A and Device B recovers, a new BFD session is established between the two routers. Device A then reselects an optimal link to forward packets.

Figure 4-7 BFD for RIP



Usage Scenario

BFD for RIP is applicable to networks that require high reliability.

Benefits

BFD for RIP improves network reliability and enables devices to rapidly detect link faults, which speeds up route convergence on RIP networks.

4.2.10 RIP NSR

Devices with a distributed architecture support RIP Non-stop Routing (NSR). RIP backs up all route data from the Active Main Board (AMB) to the Standby Main Board (SMB). Whenever the AMB fails, the SMB becomes active and takes over traffic. RIP NSR ensures that routes are synchronous between the AMB and SMB. Therefore, during the AMB/SMB switchover, the neighbor will not detect the fault on the local device.

4.2.11 RIP Authentication

As networks develop, there has been considerable growth in all types of data, voice, and video information exchanged on networks. In addition, new services, such as E-commerce, online conferencing and auctions, video on demand (VoD), and e-learning have sprung up increasingly, requiring higher information security than before. Carriers must protect data packets from being intercepted or modified by attackers and prohibit unauthorized users from accessing network resources. RIP packet authentication effectively meets these security requirements.

RIP authentication falls into the following modes:

- Simple authentication: The authenticated party adds the configured password directly to packets for authentication. This authentication mode provides the lowest password security.
- MD5 authentication: The authenticated party uses the Message Digest 5 (MD5) algorithm to generate a ciphertext password and adds it to packets for authentication. This authentication mode improves password security.
- Keychain authentication: The authenticated party configures a keychain that changes over time. This authentication mode further improves password security.
Keychain authentication improves RIP security by periodically changing the password and the encryption algorithms. For details about Keychain, see "Keychain" in NE40E Feature Description - Security.
- HMAC-SHA256 authentication: The authenticated party uses the HMAC-SHA256 algorithm to generate a ciphertext password and adds it to packets for authentication.

RIP authentication ensures network security by adding an authentication field used to encrypt a packet before sending the packet to ensure network security. After receiving a RIP packet from a remote router, the local router discards the packet if the authentication password in the packet does not match the local authentication password. This authentication mode protects the local router.

On IP networks of carriers, RIP authentication ensures the secure transmission of packets, improves the system security, and provides secure network services for carriers.

5 RIPng

About This Chapter

[5.1 Overview of RIPng](#)

[5.2 Understanding RIPng](#)

5.1 Overview of RIPng

Definition

RIP next generation (RIPng) is an extension to RIP version 2 (RIP-2) on IPv6 networks. Most RIP concepts apply to RIPng.

RIPng is a distance-vector routing protocol, which measures the distance (metric or cost) to the destination host by the hop count. In RIPng, the hop count from a device to its directly connected network is 0, and the hop count from a device to a network that is reachable through another device is 1. When the hop count is equal to or exceeds 16, the destination network or host is defined as unreachable.

To be applied on IPv6 networks, RIPng makes the following changes to RIP:

- UDP port number: RIPng uses UDP port number 521 to send and receive routing information.
- Multicast address: RIPng uses FF02::9 as the link-local multicast address of a RIPng device.
- Prefix length: RIPng uses a 128-bit (the mask length) prefix in the destination address.
- Next hop address: RIPng uses a 128-bit IPv6 address.
- Source address: RIPng uses link-local address FE80::/10 as the source address to send RIPng Update packets.

Purpose

RIPng is an extension to RIP for support of IPv6.

5.2 Understanding RIPng

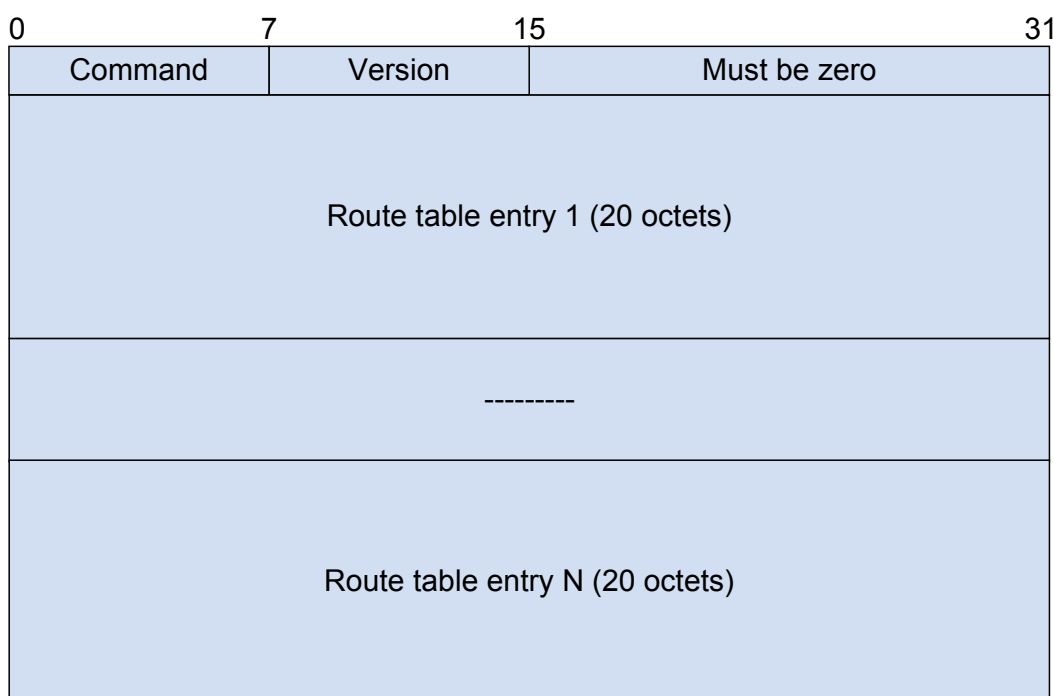
RIPng is an extension to RIPv2 on IPv6 networks and uses the same timers as RIPv2. RIPng supports split horizon, poison reverse, and triggered update, which prevents routing loops.

5.2.1 RIPng Packet Format

A RIPng packet is composed of a header and multiple route table entries (RTEs). In a RIPng packet, the maximum number of RTEs is determined by the maximum transmission unit (MTU) of an interface.

Figure 5-1 shows the basic format of a RIPng packet.

Figure 5-1 RIPng packet format



A RIPng packet contains two types of RTEs:

- Next hop RTE: It defines the IPv6 address of the next hop and is located before a group of IPv6-prefix RTEs that have the same next hop. The Metric field of a next hop RTE is always 0xFF.
- IPv6-prefix RTE: It describes the destination IPv6 address and the cost in the RIPng routing table and is located after a next hop RTE. A next hop RTE can be followed by multiple different IPv6-prefix RTEs.

Figure 5-2 shows the format of a next hop RTE.

Figure 5-2 Format of the next hop RTE

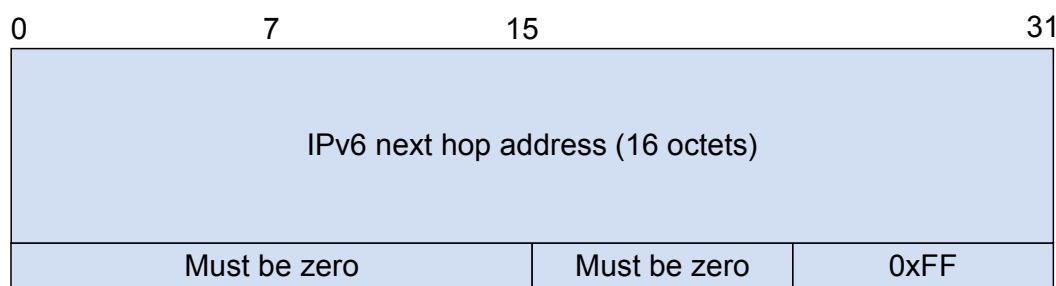
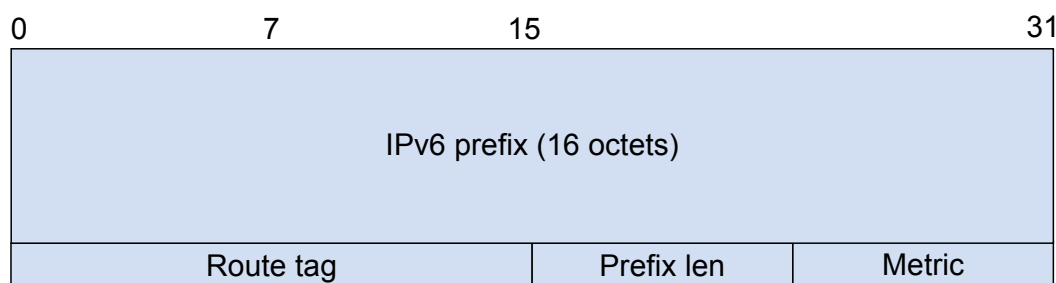


Figure 5-3 shows the format of an IPv6-prefix RTE.

Figure 5-3 Format of the IPv6-prefix RTE



5.2.2 Timers

RIPng uses the following timers:

- Update timer: This timer periodically triggers Update packet transmission. By default, the interval at which Update packets are sent is 30s. This timer is used to synchronize RIPng routes on the network.
- Age timer: If a RIPng device does not receive any Update packet from its neighbor before a route expires, the RIPng device considers the route to its neighbor unreachable.
- Garbage-collect timer: If no packet is received to update an unreachable route after the Age timer expires, this route is deleted from the RIPng routing table.
- Hold-down timer: If a RIP device receives an updated route with cost 16 from a neighbor, the route enters the holddown state, and the hold-down timer is started.

The following describes the relationship among these timers:

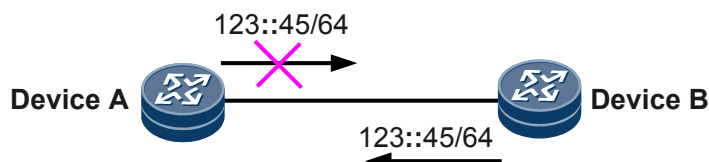
The advertisement of RIPng routing updates is periodically triggered by the update timer with default value 30 seconds. Each routing entry is associated with two timers: the Age timer and garbage-collect timer. Each time a route is learned and added to the routing table, the Age timer is started. If no Update packet is received from the neighbor within 180 seconds, the metric of the route is set to 16, and the garbage-collect timer is started. If no Update packet is received within 120 seconds, the route is deleted after the garbage-collect timer expires.

By default, hold-down timer is disabled. If you configure a hold-down timer, it starts after the system receives a route with a cost greater than 16 from its neighbor.

5.2.3 Split Horizon

Split horizon prevents a RIPng-enabled interface from sending back the routes it learns, which reduces bandwidth consumption and prevents routing loops.

Figure 5-4 Networking for split horizon

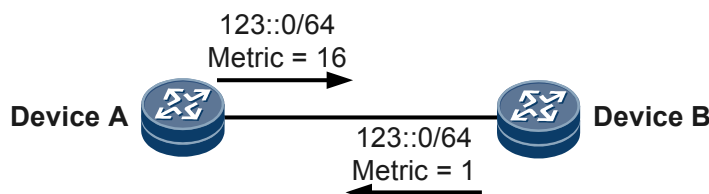


On the network shown in [Figure 5-4](#), after Device B sends a route to network 123::45 to Device A, Device A does not send the route back to Device B.

5.2.4 Poison Reverse

Poison reverse allows a RIPng-enabled interface to set the cost of the route that it learns from a neighbor to 16 (indicating that the route is unreachable) and then send the route back. After receiving this route, the neighbor can delete the useless route from its routing table, which prevents loops.

Figure 5-5 Networking for poison reverse



In [Figure 5-5](#), if poison reverse is not configured, Device B sends Device A a route learned from Device A. The cost of the route from Device A to network 123::0/64 is 1. If the route from Device A to network 123::0/64 becomes unreachable and Device B does not receive an Update packet from Device A and keeps sending Device A the route from Device A to network 123::0/64, a routing loop occurs.

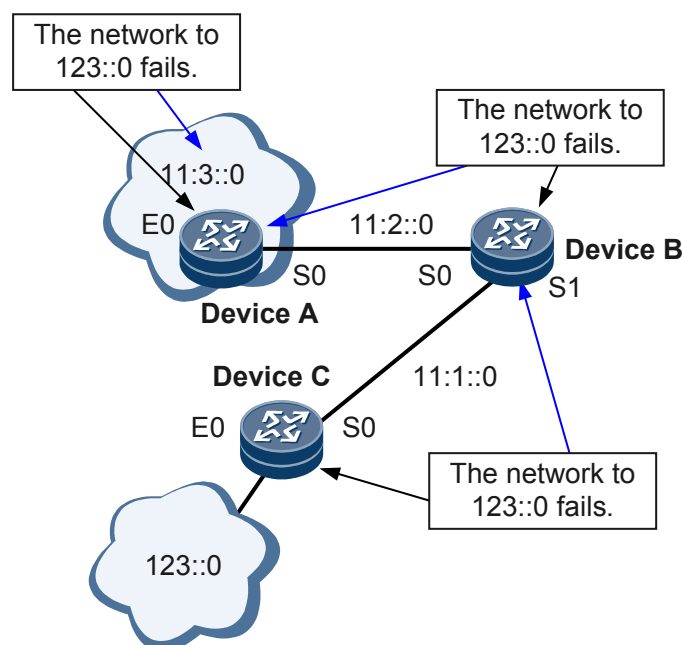
With poison reverse, after Device B receives the route from Device A, Device B sends a route unreachable message to Device A with cost 16. Device A then no longer learns the reachable route from Device B, which prevents routing loops.

If both poison reverse and split horizon are configured, only poison reverse takes effect.

5.2.5 Triggered Update

Triggered update allows a device to advertise the routing information changes immediately, which speeds up network convergence.

Figure 5-6 Networking for triggered update



In [Figure 5-6](#), if network 123::0 is unreachable, Device C learns the information first. By default, a RIPng-enabled device sends Update packets to its neighbors every 30 seconds. If Device C receives an Update packet from Device B within 30s when Device C is still waiting to send Update packets, Device C learns the incorrect route to 123::0. In this case, the next hops of the routes from Device B and Device C to 123::0 are Device C and Device B, respectively, which results in routing loops. If Device C sends an Update packet to Device B immediately after it detects a network fault, Device B can rapidly update its routing table, which prevents routing loops.

In addition, if the next hop of a route becomes unavailable due to a link failure, the local Router sets the cost of the route to 16 and then advertises the route immediately to its neighbors. This process is called route poisoning.

5.2.6 Route Summarization

Background

On large networks, the RIPng routing table of each device contains a large number of routes, which consumes lots of system resources. In addition, if a specific link connected to a device within an IP address range frequently alternates between Up and Down, route flapping occurs.

To address these problems, RIPng route summarization was introduced. With RIPng route summarization, a device summarizes routes destined for different subnets of a network segment into one route destined for one network segment and then advertises the summary route to other network segments. RIPng route summarization reduces the number of routes in the routing table, minimizes system resource consumption, and prevents route flapping.

Implementation

RIPng route summarization is interface-based. After RIPng route summarization is enabled on an interface, the interface summarizes routes based on the longest matching rule and then

advertises the summary route. The smallest metric among the specific routes for the summarization is used as the metric of the summary route.

For example, an interface has two routes: 11:11:11::24 with metric 2 and 11:11:12::34 with metric 3. After RIPng route summarization is enabled on the interface, the interface summarizes the two routes into the route 11::0/16 with metric 2 and then advertises it.

5.2.7 Multi-Process and Multi-Instance

RIPng supports multi-process and multi-instance, which simplifies network management and improves service control efficiency. Multi-process allows a set of interfaces to be associated with a specific RIPng process, which ensures that the specific RIPng process performs all the protocol operations only on this set of interfaces. Therefore, multiple RIPng processes can run on one router, and each process manages a unique set of interfaces. In addition, the routing data of each RIPng process is independent; however, processes can import routes from each other.

On routers that support VPN, each RIPng process is associated with a specific VPN instance. Therefore, all the interfaces associated with the RIPng process need to be associated with the RIPng process-related VPN instance.

5.2.8 Hot Standby

Devices of a distributed architecture support RIPng Hot Standby (HSB). In the RIPng HSB process, RIPng backs up RIPng configuration from the Active Main Board (AMB) to the Standby Main Board (SMB). Whenever the AMB fails, the SMB becomes active. After the SMB is activated, RIPng resends a request to neighbors to synchronize the route database. Therefore, RIPng is not affected.

5.2.9 IPsec Authentication

Background

As networks develop, network security has become an increasing concern. Internet Protocol Security (IPsec) authentication can be used to authenticate RIPng packets. The packets that fail to be authenticated are discarded, which prevents data transmitted based on TCP/IP from being intercepted, tampered with, or attacked.

Implementation

IPsec has an open standard architecture and ensures secure packet transmission on the Internet by encrypting packets. RIPng IPsec provides a complete set of security protection mechanisms to authenticate RIPng packets, which prevents devices from being attacked by forged RIPng packets.

IPsec includes a set of protocols that are used at the network layer to ensure data security, such as Internet Key Exchange (IKE), Authentication Header (AH), and Encapsulating Security Payload (ESP). The three protocols are described as follows:

- AH: A protocol that provides data origin authentication, data integrity check, and anti-replay protection. AH does not encrypt packets to be protected.
- ESP: A protocol that provides IP packet encryption and authentication mechanisms besides the functions provided by AH. The encryption and authentication mechanisms can be used together or independently.



AH and ESP can be used together or independently.

Benefits

RIPng IPsec offers the following benefits:

- Improves carriers' reputation and competitiveness by preventing services from being tampered with or attacked by unauthorized users.
- Ensures confidentiality and integrity of user packets.

5.2.10 RIPng NSR

Background

As networks develop, the demand for data, audio, and video services is growing, which imposes increasing requirements on IP network reliability. If an AMB/SMB switchover is performed on a device during a maintenance operation or a single point of failure occurs, routes may fail to converge, which may result in traffic loss or even a network breakdown. Non-stop routing (NSR) can address this problem and ensure uninterrupted forwarding of key services.

Related Concepts

- High availability (HA): supports data backup between the AMB and SMB.
- Non-stop forwarding (NSF): enables a node to use the GR mechanism to ensure uninterrupted transmission during an AMB/SMB switchover.
- NSR: allows a standby control plane to take over traffic from an active control plane if the active control plane fails, preventing the control planes of neighbors from detecting the fault.
- AMB and SMB: run control plane processes.

Implementation

With RIPng NSR, RIPng real-time data is synchronized between the AMB and SMB. After an AMB/SMB switchover is performed on a device, the SMB takes over services from the AMB, and neighbors are unaware of the local fault. After the switchover, the new AMB restores RIPng immediately based on the synchronized RIPng real-time data. Therefore, neighbors are unaware of the switchover as well. RIPng NSR requires synchronization of the following data:

- All configuration data, such as information about neighbors, timer parameters, and process configurations.
- Dynamic data, such as the interface parameters and state, and information about neighbors and the link state database (LSDB).



For details about NSR, see "Uninterruptible Service Technology" in *Feature Description - Reliability*.

Usage Scenario

NSR minimizes the impact of control plane faults and prevents route flapping on networks that require high reliability.

Benefits

NSR improves network reliability and ensures uninterrupted traffic forwarding.

6 OSPF

About This Chapter

[6.1 Overview of OSPF](#)

[6.2 Understanding OSPF](#)

6.1 Overview of OSPF

Definition

Open Shortest Path First (OSPF) is a link-state Interior Gateway Protocol (IGP) developed by the Internet Engineering Task Force (IETF).

OSPF version 2 (OSPFv2) is intended for IPv4. OSPF version 3 (OSPFv3) is intended for IPv6.

NOTE

In this document, OSPF refers to OSPFv2, unless otherwise stated.

Purpose

Before the emergence of OSPF, the Routing Information Protocol (RIP) was widely used as an IGP on networks. RIP is a distance-vector routing protocol. Due to its slow convergence, routing loops, and poor scalability, RIP is gradually being replaced with OSPF.

Typical IGPs include RIP, OSPF, and Intermediate System to Intermediate System (IS-IS). [Table 6-1](#) describes differences among the three typical IGPs.

Table 6-1 Differences among RIP, OSPF, and IS-IS

Item	RIP	OSPF	IS-IS
Protocol type	IP layer protocol	IP layer protocol	Link layer protocol

Item	RIP	OSPF	IS-IS
Application scope	Applies to small networks with simple architectures, such as campus networks.	Applies to medium-sized networks with several hundred routers supported, such as enterprise networks.	Applies to large networks, such as Internet service provider (ISP) networks.
Routing algorithm	Uses a distance-vector algorithm and exchanges routing information over the User Datagram Protocol (UDP).	Uses the shortest path first (SPF) algorithm to generate a shortest path tree (SPT) based on the network topology, calculates shortest paths to all destinations, and exchanges routing information over IP.	Uses the SPF algorithm to generate an SPT based on the network topology, calculates shortest paths to all destinations, and exchanges routing information over IP. The SPF algorithm runs separately in Level-1 and Level-2 databases.
Route convergence speed	Slow	Less than 1 second	Less than 1 second
Scalability	Not supported	Supported by partitioning a network into areas	Supported by defining router levels

Benefits

OSPF offers the following benefits:

- Wide application scope: OSPF applies to medium-sized networks with several hundred routers, such as enterprise networks.
- Network masks: OSPF packets can carry masks, and therefore the packet length is not limited by natural IP masks. OSPF can process variable length subnet masks (VLSMs).
- Fast convergence: When the network topology changes, OSPF immediately sends link state update (LSU) packets to synchronize the changes to the link state databases (LSDBs) of all routers in the same autonomous system (AS).
- Loop-free routing: OSPF uses the SPF algorithm to calculate loop-free routes based on the collected link status.
- Area partitioning: OSPF allows an AS to be partitioned into areas, which simplifies management. Routing information transmitted between areas is summarized, which reduces network bandwidth consumption.
- Equal-cost routes: OSPF supports multiple equal-cost routes to the same destination.
- Hierarchical routing: OSPF uses intra-area routes, inter-area routes, Type 1 external routes, and Type 2 external routes, which are listed in descending order of priority.
- Authentication: OSPF supports area-based and interface-based packet authentication, which ensures packet exchange security.

- Multicast: OSPF uses multicast addresses to send packets on certain types of links, which minimizes the impact on other devices.

6.2 Understanding OSPF

6.2.1 Basic Concepts of OSPF

This section describes the basic Open Shortest Path First (OSPF) concepts.

Router ID

A router ID is a 32-bit unsigned integer, which identifies a router in an autonomous system (AS). A router ID must exist before the router runs OSPF.

A router ID can be manually configured or automatically obtained.

If no router ID has been configured, the router automatically obtains a router ID using the following methods in descending order of priority.

1. The router preferentially selects the largest IP address from its loopback interface addresses as the router ID.
2. If no loopback interface has been configured, the router selects the largest IP address from its interface IP addresses as the router ID.

A router can obtain a router ID again only after a router ID is reconfigured for the router or an OSPF router ID is reconfigured and the OSPF process restarts.

Area

When a large number of routers run OSPF, link state databases (LSDBs) become very large and require a large amount of storage space. Large LSDBs also complicate shortest path first (SPF) computation and overload the routers. As the network grows, the network topology changes, which results in route flapping and frequent OSPF packet transmission. When a large number of OSPF packets are transmitted, bandwidth usage efficiency decreases, and each router on a network has to recalculate routes in case of any topology change.

OSPF resolves this problem by partitioning an AS into different areas. An area is regarded as a logical group, and each group is identified by an area ID. A router, not a link, resides at the border of an area. A network segment or link can belong only to one area. An area must be specified for each OSPF interface.

OSPF areas include common areas, stub areas, and not-so-stubby areas (NSSAs). [Table 6-2](#) describes these OSPF areas.

Table 6-2 OSPF areas

Area Type	Function	Notes
Common area	By default, OSPF areas are defined as common areas. Common areas include: ● Standard area: transmits intra-area, inter-area, and external routes. ● Backbone area: connects to all other OSPF areas and transmits inter-area routes. The backbone area is area 0. Routes between non-backbone areas must be forwarded through the backbone area.	● The backbone area must have all its devices connected. ● All non-backbone areas must remain connected to the backbone area.
Stub area	A stub area is a non-backbone area with only one area border router (ABR) and generally resides at the border of an AS. The ABR in a stub area does not transmit received AS external routes, which significantly decreases the number of entries in the routing table on the ABR and the amount of routing information to be transmitted. To ensure the reachability of AS external routes, the ABR generates a default route and advertises the route to non-ABRs in the stub area. A totally stub area allows only intra-area routes and ABR-advertised Type 3 link state advertisements (LSAs) carrying a default route to be advertised within the area.	● The backbone area cannot be configured as a stub area. ● An autonomous system boundary router (ASBR) cannot exist in a stub area. Therefore, AS external routes cannot be advertised within the stub area. ● A virtual link cannot pass through a stub area.
NSSA	An NSSA is similar to a stub area. An NSSA does not advertise Type 5 LSAs but can import AS external routes. ASBRs in an NSSA generate Type 7 LSAs to carry the information about the AS external routes. The Type 7 LSAs are advertised only within the NSSA. When the Type 7 LSAs reach an ABR in the NSSA, the ABR translates the Type 7 LSAs into Type 5 LSAs and floods them to the entire AS. A totally NSSA allows only intra-area routes to be advertised within the area.	● An ABR in an NSSA advertises default routes in Type 7 LSAs within the NSSA. ● All inter-area routes are advertised by ABRs. ● A virtual link cannot pass through an NSSA.

Router Type

Routers are classified as internal routers, ABRs, backbone routers, or ASBRs by location in an AS. **Figure 6-1** shows the four router types.

Figure 6-1 Router type layout

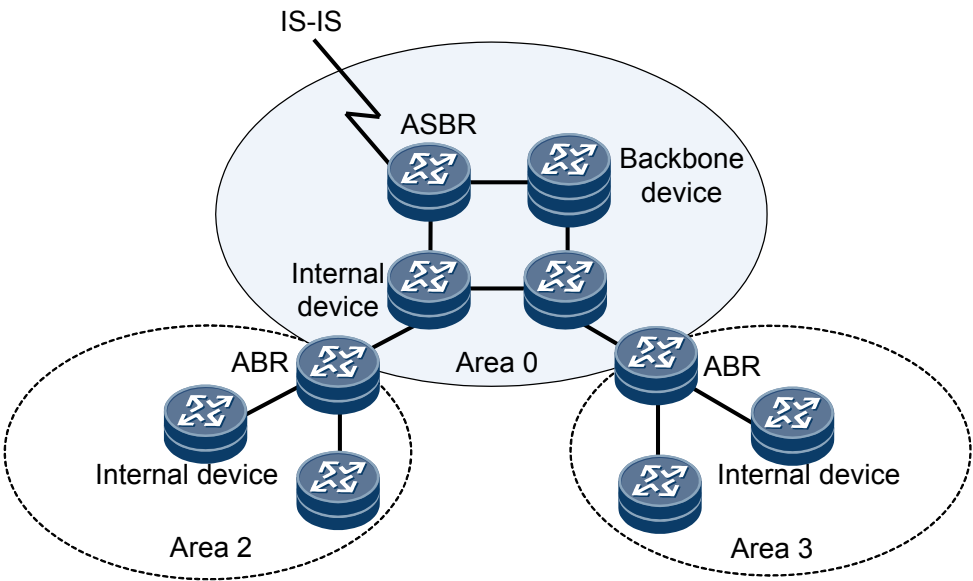


Table 6-3 describes the four router types.

Table 6-3 Router type descriptions

Router Type	Description
Internal router	All interfaces on an internal router belong to the same OSPF area.
ABR	An ABR connects the backbone area and non-backbone areas, and it can connect to the backbone area either physically or logically. An ABR can belong to two or more areas, one of which must be a backbone area .
Backbone router	Backbone routers include internal routers in the backbone area and all ABRs. At least one interface on a backbone router belongs to the backbone area.
ASBR	An ASBR exchanges routing information with other ASs. An ASBR may not reside at the boundary of an AS, and it can be an internal router or an ABR.

LSA

OSPF encapsulates routing information into LSAs for transmission. Table 6-4 describes LSAs and their functions.

Table 6-4 LSAs and their functions

LSA Type	Function
Router-LSA (Type 1)	Describes the link status and cost of a router. Router-LSAs are generated by a router and advertised within the area to which the router belongs.
Network-LSA (Type 2)	Describes the link status of all routers on the local network segment. Network-LSAs are generated by a designated router (DR) and advertised within the area to which the DR belongs.
Network-summary-LSA (Type 3)	Describes routes on a network segment of an area. Network-summary-LSAs are generated by an ABR and advertised to other areas, excluding the totally stub area and totally NSSA . For example, an ABR belongs to both area 0 and area 1, area 0 has a network segment 10.1.1.0, and area 1 has a network segment 11.1.1.0. In this case, for area 0, the ABR generates Type 3 LSAs destined for the network segment 11.1.1.0; for area 1, the ABR generates Type 3 LSAs destined for the network segment 10.1.1.0.
ASBR-summary-LSA (Type 4)	Describes routes of an area to the ASBRs of other areas. ASBR-summary-LSAs are generated by an ABR and advertised to other areas, excluding the stub area , totally stub area , NSSA , totally NSSA , and area to which the ASBR of the route generation ABR belongs.
AS-external-LSA (Type 5)	Describes AS external routes, which are advertised to all areas, excluding the stub area , totally stub area , NSSA , and totally NSSA . AS-external-LSAs are generated by an ASBR .
NSSA-LSA (Type 7)	Describes AS external routes. NSSA-LSAs are generated by an ASBR and advertised only within NSSAs.
Opaque-LSA (Type 9/Type 10/Type 11)	Provides a general mechanism for OSPF extension. Different types of LSAs are described as follows: ● Type 9 LSAs are advertised only on the network segment where the interface advertising the LSAs resides. Grace LSAs used in graceful restart (GR) are Type 9 LSAs. ● Type 10 LSAs are advertised within an OSPF area. LSAs that are used to support traffic engineering (TE) are Type 10 LSAs. ● Type 11 LSAs are advertised within an AS but have not been used in practice.

Table 6-5 describes whether a type of LSA is supported in an area.

Table 6-5 Support status of LSAs in different types of areas

Area Type	Router-LSA (Type 1)	Network-LSA (Type 2)	Network-summary-LSA (Type 3)	ASBR-summary-LSA (Type 4)	AS-external-LSA (Type 5)	NSSA-LSA (Type 7)
Common area (including standard and backbone areas)	Supported	Supported	Supported	Supported	Supported	Not supported
Stub area	Supported	Supported	Supported	Not supported	Not supported	Not supported
Totally stub area	Supported	Supported	Not supported	Not supported	Not supported	Not supported
NSSA	Supported	Supported	Supported	Not supported	Not supported	Supported
Totally NSSA	Supported	Supported	Not supported	Not supported	Not supported	Supported

Packet Type

OSPF packets are classified as Hello, Database Description (DD), Link State Request (LSR), Link State Update (LSU), or Link State Acknowledgment (LSAck) packets. [Table 6-6](#) describes OSPF packets and their functions.

Table 6-6 OSPF packets and their functions

Packet Type	Function
Hello packet	Hello packets are periodically sent to discover and maintain OSPF neighbor relationships.
DD packet	DD packets contain the summaries of LSAs in the local LSDB. DD packets are used for LSDB synchronization between two routers.
LSR packet	LSR packets are sent to OSPF neighbors to request required LSAs. A router sends LSR packets to its OSPF neighbor only after DD packets have been successfully exchanged.

Packet Type	Function
LSU packet	LSU packets are used to transmit required LSAs to OSPF neighbors.
LSAck packet	LSAck packets are used to acknowledge received LSAs.

Route Type

Route types are classified as intra-area, inter-area, Type 1 external, or Type 2 external routes. Intra-area and inter-area routes describe the network structure of an AS. Type 1 or Type 2 AS external routes describe how to select routes to destinations outside an AS.

Table 6-7 describes OSPF routes in descending order of priority.

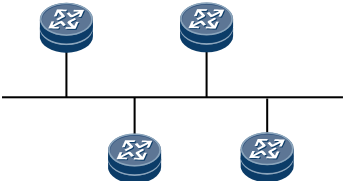
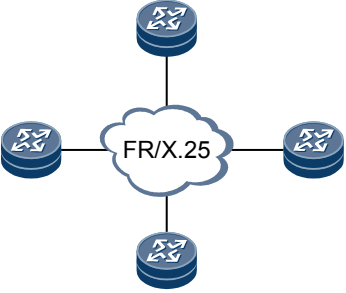
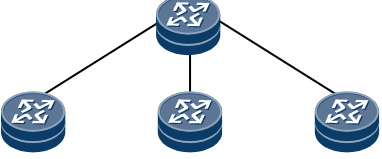
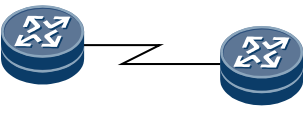
Table 6-7 OSPF routes

Route Type	Description
Intra-area route	-
Inter-area route	-
Type 1 external route	Type 1 external routes have high reliability. Cost of a Type 1 external route = Cost of the route from a router to an ASBR + Cost of the route from the ASBR to the destination When multiple ASBRs exist, the cost of each Type 1 external route equals the cost of the route from the local device to an ASBR plus the cost of the route from the ASBR to the destination. The cost is used for route selection.
Type 2 external route	Because a Type 2 external route has low reliability, its cost is considered to be much greater than the cost of any internal route to an ASBR. Cost of a Type 2 external route = Cost of the route from an ASBR to the destination If routes are imported by multiple ASBRs, the route with the smallest cost from the corresponding ASBR to its destination is selected. If the routes have the same cost from the corresponding ASBR to each route destination, the route with the smallest cost from the local router to the corresponding ASBR is selected. The cost of each Type 2 external route equals the cost of the route from the corresponding ASBR to the destination.

Network Type

Networks are classified as broadcast, non-broadcast multiple access (NBMA), point-to-multipoint (P2MP), or point-to-point (P2P) networks by link layer protocol. [Table 6-8](#) describes the network types.

Table 6-8 OSPF network classification

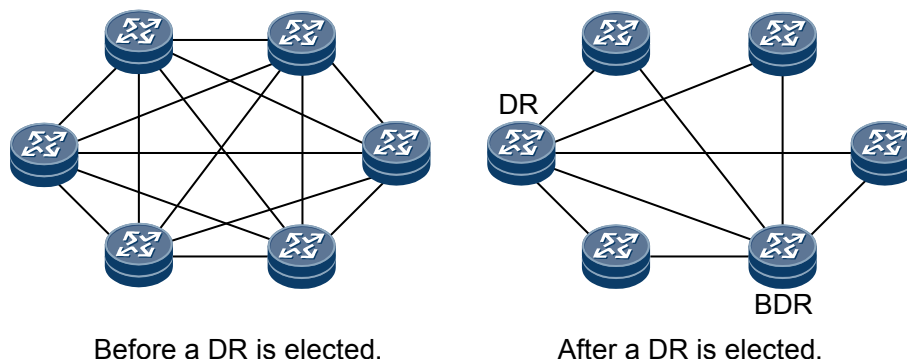
Network Type	Link Layer Protocol	Graph
Broadcast	- Ethernet - FDDI	
NBMA	X.25	
P2MP	Regardless of the link layer protocol, OSPF does not default the network type to P2MP. The network type must be manually changed to P2MP. In most cases, a non-fully connected NBMA network is changed to a P2MP network.	
P2P	- PPP - HDLC - LAPB	

DR and BDR

On broadcast or NBMA networks, any two routers need to exchange routing information. As shown in [Figure 6-2](#), n routers are deployed on the network. $n \times (n - 1)/2$ adjacencies must be established. Any route change on a router is transmitted to other routers, which wastes bandwidth resources. OSPF resolves this problem by defining a DR and a backup designated router (BDR). After a DR is elected, all routers send routing information only to the DR. Then the DR broadcasts LSAs. routers other than the DR and BDR are called DR others. The DR others establish only adjacencies with the DR and BDR and not with each other. This process

reduces the number of adjacencies established between routers on broadcast or NBMA networks.

Figure 6-2 Network topologies before and after a DR election



If the original DR fails, routers must reelect a DR and the routers except the new DR must synchronize routing information to the new DR. This process is lengthy, which may cause incorrect route calculations. A BDR is used to shorten the process. The BDR is a backup for a DR. A BDR is elected together with a DR. The BDR establishes adjacencies with all routers on the network segment and exchanges routing information with them. When the DR fails, the BDR immediately becomes a new DR. The routers need to reelect a new BDR, but this process does not affect route calculations.

The DR priority of a router interface determines its qualification for DR and BDR elections. The router interfaces with their DR priorities greater than 0 are eligible. Each router adds the elected DR to a Hello packet and sends it to other routers on the network segment. When both router interfaces on the same network segment declare that they are DRs, the router interface with a higher DR priority is elected as a DR. If the two router interfaces have the same DR priority, the router interface with a larger router ID is elected as a DR.

OSPF Multi-Process

OSPF multi-process allows multiple OSPF processes to independently run on the same router. Route exchange between different OSPF processes is similar to that between different routing protocols. A router's interface can belong only to one OSPF process.

OSPF multi-process is typically used on virtual private networks (VPNs) on which OSPF is deployed between provider edges (PEs) and customer edges (CEs). The OSPF processes on the PEs are independent of each other.

OSPF Default Route

A default route is the route whose destination address and mask are both all 0s. When no matching route is discovered, a router uses a default route to forward packets.

A default route generally applies to the following scenarios:

- An ABR in an area advertises Type 3 LSAs carrying a default route within the area. The routers in the area use the received default route to forward inter-area packets.
- An ASBR in an AS advertises Type 5 or Type 7 LSAs carrying a default route within the AS. The routers in the AS use the received default route to forward AS external packets.

OSPF routes are hierarchically managed. The priority of the default route carried in Type 3 LSAs is higher than the priority of the default route carried in Type 5 or Type 7 LSAs.

A router advertises LSAs carrying a default route by adhering to the following principles:

- A router in an area can advertise LSAs carrying a default route only when the router has an interface connected to a device outside the area.
- If a router has advertised LSAs carrying a default route, the router no longer learns the same type of LSA advertised by other routers, which carry a default route. That is, the router uses only the LSAs advertised by itself to calculate routes. The LSAs advertised by other routers are still saved in the LSDB.
- If a router must use a route to advertise LSAs carrying an external default route, the route cannot be a route learned by the local OSPF process. A router in an area uses an external default route to forward packets outside the area. If the next hops of routes in the area are routers in the area, packets cannot be forwarded outside the area.
- Before a router advertises a default route, it checks whether a neighbor in the full state is present in area 0. The router advertises a default route only when a neighbor in the full state is present in area 0. If no such a neighbor exists, the backbone area cannot forward packets and advertising a default route is meaningless. For the concept of the Full State, see [OSPF Neighbor States](#).

Table 6-9 describes the principles for advertising default routes in different areas.

Table 6-9 Principles for advertising default routes in different areas

Area Type	Advertisement Principles
Common area	By default, a router in a common area does not generate a default route. After being configured to do so, an ASBR generates a Type 5 LSA carrying a default route. The router then advertises the default route in the entire AS. If no default route is generated on the ASBR, the router does not advertise a default route.
Stub area	Type 5 LSAs cannot be advertised within a stub area. A router in the stub area must learn AS external routes from an ABR. The ABR automatically generates a Type 3 LSA carrying a default route and advertises it within the entire stub area. Then the router can learn AS external routes from the ABR.
Totally stub area	Neither Type 3 (except default Type 3 LSAs) nor Type 5 LSAs can be advertised within a totally stub area. A router in the totally stub area must learn AS external and inter-area routes from an ABR. After you configure a totally stub area, an ABR automatically generates a Type 3 LSA carrying a default route and advertises it within the entire totally stub area. Then the router can learn AS external and inter-area routes from the ABR.

Area Type	Advertisement Principles
NSSA	A small number of AS external routes learned from the ASBR in an NSSA can be imported to the NSSA. Type 5 LSAs cannot be advertised within the NSSA. When at least a neighbor in Full status and an interface that is Up exist in the backbone area, the ABR automatically generates a Type 7 LSA carrying a default route and advertises it within the entire NSSA. A small number of AS external routes can be learned from the ASBR in the NSSA, and other inter-area routes can be learned from the ABR in the NSSA. Manual configurations must be performed on the ASBR to enable the ASBR to generate a Type 7 LSA carrying a default route and advertise the LSA within the entire NSSA. An ABR does not translate Type 7 LSAs carrying a default route into Type 5 LSAs carrying a default route or flood them to the entire AS.
Totally NSSA	Neither Type 3 (except default Type 3 LSAs) nor Type 5 LSAs can be advertised within a totally NSSA. A router in the totally NSSA must learn AS external routes from an ABR. The ABR automatically generates Type 3 and Type 7 LSAs carrying a default route and advertises them to the entire totally NSSA. Then AS external and inter-area routes can be advertised within the totally NSSA.

6.2.2 OSPF Fundamentals

OSPF route calculation involves the following processes:

1. Adjacency establishment

The adjacency establishment process is as follows:

- The local and remote routers use OSPF interfaces to exchange Hello packets to establish a **neighbor relationship**.
- The local and remote routers negotiate a master/slave relationship and exchange Database Description (DD) packets.
- The local and remote routers exchange link state advertisements (LSAs) to synchronize their link state databases (LSDBs).

2. Route calculation

OSPF uses the shortest path first (SPF) algorithm to calculate routes to implement fast route convergence.

OSPF Neighbor States

To exchange routing information on an OSPF network, neighbor routers must establish adjacencies. The differences between neighbor relationships and adjacencies are described as follows:

- Neighbor relationship: After the local router starts, it uses an OSPF interface to send a Hello packet to the remote router. After the remote router receives the packet, it checks whether the parameters carried in the packet are consistent with its own parameters. If the parameters carried in the packet are consistent with its own parameters, the remote router establishes a neighbor relationship with the local router.

- Adjacency: After the local and remote routers establish a neighbor relationship, they exchange DD packets and LSAs to establish an adjacency.

OSPF has eight neighbor states: Down, Attempt, Init, 2-way, Exstart, Exchange, Loading, and Full. Down, 2-way, and Full are stable states. Attempt, Init, Exstart, Exchange, and Loading are unstable states, which last only several minutes. [Figure 6-3](#) shows the eight neighbor states.

Figure 6-3 OSPF neighbor states

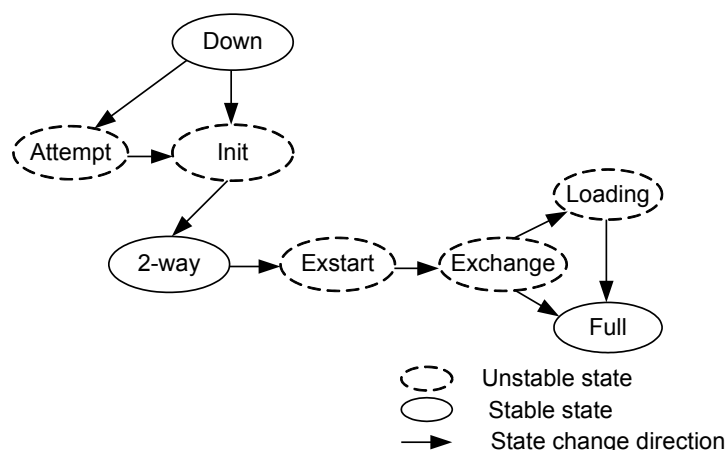


Table 6-10 OSPF neighbor states and their meanings

OSPF Neighbor State	Meaning
Down	This is the initial state of a neighbor conversation. This state indicates that a router has not received any Hello packets from its neighbors within a dead interval.
Attempt	In the Attempt state, a router periodically sends Hello packets to manually configured neighbors. NOTE This state applies only to non-broadcast multiple access (NBMA) interfaces.
Init	This state indicates that a router has received Hello packets from its neighbors but the neighbors did not receive Hello packets from the router.
2-way	This state indicates that a router has received Hello packets from its neighbors and neighbor relationships have been established between the routers. If no adjacency needs to be established, the neighbors remain in the 2-way state. If adjacencies need to be established, the neighbors enter the Exstart state.
Exstart	In the Exstart state, routers establish a master/slave relationship to ensure that DD packets are sequentially exchanged.
Exchange	In the Exchange state, routers exchange DD packets. A router uses a DD packet to describe its own LSDB and sends the packet to its neighbors.

OSPF Neighbor State	Meaning
Loading	In the Loading state, a router sends Link State Request (LSR) packets to its neighbors to request their LSAs for LSDB synchronization.
Full	In the Full state, a router establishes adjacencies with its neighbors and all LSDBs have been synchronized.

NOTE

The neighbor state of the local router may be different from that of the remote router. For example, the neighbor state of the local router is Full, but the neighbor state of the remote router is Loading.

Adjacency Establishment

Adjacencies can be established in either of the following situations:

- Two routers have established a neighbor relationship and communicate for the first time.
- The designated router (DR) or backup designated router (BDR) on a network segment changes.

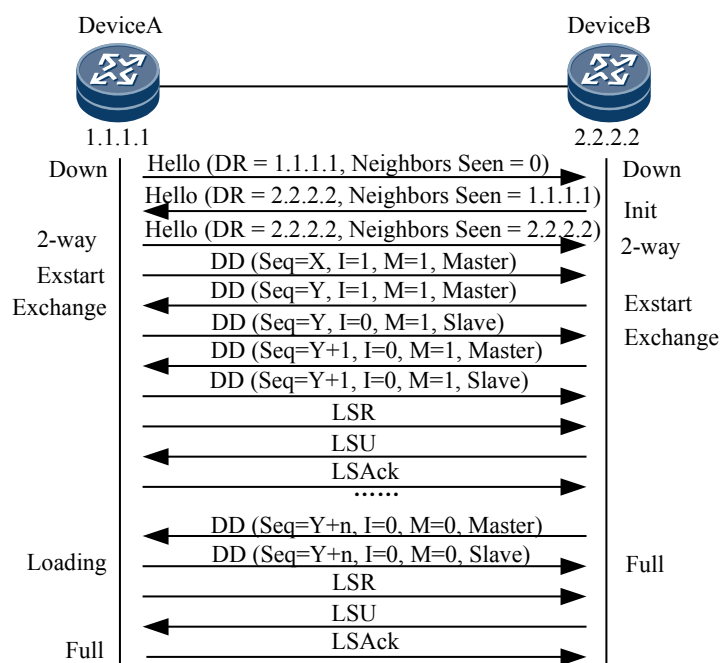
The adjacency establishment process is different on different networks.

Adjacency establishment on a broadcast network

On a broadcast network, the DR and BDR establish adjacencies with each router on the same network segment, but DR others establish only neighbor relationships.

Figure 6-4 shows the adjacency establishment process on a broadcast network.

Figure 6-4 Adjacency establishment process on a broadcast network



The adjacency establishment process on a broadcast network is as follows:

1. Neighbor relationship establishment

- a. Router A uses the multicast address 224.0.0.5 to send a Hello packet to Router B through the OSPF interface connected to a broadcast network. The packet carries the DR field of 1.1.1.1 (ID of Router A) and the Neighbors Seen field of 0. A neighbor router has not been discovered, and Router A regards itself as a DR.
- b. After Router B receives the packet, it returns a Hello packet to Router A. The returned packet carries the DR field of 2.2.2.2 (ID of Router B) and the Neighbors Seen field of 1.1.1.1 (Router A's router ID). Router A has been discovered but its router ID is less than that of Router B, and therefore Router B regards itself as a DR. Then Router B's state changes to Init.
- c. After Router A receives the packet, Router A's state changes to 2-way.

 NOTE

The following procedures are not performed for DR others on a broadcast network.

2. Master/Slave negotiation and DD packet exchange

- a. Router A sends a DD packet to Router B. The packet carries the following fields:
 - Seq field: The value x indicates the sequence number is x .
 - I field: The value 1 indicates that the packet is the first DD packet, which is used to negotiate a master/slave relationship and does not carry LSA summaries.
 - M field: The value 1 indicates that the packet is not the last DD packet.
 - MS field: The value 1 indicates that Router A declares itself a master.

To improve transmission efficiency, Router A and Router B determine which LSAs in each other's LSDB need to be updated. If one party determines that an LSA of the other party is already in its own LSDB, it does not send an LSR packet for updating the LSA to the other party. To achieve the preceding purpose, Router A and Router B first send DD packets, which carry summaries of LSAs in their own LSDBs. Each summary identifies an LSA. To ensure packet transmission reliability, a master/slave relationship must be determined during DD packet exchange. One party serving as a master uses the Seq field to define a sequence number. The master increases the sequence number by one each time it sends a DD packet. When the other party serving as a slave sends a DD packet, it adds the sequence number carried in the last DD packet received from the master to the Seq field of the packet.

- b. After Router B receives the DD packet, Router B's state changes to Exstart and Router B returns a DD packet to Router A. The returned packet does not carry LSA summaries. Because Router B's router ID is greater than Router A's router ID, Router B declares itself a master and sets the Seq field to y .
- c. After Router A receives the DD packet, it agrees that Router B is a master and Router A's state changes to Exchange. Then Router A sends a DD packet to Router B to transmit LSA summaries. The packet carries the Seq field of y and the MS field of 0. The value 0 indicates that Router A declares itself a slave.
- d. After Router B receives the packet, Router B's state changes to Exchange and Router B sends a new DD packet containing its own LSA summaries to Router A. The value of the Seq field carried in the new DD packet is changed to $y + 1$.

Router A uses the same sequence number as Router B to confirm that it has received DD packets from Router B. Router B uses the sequence number plus one to confirm that it has received DD packets from Router A. When Router B sends the last DD packet, it sets the M field of the packet to 0.

3. LSDB synchronization

- a. After Router A receives the last DD packet, it finds that many LSAs in Router B's LSDB do not exist in its own LSDB, so Router A's state changes to Loading. After Router B receives the last DD packet from Router A, Router B's state directly changes to Full, because Router B's LSDB already contains all LSAs of Router A.
- b. Router A sends an LSR packet for updating LSAs to Router B. Router B returns an LSU packet to Router A. After Router A receives the packet, it sends an LSack packet for acknowledgment.

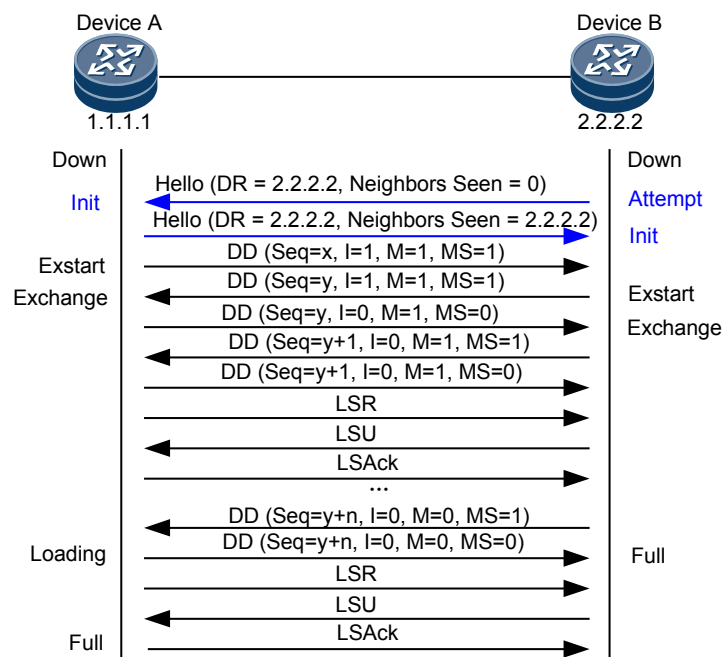
The preceding procedures continue until the LSAs in Router A's LSDB are the same as those in Router B's LSDB. Router A's state changes to Full. After Router A and Router B exchange DD packets and update all LSAs, they establish an adjacency.

Adjacency establishment on an NBMA network

The adjacency establishment process on an NBMA network is similar to that on a broadcast network. The blue part shown in [Figure 6-5](#) highlights the differences from a broadcast network.

On an NBMA network, all routers establish adjacencies only with the DR and BDR.

Figure 6-5 Adjacency establishment process on an NBMA network



The adjacency establishment process on an NBMA network is as follows:

1. Neighbor relationship establishment

- a. After Router B sends a Hello packet to a Down interface of Router A, Router B's state changes to Attempt. The packet carries the DR field of 2.2.2.2 (ID of Router B) and the Neighbors Seen field of 0. A neighbor router has not been discovered, and Router B regards itself as a DR.
- b. After Router A receives the packet, Router A's state changes to Init and Router A returns a Hello packet. The returned packet carries the DR and Neighbors Seen

fields of 2.2.2.2. Router B has been discovered but its router ID is greater than that of Router A, and therefore Router A agrees that Router B is a DR.

NOTE

The following procedures are not performed for DR others on an NBMA network.

2. Master/Slave relationship negotiation and DD packet exchange

The procedures for negotiating a master/slave relationship and exchanging DD packets on an NBMA network are the same as those on a broadcast network.

3. LSDB synchronization

The procedure for synchronizing LSDBs on an NBMA network is the same as that on a broadcast network.

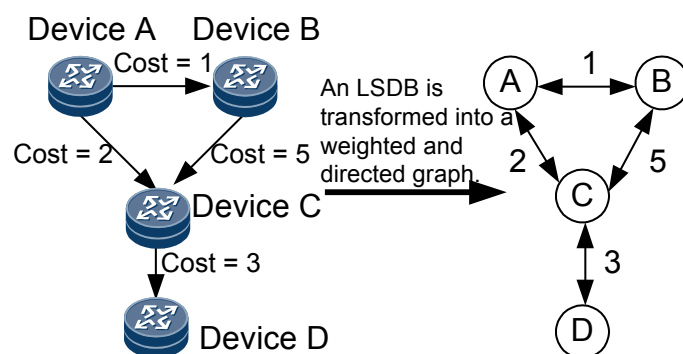
Adjacency establishment on a point-to-point (P2P)/Point-to-multipoint (P2MP) network

The adjacency establishment process on a P2P/P2MP network is similar to that on a broadcast network. On a P2P/P2MP network, however, no DR or BDR needs to be elected and DD packets are transmitted in multicast mode.

Route Calculation

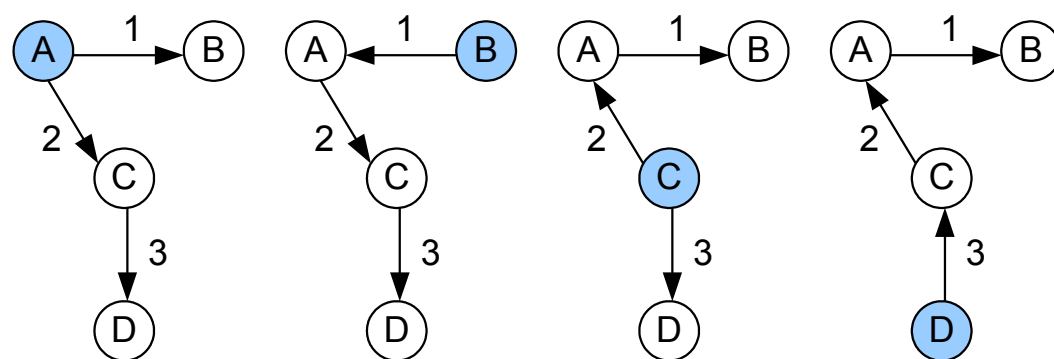
OSPF uses an LSA to describe the network topology. A Type 1 LSA describes the attributes of a link between routers. A router transforms its LSDB into a weighted, directed graph, which reflects the topology of the entire AS. All routers in the same area have the same graph. [Figure 6-6](#) shows a weighted, directed graph.

Figure 6-6 Weighted, directed graph



Based on the graph, each router uses the SPF algorithm to calculate an SPT with itself as the root. The SPT shows routes to nodes in the AS. [Figure 6-7](#) shows an SPT.

Figure 6-7 SPT



When a router's LSDB changes, the router recalculates a shortest path. Frequent SPF calculations consume a large amount of resources and affect router efficiency. Changing the interval between SPF calculations can prevent resource consumption caused by frequent LSDB changes. The default interval between SPF calculations is 5 seconds.

The route calculation process is as follows:

1. A router calculates intra-area routes.

The router uses an SPF algorithm to calculate shortest paths to other routers in an area. Type 1 and Type 2 LSAs accurately describe the network topology in an area. Based on the network topology described by a Type 1 LSA, the router calculates paths to other routers in the area.

 NOTE

If multiple equal-cost routes are produced during route calculation, the SPF algorithm retains all these routes in the LSDB.

2. The router calculates inter-area routes.

For the routers in an area, the network segment of the routes in an adjacent area is directly connected to the area border router (ABR). Because the shortest path to the ABR has been calculated in the preceding step, the routers can directly check a Type 3 LSA to obtain the shortest path to the network segment. The autonomous system boundary router (ASBR) can also be considered connected to the ABR. Therefore, the shortest path to the ASBR can also be calculated in this phase.

 NOTE

If the router performing an SPF calculation is an ABR, the router needs to check only Type 3 LSAs in the backbone area.

3. The router calculates AS external routes.

AS external routes can be considered to be directly connected to the ASBR. Because the shortest path to the ASBR has been calculated in the preceding phase, the router can check Type 5 LSAs to obtain the shortest paths to other ASs.

6.2.3 OSPF Route Control

You can use the following features to control the advertising and receiving of OSPF routes:

- **Route summarization**

Route summarization enables a router to summarize routes with the same prefix into a single route and to advertise only the summarized route to other areas. Route summarization reduces the size of a routing table and improves router performance.

- **Route filtering**

OSPF can use routing policies to filter routes. By default, OSPF does not filter routes.

- **OSPF Database Overflow**

Set the maximum number of external routes supported by the LSDB to dynamically limit the LSDB's size.

These features meet requirements for network planning and traffic management.

Route Summarization

When a large OSPF network is deployed, an OSPF routing table includes a large number of routing entries. To accelerate route lookup and simplify management, configure route

summarization to reduce the size of the OSPF routing table. If a link frequently alternates between Up and Down, the links not involved in the route summarization are not affected. This process prevents route flapping and improves network stability.

Route summarization can be carried out on an ABR or ASBR.

- ABR summarization

When an ABR transmits routing information to other areas, it generates Type 3 LSAs for each network segment. If consecutive network segments exist in this area, you can summarize these network segments into a single network segment. The ABR generates one LSA for the summarized network segment and advertises only that LSA.

- ASBR summarization

If route summarization has been configured and the local router is an ASBR, the local router summarizes imported Type 5 LSAs within the summarized address range. If an NSSA has been configured, the local router also summarizes imported Type 7 LSAs within the summarized address range.

If the local router is both an ASBR and an ABR, it summarizes Type 5 LSAs translated from Type 7 LSAs.

Route Filtering

OSPF routing policies include access control lists (ACLs), IP prefix lists, and route-policies. For details about these policies, see the section "Routing Policy" in the *NE40E Feature Description - IP Routing*.

OSPF route filtering applies in the following aspects:

- Route import

OSPF can import the routes learned by other routing protocols. A router uses a configured routing policy to filter routes and imports only the routes matching the routing policy. Only an ASBR can import routes, and therefore a routing policy for importing routes must be configured on the ASBR.

- Advertising of imported routes

A router advertises imported routes to its neighbors. Only an ASBR can import routes, and therefore a routing policy for the advertising of imported routes must be configured on the ASBR.

If OSPF imports a large number of external routes and advertises them to a device with a smaller routing table capacity, the device may restart unexpectedly. To address this problem, configure a limit on the number of LSAs generated when an OSPF process imports external routes.

- Route learning

A router uses a routing policy to filter received intra-area, inter-area, and AS external routes. The router adds only the routes matching the routing policy to its routing table. All routes can still be advertised from an OSPF routing table.

The router filters only routes calculated based on LSAs, and therefore learned LSAs are complete.

- Inter-area LSA learning

An ABR in an area can be configured to filter Type 3 LSAs advertised to the area. The ABR can advertise only Type 3 LSAs, and therefore a routing policy for inter-area LSA learning must be configured on the ABR.

During inter-area LSA learning, the ABR directly filters Type 3 LSAs advertised to the area.

- Inter-area LSA advertising

An ABR in an area can be configured to filter Type 3 LSAs advertised to other areas. The ABR can advertise only Type 3 LSAs, and therefore a routing policy for inter-area LSA advertising must be configured on the ABR.

OSPF Database Overflow

OSPF requires that devices in the same area have the same LSDB. As the number of routes increase continually, some devices cannot carry excess routing information due to limited system resources. This situation is called an OSPF database overflow.

You can configure stub areas or NSSAs to prevent resource exhaustion caused by continually increasing routing information. However, configuring stub areas or NSSAs cannot prevent an OSPF database overflow caused by a sharp increase in dynamic routes. To resolve this issue, set the maximum number of external routes supported by the LSDB to dynamically limit the LSDB's size.

NOTE

The maximum number of external routes configured for all devices in the OSPF AS must be the same.

When the number of external routes in the LSDB reaches the maximum number, the device enters the overload state and starts the overflow timer at the same time. The device automatically exits from the overflow state after the overflow timer expires. [Table 6-11](#) describes the operations performed by the device after it enters or exits from the overload state.

Table 6-11 Operations performed by the device after it enters or exits from the overload state

Phase	OSPF Processing Procedure
Staying at overload state	Deletes self-generated non-default external routes and stops advertising non-default external routes. Discards newly received non-default external routes and does not reply with a Link State Acknowledgment (LSAck) packet. Checks whether the number of external routes is still greater than the configured maximum number when the overflow timer expires. ● Restarts the timer if the number of external routes is greater than or equal to the configured maximum number. ● Exits from the overflow state if the number of external routes is less than the configured maximum number.
Exiting from the overflow state	Ends the overflow timer. Advertises non-default external routes. Accepts newly received non-default external routes and replies with LSAck packets.

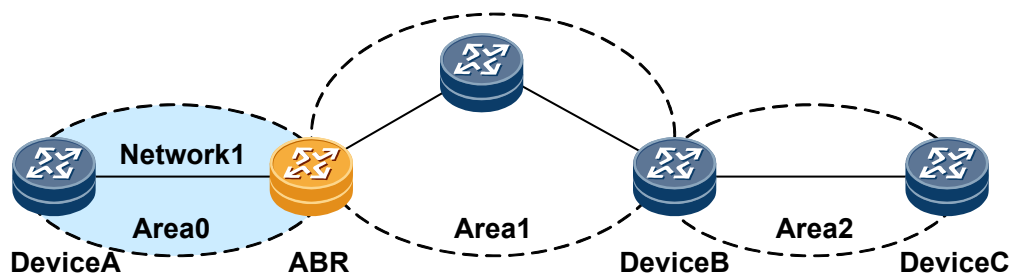
6.2.4 OSPF Virtual Link

Background

All non-backbone areas must be connected to the backbone area during OSPF deployment to ensure that all areas are reachable.

In [Figure 6-8](#), area 2 is not connected to area 0 (backbone area), and Device B is not an ABR. Therefore, Device B does not generate routing information about network 1 in area 0, and Device C does not have a route to network 1.

Figure 6-8 Non-backbone area not connected to the backbone area



Some non-backbone areas may not be connected to the backbone area. You can configure an OSPF virtual link to resolve this issue.

Related Concepts

A virtual link refers to a logical channel established between two ABRs over a non-backbone area.

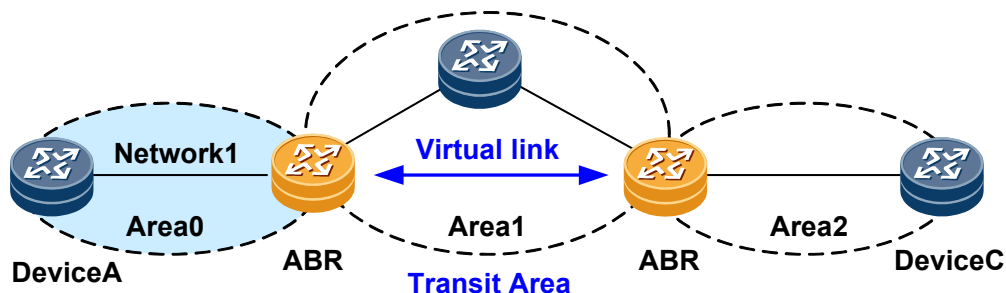
- A virtual link must be configured at both ends of the link.
- The non-backbone area involved is called a transit area.

A virtual link is similar to a point-to-point (P2P) connection established between two ABRs. You can configure interface parameters, such as the interval at which Hello packets are sent, at both ends of the virtual link as you do on physical interfaces.

Principles

In [Figure 6-9](#), two ABRs use a virtual link to directly transmit OSPF packets. The device between the two ABRs only forwards packets. Because the destination of OSPF packets is not the device, the device transparently transmits the OSPF packets as common IP packets.

Figure 6-9 OSPF virtual link

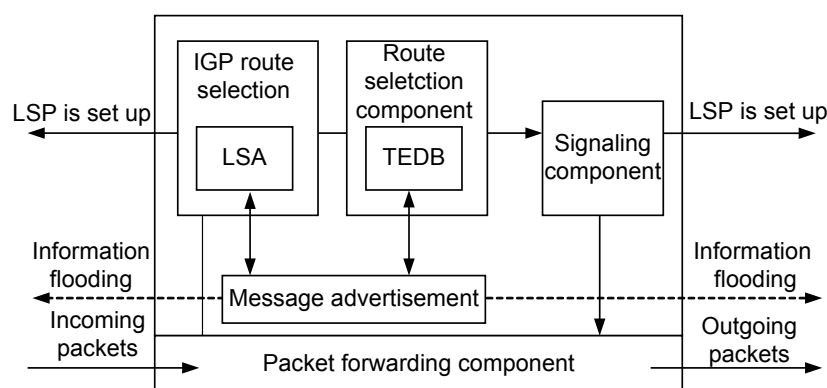


6.2.5 OSPF TE

OSPF Traffic Engineering (TE) is developed based on OSPF to support Multiprotocol Label Switching (MPLS) TE and establish and maintain TE LSPs. In the MPLS TE architecture described in "MPLS Feature Description", OSPF functions as the information advertising component, responsible for collecting and advertising MPLS TE information.

In addition to the network topology, TE needs to know network constraints, such as the bandwidth, TE metric, administrative group, and affinity attribute. However, current OSPF functions cannot meet these requirements. Therefore, OSPF introduces a new type of LSAs to advertise network constraints. Based on the network constraints, the Constraint Shortest Path First (CSPF) algorithm can calculate the path subject to specified constraints.

Figure 6-10 Overview of OSPF in the MPLS TE architecture



OSPF in the MPLS TE Architecture

In the MPLS TE architecture, OSPF functions as the information advertising component:

- Collects related information about TE.
- Floods TE information to devices in the same area.
- Uses the collected TE information to form the TE database (TEDB) so that CSPF can calculate routes.

OSPF does not care about information content or how MPLS uses the information.

TE-LSA

OSPF uses a new type of LSA (Type 10 opaque LSA) to collect and advertise TE information. Type 10 opaque LSAs contain the link status information required by TE, including the maximum link bandwidth, maximum reservable bandwidth, current reserved bandwidth, and link color. Based on the OSPF flooding mechanism, Type 10 opaque LSAs synchronize link status information among devices in an area to form a uniform TEDB for route calculation.

Interaction Between OSPF TE and CSPF

OSPF uses Type 10 LSAs to collect TE information in an area, such as the bandwidth, priority, and link metrics. After processing the collected TE information, OSPF provides it for CSPF to calculate routes.

IGP Shortcut and Forwarding Adjacency

OSPF supports IGP shortcut and forwarding adjacency. The two features allow OSPF to use a tunnel interface as an outbound interface to reach a destination.

Differences between IGP shortcut and forwarding adjacency are as follows:

- An IGP shortcut-enabled device uses a tunnel interface as an outbound interface but does not advertise the tunnel interface to neighbors. Therefore, other devices cannot use this tunnel.
- A forwarding adjacency-enabled device uses a tunnel interface as an outbound interface and advertises the tunnel interface to neighbors. Therefore, other devices can use this tunnel.
- IGP shortcut is unidirectional and needs to be configured only on the device that uses IGP shortcut.

OSPF SRLG

OSPF supports the applications of the Shared Risk Link Group (SRLG) in MPLS by obtaining information about the TE SRLG flooded among devices in an area. For details, refer to the chapter "MPLS" in this manual.

OSPF TE Tunnel Anti-Microloop

If a network fault occurs, IGP convergence is triggered. In this case, a transient forwarding status inconsistency may occur among nodes because of their different convergence rates, which poses the risk of microloops. If the outbound interface of a route before IGP convergence is a shortcut TE tunnel interface and the OSPF TE tunnel anti-microloop function is enabled, the outbound interface of the route remains unchanged during the IGP convergence. In this case, traffic is forwarded through a hot standby TE tunnel, and the forwarding process does not depend on IGP convergence on each device, preventing microloops.

6.2.6 OSPF VPN

Definition

As an extension of OSPF, OSPF VPN enables Provider Edges (PEs) and Customer Edges (CEs) in VPNs to run OSPF for interworking and use OSPF to learn and advertise routes.

Purpose

As a widely used IGP, in most cases, OSPF runs in VPNs. If OSPF runs between PEs and CEs, and PEs use OSPF to advertise VPN routes to CEs, no other routing protocols need to be configured on CEs for interworking with PEs, which simplifies management and configuration of CEs.

Running OSPF Between PEs and CEs

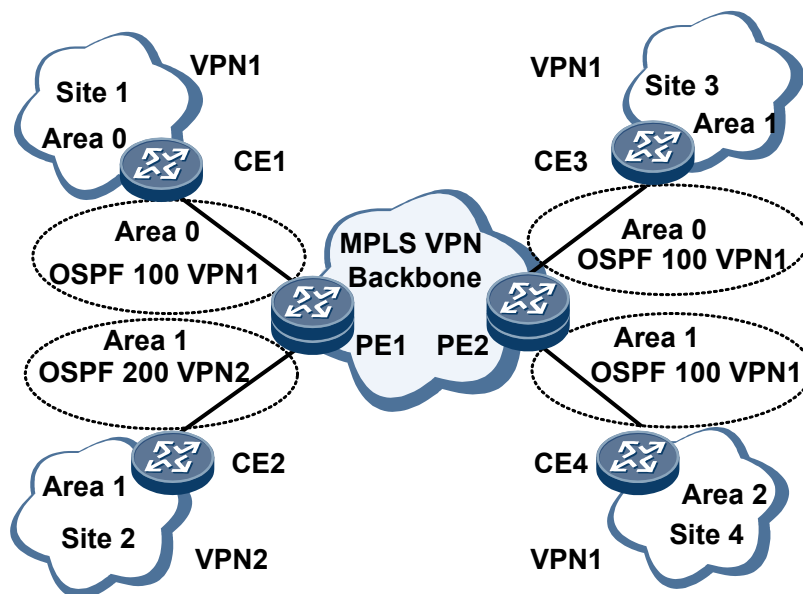
In BGP/MPLS VPN, Multi-Protocol BGP (MP-BGP) is used to transmit routing information between PEs, whereas OSPF is used to learn and advertise routes between PEs and CEs.

Running OSPF between PEs and CEs has the following benefits:

- OSPF is used in a site to learn routes. Running OSPF between PEs and CEs can reduce the number of the protocol types that CEs must support.
- Similarly, running OSPF both in a site and between PEs and CEs simplifies the work of network administrators and reduces the number of protocols that network administrators must be familiar with.
- When a network using OSPF but not VPN on the backbone network begins to use BGP/MPLS VPN, running OSPF between PEs and CEs facilitates the transition.

In **Figure 6-11**, CE1, CE3, and CE4 belong to VPN 1, and the numbers following OSPF refer to the process IDs of the multiple OSPF instances running on PEs.

Figure 6-11 Networking with OSPF running between PEs and CEs



The routes that PE1 receives from CE1 are advertised to CE3 and CE4 as follows:

1. PE1 imports OSPF routes of CE1 into BGP and converts them to BGP VPNv4 routes.
2. PE1 uses MP-BGP to advertise the BGP VPNv4 routes to PE2.
3. PE2 imports the BGP VPNv4 routes into OSPF and then advertises these routes to CE3 and CE4.

The process of advertising routes of CE4 or CE3 to CE1 is the same as the preceding process.

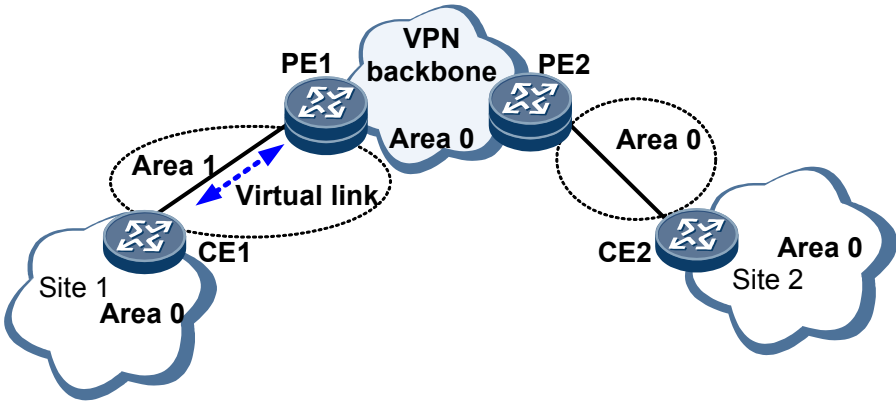
Configuring OSPF Areas Between PEs and CEs

OSPF areas between PEs and CEs can be non-backbone or backbone areas (Area 0). PEs can function only as ABRs.

In the extended application of OSPF VPN, the MPLS VPN backbone network serves as Area 0. OSPF requires that Area 0 be contiguous. Therefore, Area 0 of all VPN sites must be connected to the MPLS VPN backbone network. If a VPN site has OSPF Area 0, the PEs that CEs access must be connected to the backbone area of this VPN site through Area 0. If no physical link is available to directly connect PEs to the backbone area, a virtual link can be

deployed between the PEs and the backbone area. **Figure 6-12** shows the networking for configuring OSPF areas between PEs and CEs.

Figure 6-12 Configuring OSPF areas between PEs and CEs



A non-backbone area (Area 1) is configured between PE1 and CE1, and a backbone area (Area 0) is configured in Site 1. The backbone area in Site 1 is separated from the VPN backbone area. To ensure that the backbone areas are contiguous, a virtual link is configured between PE1 and CE1.

OSPF Domain ID

If inter-area routes are advertised between local and remote OSPF areas, these areas are considered to be in the same OSPF domain.

- Domain IDs identify domains.
- Each OSPF domain has one or more domain IDs. If more than one domain ID is available, one of the domain IDs is a primary ID, and the others are secondary IDs.
- If an OSPF instance does not have a specific domain ID, its ID is considered as null.

Before advertising the remote routes sent by BGP to CEs, PEs need to determine the type of OSPF routes (Type 3, Type 5, or Type 7) to be advertised to CEs based on domain IDs.

- If local domain IDs are the same as or compatible with remote domain IDs in BGP routes, PEs advertise Type 3 routes.
- If local domain IDs are different from or incompatible with remote domain IDs in BGP routes, PEs advertise Type 5 or Type 7 routes.

Table 6-12 Domain ID relationships and corresponding generated routes

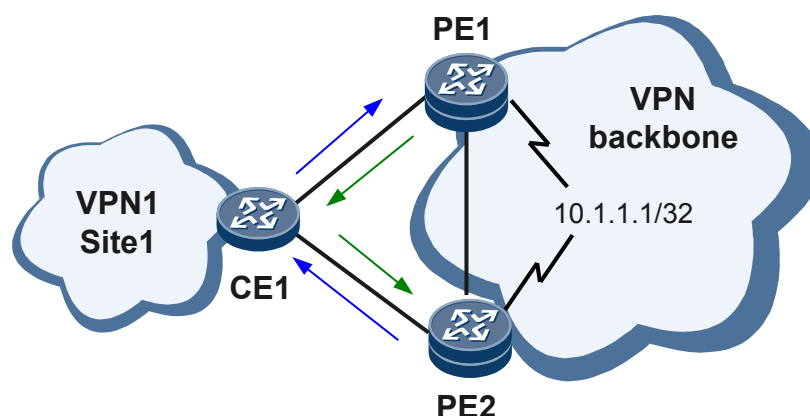
Relationship Between Local and Remote Domain IDs	Comparison Between Local and Remote Domain IDs	Type of the Generated Routes
Both the local and remote domain IDs are null.	Equal	Inter-area routes

Relationship Between Local and Remote Domain IDs	Comparison Between Local and Remote Domain IDs	Type of the Generated Routes
The remote domain ID is the same as the local primary domain ID or one of the local secondary domain IDs.	Equal	Inter-area routes
The remote domain ID is different from the local primary domain ID or any of the local secondary domain IDs.	Not equal	If the local area is a non-NSSA, external routes are generated. If the local area is an NSSA, NSSA routes are generated.

Routing Loop Prevention

Routing loops may occur between PEs and CEs when OSPF and BGP learn routes from each other.

Figure 6-13 Networking for OSPF VPN routing loops



In [Figure 6-13](#), on PE1, OSPF imports a BGP route destined for 10.1.1.1/32 and then generates and advertises a Type 5 or Type 7 LSA to CE1. Then, CE1 learns an OSPF route with 10.1.1.1/32 as the destination address and PE1 as the next hop and advertises the route to PE2. Therefore, PE2 learns an OSPF route with 10.1.1.1/32 as the destination address and CE1 as the next hop.

Similarly, CE1 also learns an OSPF route with 10.1.1.1/32 as the destination address and PE2 as the next hop. PE1 learns an OSPF route with 10.1.1.1/32 as the destination address and CE1 as the next hop.

As a result, CE1 has two equal-cost routes with PE1 and PE2 as next hops respectively, and the next hop of the routes from PE1 and PE2 to 10.1.1.1/32 is CE1, which leads to a routing loop.

In addition, the priority of an OSPF route is higher than that of a BGP route. Therefore, on PE1 and PE2, BGP routes to 10.1.1.1/32 are replaced with the OSPF route, and the OSPF

route with 10.1.1.1/32 as the destination address and CE1 as the next hop is active in the routing tables of PE1 and PE2.

The BGP route is inactive, and therefore, the LSA generated when this route is imported by OSPF is deleted, which causes the OSPF route to be withdrawn. As a result, no OSPF route exists in the routing table, and the BGP route becomes active again. This cycle causes route flapping.

OSPF VPN provides a few solutions to routing loops, as described in [Table 6-13](#).

Table 6-13 Routing loop prevention measures

Feature	Definition	Function
DN-bit	It is a flag bit used by OSPF multi-instance processes to prevent routing loops.	After OSPF multi-instance is configured on the router (a PE or an MCE), the router sets the DN-bit of generated Type 3, Type 5, or Type 7 LSAs to 1 and retains the DN-bit (0) of other LSAs. When calculating routes, the OSPF multi-instance process on the router ignores LSAs with DN-bit 1, which prevents the router from receiving the LSAs that are advertised by itself. In Figure 6-13, PE1 generates Type 3, Type 5, or Type 7 LSAs, sets their DN-bit to 1, and advertises these LSAs to CE1. After receiving these LSAs, CE1 forwards them to PE2. Upon reception of these LSAs, PE2 ignores them and does not forward them back to PE1, which prevents a routing loop.
VPN route tag	The VPN route tag is carried in Type 5 or Type 7 LSAs generated by PEs based on the received BGP VPN route. It is not carried in BGP extended community attributes. The VPN route tag is valid only on PEs that receive BGP routes and generate OSPF LSAs.	When a PE detects that the VPN route tag in the incoming LSA is the same as the local route tag, the PE ignores the LSA, which prevents routing loops.
Default route	It is a route whose destination IP address and mask are both 0.	Default routes are used to forward traffic from CEs or sites where CEs reside to the VPN backbone network.

Disabling Routing Loop Prevention

NOTICE

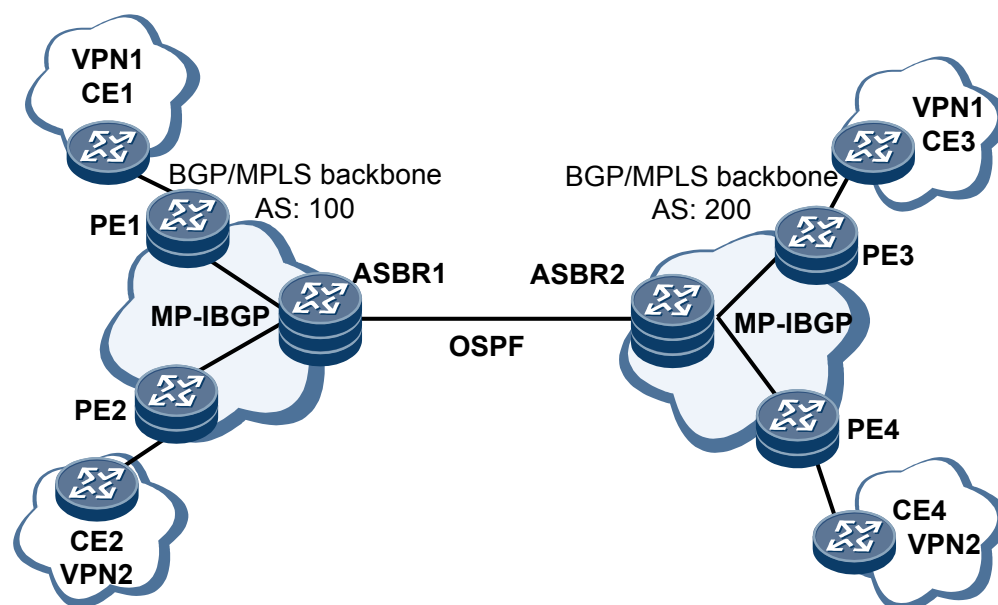
Exercise caution when disabling routing loop prevention because it may cause routing loops.

During BGP or OSPF route exchanges, routing loop prevention prevents OSPF routing loops in VPN sites.

In the inter-AS VPN Option A scenario, if OSPF runs between ASBRs to transmit VPN routes, the remote ASBR may fail to learn the OSPF routes sent by the local ASBR due to the routing loop prevention mechanism.

In [Figure 6-14](#), inter-AS VPN Option A is deployed with OSPF running between PE1 and CE1. CE1 sends VPN routes to CE2.

Figure 6-14 Networking for inter-AS VPN Option A



1. PE1 learns routes to CE1 using the OSPF process in a VPN instance, imports these routes into MP-BGP, and sends the MP-BGP routes to ASBR1.
2. After receiving the MP-BGP routes, ASBR1 imports the routes into the OSPF process in a VPN instance and generates Type 3, Type 5, or Type 7 LSAs carrying DN bit 1.
3. ASBR2 uses OSPF to learn these LSAs and checks the DN bit of each LSA. After learning that the DN bit in each LSA is 1, ASBR2 does not add the routes carried in these LSAs to its routing table.

The routing loop prevention mechanism prevents ASBR2 from learning the OSPF routes sent from ASBR1. As a result, CE1 cannot communicate with CE3.

To address the preceding problem, use either of the following methods:

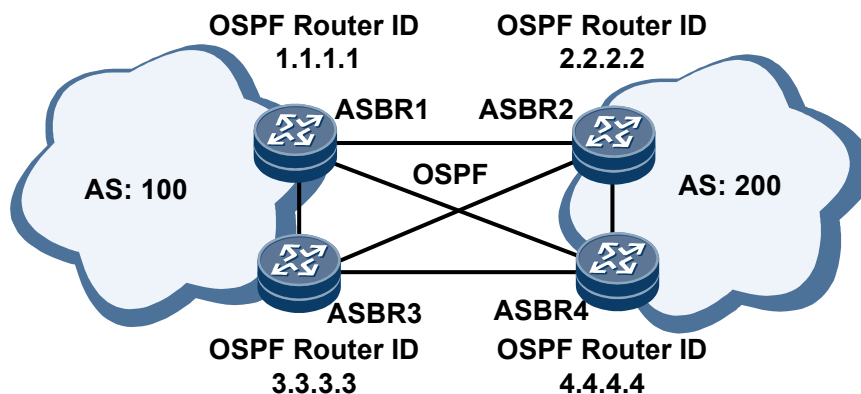
- Disable the device from setting the DN bit to 1 in the LSAs when importing BGP routes into OSPF. For example, if ASBR1 does not set the DN bit to 1 when importing MP-BGP routes into OSPF. After ASBR2 receives these routes and finds that the DN bit in the LSAs carrying these routes is 0, ASBR2 will add the routes to its routing table.
- Disable the device from checking the DN bit after receiving LSAs. For example, ASBR1 sets the DN bit to 1 in LSAs when importing MP-BGP routes into OSPF. ASBR2, however, does not check the DN bit after receiving these LSAs.

The preceding methods can be used based on specific types of LSAs. You can configure a sender to determine whether to set the DN bit to 1 or configure a receiver to determine whether to check the DN bit in the Type 3 LSAs based on the router ID of the device that generates the Type 3 LSAs.

In the inter-AS VPN Option A scenario shown in [Figure 6-15](#), the four ASBRs are fully meshed and run OSPF. ASBR2 may receive the Type 3, Type 5, or Type 7 LSAs generated on ASBR4. ASBR2 denies the Type 5 or Type 7 LSAs, because the VPN route tags carried in the LSAs are the same as the default VPN route tag of the OSPF process on ASBR2. If ASBR2 is disabled from checking the DN bit in the LSAs, ASBR2 accepts the Type 3 LSAs, and routing loops may occur.

To address the routing loop problem caused by Type 3 LSAs, ASBR2 can be disabled from checking the DN bit in the Type 3 LSAs generated by devices with router ID 1.1.1.1 and router ID 3.3.3.3. After the configuration is complete, if ASBR2 receives Type 3 LSAs sent by ASBR4 with router ID 4.4.4.4, ASBR2 checks the DN bit and denies these Type 3 LSAs because the DN bit is set to 1.

Figure 6-15 Networking for fully meshed ASBRs in the inter-AS VPN Option A scenario

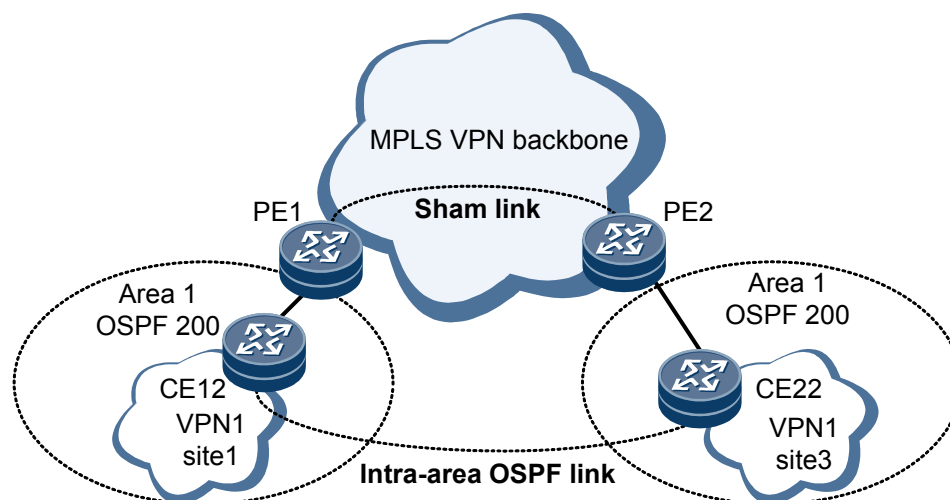


Sham Link

OSPF sham links are unnumbered P2P links between two PEs over an MPLS VPN backbone network.

Generally, BGP extended community attributes carry routing information over the MPLS VPN backbone between BGP peers. OSPF running on the other PE can use the routing information to generate inter-area routes from PEs to CEs.

Figure 6-16 OSPF Sham link



In **Figure 6-16**, if an intra-area OSPF link exists between the network segments of local and remote CEs. Routes that pass through the intra-area route link and have higher priorities than inter-area routes that pass through the MPLS VPN backbone network. As a result, VPN traffic is always forwarded through the intra-area route instead of the backbone network. To prevent such a problem, an OSPF sham link can be established between PEs so that the routes that pass through the MPLS VPN backbone network also become OSPF intra-area routes and take precedence.

- A sham link is a link between two VPN instances. Each VPN instance contains the address of an end-point of a sham link. The address is a loopback address with the 32-bit mask in the VPN address space on the PE.
- After a sham link is established between two PEs, the PEs become neighbors on the sham link and exchange routing information.
- A sham link functions as a P2P link within an area. Users can select a route from the sham link and intra-area route link by adjusting the metric.

Multi-VPN-Instance CE

OSPF multi-instance generally runs on PEs. Devices that run OSPF multi-instance within user LANs are called Multi-VPN-Instance CEs (MCEs).

Compared with OSPF multi-instance running on PEs, MCEs have the following characteristics:

- MCEs do not need to support OSPF-BGP association.
- MCEs establish one OSPF instance for each service. Different virtual CEs transmit different services, which ensures LAN security at a low cost.
- MCEs implement different OSPF instances on a CE. The key to implementing MCEs is to disable loop detection and calculate routes directly. MCEs also use the received LSAs with the DN-bit 1 for route calculation.

6.2.7 OSPF NSSA

Background

As defined in OSPF, stub areas cannot import external routes. This mechanism prevents external routes from consuming the bandwidth and storage resources of routers in stub areas. If you need to both import external routes and prevent resource consumption caused by external routes, you can configure not-so-stubby areas (NSSAs).

There are many similarities between NSSAs and stub areas. However, different from stub areas, NSSAs can import AS external routes into the OSPF AS and advertise the imported routes in the OSPF AS without learning external routes from other areas on the OSPF network.

Related Concepts

- **N-bit**

A router uses the N-bit carried in a Hello packet to identify the area type that it supports. The same area type must be configured for all routers in an area. If routers have different area types, they cannot establish OSPF neighbor relationships. Some vendors' devices do not comply with standard protocols, but the N-bit is also set in OSPF Database Description (DD) packets. You can manually set the N-bit on a router to interwork with the vendors' devices.

- **Type 7 LSA**

Type 7 LSAs, which describe imported external routes, are introduced to support NSSAs. Type 7 LSAs are generated by an ASBR in an NSSA and advertised only within the NSSA. After an ABR in an NSSA receives Type 7 LSAs, it selectively translates Type 7 LSAs into Type 5 LSAs to advertise external routes to other areas on an OSPF network.

Principles

To advertise external routes imported by an NSSA to other areas, a translator must translate Type 7 LSAs into Type 5 LSAs. Notes for an NSSA are as follows:

- By default, the translator is the ABR with the largest router ID in the NSSA.
- The propagate bit (P-bit) is used to notify a translator whether Type 7 LSAs need to be translated.
- Only Type 7 LSAs with the P-bit set and a non-zero forwarding address (FA) can be translated into Type 5 LSAs. An FA indicates that packets to a destination address will be forwarded to the address specified by the FA.

 NOTE

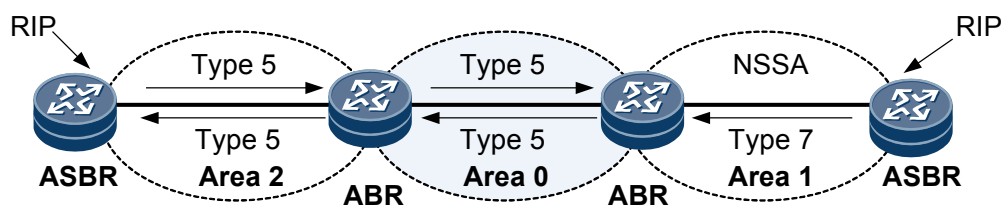
FA indicates that the packet to a specific destination address is to be forwarded to the address specified by.

The loopback interface address in an area is preferentially selected as the FA. If no loopback interface exists, the address of the interface that is Up and has the largest logical index in the area is selected as the FA.

- The P-bit is not set for default routes in Type 7 LSAs generated by an ABR.

Figure 6-17 shows an NSSA.

Figure 6-17 NSSA



Advantages

Multiple ABRs may be deployed in an NSSA. To prevent routing loops caused by default routes, ABRs do not calculate the default routes advertised by each other.

6.2.8 OSPF Local MT

Background

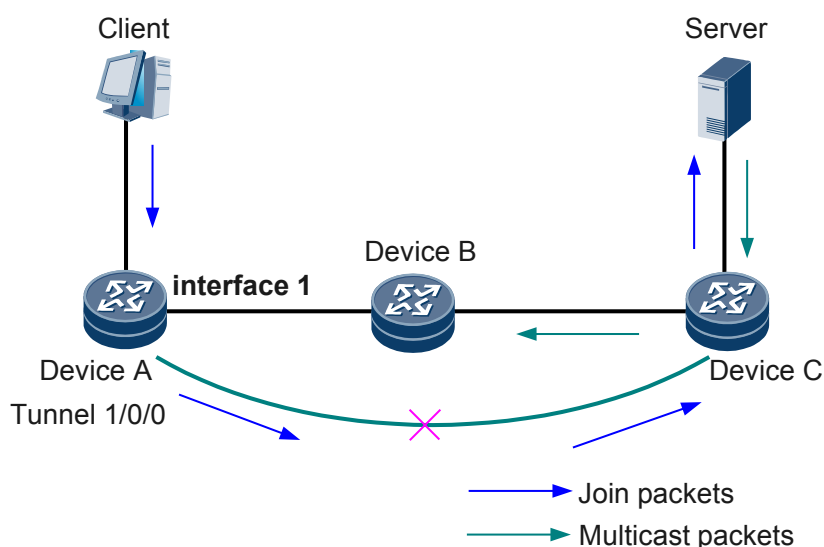
When multicast and an IGP Shortcut-enabled MPLS TE tunnel are configured on a network, the outbound interface of the route calculated by an IGP may not be a physical interface but a TE tunnel interface. The TE tunnel interface on the Device sends multicast Join packets over a unicast route to the multicast source address. The multicast Join packets are transparent to the Device through which the TE tunnel passes. As a result, the Device through which the TE tunnel passes cannot generate multicast forwarding entries.

To resolve the problem, configure OSPF local multicast topology (MT) to create a multicast routing table for multicast packet forwarding.

Principles

On the network shown in [Figure 6-18](#), multicast and an IGP Shortcut-enabled MPLS TE tunnel are configured, and the TE tunnel passes through Device B. As a result, Device B cannot generate multicast forwarding entries.

Figure 6-18 OSPF Local MT



Since the TE tunnel is unidirectional, multicast data packets from the multicast source are sent to a physical interface of Device B. Device B discards these packets, because Device B has no multicast forwarding entry. As a result, services are interrupted.

After local MT is enabled, if the outbound interface of a calculated route is an IGP Shortcut-enabled TE tunnel interface, the route management (RM) module creates an independent Multicast IGP (MIGP) routing table for the multicast protocol, calculates a physical outbound interface for the route, and adds the route to the MIGP routing table. Multicast packets are then forwarded along this route.

After receiving multicast Join packets from Client, interface 1 on Device A forwards these packets to Device B. With local MT enabled, Device B can generate multicast forwarding entries.

6.2.9 BFD for OSPF

Definition

Bidirectional Forwarding Detection (BFD) is a mechanism to detect communication faults between forwarding engines.

To be specific, BFD detects the connectivity of a data protocol along a path between two systems. The path can be a physical link, a logical link, or a tunnel.

In BFD for OSPF, a BFD session is associated with OSPF. The BFD session quickly detects a link fault and then notifies OSPF of the fault, which speeds up OSPF's response to network topology changes.

Purpose

A link fault or a topology change causes routers to recalculate routes. Routing protocol convergence must be as quick as possible to improve network availability. Link faults are inevitable, and therefore a solution must be provided to quickly detect faults and notify routing protocols.

BFD for Open Shortest Path First (OSPF) associates BFD sessions with OSPF. After BFD for OSPF is configured, BFD quickly detects link faults and notifies OSPF of the faults. BFD for OSPF accelerates OSPF response to network topology changes.

Table 6-14 describes OSPF convergence speeds before and after BFD for OSPF is configured.

Table 6-14 OSPF convergence speeds before and after BFD for OSPF is configured

Item	Link Fault Detection Mechanism	Convergence Speed
BFD for OSPF is not configured.	An OSPF Dead timer expires.	Second-level
BFD for OSPF is configured.	A BFD session goes Down.	Millisecond-level

Principles

Figure 6-19 BFD for OSPF

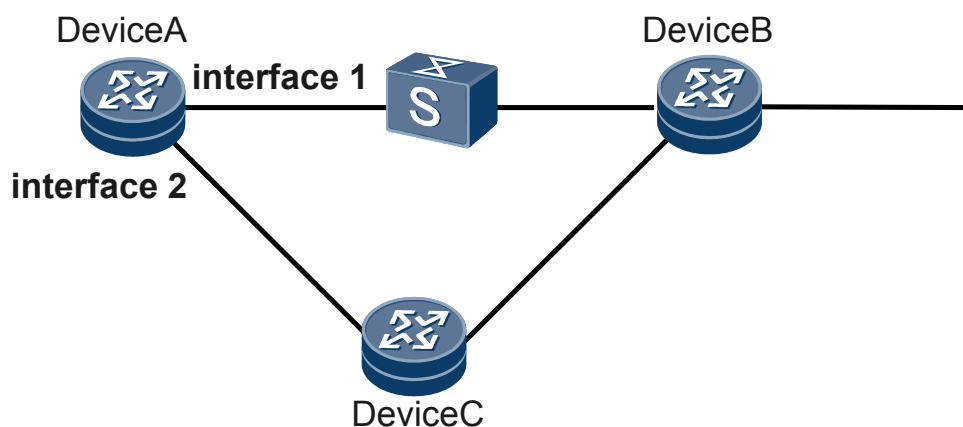


Figure 6-19 shows a typical network topology with BFD for OSPF configured. The principles of BFD for OSPF are described as follows:

1. OSPF neighbor relationships are established between these three routers.
2. After a neighbor relationship becomes Full, a BFD session is established.
3. The outbound interface on Device A connected to Device B is interface 1. If the link between Device A and Device B fails, BFD detects the fault and then notifies Device A of the fault.
4. Device A processes the event that a neighbor relationship goes Down and recalculates routes. The new route passes through Device C and reaches Device A, with interface 2 as the outbound interface.

6.2.10 OSPF GTSM

Definition

Generalized TTL security mechanism (GTSM) is a mechanism that protects services over the IP layer by checking whether the TTL value in an IP packet header is within a pre-defined range.

Purpose

On networks, attackers may simulate OSPF packets and keep sending them to a device. After receiving these packets, the device directly sends them to the control plane for processing without checking their validity if the packets are destined for the device. As a result, the control plane is busy processing these packets, resulting in high CPU usage.

GTSM is used to protect the TCP/IP-based control plane against CPU-utilization attacks, such as CPU-overload attacks.

Principles

GTSM-enabled devices check the TTL value in each received packet based on a configured policy. The packets that fail to pass the policy are discarded or sent to the control plane, which

prevents the devices from possible CPU-utilization attacks. A GTSM policy involves the following items:

- Source address of the IP packet sent to the device
- VPN instance to which the packet belongs
- Protocol number of the IP packet (89 for OSPF, and 6 for BGP)
- Source port number and destination port number of protocols above TCP/UDP
- Valid TTL range

GTSM is implemented as follows:

- For directly connected OSPF neighbors, the TTL value of the unicast protocol packets to be sent is set to 255.
- For multi-hop neighbors, a reasonable TTL range is defined.

The applicability of GTSM is as follows:

- GTSM takes effect on unicast packets rather than multicast packets. This is because the TTL value of multicast packets can only be 255, and therefore GTSM is not needed to protect against multicast packets.
- GTSM does not support tunnel-based neighbors.

6.2.11 OSPF Smart-discover

Definition

routers periodically send Hello packets through OSPF interfaces. By exchanging Hello packets, the routers establish and maintain the neighbor relationship, and elect the DR and the Backup Designated Router (BDR) on the multiple-access network (broadcast or NBMA network). OSPF uses a Hello timer to control the interval at which Hello packets are sent. A router can send Hello packets again only after the Hello timer expires. Neighbors keep waiting to receive Hello packets until the Hello timer expires. This process delays the establishment of OSPF neighbor relationships or election of the DR and the BDR.

Enabling Smart-discover can solve the preceding problem.

Table 6-15 Processing differences with and without Smart-discover

With or Without Smart-discover	Processing
Without Smart-discover	● Hello packets are sent only when the Hello timer expires. ● Hello packets are sent at the Hello interval. ● Neighbors keep waiting to receive Hello packets within the Dead interval.
With Smart-discover	● Hello packets are sent directly regardless of whether the Hello timer expires. ● Neighbors can receive packets and change the state immediately.

Principles

In the following situations, Smart-discover-enabled interfaces can send Hello packets to neighbors regardless of whether the Hello timer expires:

- On broadcast or NBMA networks, neighbor relationships can be established and a DR and a BDR can be elected rapidly.
 - The neighbor status becomes 2-way for the first time or returns to Init from 2-way or a higher state.
 - The interface status of the DR or BDR on a multiple-access network changes.
- On P2P or P2MP networks, neighbor relationships can be established rapidly. The establishment of neighbor relationships on a P2P or P2MP network is the same as that on a broadcast or NBMA network.

6.2.12 OSPF-BGP Synchronization

Background

When a new device is deployed on a network or a device is restarted, network traffic may be lost during BGP route convergence because IGP routes converge more quickly than BGP routes.

OSPF-BGP synchronization can address this problem.

Purpose

If a backup link exists, BGP traffic may be lost during traffic switchback because BGP routes converge more slowly than OSPF routes do.

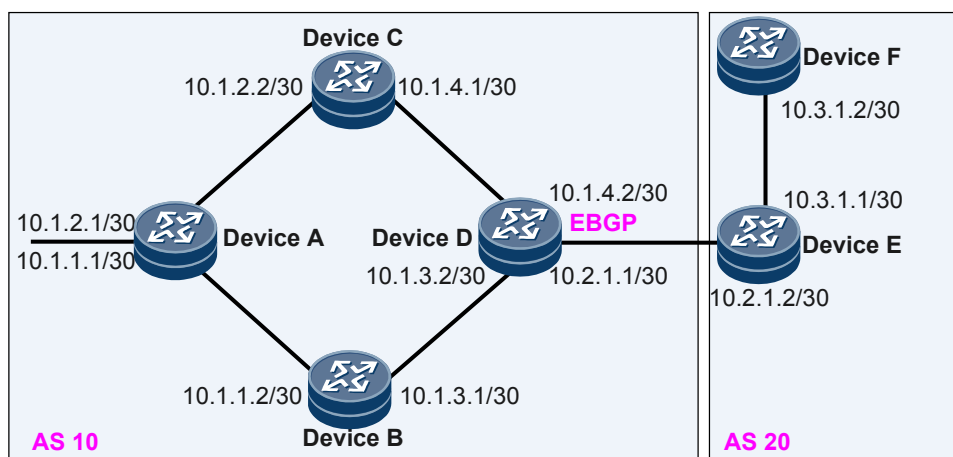
In [Figure 6-20](#), Device A, Device B, Device C, and Device D run OSPF and establish IBGP connections. Device C functions as the backup of Device B. When the network is stable, BGP and OSPF routes converge completely on the router.

In most cases, traffic from Device A to 10.3.1.0/30 passes through Device B. If Device B fails, traffic is switched to Device C. After Device B recovers, traffic is switched back to Device B. During this process, packet loss occurs.

Consequently, convergence of OSPF routes is complete whereas BGP route convergence is still going on. As a result, Device B does not have the route to 10.3.1.0/30.

When packets from Device A to 10.3.1.0/30 reach Device B, Device B discards them because Device B does not have the route to 10.3.1.0/30.

Figure 6-20 Networking for OSPF-BGP synchronization



Principles

If OSPF-BGP synchronization is configured on a device, the device remains as a stub router during the set synchronization period. During this period, the link metric in the LSA advertised by the device is set to the maximum value (65535), instructing other OSPF routers not to use it as a transit router for data forwarding.

In [Figure 6-20](#), OSPF-BGP synchronization is enabled on Router B. In this situation, before BGP route convergence is complete, Device A keeps forwarding data through Device C rather than Device B until BGP route convergence on Device B is complete.

6.2.13 LDP-IGP Synchronization

Background

LDP-IGP synchronization enables the LDP status and the IGP status to go Up simultaneously, which helps minimize traffic interruption time if a fault occurs.

On a network with both primary and backup LDP LSPs, if the primary LDP LSP fails, both IGP and LSP traffic switches to the standby link. When the primary LSP recovers, IGP traffic is switched back to the primary LSP earlier than LDP traffic because IGP route convergence is faster than LDP convergence. As a result, LSP traffic is lost because the primary LSP is not reestablished. When the LDP session between nodes on the active fails, the LSP on the link is deleted, but the IGP continues to use the active link. As a result, LSP traffic cannot be switched to the standby link, and packets are lost continuously.

NOTE

LDP-IGP synchronization supports OSPFv2 and IS-IS IPv4.

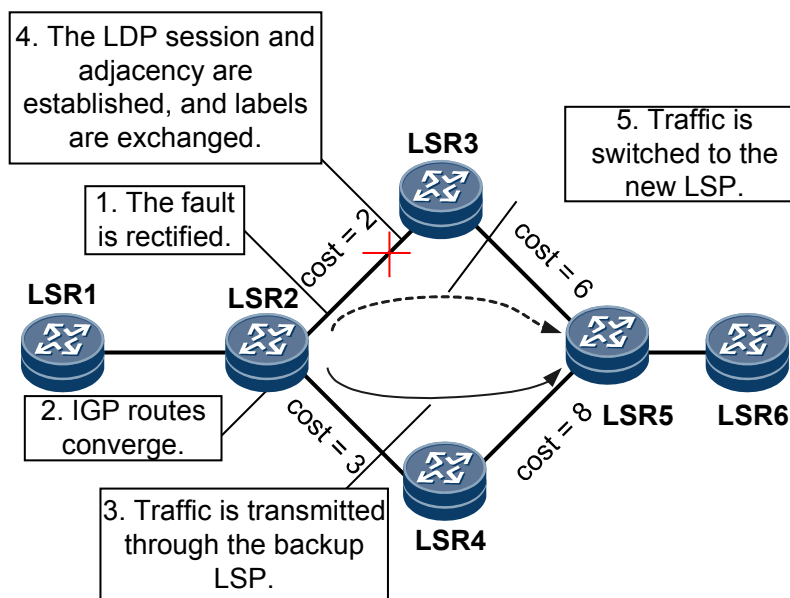
On a network enabled with LDP-IGP synchronization, an IGP keeps advertising the maximum cost of an IGP route over the new active link to delay IGP route convergence until LDP converges. Traffic keeps switching back and forth between the standby and active links. The backup LSP is torn down after the LSP on the active link is established.

LDP-IGP synchronization involves the following timers:

- Hold-max-cost timer
- Delay timer

Implementation

Figure 6-21 Switchback in LDP-IGP synchronization



- In [Figure 6-21](#), a network has both an active and standby link. When the active link recovers from any fault, traffic is switched from the standby link to the active link. During the traffic switchback, the backup LSP cannot be used, and a new LSP cannot be set up over the active link once IGP route convergence is complete. This causes a traffic interruption for a short period of time. To help prevent this problem, LDP-IGP synchronization can be configured to delay the IGP route switchback until LDP converges. The backup LSP is not deleted and continues forwarding traffic until an LSP over the active link is established. The process of LDP-IGP synchronization is as follows:
 - A link recovers from a fault.
 - An LDP session is set up between LSR2 and LSR3. The IGP advertises the maximum cost of the active link to delay the IGP route switchback.
 - Traffic is still forwarded along the backup LSP.
 - Once set up, the LDP session transmits Label Mapping messages and advertises the IGP to start synchronization.
 - The IGP advertises the normal cost of the active link, and its routes converge on the original forwarding path. The LSP is reestablished and delivers entries to the forwarding table. The whole process usually takes milliseconds.
- If an LDP session between two nodes on an active link fails, the LSP on the active link is torn down, but the IGP route for the active link is reachable. In this case, traffic fails to switch from the primary LSP to a backup LSP and is discarded. To prevent this problem, LDP-IGP synchronization can be configured so that after an LDP session fails, LDP notifies an IGP of the failure. The IGP advertises the maximum cost of the failed link,

which enables the route to switch from the active link to the standby link. In addition to the LSP switchover from the primary LSP to the backup LSP, LDP-IGP synchronization is implemented. The process of LDP-IGP synchronization is as follows:

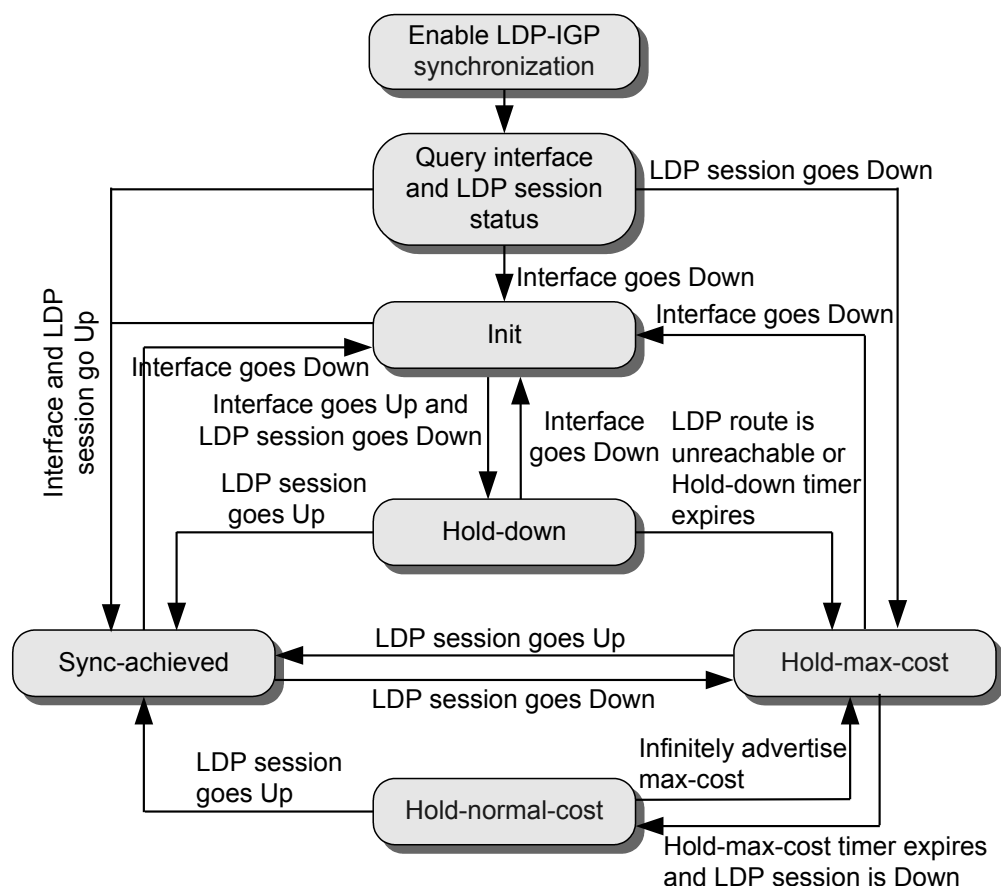
- An LDP session between two nodes on an active link fails.
- LDP notifies an IGP of failure in the LDP session over which the primary LSP is established. The IGP then advertises the maximum cost along the active link.
- The IGP route switches to the standby link.
- A backup LSP is set up over the standby link and then forwarding entries are delivered.

To prevent an LDP session from failing to be established, you can set the value of a Hold-max-cost timer to always advertise the maximum cost, which enables traffic to be transmitted along the backup link before the LDP session is reestablished on the active link.

- LDP-IGP synchronization state machine

After LDP-IGP synchronization is enabled on an interface, the LDP-IGP synchronization state machine operates based on the flowchart shown in [Figure 6-22](#).

Figure 6-22 LDP-IGP synchronization status transition



NOTE

Note differences when different IGP protocols are used:

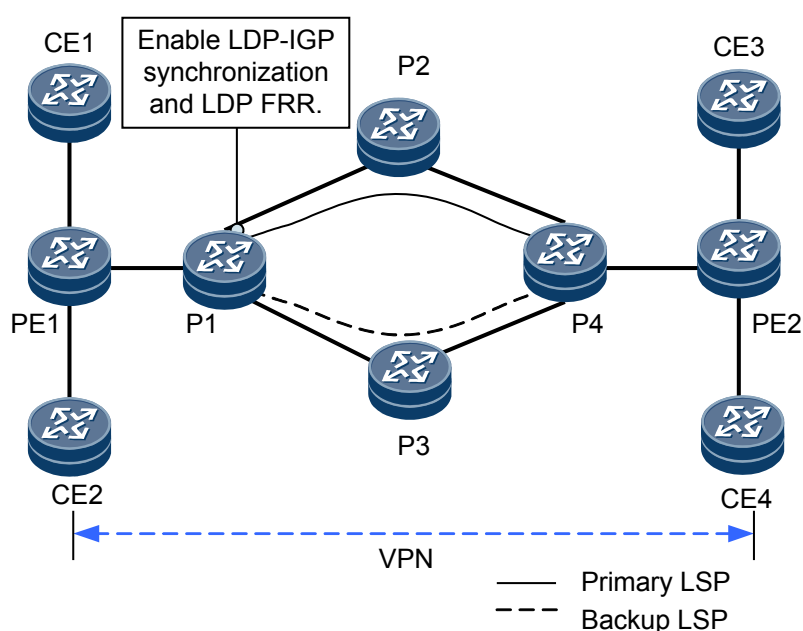
- When OSPF is used, the status transits based on the flowchart shown in [Figure 6-22](#).
- When IS-IS is used, the Hold-normal-cost state does not exist. After the Hold-max-cost timer expires, IS-IS advertises the actual link cost, but the Hold-max-cost state is displayed even though this state is nonexistent.

Usage Scenario

[Figure 6-23](#) shows an LDP-IGP synchronization scenario.

On the network shown in [Figure 6-23](#), an active link and a standby link are established. LDP-IGP synchronization and LDP FRR are deployed.

Figure 6-23 LDP-IGP synchronization scenario



Benefits

Packet loss is reduced during an active/standby link switchover, improving network reliability.

6.2.14 OSPF Fast Convergence

OSPF fast convergence is an extended feature of OSPF to speed up the convergence of routes. It includes the following components:

- Partial Route Calculation (PRC): calculates only those routes which have changed when the network topology changes.
- An OSPF intelligent timer: can dynamically adjust its value based on the user's configuration and the interval at which an event is triggered, such as the route calculation interval, which ensures rapid and stable network operation.

OSPF intelligent timer uses the exponential backoff technology so that the value of the timer can reach the millisecond level.

PRC

When a node changes on the network, this algorithm is used to recalculate all routes. The PRC calculation takes a long time and consumes too many CPU resources, which affects the convergence speed.

In route calculation, a leaf represents a route, and a node represents a router. Either an SPT or a leaf change causes a route change. The SPT change is irrelevant to the leaf change. PRC processes routing information as follows:

- If the SPT changes, PRC processes the routing information of all leaves on a changed node.
- If the SPT remains unchanged, PRC does not process the routing information on any node.
- If a leaf changes, PRC processes the routing information on the leaf only.
- If a leaf remains unchanged, PRC does not process the routing information on any leaf.

OSPF Intelligent Timer

On an unstable network, routes are calculated frequently, which consumes a great number of CPU resources. In addition, LSAs that describe the unstable topology are generated and transmitted on the unstable network. Frequently processing such LSAs affects the rapid and stable operation of the entire network.

To speed up route convergence on the entire network, the OSPF intelligent timer controls route calculation, LSA generation, and LSA receiving.

The OSPF intelligent timer works as follows:

- On a network where routes are calculated repeatedly, the OSPF intelligent timer dynamically adjusts the route calculation based on user's configuration and the exponential backoff technology. The number of route calculation times and the CPU resource consumption are decreased. Routes are calculated after the network topology stabilizes.
- On an unstable network, if a router generates or receives LSAs due to frequent topology changes, the OSPF intelligent timer can dynamically adjust the interval. No LSAs are generated or processed within an interval, which prevents invalid LSAs from being generated and advertised on the entire network.

6.2.15 OSPF Neighbor Relationship Flapping Suppression

OSPF neighbor relationship flapping suppression works by delaying OSPF neighbor relationship reestablishment or setting the link cost to the maximum value (65535).

Background

If an interface carrying OSPF services alternates between Up and Down, OSPF neighbor relationship flapping occurs on the interface. During the flapping, OSPF frequently sends Hello packets to reestablish the neighbor relationship, synchronizes LSDBs, and recalculates routes. In this process, a large number of packets are exchanged, adversely affecting neighbor relationship stability, OSPF services, and other OSPF-dependent services, such as LDP and

BGP. OSPF neighbor relationship flapping suppression can address this problem by delaying OSPF neighbor relationship reestablishment or preventing service traffic from passing through flapping links.

Related Concepts

Flapping-event: reported when the status of a neighbor relationship on an interface last changes from Full to a non-Full state. The flapping-event triggers flapping detection.

Flapping-count: number of times flapping has occurred.

Detecting-interval: detection interval. The interval is used to determine whether to trigger a valid flapping_event.

Threshold: flapping suppression threshold. When the flapping_count reaches or exceeds **threshold**, flapping suppression takes effect.

Resume-interval: interval for exiting from OSPF neighbor relationship flapping suppression. If the interval between two successive valid flapping_events is longer than **resume-interval**, the flapping_count is reset.

Implementation

Flapping detection

Each OSPF interface on which OSPF neighbor relationship flapping suppression is enabled starts a flapping counter. If the interval between two successive neighbor status changes from Full to a non-Full state is shorter than **detecting-interval**, a valid flapping_event is recorded, and the flapping_count increases by 1. When the flapping_count reaches or exceeds **threshold**, flapping suppression takes effect. If the interval between two successive neighbor status changes from Full to a non-Full state is longer than **resume-interval**, the flapping_count is reset.

The **detecting-interval**, **threshold**, and **resume-interval** are configurable.

NOTE

The value of **resume-interval** must be greater than that of **detecting-interval**.

Flapping suppression

Flapping suppression works in either Hold-down or Hold-max-cost mode.

- **Hold-down mode:** In the case of frequent flooding and topology changes during neighbor relationship establishment, interfaces prevent neighbor relationship reestablishment during Hold-down suppression, which minimizes LSDB synchronization attempts and packet exchanges.
- **Hold-max-cost mode:** If the traffic forwarding path changes frequently, interfaces use 65535 as the cost of the flapping link during Hold-max-cost suppression, which prevents traffic from passing through the flapping link.

Flapping suppression can also work first in Hold-down mode and then in Hold-max-cost mode.

By default, the Hold-max-cost mode takes effect. The mode and suppression duration can be changed manually.

If an attack causes frequent neighbor relationship flapping, Hold-down mode can minimize the impact of the attack.

 NOTE

When an interface enters the flapping suppression state, all neighbor relationships on the interface enter the state accordingly.

Exiting from flapping suppression

Interfaces exit from flapping suppression in the following scenarios:

- The suppression timer expires.
- The corresponding OSPF process is reset.
- An OSPF neighbor is reset.
- A command is run to exit from flapping suppression.

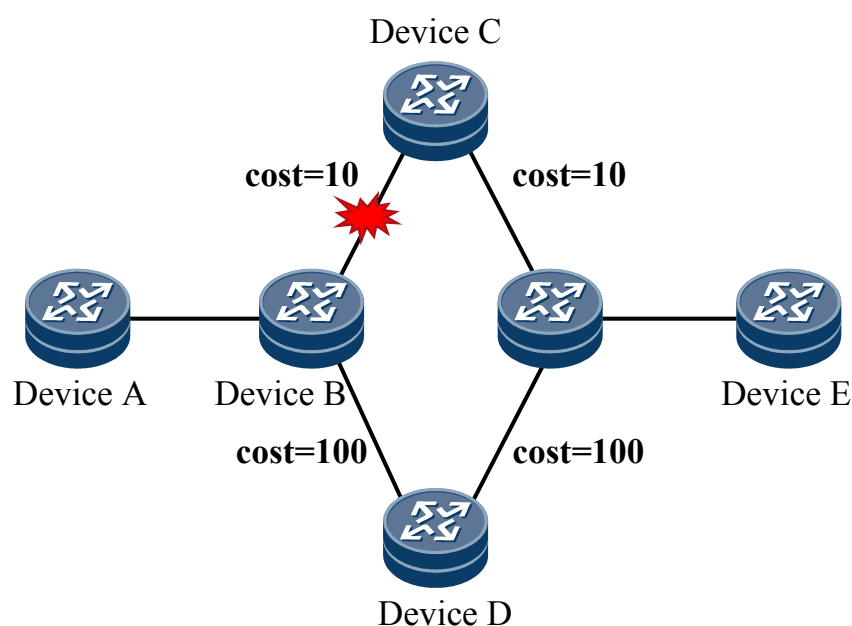
Typical Scenarios

Basic scenario

In [Figure 6-24](#), the traffic forwarding path is Device A -> Device B -> Device C -> Device E before a link failure occurs. After the link between Device B and Device C fails, the forwarding path switches to Device A -> Device B -> Device D -> Device E. If the neighbor relationship between Device B and Device C frequently flaps at the early stage of the path switchover, the forwarding path will be switched frequently, causing traffic loss and affecting network stability. If the neighbor relationship flapping meets suppression conditions, flapping suppression takes effect.

- If flapping suppression works in Hold-down mode, the neighbor relationship between Device B and Device C is prevented from being reestablished during the suppression period, in which traffic is forwarded along the path Device A -> Device B -> Device D -> Device E.
- If flapping suppression works in Hold-max-cost mode, 65535 is used as the cost of the link between Device B and Device C during the suppression period, and traffic is forwarded along the path Device A -> Device B -> Device D -> Device E.

Figure 6-24 Flapping suppression in a basic scenario



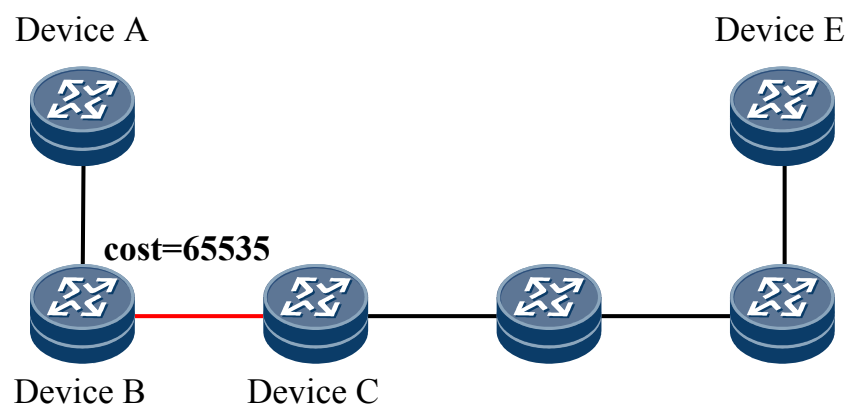
Single-forwarding path scenario

When only one forwarding path exists on the network, the flapping of the neighbor relationship between any two devices on the path will interrupt traffic forwarding. In [Figure 6-25](#), the traffic forwarding path is Device A -> Device B -> Device C -> Device E. If the neighbor relationship between Device B and Device C flaps, and the flapping meets suppression conditions, flapping suppression takes effect. However, if the neighbor relationship between Device B and Device C is prevented from being reestablished, the whole network will be divided. Therefore, Hold-max-cost mode (rather than Hold-down mode) is recommended. If flapping suppression works in Hold-max-cost mode, 65535 is used as the cost of the link between Device B and Device C during the suppression period. After the network stabilizes and the suppression timer expires, the link is restored.

NOTE

By default, the Hold-max-cost mode takes effect.

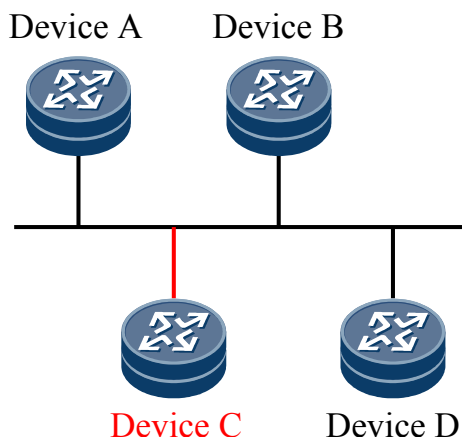
Figure 6-25 Flapping suppression in a single-forwarding path scenario



Broadcast scenario

In [Figure 6-26](#), four devices are deployed on the same broadcast network using switches, and the devices are broadcast network neighbors. If Device C flaps due to a link failure, and Device A and Device B were deployed at different time (Device A was deployed earlier for example) or the flapping suppression parameters on Device A and Device B are different, Device A first detects the flapping and suppresses Device C. Consequently, the Hello packets sent by Device A do not carry Device C's router ID. However, Device B has not detected the flapping yet and still considers Device C a valid node. As a result, the DR candidates identified by Device A are Device B and Device D, whereas the DR candidates identified by Device B are Device A, Device C, and Device D. Different DR candidates result in a different DR election result, which may lead to route calculation errors. To prevent this problem in scenarios where an interface has multiple neighbors, such as on a broadcast, P2MP, or NBMA network, all neighbors on the interface are suppressed when the status of a neighbor relationship last changes to **ExStart** or **Down**. Specifically, if Device C flaps, Device A, Device B, and Device D on the broadcast network are all suppressed. After the network stabilizes and the suppression timer expires, Device A, Device B, and Device D are restored to normal status.

Figure 6-26 Flapping suppression on a broadcast network



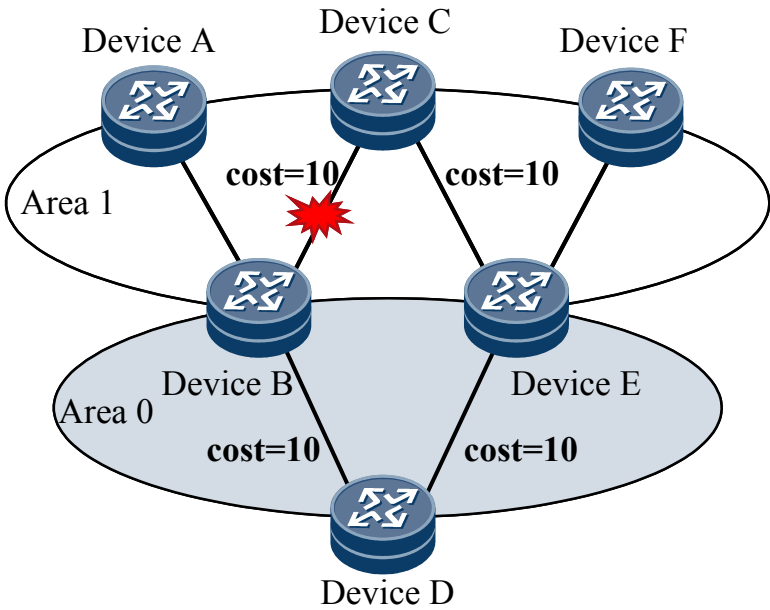
Multi-area scenario

In [Figure 6-27](#), Device A, Device B, Device C, Device E, and Device F are connected in area 1, and Device B, Device D, and Device E are connected in backbone area 0. Traffic from Device A to Device F is preferentially forwarded along an intra-area route, and the forwarding path is Device A -> Device B -> Device C -> Device E -> Device F. When the neighbor relationship between Device B and Device C flaps and the flapping meets suppression conditions, flapping suppression takes effect in the default mode (Hold-max-cost). Consequently, 65535 is used as the cost of the link between Device B and Device C. However, the forwarding path remains unchanged because intra-area routes take precedence over inter-area routes during route selection according to OSPF route selection rules. To prevent traffic loss in multi-area scenarios, configure Hold-down mode to prevent the neighbor relationship between Device B and Device C from being reestablished during the suppression period. During this period, traffic is forwarded along the path Device A -> Device B -> Device D -> Device E -> Device F.

NOTE

By default, the Hold-max-cost mode takes effect. The mode can be changed to Hold-down manually.

Figure 6-27 Flapping suppression in a multi-area scenario



Scenario with both LDP-IGP synchronization and OSPF neighbor relationship flapping suppression configured

In [Figure 6-28](#), if the link between PE1 and P1 fails, an LDP LSP switchover is implemented immediately, causing the original LDP LSP to be deleted before a new LDP LSP is established. To prevent traffic loss, LDP-IGP synchronization needs to be configured. With LDP-IGP synchronization, 65535 is used as the cost of the new LSP to be established. After the new LSP is established, the original cost takes effect. Consequently, the original LSP is deleted, and LDP traffic is forwarded along the new LSP.

LDP-IGP synchronization and OSPF neighbor relationship flapping suppression work in either Hold-down or Hold-max-cost mode. If both functions are configured, Hold-down mode takes precedence over Hold-max-cost mode, followed by the configured link cost. [Table 6-16](#) lists the suppression modes that take effect in different situations.

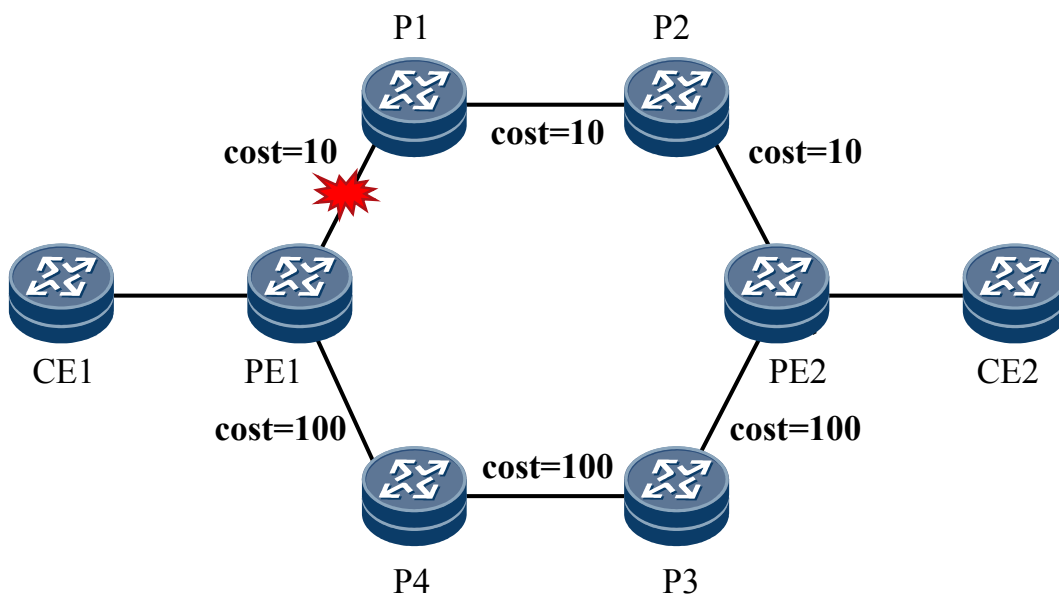
Table 6-16 Principles for selecting the suppression modes that take effect in different situations

LDP-IGP Synchronization/ OSPF Neighbor Relationship Flapping Suppression Mode	LDP-IGP Synchronization Hold-down Mode	LDP-IGP Synchronization Hold-max-cost Mode	Exited from LDP-IGP Synchronization Suppression
OSPF Neighbor Relationship Flapping Suppression Hold-down Mode	Hold-down	Hold-down	Hold-down

OSPF Neighbor Relationship Flapping Suppression Hold-max-cost Mode	Hold-down	Hold-max-cost	Hold-max-cost
Exited from OSPF Neighbor Relationship Flapping Suppression	Hold-down	Hold-max-cost	Exited from LDP-IGP synchronization and OSPF neighbor relationship flapping suppression

For example, the link between PE1 and P1 frequently flaps in [Figure 6-28](#), and both LDP-IGP synchronization and OSPF neighbor relationship flapping suppression are configured. In this case, the suppression mode is selected based on the preceding principles. No matter which mode (Hold-down or Hold-max-cost) is selected, the forwarding path is PE1 -> P4 -> P3 -> PE2.

Figure 6-28 Scenario with both LDP-IGP synchronization and OSPF neighbor relationship flapping suppression configured



Scenario with both bit-error-triggered protection switching and OSPF neighbor relationship flapping suppression configured

If a link has poor link quality, services transmitted along it may be adversely affected. If bit-error-triggered protection switching is configured and the bit error rate (BER) along a link exceeds a specified value, a bit error event is reported, and 65535 is used as the cost of the link, triggering route reselection. Consequently, service traffic is switched to the backup link. If both bit-error-triggered protection switching and OSPF neighbor relationship flapping suppression are configured, they both take effect. Hold-down mode takes precedence over Hold-max-cost mode, followed by the configured link cost.

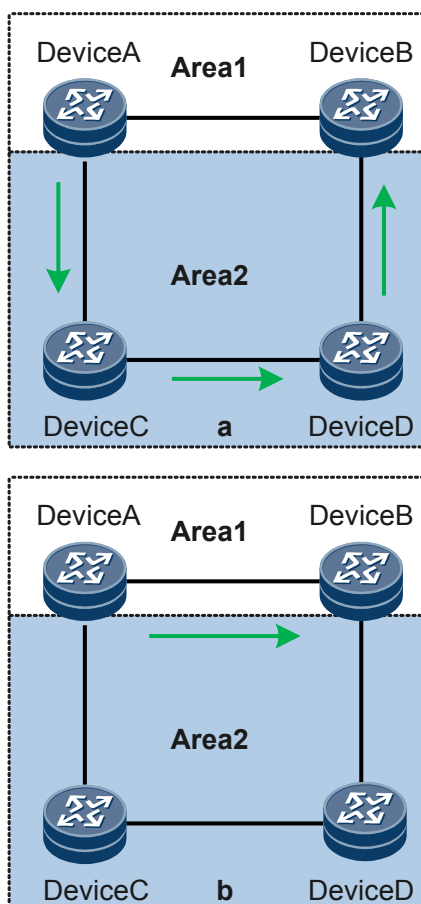
6.2.16 OSPF Multi-Area Adjacency

Background

In OSPF, intra-area links take precedence over inter-area links during route selection even when the inter-area links are shorter than the intra-area links. Each OSPF interface belongs to only one area. As a result, even when a high-speed link exists in an area, traffic of another area cannot be forwarded along the link. A common method used to solve this problem is to configure multiple sub-interfaces and add them to different areas. However, this method has a defect that an independent IP address needs to be configured for each sub-interface and then is advertised, which increases the total number of routes. In this situation, OSPF multi-area adjacency is introduced.

OSPF multi-area adjacency allows an OSPF interface to be multiplexed by multiple areas so that a link can be shared by the areas.

Figure 6-29 Traffic forwarding paths before and after OSPF multi-area adjacency is enabled



In **Figure 6-29**, the link between Device A and Device B in area 1 is a high-speed link.

In **Figure 6-29** a, OSPF multi-area adjacency is disabled on Device A and Device B, and traffic from Device A to Device B in area 2 is forwarded along the low-speed link of Device A -> Device C -> Device D -> Device B.

In **Figure 6-29** b, OSPF multi-area adjacency is enabled on Device A and Device B, and their multi-area adjacency interfaces belong to area 2. In this case, traffic from Device A to Device B in area 2 is forwarded along the high-speed link of Device A -> Device B.

OSPF multi-area adjacency has the following advantages:

- Allows interface multiplexing, which reduces OSPF interface resource usage in multi-area scenarios.
- Allows link multiplexing, which prevents a traffic detour to low-speed links and optimizes the OSPF network.

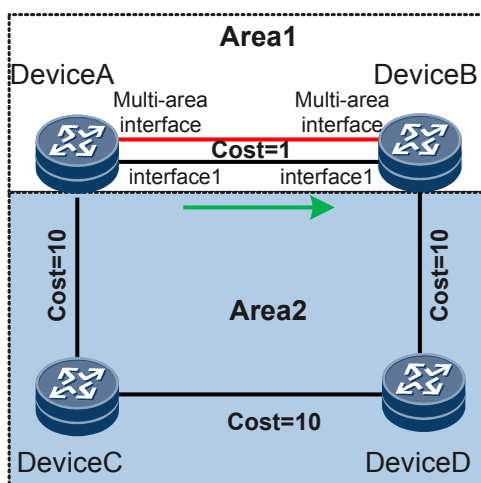
Related Concepts

Multi-area adjacency interface: indicates the OSPF logical interface created when OSPF multi-area adjacency is enabled on an OSPF-capable interface (main OSPF interface). The multi-area adjacency interface is also referred to as a secondary OSPF interface. The multi-area adjacency interface has the following characteristics:

- The multi-area adjacency interface and the main OSPF interface belong to different OSPF areas.
- The network type of the multi-area adjacency interface must be P2P. The multi-area adjacency interface runs an independent interface state machine and neighbor state machine.
- The multi-area adjacency interface and the main OSPF interface share the same interface index and packet transmission channel. Whether the multi-area adjacency interface or the main OSPF interface is selected to forward an OSPF packet is determined by the area ID carried in the packet header and related configuration.
- If the interface is P2P, its multi-area adjacency interface sends packets through multicast.
- If the interface is not P2P, its multi-area adjacency interface sends packets through unicast.

Principles

Figure 6-30 Networking for OSPF multi-area adjacency



In **Figure 6-30**, the link between Device A and Device B in area 1 is a high-speed link. In area 2, traffic from Device A to Device B is forwarded along the low-speed link of Device A -

> Device C -> Device D -> Device B. If you want the traffic from Device A to Device B in area 2 to be forwarded along the high-speed link of Device A -> Device B, deploy OSPF multi-area adjacency.

Specifically, configure OSPF multi-area adjacency on the main interfaces of Device A and Device B to create multi-area adjacency interfaces. The multi-area adjacency interfaces belong to area 2.

1. An OSPF adjacency is established between Device A and Device B. For details about the establishment process, see [Adjacency Establishment](#).
2. Route calculation is implemented. For details about the calculation process, see [Route Calculation](#).

The optimal path in area 2 obtained by OSPF through calculation is the high-speed link of Device A -> Device B. In this case, the high-speed link is shared by area 1 and area 2.

6.2.17 OSPF NSR

Non-Stop Routing (NSR) is a routing technique that prevents neighboring routers from detecting faults on the control plane of a device with a backup control plane. With NSR, when the control plane of the device fails, the neighbor relationship set up through specific routing protocols, MPLS, and other protocols that carry services is uninterrupted.

As networks develop and expand, carriers place increasingly higher requirements on the reliability of IP networks. NSR was introduced as a high availability (HA) solution to ensure that services transmitted by a device were unaffected by hardware or software failures on the device.

OSPF NSR synchronizes the protocol data on the master main control board to the slave main control board in real time. When the master main control board fails or needs to be upgraded, the slave main control board immediately takes over services from the master main control board, and the neighbor remains unaware the fault occurred. Real-time data synchronization between the master and slave main control boards through OSPF NSR is implemented as follows:

- OSPF backs up configuration data and dynamic data, including information about interfaces, neighbors, and LSDBs.
- OSPF does not back up routes, shortest path trees (SPTs), or Traffic Engineering Databases (TEDBs). Such information can be restored using the source data in the database backup process.
- When the master/slave main control board switchover occurs, the new master main control board restores the operations data and takes over services from the former master main control board. Throughout this process, the neighbor remains unaware of the fault.

6.2.18 OSPF IP FRR

With OSPF IP fast reroute (FRR), a device pre-computes alternate next hops and stores them in the IP routing table. If a primary link fails, the device switches the traffic to a backup link.

Background

As networks develop, voice over IP (VoIP) and online video services pose higher requirements for real-time transmission. Nevertheless, if a primary link fails, OSPF-enabled devices need to perform multiple operations, including detecting the fault, updating the link-state advertisement (LSA), flooding the LSA, calculating routes, and delivering forward

information base (FIB) entries before switching traffic to a new link. This process takes a much longer time, the minimum delay to which users are sensitive. As a result, the requirements for real-time transmission cannot be met. OSPF IP FRR can solve this problem. OSPF IP FRR conforms to dynamic IP FRR defined by standard protocols. With OSPF IP FRR, devices can switch traffic from a faulty primary link to a backup link, protecting against a link or node failure.

Major FRR techniques include loop-free alternate (LFA), U-turn, Not-Via, Remote LFA, and MRT, among which OSPF supports only LFA and Remote LFA.

Related Concepts

OSPF IP FRR

OSPF IP FRR refers to a mechanism in which a device uses the loop-free alternate (LFA) algorithm to compute the next hop of a backup link and stores the next hop together with the primary link in the forwarding table. If the primary link fails, the device switches the traffic to the backup link before routes are converged on the control plane. This mechanism keeps the traffic interruption duration and minimizes the impacts.

OSPF IP FRR policy

An OSPF IP FRR policy can be configured to filter alternate next hops. Only the alternate next hops that match the filtering rules of the policy can be added to the IP routing table.

LFA algorithm

A device uses the shortest path first (SPF) algorithm to calculate the shortest path from each neighbor with a backup link to the destination node. The device then uses the inequalities defined in standard protocols and the LFA algorithm to calculate the next hop of the loop-free backup link that has the smallest cost of the available shortest paths.

Remote LFA

LFA FRR cannot be used to calculate alternate links on large-scale networks, especially on ring networks. Remote LFA FRR addresses this problem by calculating a PQ node and establishing a tunnel between the source node of a primary link and the PQ node. If the primary link fails, traffic can be automatically switched to the tunnel, which improves network reliability.

P space

P space consists of the nodes through which the shortest path trees (SPTs) with the source node of a primary link as the root are reachable without passing through the primary link.

Extended P space

Extended P space consists of the nodes through which the SPTs with neighbors of a primary link's source node as the root are reachable without passing through the primary link.

Q space

Q space consists of the nodes through which the SPTs with the destination node of a primary link as the root are reachable without passing through the primary link.

PQ node

A PQ node exists both in the extended P space and Q space and is used by Remote LFA as the destination of a protection tunnel.

OSPF LFA FRR

OSPF IP FRR guarantees protection against either a link failure or a node-and-link failure. The link-and-node protection takes precedence over the link protection.

Link protection

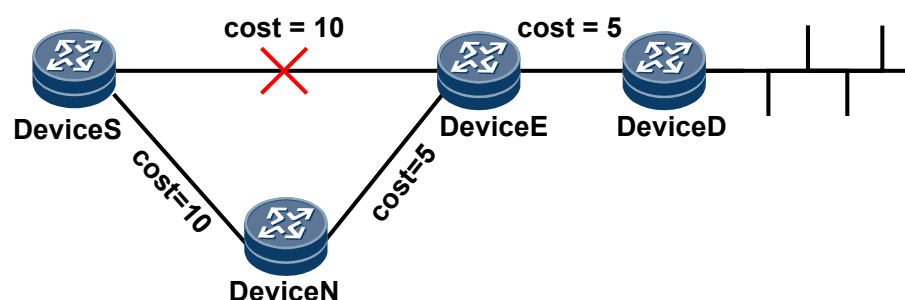
Link protection takes effect when the traffic to be protected flows along a specified link.

In **Figure 6-31**, traffic flows from Device S to Device D. The primary link is Device S->Device E->Device D, and the backup link is Device S->Device N->Device E->Device D. The link costs meet the inequality: $\text{Distance_opt}(N, D) < \text{Distance_opt}(N, S) + \text{Distance_opt}(S, D)$. With OSPF IP FRR, Device S switches the traffic to the backup link if the primary link fails, keeping the traffic interruption duration.

NOTE

$\text{Distance_opt}(X, Y)$ indicates the shortest link from X to Y. S stands for a source node, E for the faulty node, N for a node along a backup link, and D for a destination node.

Figure 6-31 OSPF IP FRR link protection



Node-and-link protection

Node-and-link protection takes effect when the traffic to be protected.

In **Figure 6-32**, traffic flows from Device S to Device D. The primary link is Device S->Device E->Device D, and the backup link is Device S->Device N->Device D. The preceding inequalities are met. With OSPF IP FRR, Device S switches the traffic to the backup link if the primary link fails, keeping the traffic interruption duration.

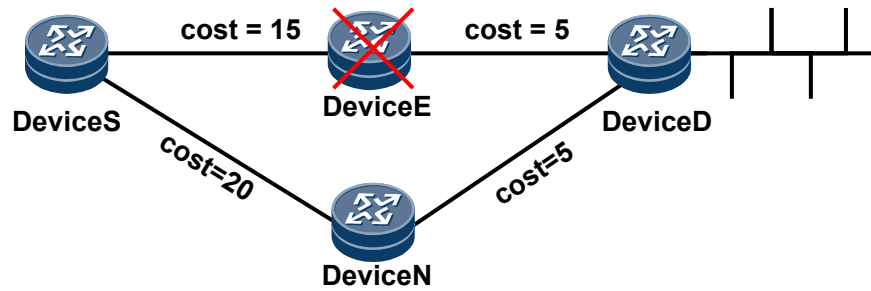
The traffic to be protected flows along a specified link and node and the following conditions are met:

- The link costs meet the inequality: $\text{Distance_opt}(N, D) < \text{Distance_opt}(N, S) + \text{Distance_opt}(S, D)$.
- The interface costs meet the inequality: $\text{Distance_opt}(N, D) < \text{Distance_opt}(N, E) + \text{Distance_opt}(E, D)$.

NOTE

$\text{Distance_opt}(X, Y)$ indicates the shortest link from X to Y. S stands for a source node, E for the faulty node, N for a node along a backup link, and D for a destination node.

Figure 6-32 OSPF IP FRR node-and-link protection

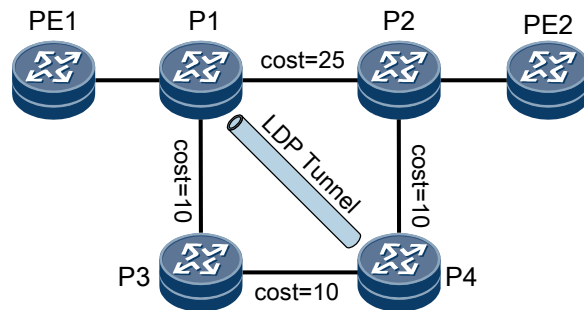


OSPF Remote LFA FRR

Similar to LFA FRR, Remote LFA protects against both link and node-and-link failures. The following example shows how Remote LFA works to protect against link failures:

In **Figure 6-33**, traffic flows through PE1 -> P1 -> P2 -> PE2, and the primary link is between P1 and P2. Remote LFA calculates a PQ node (P4) and establishes a Label Distribution Protocol (LDP) tunnel between P1 and P4. If P1 detects a failure on the primary link, P1 encapsulates packets into MPLS packets and forwards MPLS packets to P4. After receiving the packets, P4 removes the MPLS label from them and searches the IP routing table for a next hop to forward the packets to PE2. Remote LFA ensures uninterrupted traffic forwarding.

Figure 6-33 Networking for Remote LFA



On the network shown in **Figure 6-33**, Remote LFA calculates the PQ node as follows:

1. Calculates the SPTs with all neighbors of P1 as roots. The nodes through which the SPTs are reachable without passing through the primary link form an extended P space. The extended P space in this example is {PE1, P1, P3, P4}.
2. Calculates the SPTs with P2 as the root and obtains the Q space {PE2, P4}.
3. Selects the PQ node (P4) that exists both in the extended P space and Q space.

OSPF anti-microloop

In **Figure 6-33**, OSPF remote LFA FRR is enabled, the primary link is PE1 -> P1 -> P2 -> PE2, and the backup link is PE1 -> P1 -> P3 -> P4 -> P2 -> PE2, and the link P1 -> P3 -> P4 is an LDP tunnel. If the primary link fails, traffic is switched to the backup link, and then another round of the new primary link calculation begins. Specifically, after P1 completes route convergence, its next hop becomes P3. However, the route convergence on P3 is slower than that on P1, and P3's next hop is still P1. As a result, a temporary loop occurs between P1

and P3. OSPF anti-microloop can address this problem by delaying P1 from switching its next hop until the next hop of P3 becomes P4. Then traffic is switched to the new primary link (PE1 -> P1 -> P3 -> P4 -> P2 -> PE2), and on the link P1 -> P3 -> P4, traffic is forwarded based on IP routes.

NOTE

OSPF anti-microloop applies only to OSPF remote LFA FRR.

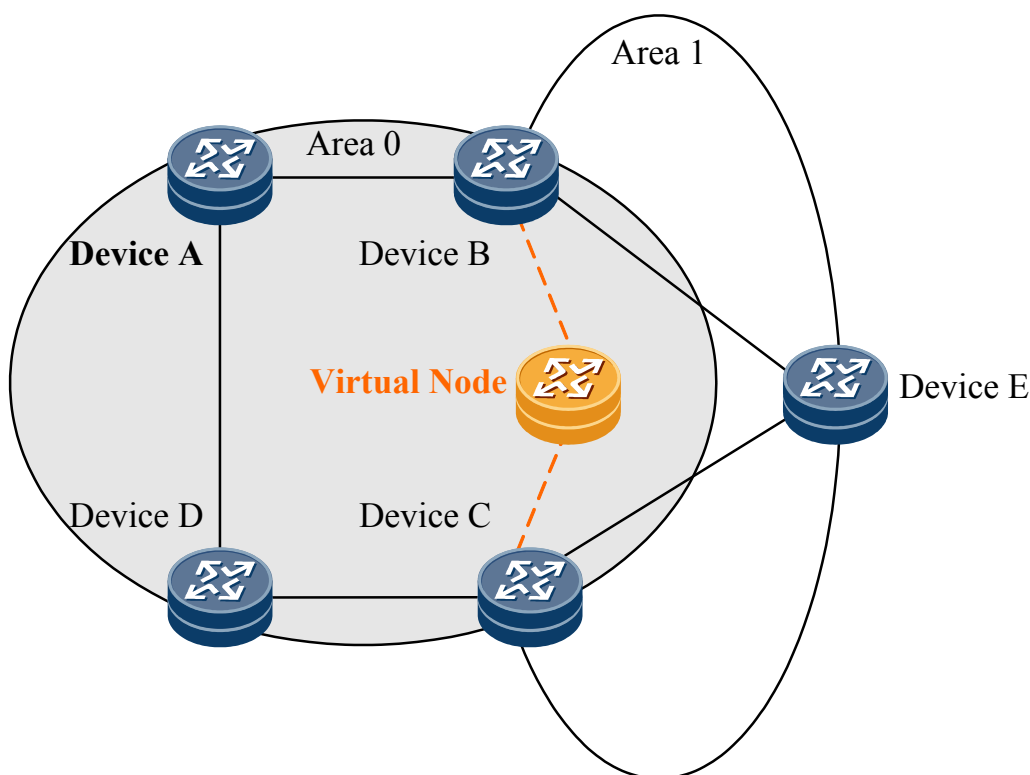
OSPF FRR in a Multi-Source Routing Scenario

Both OSPF LFA FRR and OSPF remote LFA FRR use the SPF algorithm to calculate the shortest path from each neighbor (root node) that provides a backup link to the destination node and store the node-based backup next hop, which applies to single-source routing scenarios. As networks are increasingly diversified, two ABRs or ASBRs are deployed to improve network reliability. In this case, OSPF FRR in a multi-source routing scenario is needed.

NOTE

In a multi-source routing scenario, OSPF FRR is implemented by calculating the Type 3 LSAs advertised by ABRs of an area for intra-area, inter-area, ASE, or NSSA routing. Inter-area routing is used as an example to describe how OSPF FRR in a multi-source routing scenario works.

Figure 6-34 OSPF FRR in a multi-route source scenario



In [Figure 6-34](#), Device B and Device C function as ABRs to forward area 0 and area 1 routes. Device E advertises an intra-area route. Upon receipt of the route, Device B and Device C translate it to a Type 3 LSA and flood the LSA to area 0. After OSPF FRR is enabled on Device A, Device A considers Device B and Device C as its neighbors. Without a fixed neighbor as the root node, Device A fails to calculate FRR backup next hop. To address this problem, a virtual node is simulated between Device B and Device C and used as the root

node of Device A, and Device A uses the LFA or remote LFA algorithm to calculate the backup next hop. This solution converts multi-source routing into single-source routing.

For example, both Device B and Device C advertise the 100.1.1.0/24 route. After Device A receives the route, it fails to calculate a backup next hop for the route due to a lack of a fixed root node. To address this problem, a virtual node is simulated between Device B and Device C and used as the root node of Device A. The cost of the Device B-virtual node link is 0, and the cost of the Device C-virtual node link is 5. The cost of the virtual node-Device B or Device C link is the maximum value (65535). If the virtual node advertises the 100.1.1.0/24 route, it will use the smaller cost of the routes advertised by Device B and Device C as the cost of the route. Device A is configured to consider Device B and Device C as invalid sources of the 100.1.1.0/24 route and use the LFA or remote LFA algorithm to calculate the backup next hop for the route, with the virtual node as the root node.

In a multi-source routing scenario, OSPF FRR can use the LFA or remote LFA algorithm. When OSPF FRR uses the remote LFA algorithm, PQ node selection has the following restrictions:

- An LDP LSP will be established between a faulty node and a PQ node, and a virtual node in a multi-source routing scenario cannot transmit traffic through LDP LSPs. As a result, the virtual node cannot be selected as a PQ node.
- The destination node is not used as a PQ node. After a virtual node is added to a multi-source routing scenario, the destination node becomes the virtual node. As a result, the nodes directly connected to the virtual node cannot be selected as PQ nodes.

Derivative Functions

If you bind a Bidirectional Forwarding Detection (BFD) session with OSPF IP FRR, the BFD session goes Down if BFD detects a link fault. If the BFD session goes Down, OSPF IP FRR is triggered to switch traffic from the faulty link to the backup link, which minimizes the loss of traffic.

6.2.19 OSPF Authentication

OSPF authentication encrypts OSPF packets by adding the authentication field to packets to ensure network security. When a local device receives OSPF packets from a remote device, the local device discards the packets if the authentication passwords carried in these packets do not match the local one, which protects the local device from potential attacks.

In terms of the packet type, the authentication is classified as follows:

- Area authentication
Area authentication is configured in the OSPF area view and applies to packets received by all interfaces in the OSPF area.
- Interface authentication
Interface authentication is configured in the interface view and applies to all packets received by the interface.

In terms of packet the authentication modes, the authentication is classified as follows:

- Non-authentication
Authentication is not required.
- Simple authentication

The authenticated party directly adds the configured password to packets for authentication. This authentication mode provides the lowest password security.

- MD5 authentication

The authenticated party encrypts the configured password using a Message Digest 5 (MD5) algorithm and adds the ciphertext password to packets for authentication. This authentication mode improves password security. The supported MD5 algorithms include MD5 and HMAC-MD5.

- Keychain authentication

A keychain consists of multiple authentication keys, each of which contains an ID and a password. Each key has the lifetime. Keychain dynamically selects the authentication key based on the lifetime. A keychain can dynamically select the authentication key to enhance attack defense.

Keychain dynamically changes algorithms and keys, which improves OSPF security. For detailed information about Keychain, see the chapter "Keychain" in HUAWEI NetEngine40E Feature Description - Security.

- HMAC-SHA256 authentication

A password is encrypted using the HMAC-SHA256 algorithm before it is added to the packet, which improves password security.

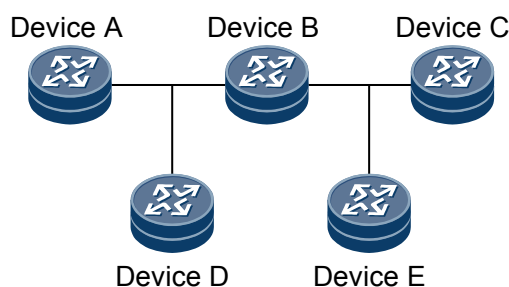
OSPF carries authentication types in packet headers and authentication information in packet tails.

The authentication types include:

- 0: non-authentication
- 1: simple authentication
- 2: Ciphertext authentication

Usage Scenario

Figure 6-35 OSPF authentication on a broadcast network



The configuration requirements are as follows:

- The interface authentication configurations must be the same on all devices on the same network so that OSPF neighbor relationships can be established.
- The area authentication configurations must be the same on all devices in the same area.

6.2.20 OSPF Packet Format

The OSPF protocol number is 89. OSPF packets are encapsulated into IP packets. OSPF packets are classified into the following types:

- **Hello packet**
- **Database Description (DD) packet**
- **Link State Request (LSR) packet**
- **Link State Update (LSU) packet**
- **Link State Acknowledgment (LSAck) packet**

Packet Header Format

The five types of OSPF packets have the same packet header format. The length of an OSPF packet header is 24 bytes. **Figure 6-36** shows an OSPF packet header.

Figure 6-36 OSPF packet header

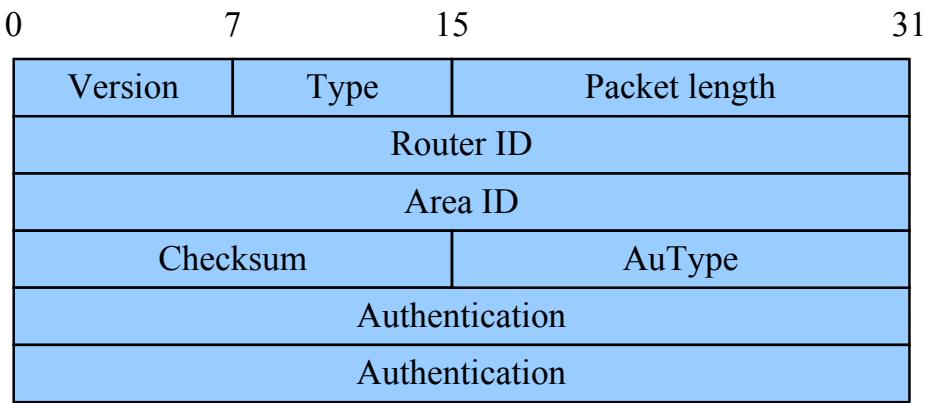


Table 6-17 describes packet header fields.

Table 6-17 Packet header fields

Field	Length	Description
Version	8 bits	OSPF version number. For OSPFv2, the value is 2.
Type	8 bits	OSPF packet type. The values are as follows: ● 1: Hello packet ● 2: DD packet ● 3: LSR packet ● 4: LSU packet ● 5: LSAck packet
Packet length	16 bits	Length of the OSPF packet with the packet header, in bytes.
Router ID	32 bits	ID of the router that sends the OSPF packet.
Area ID	32 bits	ID of the area to which the router that sends the OSPF packet belongs.

Field	Length	Description
Checksum	16 bits	Checksum of the OSPF packet that does not carry the Authentication field.
AuType	16 bits	Authentication type. The values are as follows: ● 0: non-authentication ● 1: simple authentication ● 2: message digest algorithm 5 (MD5) authentication
Authentication	64 bits	This field has different meanings for different AuType values: ● 0: This field is not defined. ● 1: This field defines password information. ● 2: This field contains the key ID, MD5 authentication data length, and sequence number.

 NOTE

MD5 authentication data is added to an OSPF packet and is not included in the Authentication field.

Hello Packet

Hello packets are commonly used packets, which are periodically sent by OSPF interfaces to establish and maintain neighbor relationships. A Hello packet includes information about the designated router (DR), backup designated router (BDR), timers, and known neighbors.

[Figure 6-37](#) shows the format of a Hello packet.

Figure 6-37 Format of a Hello packet

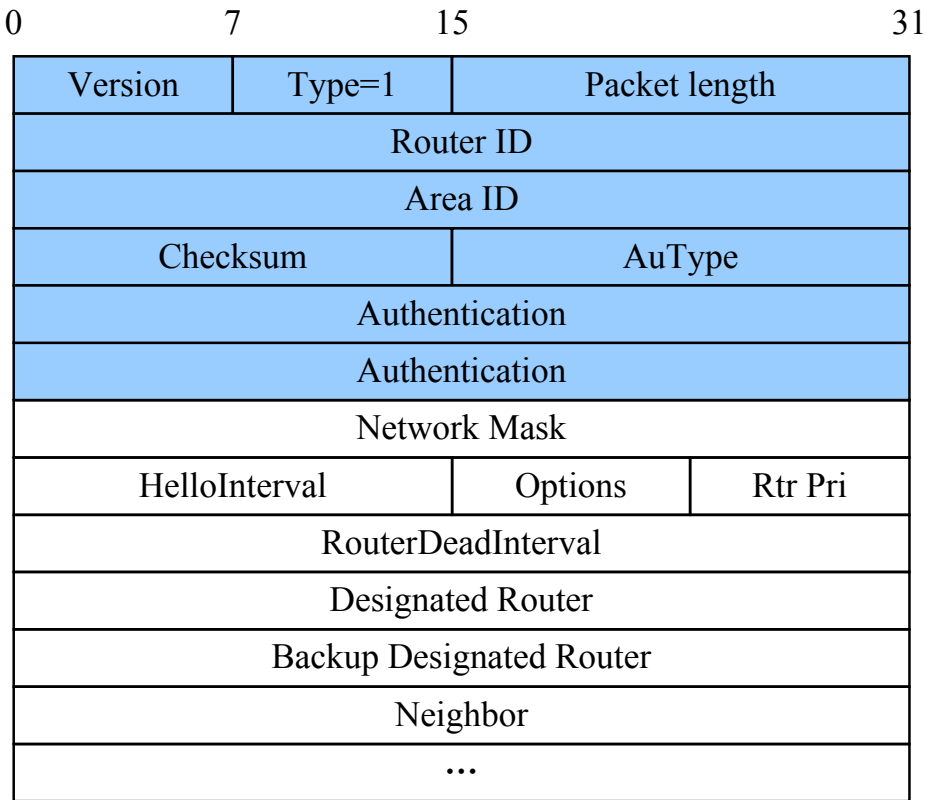


Table 6-18 describes Hello packet fields.

Table 6-18 Hello packet fields

Field	Length	Description
Network Mask	32 bits	Mask of the network on which the interface that sends the Hello packet resides.
HelloInterval	16 bits	Interval at which Hello packets are sent.
Options	8 bits	The values are as follows: ● E: Type 5 link state advertisements (LSAs) are flooded. ● MC: IP multicast packets are forwarded. ● N/P: Type 7 LSAs are processed. ● DC: On-demand links are processed.
Rtr Pri	8 bits	DR priority. The default value is 1. NOTE If the DR priority of a router interface is set to 0, the interface cannot participate in a DR or BDR election.
RouterDeadInterval	32 bits	Dead interval. If a router does not receive any Hello packets from its neighbors within a specified dead interval, the neighbors are considered Down.

Field	Length	Description
Designated Router	32 bits	Interface address of the DR.
Backup Designated Router	32 bits	Interface address of the BDR.
Neighbor	32 bits	Router ID of the neighbor.

Table 6-19 lists the address types, interval types, and default intervals used when Hello packets are transmitted on different networks.

Table 6-19 Hello packet characteristics for various network types

Network Type	Address Type	Interval Type	Default Interval
Broadcast	Multicast address	HelloInterval.	10 seconds
Non-broadcast multiple access (NBMA)	Unicast address	● HelloInterval is used by the DR, BDR, and router that can become a DR. ● PollInterval is used when neighbors become Down, and HelloInterval is used in other cases.	30 seconds for HelloInterval 120 seconds for PollInterval
Point-to-point (P2P)	Multicast address	HelloInterval.	10 seconds
Point-to-multipoint (P2MP)	Multicast address	HelloInterval.	30 seconds

NOTE

To establish neighbor relationships between routers on the same network segment, set the same HelloInterval, PollInterval, and RouterDeadInterval values for the routers. PollInterval applies only to NBMA networks.

DD Packet

During an adjacency initialization, two routers use DD packets to describe their own link state databases (LSDBs) for LSDB synchronization. A DD packet contains the header of each LSA in an LSDB. An LSA header uniquely identifies an LSA. The LSA header occupies only a small portion of the LSA, which reduces the amount of traffic transmitted between routers. A neighbor can use the LSA header to check whether it already has the LSA. When two routers

exchange DD packets, one functions as the master, and the other functions as the slave. The master defines a start sequence number and increases the sequence number by one each time it sends a DD packet. After the slave receives a DD packet, it uses the sequence number carried in the DD packet for acknowledgment.

Figure 6-38 shows the format of a DD packet.

Figure 6-38 Format of a DD packet

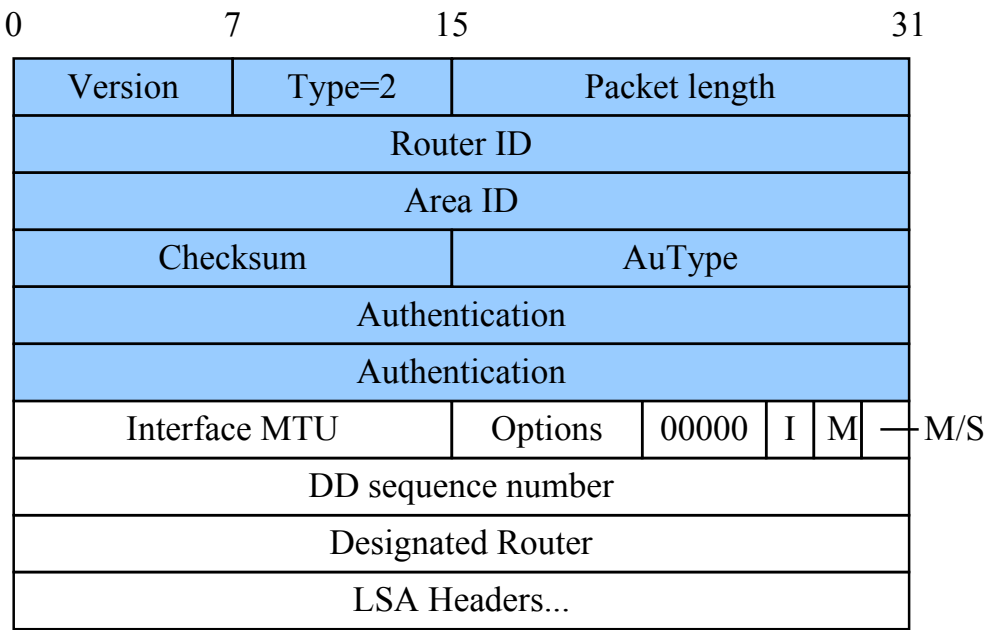


Table 6-20 describes DD packet fields.

Table 6-20 DD packet fields

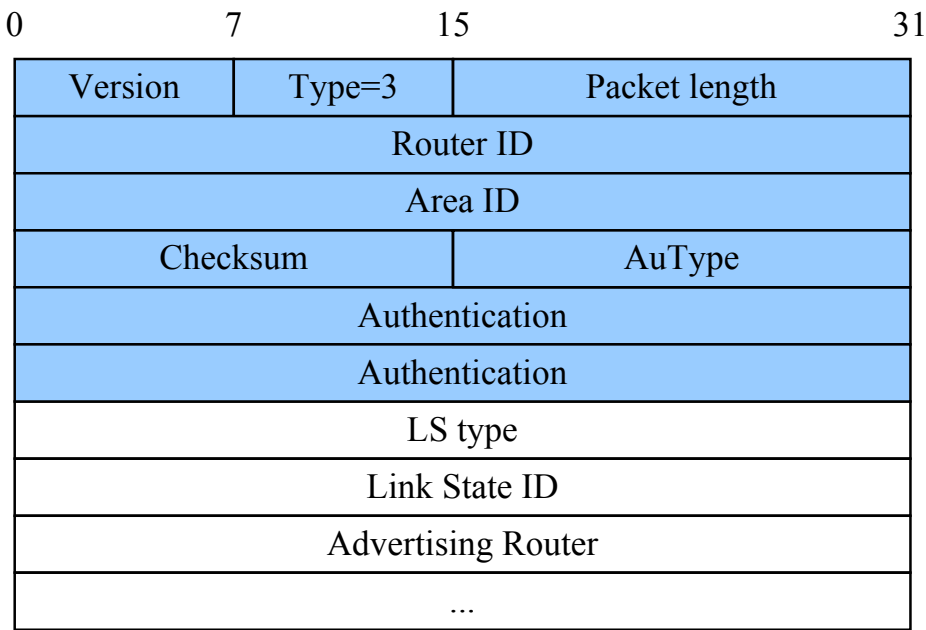
Field	Length	Description
Interface MTU	16 bits	Maximum length of the DD packet sent by the interface with packet fragmentation disabled.
Options	8 bits	The values are as follows: ● E: Type 5 LSAs are flooded. ● MC: IP multicast packets are forwarded. ● N/P: Type 7 LSAs are processed. ● DC: On-demand links are processed.
I	1 bit	If the DD packet is the first packet among multiple consecutive DD packets sent by a router, this field is set to 1. In other cases, this field is set to 0.
M (More)	1 bit	If the DD packet is the last packet among multiple consecutive DD packets sent by a router, this field is set to 0. In other cases, this field is set to 1.

Field	Length	Description
M/S (Master/ Slave)	1 bit	When two routers exchange DD packets, they negotiate a master/slave relationship. The router with a larger router ID becomes the master. If this field is set to 1, the DD packet is sent by the master.
DD sequence number	32 bits	Sequence number of the DD packet. The master and slave use the sequence number to ensure that DD packets are correctly transmitted.
LSA Headers	-	LSA header information included in the DD packet.

LSR Packet

After two routers exchange DD packets, they send LSR packets to request each other's LSAs. The LSR packets contain the summaries of the requested LSAs. [Figure 6-39](#) shows the format of an LSR packet.

Figure 6-39 Format of an LSR packet



[Table 6-21](#) describes LSR packet fields.

Table 6-21 LSR packet fields

Field	Length	Description
LS type	32 bits	Type of the LSA.

Field	Length	Description
Link State ID	32 bits	This field together with the LS type field describes an LSA in an AS.
Advertising Router	32 bits	Router ID of the router that generates the LSA.

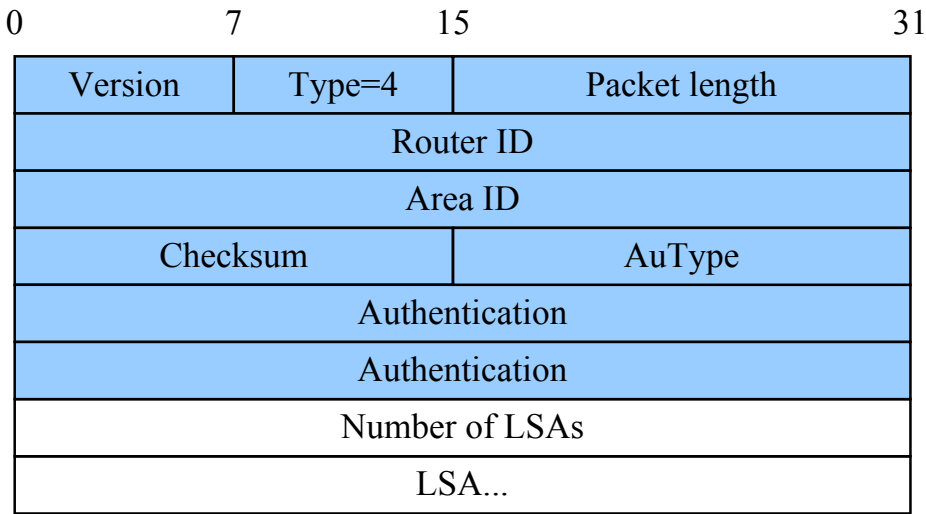
 **NOTE**

The LS type, Link State ID, and Advertising Router fields can uniquely identify an LSA. If two LSAs have the same LS type, Link State ID, and Advertising Router fields, a router uses the LS sequence number, LS checksum, and LS age fields to obtain a required LSA.

LSU Packet

A router uses an LSU packet to transmit LSAs requested by its neighbors or to flood its own updated LSAs. The LSU packet contains a set of LSAs. For multicast and broadcast networks, LSU packets are multicast to flood LSAs. To ensure reliable LSA flooding, a router uses an LSAck packet to acknowledge the LSAs contained in an LSU packet that is received from a neighbor. If an LSA fails to be acknowledged, the router retransmits the LSA to the neighbor. [Figure 6-40](#) shows the format of an LSU packet.

Figure 6-40 Format of an LSU packet



[Table 6-22](#) describes the LSU packet field.

Table 6-22 LSU packet field

Field	Length	Description
Number of LSAs	32 bits	Number of LSAs contained in the LSU packet

LSAck Packet

A router uses an LSAck packet to acknowledge the LSAs contained in a received LSU packet. The LSAs can be acknowledged using LSA headers. LSAck packets can be transmitted over different links in unicast or multicast mode. **Figure 6-41** shows the format of an LSAck packet.

Figure 6-41 Format of an LSAck packet

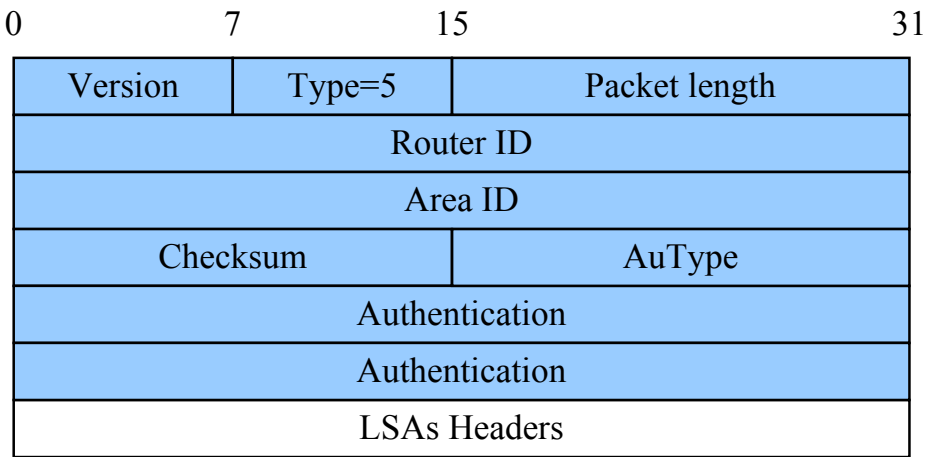


Table 6-23 describes the LSAck packet field.

Table 6-23 LSAck packet field

Field	Length	Description
LSAs Headers	Determine d by the header length of the LSA to be acknowledged.	This field is used to acknowledge an LSA.

6.2.21 OSPF LSA Format

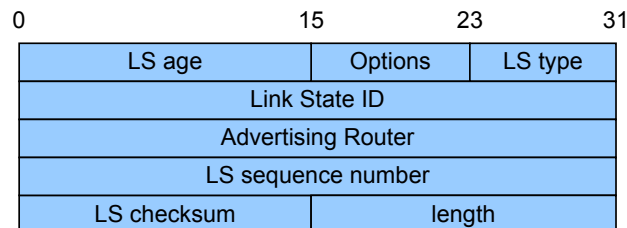
Each router in an AS generates one or more types of LSAs, depending on the router's type. Multiple LSAs form an LSDB. OSPF encapsulates routing information into LSAs for transmission. Commonly used LSAs include:

- Router-LSAs (Type 1)
- Network-LSAs (Type 2)
- Summary-LSAs, including network-summary-LSAs (Type 3) and ASBR-summary-LSAs (Type 4)
- AS-external-LSAs (Type 5)

LSA Header Format

All LSAs have the same header. [Figure 6-42](#) shows an LSA header.

Figure 6-42 LSA header



[Table 6-24](#) describes LSA header fields.

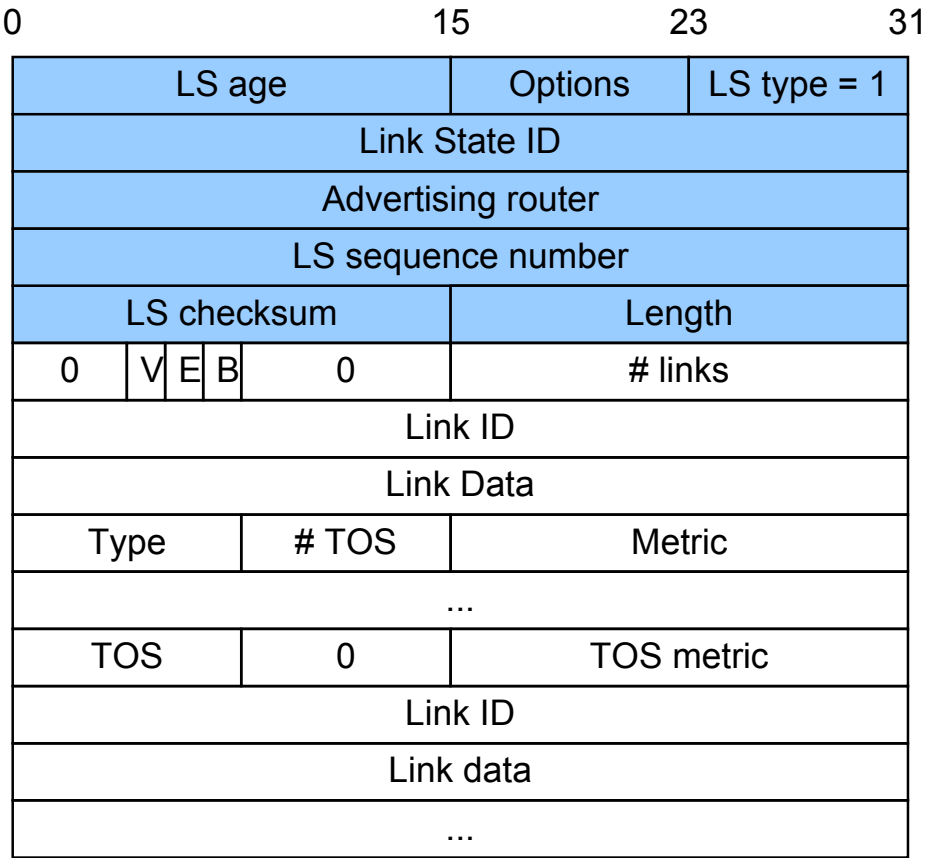
Table 6-24 LSA header fields

Field	Length	Description
LS age	16 bits	Time that elapses after an LSA is generated, in seconds. The value of this field continually increases regardless of whether the LSA is transmitted over a link or saved in an LSDB.
Options	8 bits	The values are as follows: ● E: Type 5 LSAs are flooded. ● MC: IP multicast packets are forwarded. ● N/P: Type 7 LSAs are processed. ● DC: On-demand links are processed.
LS type	8 bits	Type of the LSA. The values are as follows: ● Type1: Router-LSA ● Type2: Network-LSA ● Type3: Network-summary-LSA ● Type4: ASBR-summary-LSA ● Type5: AS-external-LSA ● Type7: NSSA-LSA
Link State ID	32 bits	This field together with the LS type field describes an LSA in an AS.
Advertising Router	32 bits	Router ID of the router that generates the LSA.
LS sequence number	32 bits	Sequence number of the LSA. Neighbors can use this field to identify the latest LSA.
LS checksum	16 bits	Checksum of all fields except the LS age field.
length	16 bits	Length of the LSA including the LSA header, in bytes.

Router-LSA

A router-LSA describes the link status and cost of a router. Router-LSAs are generated by a router and advertised within the area to which the router belongs. [Figure 6-43](#) shows the format of a router-LSA.

Figure 6-43 Format of a router-LSA



[Table 6-25](#) describes router-LSA fields.

Table 6-25 Router-LSA fields

Field	Length	Description
Link State ID	32 bits	Router ID of the router that generates the LSA.
V (Virtual Link)	1 bit	If the router that generates the LSA is located at one end of a virtual link, this field is set to 1. In other cases, this field is set to 0.
E (External)	1 bit	If the router that generates the LSA is an autonomous system boundary router (ASBR), this field is set to 1. In other cases, this field is set to 0.

Field	Length	Description
B (Border)	1 bit	If the router that generates the LSA is an area border router (ABR), this field is set to 1. In other cases, this field is set to 0.
# links	16 bits	Number of links and interfaces described in the LSA, including all links and interfaces in the area to which the router belongs.
Link ID	32 bits	Object to which the router is connected. Its meanings are as follows: ● 1: router ID ● 2: interface IP address of the designated router (DR) ● 3: network segment or subnet number ● 4: router ID of the neighbor on a virtual link
Link Data	32 bits	Link data. Its meanings are as follows: ● 1: interface index ● 3: subnet mask ● 2 and 4: interface address of the router
Type	8 bits	Type of the router link. The values are as follows: ● 1: The router is connected to another router in point-to-point (P2P) mode. ● 2: The router is connected to a transport network. ● 3: The router is connected to a stub network. ● 4: The router is connected to another router over a virtual link.
# ToS	8 bits	Number of types of service (ToSs).
metric	16 bits	Cost of the link.
ToS	8 bits	Type of service.
ToS metric	16 bits	Metric for the specified ToS.

Network-LSA

A network-LSA describes the link status of all routers on the local network segment. Network-LSAs are generated by a DR on a broadcast or non-broadcast multiple access (NBMA) network and advertised within the area to which the DR belongs. [Figure 6-44](#) shows the format of a network-LSA.

Figure 6-44 Format of a network-LSA

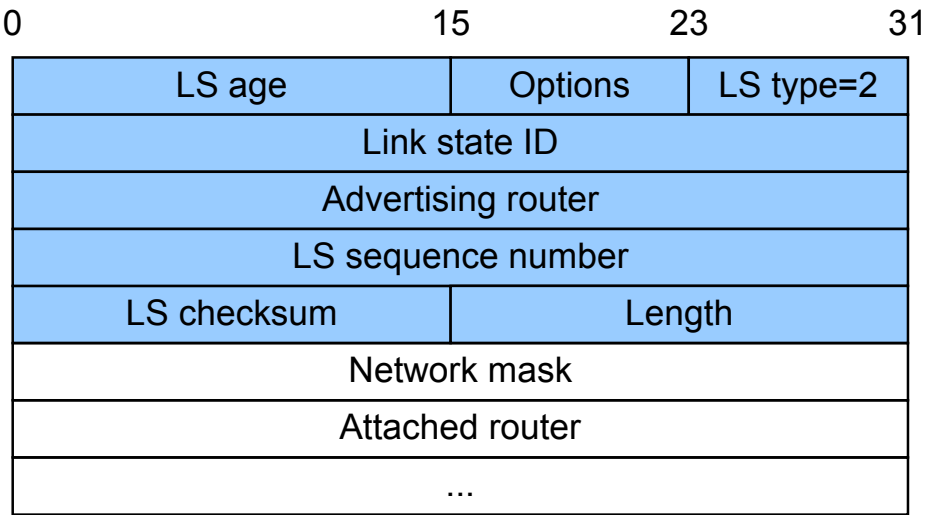


Table 6-26 describes network-LSA fields.

Table 6-26 Network-LSA fields

Field	Length	Description
Link State ID	32 bits	Interface IP address of the DR
Network Mask	32 bits	Mask of the broadcast or NBMA network
Attached Router	32 bits	Router IDs of all routers on the broadcast or NBMA network, including the router ID of the DR

Summary-LSA

A network-summary-LSA describes routes on a network segment in an area. The routes are advertised to other areas.

An ASBR-summary-LSA describes routes to the ASBR in an area. The routes are advertised to all areas except the area to which the ASBR belongs.

The two types of summary-LSAs have the same format and are generated by an ABR. Figure 6-45 shows the format of a summary-LSA.

Figure 6-45 Format of a summary-LSA

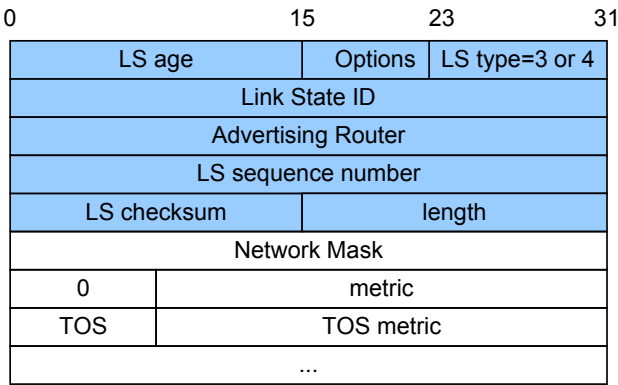


Table 6-27 describes network-summary-LSA fields.

Table 6-27 Network-summary-LSA fields

Field	Length	Description
Link State ID	32 bits	Advertised network address
Network Mask	32 bits	Mask of the broadcast or NBMA network
metric	24 bits	Cost of the route to the destination address
ToS	8 bits	Type of service
ToS metric	24 bits	Metric for the specified ToS

NOTE

When a default route is advertised, both the Link State ID and Network Mask fields are set to 0.0.0.0.

Table 6-28 describes ASBR-summary-LSA fields.

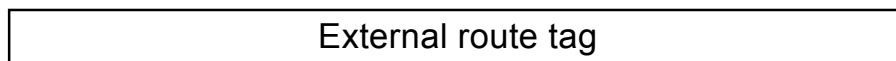
Table 6-28 ASBR-summary-LSA fields

Field	Length	Description
Link State ID	32 bits	Router ID of the ASBR
Network Mask	32 bits	Set to 0.0.0.0
metric	24 bits	Cost of the route to the destination address
ToS	8 bits	Type of service
ToS metric	24 bits	Metric for the specified ToS

AS-External-LSA

An AS-external-LSA describes AS external routes. AS-external-LSAs are generated by an ASBR. Among the five types of LSAs, only AS-external-LSAs can be advertised to all areas except stub areas and not-so-stubby areas (NSSAs). [Figure 6-46](#) shows the format of an AS-external-LSA.

Figure 6-46 Format of an AS-external-LSA



[Table 6-29](#) describes AS-external-LSA fields.

Table 6-29 AS-external-LSA fields

Field	Length	Description
Link State ID	32 bits	Advertised network address.
Network Mask	32 bits	Mask of the advertised destination address.
E	1 bit	Type of the external route. The values are as follows: ● 0: Type 1 external route ● 1: Type 2 external route
metric	24 bit	Cost of the route to the destination address.
Forwarding Address	32 bits	Packets destined for the advertised destination address are forwarded to the address specified by this field.
External Route Tag	32 bits	Tag added to the external route. This field can be used to manage external routes. OSPF itself does not use this field.
ToS	8 bits	Type of service.
ToS metric	24 bits	Metric for the specified ToS.

 NOTE

When AS-external-LSAs are used to advertise default routes, both the Link State ID and Network Mask fields are set to 0.0.0.0.

7 OSPFv3

About This Chapter

[7.1 Overview of OSPFv3](#)

[7.2 Understanding OSPFv3](#)

7.1 Overview of OSPFv3

Definition

Open Shortest Path First (OSPF), developed by the Internet Engineering Task Force (IETF), is a link-state Interior Gateway Protocol (IGP).

At present, OSPF Version 2 (OSPFv2) is used for IPv4, whereas OSPF Version 3 (OSPFv3), developed on the basis of OSPFv2, is used for IPv6.

Purpose

The primary purpose of OSPFv3 is to develop a routing protocol independent of any specific network layer. The internal OSPFv3 router information is redesigned to achieve this purpose.

7.2 Understanding OSPFv3

7.2.1 OSPFv3 Fundamentals

Running on IPv6, OSPFv3 is an independent routing protocol that is developed on the basis of OSPFv2.

- OSPFv3 and OSPFv2 are the same in terms of the working principles of the Hello packet, state machine, link-state database (LSDB), flooding, and route calculation.
- OSPFv3 packets are encapsulated into IPv6 packets and can be transmitted in unicast or multicast mode.

OSPFv3 Packet Types

Packet Type	Function
Hello packet	Hello packets are sent periodically to discover and maintain OSPFv3 neighbor relationships.
Database Description (DD) packet	DD packets contain the summary of the local LSDB and are exchanged between two OSPFv3 routers to update the LSDBs.
Link State Request (LSR) packet	LSR packets are sent to the neighbor to request the required LSAs. An OSPFv3 device sends LSR packets to its neighbor only after they exchange DD packets.
Link State Update (LSU) packet	LSU packets carry the LSAs required by neighbors.
Link State Acknowledgment (LSAck) packet	LSAck packets acknowledge the receipt of an LSA.

LSA Types

LSA Type	Description
Router-LSA (Type 1)	Describes the link status and link cost of a router. It is generated by every router and advertised in the area to which the router belongs.
Network-LSA (Type 2)	Describes the link status of all routers on the local network segment. Network-LSAs are generated by a designated router (DR) and advertised in the area to which the DR belongs.
Inter-Area-Prefix-LSA (Type 3)	Describes routes to a specific network segment in an area. Inter-Area-Prefix-LSAs are generated on the Area Border Router (ABR) and sent to related areas.
Inter-Area-Router-LSA (Type 4)	Describes routes to an Autonomous System Boundary Router (ASBR). Inter-Area-Router-LSAs are generated by an ABR and advertised to all related areas except the area to which the ASBR belongs.
AS-external-LSA (Type 5)	Describes routes to a destination outside the AS. AS-external-LSAs are generated by an ASBR and advertised to all areas except stub areas and NSSAs.
NSSA LSA (Type 7)	Describes routes to a destination outside the AS. It is generated by an ASBR and advertised in NSSAs only.
Link-LSA (Type 8)	Describes the link-local address and IPv6 address prefix associated with the link and the link option set in the network LSA. Link LSAs are transmitted only on the link.

LSA Type	Description
Intra-Area-Prefix-LSA (Type 9)	Each device or DR generates one or more intra-area prefix LSAs and transmits it in the local area. An intra-area prefix LSA generated by a device describes the IPv6 address prefix associated with the router LSA. An intra-area prefix LSA generated by a DR describes the IPv6 address prefix associated with the network LSA.

Router Types

Figure 7-1 Router types

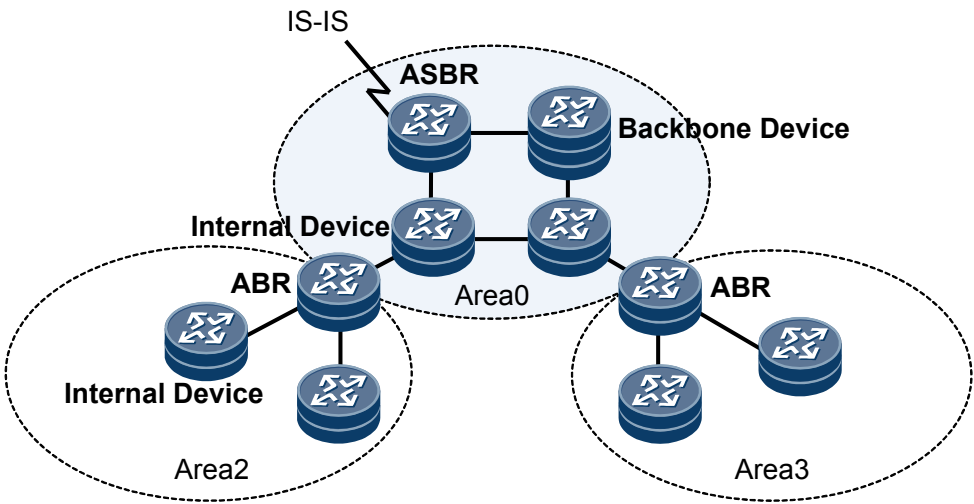


Table 7-1 Router types and descriptions

Router Type	Description
Internal router	All interfaces on an internal router belong to the same OSPFv3 area.
Area border router (ABR)	An ABR belongs to two or more areas, one of which must be the backbone area. An ABR is used to connect the backbone area and non-backbone areas. It can be physically or logically connected to the backbone area.

Router Type	Description
Backbone router	At least one interface on a backbone router belongs to the backbone area. Internal routers in Area 0 and all ABRs are backbone routers.
AS boundary router (ASBR)	An ASBR exchanges routing information with other ASs. An ASBR does not necessarily reside on the border of an AS. It can be an internal router or an ABR. An OSPFv3 device that has imported external routing information will become an ASBR.

OSPFv3 Route Types

Inter-area routes and intra-area routes describe the network structure of an AS. External routes describe how to select a route to the destination outside an AS. OSPFv3 classifies the imported AS external routes into Type 1 routes and Type 2 routes.

Table 7-2 lists route types in descending order of priority.

Table 7-2 Types of OSPFv3 routes

Route Type	Description
Intra area route	Indicates routes within an area.
Inter area route	Indicates routes between areas.
Type 1 external route	Type 1 external routes have high reliability. Cost of a Type 1 external route = Cost of the route from a local router to an ASBR + Cost of the route from the ASBR to the destination of the Type 1 external route
Type 2 external route	Type 2 external routes have low reliability, and therefore OSPFv3 considers the cost of the route from an ASBR to the destination of a Type 2 external route to be much greater than the cost of any internal route to the ASBR. Cost of a Type 2 external route = Cost of the route from the ASBR to the destination of the Type 2 external route

Area

When a large number of routers run OSPFv3, LSDBs become very large and require a large amount of storage space. Large LSDBs also complicate shortest path first (SPF) computation and are computationally intensive for the routers. Network expansion causes the network topology to change, which results in route flapping and frequent OSPFv3 packet transmission. When a large number of OSPFv3 packets are transmitted on the network, bandwidth usage efficiency decreases. Each change in the network topology causes all routers on the network to recalculate routes.

OSPFv3 resolves this problem by partitioning an AS into different areas. An area is regarded as a logical group, and each group is identified by an area ID. A router, not a link, resides at the border of an area. A network segment or link can belong only to one area. An area must be specified for each OSPFv3 interface.

OSPFv3 areas include common areas, stub areas, and not-so-stubby areas (NSSAs), as described in [Table 7-3](#).

Table 7-3 OSPFv3 areas

Area Type	Function	Notes
Common area	OSPFv3 areas are common areas by default. Common areas include standard areas and backbone areas. ● Standard area: transmits intra-area, inter-area, and external routes. ● Backbone area: connects to all other OSPFv3 areas and transmits inter-area routes. The backbone area is represented by area 0. Routes between non-backbone areas must be forwarded through the backbone area.	● In the backbone area, all devices must be connected. ● All non-backbone areas must be connected to the backbone area.
Stub area	A stub area is a non-backbone area with only one ABR and generally resides at the border of an AS. The ABR in a stub area does not transmit received AS external routes, which significantly decreases the number of entries in the routing table on the ABR and the amount of routing information to be transmitted. To ensure the reachability of AS external routes, the ABR in the stub area generates a default route and advertises the route to non-ABRs in the stub area. A totally stub area allows only intra-area routes and ABR-advertised Type 3 LSAs carrying a default route to be advertised within the area.	● The backbone area cannot be configured as a stub area. ● An ASBR cannot exist in a stub area. Therefore, AS external routes cannot be advertised within the stub area.
NSSA	An NSSA is similar to a stub area. An NSSA does not advertise Type 5 LSAs but can import AS external routes. ASBRs in an NSSA generate Type 7 LSAs to carry the information about the AS external routes. The Type 7 LSAs are advertised only within the NSSA. When the Type 7 LSAs reach an ABR in the NSSA, the ABR translates the Type 7 LSAs into Type 5 LSAs and floods them to the entire AS. A totally NSSA area allows only intra-area routes to be advertised within the area.	● ABRs in an NSSA advertise Type 3 LSAs carrying a default route within the NSSA. All inter-area routes are advertised by ABRs.

Network Types Supported by OSPFv3

OSPFv3 classifies networks into the following types based on link layer protocols.

Table 7-4 Types of OSPFv3 networks

Network Type	Description
Broadcast	OSPFv3 considers networks with Ethernet or Fiber Distributed Data Interface (FDDI) as the link layer protocol as broadcast networks by default. On a broadcast network: ● Hello packets, LSU packets, and LSAck packets are usually transmitted in multicast mode. FF02::5 is an IPv6 multicast address reserved for an OSPFv3 device. FF02::6 is an IPv6 multicast address reserved for an OSPFv3 DR or backup designated router (BDR). ● DD and LSR packets are transmitted in unicast mode.
Non-broadcast Multiple Access (NBMA)	OSPFv3 considers networks with X.25 as the link layer protocol as NBMA networks by default. On an NBMA network, protocol packets, such as Hello packets, DD packets, LSR packets, LSU packets, and LSAck packets are sent in unicast mode.
Point-to-Multipoint (P2MP)	No network is a P2MP network by default, no matter what type of link layer protocol is used on the network. A non-fully meshed NBMA network can be changed to a P2MP network. On a P2MP network: ● Hello packets are transmitted in multicast mode using the multicast address FF02::5. ● Other types of protocol packets, such as DD packets, LSR packets, LSU packets, and LSAck packets are sent in unicast mode.
Point-to-point (P2P)	OSPFv3 considers networks with PPP, HDLC, or LAPB as the link layer protocol to be P2P networks by default. On a P2P network, protocol packets, such as Hello packets, DD packets, LSR packets, LSU packets, and LSAck packets are sent in multicast mode using the multicast address FF02::5.

Stub Area

Stub areas are specific areas where ABRs do not flood received AS external routes. In stub areas, routers maintain fewer routing entries and less routing information than the routers in other areas.

Configuring a stub area is optional. Not every area can be configured as a stub area, because a stub area is usually a non-backbone area with only one ABR and is located at the AS border.

To ensure the reachability of the routes to destinations outside an AS, the ABR in the stub area generates a default route and advertises the route to the non-ABRs in the same stub area.

Note the following points when configuring a stub area:

- The backbone area cannot be configured as a stub area.
- Configure stub area attributes on all routers in the area to be configured as a stub area.
- No ASBRs are allowed in the area to be configured as a stub area because AS external routes cannot be transmitted in the stub area.

NSSA

Stub areas cannot import or transmit external routes, which prevents a large number of external routes from consuming the bandwidth and storage resources of routers in the Stub areas. If you need to import external routes to an area and prevent these routes from consuming resources, configure the area as a not-so-stubby area (NSSA).

Derived from stub areas, NSSAs resemble stub areas in many ways. Different from stub areas, NSSAs can import AS external routes and advertise them within the entire AS, without learning external routes from other areas in the AS.

To advertise external routes imported by an NSSA to other areas on the OSPFv3 network, a translator must translate Type 7 LSAs into Type 5 LSAs.

NOTE

- The propagate bit (P-bit) is used to notify a translator whether Type 7 LSAs need to be translated.
- By default, the translator is the ABR with the largest router ID in the NSSA.
- The P-bit is not set for Type 7 LSAs generated by an ABR.

OSPFv3 Route Summarization

Routes with the same IPv6 prefix can be summarized into one route. On a large-scale OSPFv3 network, route lookup may slow down because of the large size of the routing table. To reduce the routing table size and simplify management, configure route summarization. With route summarization, if a link connected to a device within an IPv6 address range that has been summarized alternates between Up and Down, the link status change is not advertised to the devices beyond the IPv6 address range. This prevents route flapping and improves network stability.

OSPFv3 route summarization is classified as follows:

- Route summarization on an ABR
An ABR can summarize routes with the same prefix into one route and advertise the summarized route to other areas.
When an ABR transmits routing information to other areas, it generates Type 3 LSAs for each network segment. If contiguous segments exist in this area, these segments can be summarized into one segment so that the ABR sends only one summarized LSA.
- Route summarization on an ASBR
An ASBR can summarize imported routes with the same prefix into one route and then advertise the summarized route to other areas.
With route summarization, an ASBR summarizes imported Type 5 LSAs within the summarized address range. After route summarization, the ASBR does not generate a separate Type 5 LSA for each specific prefix within the configured range. Instead, the

ASBR generates a Type 5 LSA for only the summarized prefix. In an NSSA, an ASBR summarizes multiple imported Type 7 LSAs within the summarized address range into one Type 7 LSA.

OSPFv3 Multi-process

OSPFv3 supports multi-process. Multiple OSPFv3 processes can independently run on the same router. Route exchange between different OSPFv3 processes is similar to that between different routing protocols.

7.2.2 Comparison Between OSPFv3 and OSPFv2

OSPFv3 and OSPFv2 are the same in the following aspects:

- Network types and interface types
- Interface state machines and neighbor state machines
- LSDB
- Flooding mechanism
- Five types of packets: Hello, DD, LSR, LSU, and LSAck packets
- Route calculation

OSPFv3 and OSPFv2 differ as follows:

- In OSPFv3, only LSUs contain IP addresses.
- OSPFv3 uses IPv6 which is based on links rather than network segments.
Therefore, the interfaces on which OSPFv3 is to be configured must be on the same link rather than in the same network segment. In addition, the interfaces can establish OSPFv3 sessions without IPv6 global addresses.
- OSPFv3 does not depend on IP addresses.
OSPFv3 separates topology calculation from IP addresses. Specifically, OSPFv3 can calculate the OSPFv3 topology without IPv6 global addresses which only apply to virtual link interfaces and packet forwarding.
- OSPFv3 packets and the LSA format change.
 - OSPFv3 packets do not contain IP addresses.
 - OSPFv3 router LSAs and network LSAs do not contain IP addresses, which are advertised through link LSAs and intra-area prefix LSAs.
 - In OSPFv3, router IDs, area IDs, and LSA link state IDs no longer indicate IP addresses, but the IPv4 address format is still reserved.
 - Neighbors are identified by router IDs instead of IP addresses on broadcast, NBMA, or P2MP networks.
- Information about the flooding scope is added to OSPFv3 LSAs.
Information about the flooding scope is added to the LSA Type field of OSPFv3 LSAs. Therefore, OSPFv3 routers can process LSAs of unidentified types more flexibly.
 - OSPFv3 can store or flood unidentified packets, whereas OSPFv2 discards unidentified packets.
 - In OSPFv3, unknown LSAs with 1 as the U flag bit can be flooded, and the flooding scope of such LSAs is specified by the LSAs.

For example, Device A and Device B can identify LSAs of a certain type. Device A and Device B are connected through Device C which, however, cannot identify these LSAs.

If Device A floods such LSA to Device C, Device C can still flood the received LSAs to Device B although Device C does not identify these LSAs. Device B then processes these LSAs.

If OSPFv2 is run, Device C discards the unidentified LSAs. As a result, these LSAs cannot reach Device B.

- OSPFv3 supports multi-process on a link.

In OSPFv2, one physical interface can be bound to only one multi-instance. In OSPFv3, one physical interface can be bound to multiple multi-instances that are identified by different instance IDs. In these OSPFv3 multi-instances running on one physical interface, neighbor relationships are established separately, sharing resources on the same link.

- OSPFv3 uses IPv6 link-local addresses.

IPv6 implements neighbor discovery and automatic configuration based on link-local addresses. Routers running IPv6 do not forward IPv6 packets whose destination address is a link-local address, and those packets can only be exchanged on the same link. The unicast link-local address starts from FE80/10.

As a routing protocol running on IPv6, OSPFv3 also uses link-local addresses to maintain neighbor relationships and update LSDBs. Except Vlink interfaces, all OSPFv3 interfaces use link-local addresses as the source address and the next hop to transmit OSPFv3 packets.

The advantages are as follows:

- OSPFv3 can calculate the topology without global IPv6 addresses.
- The packets flooded on a link are not transmitted to other links, which prevents unnecessary flooding and saves bandwidth.

- OSPFv3 supports two new LSAs.

- Link LSA: A device floods a link LSA on the link where it resides to advertise its link-local address and the configured global IPv6 address.
- Intra-area prefix LSA: A device advertises an intra-area prefix LSA in the local OSPF area to inform the other routers in the area or the network (either a broadcast network or an NBMA network) of its IPv6 global address.

- OSPFv3 identifies neighbors based on Router IDs only.

On broadcast, NBMA, and P2MP networks, OSPFv2 identifies neighbors based on IPv4 addresses of interfaces.

OSPFv3 identifies neighbors based on Router IDs only.

7.2.3 BFD for OSPFv3

Definition

Bidirectional Forwarding Detection (BFD) is a mechanism to detect communication faults between forwarding engines.

To be specific, BFD detects the connectivity of a data protocol along a path between two systems. The path can be a physical link, a logical link, or a tunnel.

In BFD for OSPFv3, a BFD session is associated with OSPFv3. The BFD session quickly detects a link fault and then notifies OSPFv3 of the fault, which speeds up OSPFv3's response to network topology changes.

Purpose

A link fault or a topology change causes routers to recalculate routes. Routing protocol convergence must be as quick as possible to improve network availability. Link faults are inevitable, and therefore a solution must be provided to quickly detect faults and notify routing protocols.

BFD for Open Shortest Path First version 3 (OSPFv3) associates BFD sessions with OSPFv3. After BFD for OSPFv3 is configured, BFD quickly detects link faults and notifies OSPFv3 of the faults. BFD for OSPFv3 accelerates OSPFv3 response to network topology changes.

Table 7-5 describes OSPFv3 convergence speeds before and after BFD for OSPFv3 is configured.

Table 7-5 OSPFv3 convergence speeds before and after BFD for OSPFv3 is configured

Item	Link Fault Detection Mechanism	Convergence Speed
BFD for OSPFv3 is not configured.	An OSPFv3 Dead timer expires.	Second-level
BFD for OSPFv3 is configured.	A BFD session goes Down.	Millisecond-level

Principles

Figure 7-2 BFD for OSPFv3

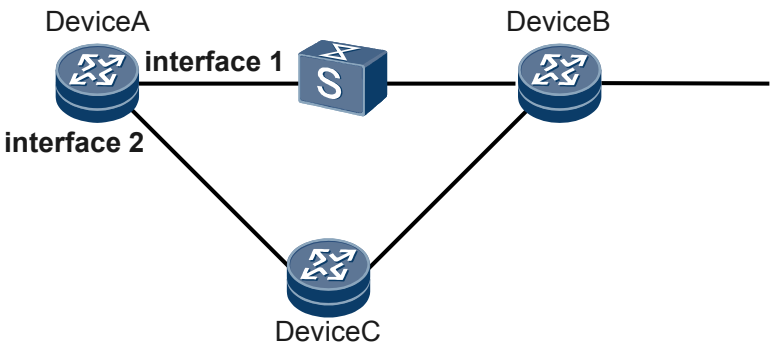


Figure 7-2 shows a typical network topology with BFD for OSPFv3 configured. The principles of BFD for OSPFv3 are described as follows:

- OSPFv3 neighbor relationships are established between these three routers.
- After a neighbor relationship becomes Full, a BFD session is established.
- The outbound interface on Device A connected to Device B is interface 1. If the link between Device A and Device B fails, BFD detects the fault and then notifies Device A of the fault.

4. Device A processes the event that a neighbor relationship has become Down and recalculates routes. The new route passes through Device C and reaches Device B, with interface 2 as the outbound interface.

7.2.4 Priority-based Convergence

Priority-based OSPFv3 convergence ensures that specific routes are converged first in the case of a great number of routes. Different routes can be set with different convergence priorities.

A higher convergence priority can be configured for routes over which key services are transmitted so that these routes can converge first, which minimizes the impact on key services.

7.2.5 OSPFv3 Auto FRR

OSPFv3 Auto Fast Reroute (FRR) is dynamic IP FRR in which a device pre-computes alternate next hops and stores them in the IP routing table. If a primary link fails, the device switches the traffic to a backup link.

OSPFv3 Auto FRR complies with standard protocols. With OSPFv3 Auto FRR, devices can switch traffic from a faulty primary link to a backup link, protecting against a link or node failure.

Background

As networks develop, Voice over Internet Protocol (VoIP) and online video services pose higher requirements for real-time transmission. Nevertheless, if a primary link fails, OSPFv3-enabled devices need to perform multiple operations, including detecting the fault, updating the link-state advertisement (LSA), flooding the LSA, calculating routes, and delivering forward information base (FIB) entries before switching traffic to a new link. This process takes a much longer time, the minimum delay to which users are sensitive. As a result, the requirements for real-time transmission cannot be met.

Principles

With OSPFv3 IP FRR, a device uses the loop-free alternate (LFA) algorithm to compute the next hop of a backup link and stores the next hop together with the primary link in the forwarding table. If the primary link fails, the device switches the traffic to the backup link before routes are converged on the control plane. This mechanism keeps the traffic interruption duration and minimizes the impacts. The NE40E supports OSPFv3 Auto FRR.

A device uses shortest path first (SPF) algorithm to calculate the shortest path from each neighbor that can provide a backup link to the destination node. The device then uses the inequalities defined in standard protocols and the LFA algorithm to calculate the next hop of the loop-free backup link that has the smallest cost of the available shortest paths.

An OSPFv3 Auto FRR policy is used to filter alternate next hops. Only the alternate next hops that match the filtering rules of the policy can be added to the IP routing table. Users can configure a desired OSPF IP FRR policy to filter alternate next hops.

If a Bidirectional Forwarding Detection (BFD) session is bound to OSPFv3 Auto FRR, the BFD session goes Down if BFD detects a link fault. If the BFD session goes Down, OSPFv3 Auto FRR is triggered on the interface to switch traffic from the faulty link to the backup link, which minimizes the loss of traffic.

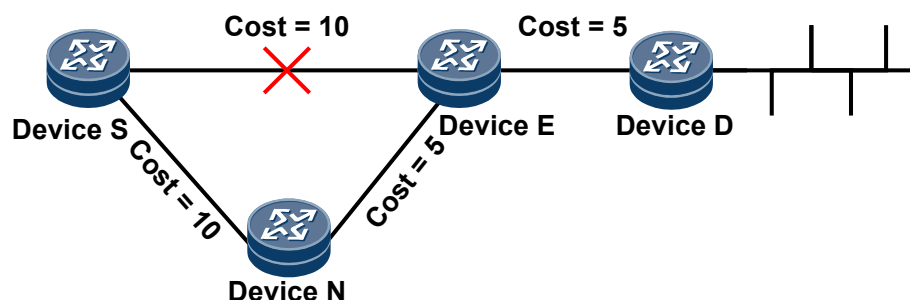
Usage Scenario

OSPFv3 Auto FRR guarantees protection against either a link failure or a node-and-link failure. Distance_opt (X,Y) indicates the shortest path between node X and node Y.

- **Link protection:** Link protection takes effect when the traffic to be protected flows along a specified link and the link costs meet the inequality: $\text{Distance_opt}(N, D) < \text{Distance_opt}(N, S) + \text{Distance_opt}(S, D)$.
 - S: source node
 - N: node along a backup link
 - D: destination node

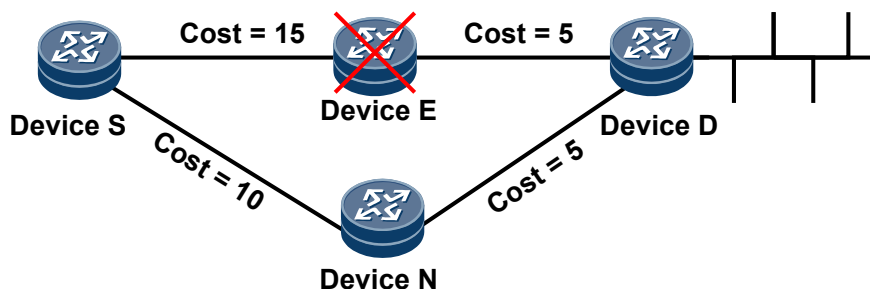
On the network shown in [Figure 7-3](#), traffic flows from Device S to Device D. The link cost satisfies the link protection inequality. If the primary link (Device S -> Device E -> Device D) fails, Device S switches the traffic to the backup link (Device S -> Device N -> Device E -> Device D), keeping the traffic interruption duration.

Figure 7-3 Networking for OSPFv3 Auto FRR link protection



- **Link-and-node protection:** [Figure 7-4](#) shows a networking for link-and-node protection. The link-and-node protection takes precedence over the link protection. Link-and-node protection must satisfy the following conditions:
 - The link cost must satisfy the inequality: $\text{Distance_opt}(N, D) < \text{Distance_opt}(N, S) + \text{Distance_opt}(S, D)$.
 - The interface cost of the Device must satisfy the inequality: $\text{Distance_opt}(N, D) < \text{Distance_opt}(N, E) + \text{Distance_opt}(E, D)$.
- E: faulty node
N: node on the backup link
D: destination node

Figure 7-4 Networking for OSPFv3 Auto FRR link-and-node protection



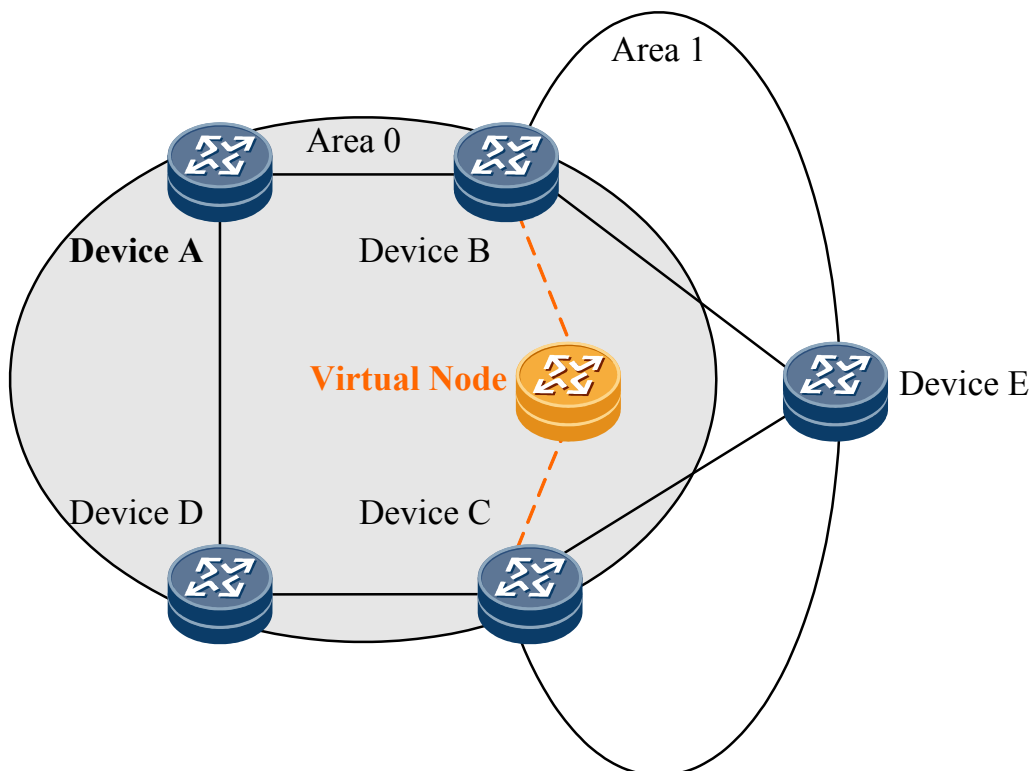
OSPFv3 FRR in a Multi-Source Routing Scenario

OSPFv3 LFA FRR uses the SPF algorithm to calculate the shortest path from each neighbor (root node) that provides a backup link to the destination node and store the node-based backup next hop, which applies to single-source routing scenarios. As networks are increasingly diversified, two ABRs or ASBRs are deployed to improve network reliability. In this case, OSPFv3 FRR in a multi-source routing scenario is needed.

NOTE

In a multi-source routing scenario, OSPFv3 FRR is implemented by calculating the Type 3 LSAs advertised by ABRs of an area for intra-area, inter-area, ASE routing. Inter-area routing is used as an example to describe how OSPFv3 FRR in a multi-source routing scenario works.

Figure 7-5 OSPFv3 FRR in a multi-route source scenario



In [Figure 7-5](#), Device B and Device C function as ABRs to forward area 0 and area 1 routes. Device E advertises an intra-area route. Upon receipt of the route, Device B and Device C translate it to a Type 3 LSA and flood the LSA to area 0. After OSPFv3 FRR is enabled on Device A, Device A considers Device B and Device C as its neighbors. Without a fixed neighbor as the root node, Device A fails to calculate FRR backup next hop. To address this problem, a virtual node is simulated between Device B and Device C and used as the root node of Device A, and Device A uses the LFA algorithm to calculate the backup next hop. This solution converts multi-source routing into single-source routing.

For example, both Device B and Device C advertise the 2001:DB8:1::1/64 route. After Device A receives the route, it fails to calculate a backup next hop for the route due to a lack of a fixed root node. To address this problem, a virtual node is simulated between Device B and Device C and used as the root node of Device A. The cost of the Device B-virtual node link is 0, and the cost of the Device C-virtual node link is 5. The cost of the virtual node-Device B or Device C link is the maximum value (65535). If the virtual node advertises the

2001:DB8:1::1/64 route, it will use the smaller cost of the routes advertised by Device B and Device C as the cost of the route. Device A is configured to consider Device B and Device C as invalid sources of the 2001:DB8:1::1/64 route and use the LFA algorithm to calculate the backup next hop for the route, with the virtual node as the root node.

7.2.6 OSPFv3 GR

NOTE

The NE40E can be configured as a GR helper rather than a GR restarter.

Graceful restart (GR) is a technology used to ensure proper traffic forwarding, especially the forwarding of key services, during the restart of routing protocols.

Without GR, the master/slave main control board switchover due to various reasons leads to transient service interruption, and as a result, route flapping occurs on the whole network. Such route flapping and service interruption are unacceptable on large-scale networks, especially carrier networks.

GR is one of the high availability (HA) technologies which comprise a series of comprehensive technologies, such as fault-tolerant redundancy, link protection, faulty node recovery, and traffic engineering technologies. As a fault-tolerant redundancy technology, GR is widely used to ensure non-stop forwarding of key data during the master/slave main control board switchovers and system upgrade.

In GR mode, the forwarding plane continues data forwarding during a restart, and operations on the control plane, such as re-establishment of neighbor relationships and route calculation, do not affect the forwarding plane, preventing service interruptions caused by route flapping and improving network reliability.

Comparison Between Master/Slave Main Control Board Switchovers with and Without GR

Table 7-6 Comparison between master/slave main control board switchovers with and without GR

Master/Slave Main Control Board Switchovers Without GR	Master/Slave Main Control Board Switchovers with GR
● OSPFv3 neighbor relationships are reestablished. ● Routes are recalculated. ● FIB entries change. ● The entire network detects route changes, and route flapping occurs for a short period of time. ● Packets are lost during forwarding, and services are interrupted.	● OSPFv3 neighbor relationships are reestablished. ● Routes are recalculated. ● FIB entries remain unchanged. ● Except the neighbors of the router on which a master/slave main control board switchover occurs, other routers do not detect route changes. ● No packets are lost during forwarding, and services are not affected.

7.2.7 OSPFv3 VPN

Definition

As an extension to OSPFv3, OSPFv3 VPN multi-instance enables Provider Edges (PEs) and Customer Edges (CEs) in VPN networks to run OSPFv3 for interworking and use OSPFv3 to learn and advertise routes.

Purpose

As a widely used IGP, in most cases, OSPFv3 runs in VPNs. If OSPFv3 runs between PEs and CEs, and PEs use OSPFv3 to advertise VPN routes to CEs, no other routing protocols need to be configured on CEs for interworking with PEs, which simplifies management and configuration of CEs.

Running OSPFv3 Between PEs and CEs

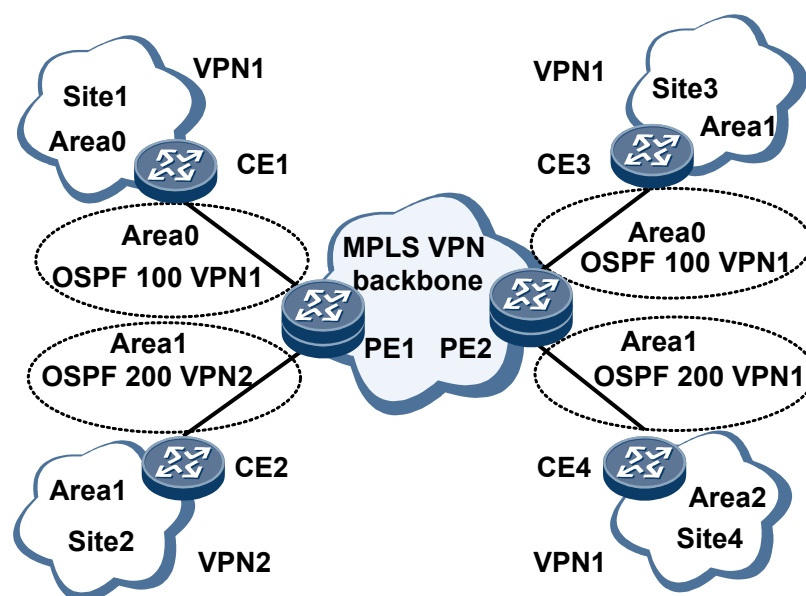
In BGP/MPLS VPN, Multi-Protocol BGP (MP-BGP) is used to transmit routing information between PEs, whereas OSPFv3 is used to learn and advertise routes between PEs and CEs.

Running OSPFv3 between PEs and CEs features the following benefits:

- OSPFv3 is used in a site to learn routes. Running OSPFv3 between PEs and CEs can reduce the number of protocol types supported by CEs.
- Similarly, running OSPFv3 both in a site and between PEs and CEs simplifies the work of network administrators and reduces the number of protocols that network administrators must be familiar with.
- When the network, which originally uses OSPFv3 but not VPN on the backbone network begins to use BGP/MPLS VPN, running OSPFv3 between PEs and CEs facilitates the transition.

In [Figure 7-6](#), CE1, CE3, and CE4 belong to VPN 1, and the numbers following OSPFv3 indicate the process IDs of the multiple OSPFv3 instances running on PEs.

Figure 7-6 Running OSPFv3 between PEs and CEs



CE1 advertises routes to CE3 and CE4 as follows:

1. PE1 imports OSPF routes of CE1 into BGP and forms BGP VPNv4 routes.
2. PE1 uses MP-BGP to advertise the BGP VPNv4 routes to PE2.
3. PE2 imports the BGP VPNv6 routes into OSPFv3 and then advertises these routes to CE3 and CE4.

The process of advertising routes of CE4 or CE3 to CE1 is the same as the preceding process.

OSPFv3 Domain ID

If inter-area routes are advertised between local and remote OSPFv3 areas, these areas are considered to be in the same OSPFv3 domain.

- Domain IDs identify domains.
- Each OSPFv3 domain has one or more domain IDs. If more than one domain ID is available, one of the domain IDs is a primary ID, and the others are secondary IDs.
- If an OSPFv3 instance does not have specific domain IDs, its ID is considered as null.

Before advertising the remote routes sent by BGP to CEs, PEs need to determine the type of OSPFv3 routes (Type 3 or Type 5) to be advertised to CEs based on the domain IDs.

- If local domain IDs are the same as or compatible with remote domain IDs in BGP routes, PEs advertise Type 3 routes.
- If local domain IDs are different from or incompatible with remote domain IDs in BGP routes, PEs advertise Type 5 or Type 7 routes.

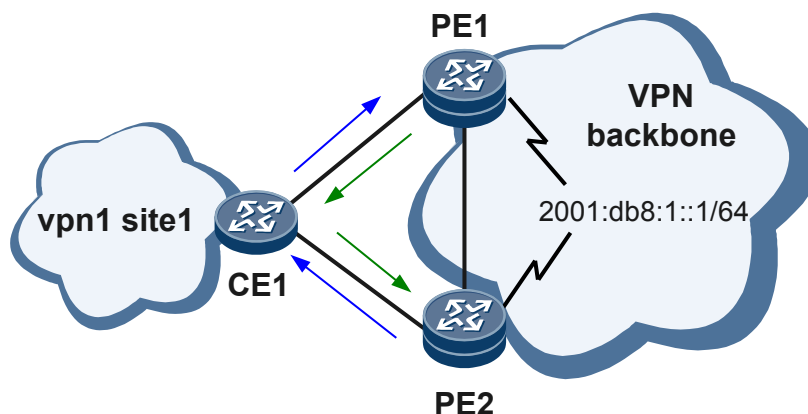
Table 7-7 Domain ID relationships and corresponding generated routes

Relationship Between Local and Remote Domain IDs	Type of the Generated Routes
Both are null.	Inter-area routes
The remote domain ID equals the local primary domain ID or one of the local secondary domain IDs.	Inter-area routes
The remote domain ID is different from the local primary domain ID or any of the local secondary domain IDs.	If the local area is a non-NSSA, external routes are generated. If the local area is an NSSA, NSSA routes are generated.

Routing Loop Prevention

Routing loops may occur between PEs and CEs when OSPFv3 and BGP learn routes from each other.

Figure 7-7 OSPFv3 VPN routing loops



In [Figure 7-7](#), on PE1, OSPFv3 imports a BGP route destined for 2001:db8:1::1/64 and then generates and advertises a Type 5 or Type 7 LSA to CE1. Then, CE1 learns an OSPFv3 route with 2001:db8:1::1/64 as the destination address and PE1 as the next hop and advertises the route to PE2. Therefore, PE2 learns an OSPFv3 route with 2001:db8:1::1/64 as the destination address and CE1 as the next hop.

Similarly, CE1 also learns an OSPFv3 route with 2001:db8:1::1/64 as the destination address and PE2 as the next hop. PE1 learns an OSPF route with 2001:db8:1::1/64 as the destination address and CE1 as the next hop.

As a result, CE1 has two equal-cost routes with PE1 and PE2 as next hops respectively, and the next hops of the routes from PE1 and PE2 to 2001:db8:1::1/64 are CE1, which leads to a routing loop.

In addition, the priority of an OSPFv3 route is higher than that of a BGP route. Therefore, on PE1 and PE2, BGP routes to 2001:db8:1::1/64 are replaced with the OSPFv3 route, and the OSPFv3 route with 2001:db8:1::1/64 as the destination address and CE1 as the next hop is active in the routing tables of PE1 and PE2.

The BGP route is inactive, and therefore, the LSA generated when this route is imported by OSPFv3 is deleted, which causes the OSPFv3 route to be withdrawn. As a result, no OSPFv3 route exists in the routing table, and the BGP route becomes active again. This cycle causes route flapping.

OSPFv3 VPN provides a few solutions to routing loops, as described [Table 7-8](#).

Table 7-8 Routing loop prevention measures

Feature	Definition	Function
DN-bit	It is a flag bit used by OSPFv3 multi-instance processes to prevent routing loops.	When advertising the generated Type 3, Type 5, or Type 7 LSAs to CEs, PEs set the DN-bit of these LSAs to 1. PEs retain the DN-bit (0) of other LSAs. When calculating routes, the OSPFv3 multi-instance process of a PE ignores LSAs with DN-bit 1, which prevents the PE from receiving the LSAs that are advertised by itself.
VPN route tag	The VPN route tag is carried in Type 5 or Type 7 LSAs generated by PEs based on the received BGP VPN route. It is not carried in BGP extended community attributes. The VPN route tag is valid only on the PEs that receive BGP routes and generate OSPFv3 LSAs.	When a PE detects that the VPN route tag in the incoming LSA is the same as that in the local LSA, the PE ignores this LSA, which prevents routing loops.
Default route	It is a route whose destination IP address and mask are both 0.	PEs do not calculate default routes. Default routes are used to forward the traffic from CEs or the sites where CEs reside to the VPN backbone network.

Multi-VPN-Instance CE

OSPFv3 multi-instance generally runs on PEs. Devices that run OSPFv3 multi-instance within user LANs are called Multi-VPN-Instance CEs (MCEs).

Compared with OSPFv3 multi-instance running on PEs, MCEs have the following characteristics:

- MCEs do not need to support OSPFv3-BGP association.
- MCEs establish one OSPFv3 instance for each service. Different virtual CEs transmit different services, which ensures LAN security at a low cost.
- MCEs implement different OSPFv3 instances on a CE. The key to implementing MCEs is to disable loop detection and calculate routes directly. MCEs also use the received LSAs with the DN-bit 1 for route calculation.

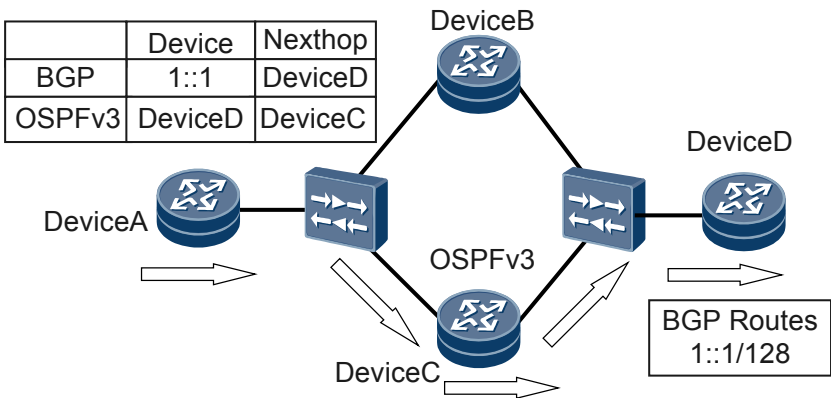
7.2.8 OSPFv3-BGP Association

When a new device is deployed on a network or a device is restarted, network traffic may be lost during BGP route convergence because IGP routes converge more quickly than BGP routes. OSPFv3-BGP association can address this problem.

After a device on a BGP network recovers from a fault, BGP convergence is performed again and packet loss may occur during the convergence.

In [Figure 7-8](#), traffic from Device A to Device D through Device C traverses a BGP network.

Figure 7-8 Traffic traversing a BGP network

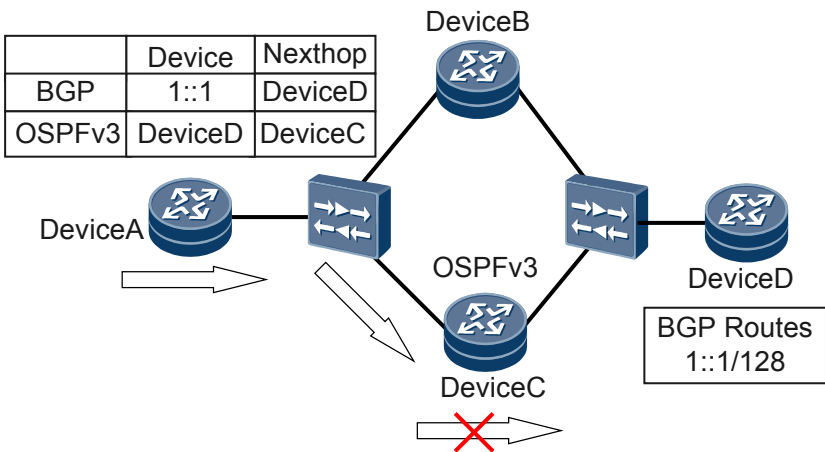


If Device C fails, traffic is switched to Device B after rerouting. Packets are lost when Device C recovers.

Because OSPFv3 route convergence is faster than BGP route convergence, OSPFv3 convergence is complete whereas BGP route convergence is still going on when Device C recovers. The next hop of the route from Device A to Device D is Device C, which, however, does not know the route to Device D since BGP convergence on Device C is not complete.

Therefore, Device C discards the packets destined for Device D after receiving them from Device A, as shown in [Figure 7-9](#).

Figure 7-9 Packet loss during a device restart without OSPFv3-BGP association



OSPFv3-BGP Association Process

When a device with OSPFv3-BGP association restarts, the device sets the weight to the largest value (65535) in LSAs, instructing other OSPFv3 routers not to use it as a transit router for data forwarding. BGP routes, however, can still reach the device.

7.2.9 OSPFv3 Authentication

OSPFv3 IPsec Authentication

The rapid development of networks poses higher requirements for network security. Routing protocol packets that are transmitted on networks may be intercepted, changed, or forged, and packet attacks may cause network interruption. Therefore, packets need to be protected.

Standard protocols do not define any authentication mechanisms for OSPFv3. Therefore, OSPFv3 packets do not carry any authentication information.

Standard protocol defines the use of the IP Security (IPsec) mechanism to authenticate OSPFv3 packets.

The IPsec protocol family, which consists of a series of protocols defined by the Internet Engineering Task Force (IETF), provides high-quality, interoperable, and cryptology-based security for IP packets.

By encrypting data and authenticating the data source at the IP layer, communicating parties can ensure confidentiality, data integrity, data source authentication, and anti-replay for the data transmitted across the network.

NOTE

- Confidentiality: The data is encrypted and transmitted in cipher text.
- Data integrity: Received packets are authenticated to check whether they have been modified.
- Data authentication: The data source is authenticated to ensure that the data is sent from a real sender.
- Anti-replay: The attacks from malicious users who repeatedly send obtained data packets are prevented. Specifically, the receiver rejects old or repeated data packets.

IPsec adopts two security protocols: Authentication Header (AH) security and Encapsulating Security Payload (ESP):

- AH: A protocol that provides data origin authentication, data integrity check, and anti-replay protection. AH does not encrypt packets to be protected.

AH data is carried in the following fields:

- IP version
- Header length
- Packet length
- Identification
- Protocol
- Source and destination addresses
- Options

- ESP: A protocol that provides IP packet encryption and authentication mechanisms besides the functions provided by AH. The encryption and authentication mechanisms can be used together or independently.

OSPFv3 Authentication Trailer

Prior to the OSPFv3 Authentication Trailer, OSPFv3 can use only IPSec for authentication. However, on some special networks, a mobile ad hoc network (MANET) for example, IPSec is difficult to deploy and maintain. To address this problem, standard protocol introduces Authentication Trailer for OSPFv3, which provides another approach for OSPFv3 to implement authentication.

In OSPFv3 authentication, an authentication field is added to each OSPFv3 packet for encryption. When a local device receives an OSPFv3 packet from a remote device, the local device discards the packet if the authentication password carried in the packet is different from the local one, which protects the local device against potential attacks. Therefore, OSPFv3 authentication improves network security.

Based on the applicable scope, OSPFv3 authentication is classified as follows:

- Area authentication
Area authentication is configured in the OSPFv3 area view and applies to packets received by all interfaces in an OSPF area.
- Process authentication
Process authentication is configured in the OSPFv3 view and applies to all packets in an OSPF process.
- Interface authentication
Interface authentication is configured in the interface view and applies to all packets received by the interface.

OSPFv3 uses HMAC-SHA256 to authenticate packets. In HMAC-SHA256 authentication, a password is encrypted using the HMAC-SHA256 algorithm before being added to a packet, which improves password security.

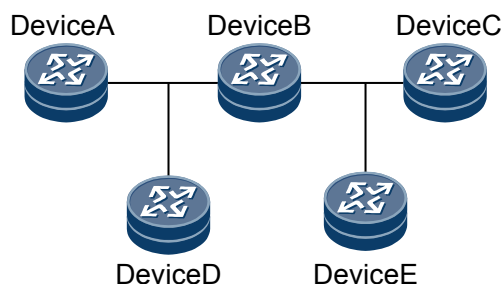
Each OSPFv3 packet carries an authentication type in the header and authentication information in the tail.

The authentication types are as follows:

- 1: simple authentication
- 2: ciphertext authentication

Networking Application of OSPFv3 Authentication Trailer

Figure 7-10 OSPFv3 authentication trailer on a broadcast network



The configuration requirements are as follows:

- Interface authentication configurations must be the same on all devices of the same network so that OSPFv3 neighbor relationships can be established.
- Area authentication configurations must be the same on all devices in the same area.

7.2.10 OSPFv3 Neighbor Relationship Flapping Suppression

OSPFv3 neighbor relationship flapping suppression works by delaying OSPFv3 neighbor relationship reestablishment or setting the link cost to the maximum value (65535).

Background

If the status of an interface carrying OSPFv3 services alternates between Up and Down, OSPFv3 neighbor relationship flapping occurs on the interface. During the flapping, OSPFv3 frequently sends Hello packets to reestablish the neighbor relationship, synchronizes LSDBs, and recalculates routes. In this process, a large number of packets are exchanged, adversely affecting neighbor relationship stability, OSPFv3 services, and other OSPFv3-dependent services, such as LDP and BGP. OSPFv3 neighbor relationship flapping suppression can address this problem by delaying OSPFv3 neighbor relationship reestablishment or preventing service traffic from passing through flapping links.

Related Concepts

Flapping_event: reported when the status of a neighbor relationship on an interface last changes from Full to a non-Full state. The flapping_event triggers flapping detection.

Flapping_count: number of times flapping has occurred.

Detect-interval: detection interval. The interval is used to determine whether to trigger a valid flapping_event.

Threshold: flapping suppression threshold. When the flapping_count reaches or exceeds **threshold**, flapping suppression takes effect.

Resume-interval: interval for exiting from OSPFv3 neighbor relationship flapping suppression. If the interval between two successive valid flapping_events is longer than **resume-interval**, the flapping_count is reset.

Implementation

Flapping detection

Each OSPFv3 interface on which OSPFv3 neighbor relationship flapping suppression is enabled starts a flapping counter. If the interval between two successive neighbor status changes from Full to a non-Full state is shorter than **detecting-interval**, a valid flapping_event is recorded, and the flapping_count increases by 1. When the flapping_count reaches or exceeds **threshold**, flapping suppression takes effect. If the interval between two successive neighbor status changes from Full to a non-Full state is longer than **resume-interval**, the flapping_count is reset.

The **detecting-interval**, **threshold**, and **resume-interval** are configurable.

NOTE

The value of **resume-interval** must be greater than that of **detecting-interval**.

Flapping suppression

Flapping suppression works in either Hold-down or Hold-max-cost mode.

- **Hold-down mode:** In the case of frequent flooding and topology changes during neighbor relationship establishment, interfaces prevent neighbor relationships from being reestablished during the suppression period, which minimizes LSDB synchronization attempts and packet exchanges.
- **Hold-max-cost mode:** If the traffic forwarding path changes frequently, interfaces use 65535 as the cost of the flapping link during Hold-max-cost suppression, which prevents traffic from passing through the flapping link.

Flapping suppression can also work first in Hold-down mode and then in Hold-max-cost mode.

By default, the Hold-max-cost mode takes effect. The mode and suppression duration can be changed manually.

If an attack causes frequent neighbor relationship flapping, Hold-down mode can minimize the impact of the attack.

NOTE

When an interface enters the flapping suppression state, all neighbor relationships on the interface enter the state accordingly.

Exiting from flapping suppression

Interfaces exit from flapping suppression in the following scenarios:

- The suppression timer expires.
- The corresponding OSPFv3 process is reset.
- An OSPF neighbor is reset.
- A command is run to exit from flapping suppression.

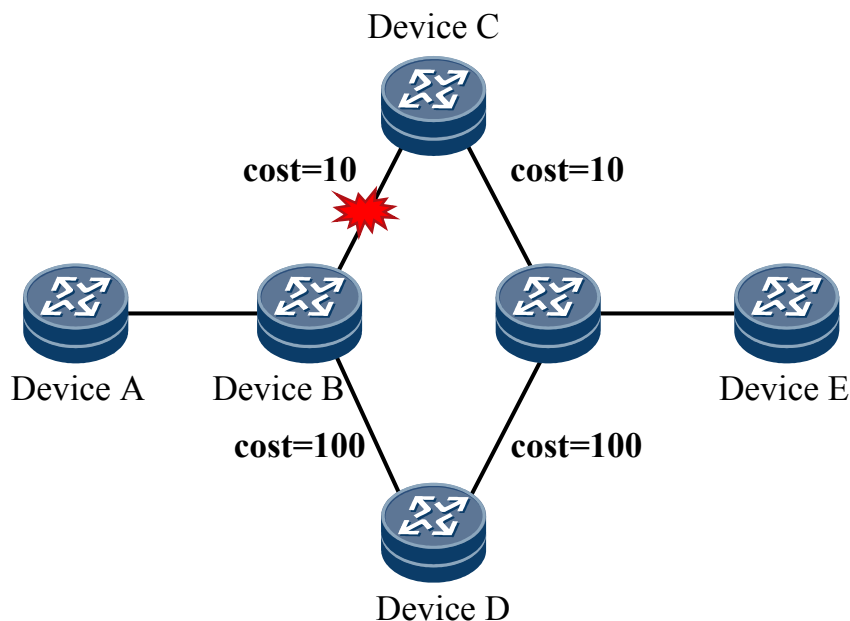
Typical Scenarios

Basic scenario

In [Figure 7-11](#), the traffic forwarding path is Device A -> Device B -> Device C -> Device E before a link failure occurs. After the link between Device B and Device C fails, the forwarding path switches to Device A -> Device B -> Device D -> Device E. If the neighbor relationship between Device B and Device C frequently flaps at the early stage of the path switchover, the forwarding path will be switched frequently, causing traffic loss and affecting network stability. If the neighbor relationship flapping meets suppression conditions, flapping suppression takes effect.

- If flapping suppression works in Hold-down mode, the neighbor relationship between Device B and Device C is prevented from being reestablished during the suppression period, in which traffic is forwarded along the path Device A -> Device B -> Device D -> Device E.
- If flapping suppression works in Hold-max-cost mode, 65535 is used as the cost of the link between Device B and Device C during the suppression period, and traffic is forwarded along the path Device A -> Device B -> Device D -> Device E.

Figure 7-11 Flapping suppression in a basic scenario



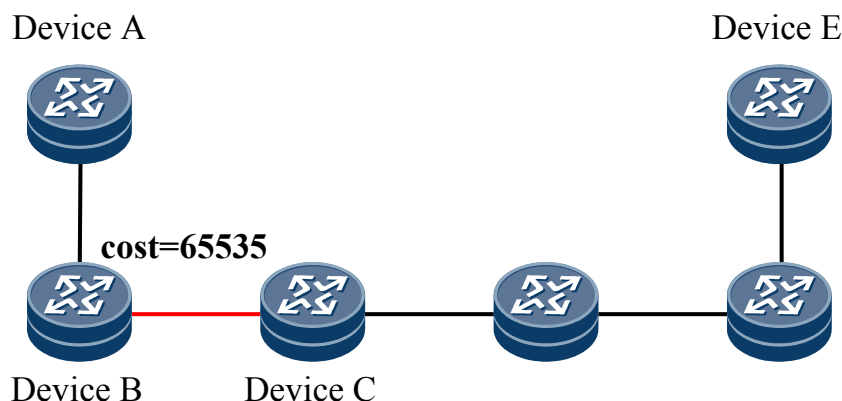
Single-forwarding path scenario

When only one forwarding path exists on the network, the flapping of the neighbor relationship between any two devices on the path will interrupt traffic forwarding. In [Figure 7-12](#), the traffic forwarding path is Device A -> Device B -> Device C -> Device E. If the neighbor relationship between Device B and Device C flaps, and the flapping meets suppression conditions, flapping suppression takes effect. However, if the neighbor relationship between Device B and Device C is prevented from being reestablished, the whole network will be divided. Therefore, Hold-max-cost mode (rather than Hold-down mode) is recommended. If flapping suppression works in Hold-max-cost mode, 65535 is used as the cost of the link between Device B and Device C during the suppression period. After the network stabilizes and the suppression timer expires, the link is restored.

NOTE

By default, the Hold-max-cost mode takes effect.

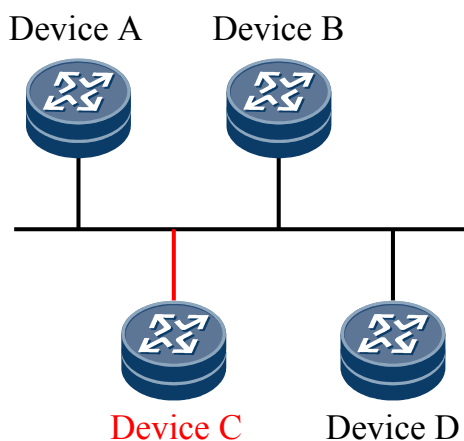
Figure 7-12 Flapping suppression in a single-forwarding path scenario



Broadcast scenario

In **Figure 7-13**, four devices are deployed on the same broadcast network using switches, and the devices are broadcast network neighbors. If Device C flaps due to a link failure, and Device A and Device B were deployed at different time (Device A was deployed earlier for example) or the flapping suppression parameters on Device A and Device B are different, Device A first detects the flapping and suppresses Device C. Consequently, the Hello packets sent by Device A do not carry Device C's router ID. However, Device B has not detected the flapping yet and still considers Device C a valid node. As a result, the DR candidates identified by Device A are Device B and Device D, whereas the DR candidates identified by Device B are Device A, Device C, and Device D. Different DR candidates result in a different DR election result, which may lead to route calculation errors. To prevent this problem in scenarios where an interface has multiple neighbors, such as on a broadcast, P2MP, or NBMA network, all neighbors on the interface are suppressed when the status of a neighbor relationship last changes to **ExStart** or **Down**. Specifically, if Device C flaps, Device A, Device B, and Device D on the broadcast network are all suppressed. After the network stabilizes and the suppression timer expires, Device A, Device B, and Device D are restored to normal status.

Figure 7-13 Flapping suppression on a broadcast network



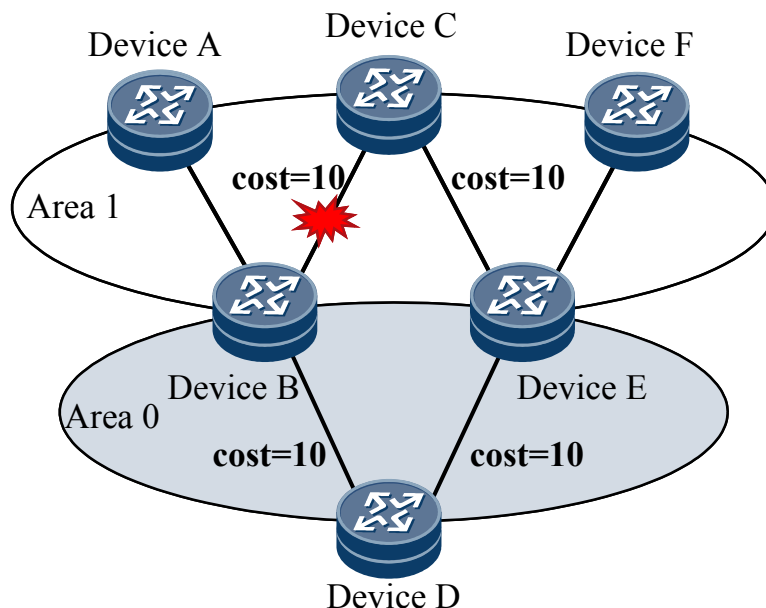
Multi-area scenario

In **Figure 7-14**, Device A, Device B, Device C, Device E, and Device F are connected in area 1, and Device B, Device D, and Device E are connected in backbone area 0. Traffic from Device A to Device F is preferentially forwarded along an intra-area route, and the forwarding path is Device A -> Device B -> Device C -> Device E -> Device F. When the neighbor relationship between Device B and Device C flaps and the flapping meets suppression conditions, flapping suppression takes effect in the default mode (Hold-max-cost). Consequently, 65535 is used as the cost of the link between Device B and Device C. However, the forwarding path remains unchanged because intra-area routes take precedence over inter-area routes during route selection according to OSPFv3 route selection rules. To prevent traffic loss in multi-area scenarios, configure Hold-down mode to prevent the neighbor relationship between Device B and Device C from being reestablished during the suppression period. During this period, traffic is forwarded along the path Device A -> Device B -> Device D -> Device E -> Device F.

NOTE

By default, the Hold-max-cost mode takes effect. The mode can be changed to Hold-down manually.

Figure 7-14 Flapping suppression in a multi-area scenario



7.2.11 OSPFv3 Flush

Background

If network-wide OSPFv3 LSAs are flushed, network stability will be adversely affected. In this case, source tracing must be implemented to locate the root cause of the fault immediately to minimize the impact. However, OSPFv3 itself does not support source tracing. A conventional solution is isolation node by node until the faulty node is located. The solution is complex and time-consuming. Therefore, a fast source tracing method is required. In this case, OSPFv3 flush LSA source tracing is introduced, which allows maintenance personnel to locate the faulty source on any device on the network.

Related Concepts

OSPFv3 flush LSA source tracing

A mechanism that helps locate the device that flushes LSAs. The feature has the following characteristics:

- Uses a new UDP port. Source tracing packets are carried by UDP packets, and the UDP packets also carry the OSPFv3 LSAs flushed by the current device and are flooded hop by hop based on the OSPFv3 topology.
- Forwards packets along UDP channels which are independent of the channels used to transmit OSPFv3 packets. Therefore, OSPFv3 flush LSA source tracing supports incremental deployment. In addition, source tracing does not affect the devices with the related UDP port disabled.
- Supports query of the node that flushed LSAs on any of the devices after OSPFv3 flush LSA source tracing packets are flooded on the network, which speeds up fault locating and faulty node isolation.
- Is Huawei proprietary.

Flush

Network-wide OSPFv3 LSAs are deleted.

PS-Hello packets

Packets used to negotiate the OSPFv3 flush LSA source tracing capability between OSPFv3 neighbors.

PS-LSA

When a device flushes an OSPFv3 LSA, it generates a PS-LSA carrying information about the device and brief information about the OSPFv3 LSA.

PS-LSU packets

OSPFv3 flush LSA source tracing packets that carry PS-LSAs.

PS-LSU ACK packets

Acknowledgment packets used to improve OSPFv3 flush LSA source tracing packets.

OSPFv3 flush LSA source tracing port

ID of the UDP port that receives and sends OSPFv3 flush LSA source tracing packets. The default port ID is 50133, which is configurable.

Implementation

The implementation of OSPFv3 flush LSA source tracing is as follows:

1. Source tracing capability negotiation

After an OSPFv3 neighbor relationship is established between two devices, they need to negotiate the source tracing capability through PS-Hello packets.

2. PS-LSA generation and flooding

When a device flushes an OSPFv3 LSA, it generates a PS-LSA carrying information about the device and brief information about the OSPFv3 LSA, adds the PS-LSA to a PS-LSU packet, and floods the PS-LSU packet to source tracing-capable neighbors. The PS-LSU packet is used to locate the faulty source.

NOTE

Only Router-LSAs, Network-LSAs, and Inter-Area-Router-LSAs can be flushed. Therefore, a device generates a PS-LSA only when it flushes a Router-LSA, Network-LSA, or Inter-Area-Router-LSA.

Source Tracing Capability Negotiation

OSPFv3 flush LSA source tracing uses UDP to carry source tracing packets and listens to the UDP port which is used to receive and send source tracing packets. OSPFv3 flush LSA source tracing is Huawei proprietary. If a source tracing-capable Huawei device sends source tracing packets to a source tracing-incapable Huawei device or non-Huawei device, the source tracing-capable Huawei device may be incorrectly identified as an attacker. Therefore, the source tracing capability need to be negotiated between the devices. In addition, the source tracing-capable device needs to help the source tracing-incapable device to send source tracing information, which also requires negotiation.

Source tracing capability negotiation depends on OSPFv3 neighbor relationships. Specifically, after an OSPFv3 neighbor relationship is established, the local device initiates source tracing capability negotiation based on the IP address of the neighbor. [Figure 7-15](#) shows the negotiation process.

Figure 7-15 Source tracing capability negotiation

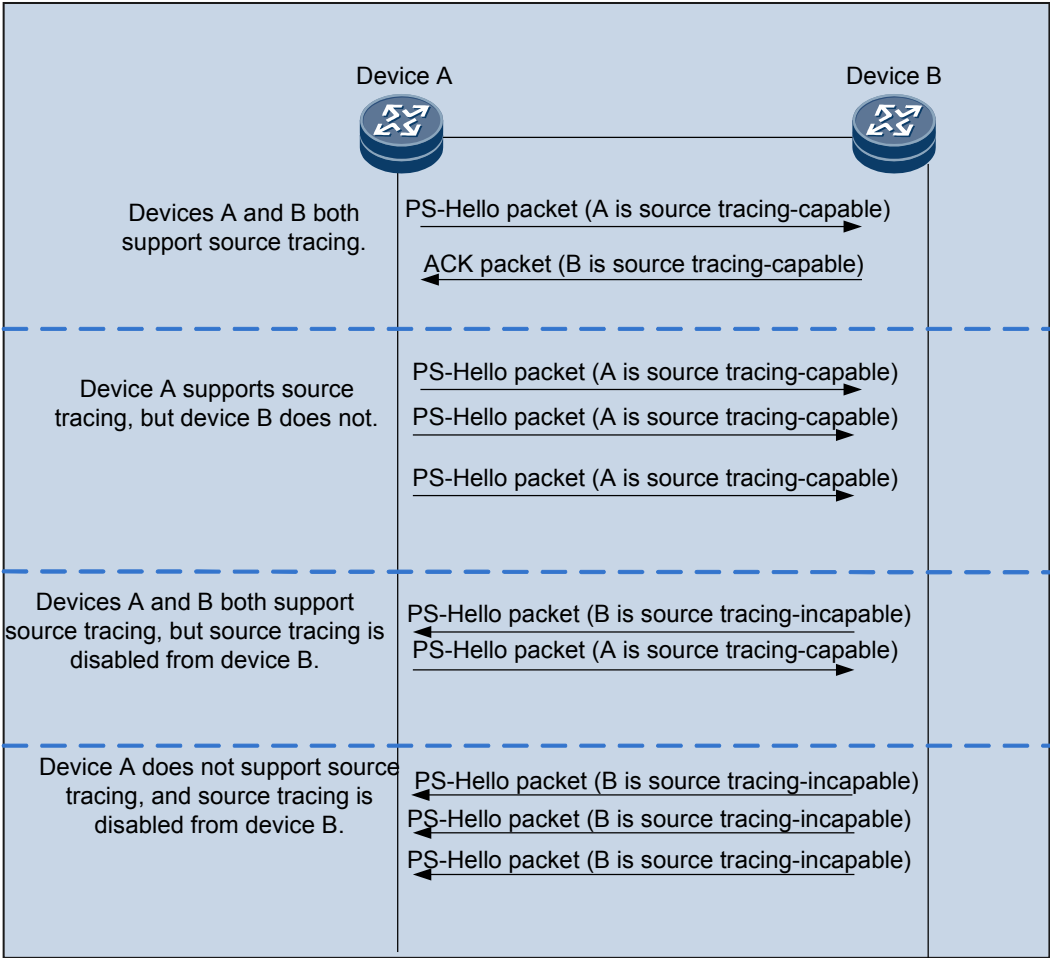


Table 7-9 Source tracing capability negotiation

Whether Source Tracing Is Supported	Source Tracing Capability Negotiation Process
Devices A and B both support source tracing.	1. Device A sends a PS-Hello packet to notify its source tracing capability to device B. 2. Upon reception of the PS-Hello packet, device B sets the source tracing field for device A and replies with an ACK packet to notify its source tracing capability to device A. 3. Upon reception of the ACK packet, device A sets the source tracing field for device B.

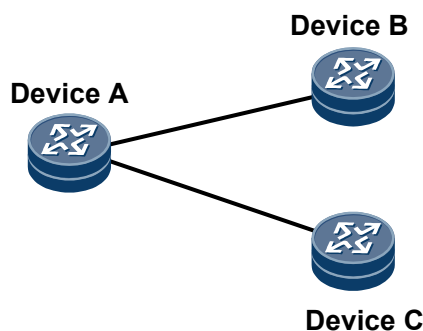
Whether Source Tracing Is Supported	Source Tracing Capability Negotiation Process
Devices A supports source tracing, but device B does not.	1. Device A sends a PS-Hello packet to notify its source tracing capability to device B. 2. Device A fails to receive an ACK packet from device B within 10s and retransmits the PS-Hello packet. Device A can retransmit the PS-Hello packet twice at most. After device A fails to receive an ACK packet from device B after the PS-Hello packet is retransmitted twice, device A considers that device B does not support source tracing.
Devices A and B both support source tracing, but source tracing is disabled from device B.	1. After source tracing is disabled from device B, device B sends a PS-Hello packet to notify its source tracing incapability to device A. 2. Upon reception of the PS-Hello packet, device A replies with an ACK packet that carries the source tracing capability. 3. Upon reception of the ACK packet, device B considers the capability negotiation complete and disables the UDP port.
Devices A does not support source tracing, and source tracing is disabled from device B.	1. After source tracing is disabled from device B, device B sends a PS-Hello packet to notify its source tracing incapability to device A. 2. Device B fails to receive an ACK packet from device A within 10s and retransmits the PS-Hello packet. Device B can retransmit the PS-Hello packet twice at most. After device B fails to receive an ACK packet from device B after the PS-Hello packet is retransmitted twice, device A considers the capability negotiation complete and disables the UDP port.

PS-LSA Generation and Flooding

PS-LSA: carries information about the node that flushed OSPFv3 LSAs.

- If a device flushes an OSPFv3 LSA, it generates and floods a PS-LSA to all its neighbors.
- If a device receives a flush LSA from a source tracing-incapable neighbor, the device generates and floods a PS-LSA to all its neighbors. If a device receives the same flush LSA (with the same LSID and sequence number) from more than one source tracing-incapable neighbor, the device generates only one PS-LSA.
- If a device flushes a Router-LSA, Network-LSA, or Inter-Area-Router-LSA, it generates a PS-LSA, adds the PS-LSA to a PS-LSU packet, and floods the PS-LSU packet to all source tracing-capable neighbors.

Figure 7-16 PS-LSA generation rules



PS-LSA generation rules

- When device A flushes a Router-LSA, Network-LSA, or Inter-Area-Router-LSA, it generates a PS-LSA in which the **Flush Router** field is its router ID and the **Neighbor Router** field is 0, and adds the PS-LSA to the queue where packets are to be sent to all source tracing-capable neighbors.
- After device A receives the flush LSA from device B (source tracing incapable), device A generates a PS-LSA in which the **Flush Router** field is its router ID and the **Neighbor Router** field is the router ID of device B, and adds the PS-LSA to the queue where packets are to be sent to all source tracing-capable neighbors.
- After device A receives the flush LSA from device B, followed by the same flush LSA sent by device C, device A generates a PS-LSA in which the **Flush Router** field is its router ID and the **Neighbor Router** field is the router ID of device B, and adds the PS-LSA to the queue where packets are to be sent to all source tracing-capable neighbors. Device A does not generate a PS-LSA in response to the flush LSA sent by device C.

PS-LSU packet sending rules

- During neighbor relationship establishment, a device initializes the sequence number of the PS-LSU packet of the neighbor. When the device replies with a PS-LSU packet, it adds the sequence number of the PS-LSU packet of the neighbor. During PS-LSU packet retransmission, the sequence number remains unchanged. After the device receives a PS-LSU ACK packet with the same sequence number, it increases the sequence number of the neighbor's PS-LSU packet by 1.
- The neighbor manages the PS-LSA sending queue. When a PS-LSA is added to the queue which was empty, the neighbor starts a timer. After the timer expires, the neighbor adds the PS-LSAs in the queue to a PS-LSU packet, sends the packet to its neighbor, and starts another timer to wait for a PS-LSU ACK packet.
- After the PS-LSU ACK timer expires, the PS-LSU packet is retransmitted.
- When the device receives a PS-LSU ACK packet with a sequence number same as that in the neighbor record, the device clears PS-LSAs from the neighbor queue, and sends another PS-LSU packet after the timer expires.
 - If the sequence number of a received PS-LSU ACK packet is less than that in the neighbor record, the device ignores the packet.
 - If the sequence number of a received PS-LSU ACK packet is greater than that in the neighbor record, the device discards the packet.

NOTE

PS-LSU packet sending is independent among neighbors.

PS-LSU packet receiving rules

- When a device receives a PS-LSU packet from a neighbor, the neighbor records the sequence number of the packet and replies with a PS-LSU ACK packet.
- When the device receives a PS-LSU packet with the sequence number same as that in the neighbor record, the device discards the PS-LSU packet.
- After the device parses a PS-LSU packet, it adds the PS-LSA in the packet to the LSDB. The device also checks whether the PS-LSA is newer than the corresponding PS-LSA in the LSDB.
 - If the PS-LSA is newer, the device floods it to other neighbors.
 - If the PS-LSA is the same as the corresponding PS-LSA in the LSDB, the device does not process the received PS-LSA.
 - If the PS-LSA is older, the device floods the corresponding PS-LSA in the LSDB to the neighbor.
- If the device receives a PS-LSU packet from a neighbor and the neighbor does not support source tracing, the device modifies the neighbor status as source tracing capable.

Source Tracing Security

OSPFv3 flush LSA source tracing uses a UDP port to receive and send source tracing packets. Therefore, the security of the port must be taken into consideration.

OSPFv3 flush LSA source tracing inevitably increases packet receiving and sending workload and intensifies bandwidth pressure. To minimize the impact on OSPFv3, the number of source tracing packets must be controlled.

The following security measures are available:

Table 7-10 Security measures for source tracing

Security Measures for Source Tracing	Principles
Authentication	Source tracing is embedded in OSPFv3 and uses OSPFv3 authentication parameters to authenticate packets.
GTSM	GTSM is a security mechanism that checks whether the time to live (TTL) value in each received IP packet header is within a pre-defined range. OSPFv3 flush LSA source tracing packets can be flooded as far as one hop. ● When a device sends a packet, it sets the TTL of the packet to 255. ● If the TTL is not 254 when the packet is received, the packet will be discarded.

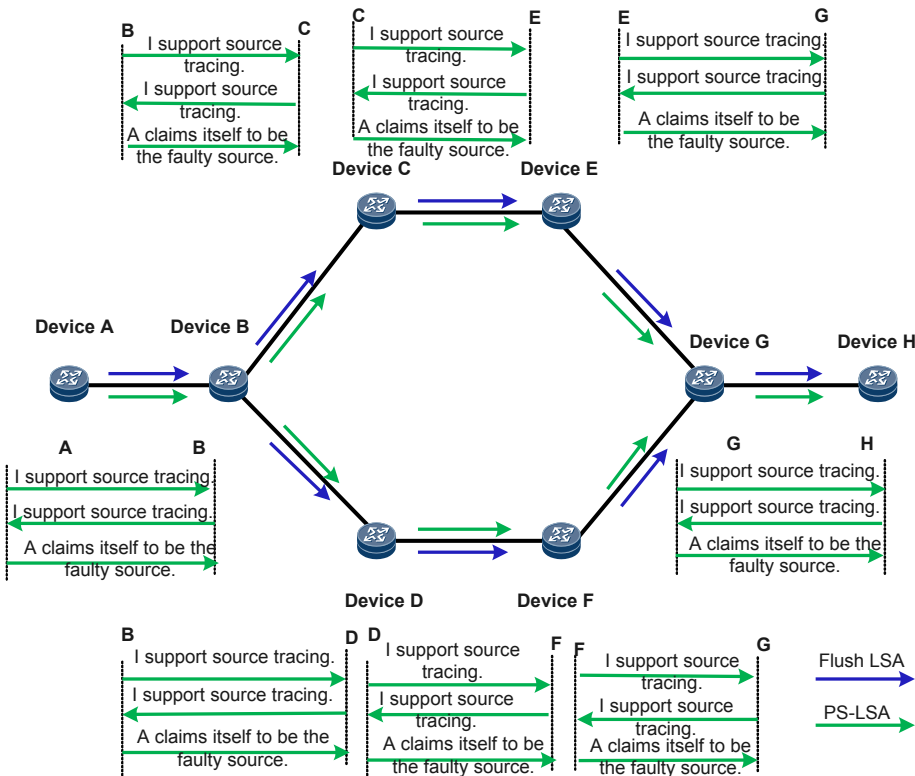
Security Measures for Source Tracing	Principles
Cpu-car	Interface boards can check the packets to be sent to the CPU for processing and prevent the main control board from being overloaded by a large number of packets that are sent to the CPU. OSPFv3 flush LSA source tracing needs to apply for an independent CAR channel and has small committed information rate (CIR) and committed burst size (CBS) values configured.

Typical Scenarios

Scenario where all nodes support source tracing

All nodes on the network support source tracing, and node A is the faulty source. [Figure 7-17](#) shows the networking.

Figure 7-17 Scenario where all nodes support source tracing

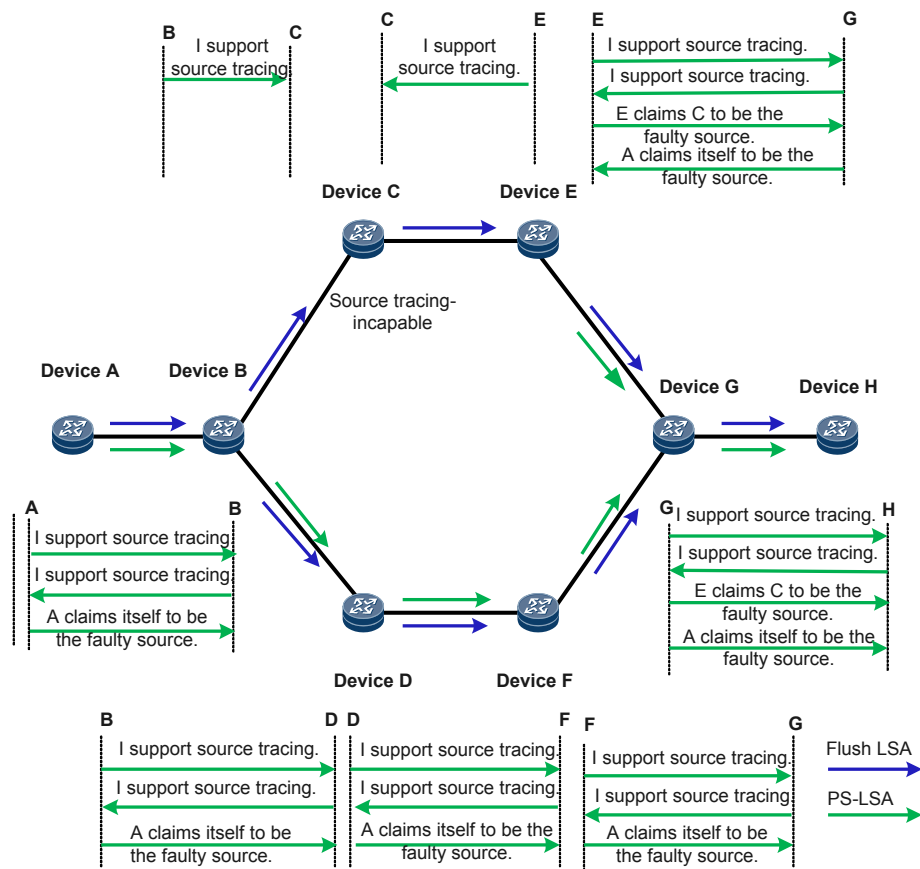


When device A flushes an OSPFv3 LSA, it generates a PS-LSA that carries device A information and brief information about the flush LSA. Then the PS-LSA is flooded on the network hop by hop. After the fault occurs, maintenance personnel can log in to any node on the network to locate device A that keeps sending flush LSAs and isolate device A from the network.

Scenario where source tracing-incapable nodes are not isolated from source tracing-capable nodes

All nodes on the network except device C support source tracing, and device A is the faulty source. In this case, the PS-LSA can be flooded on the entire network. **Figure 7-18** shows the networking.

Figure 7-18 Scenario where source tracing-incapable nodes are not isolated from source tracing-capable nodes



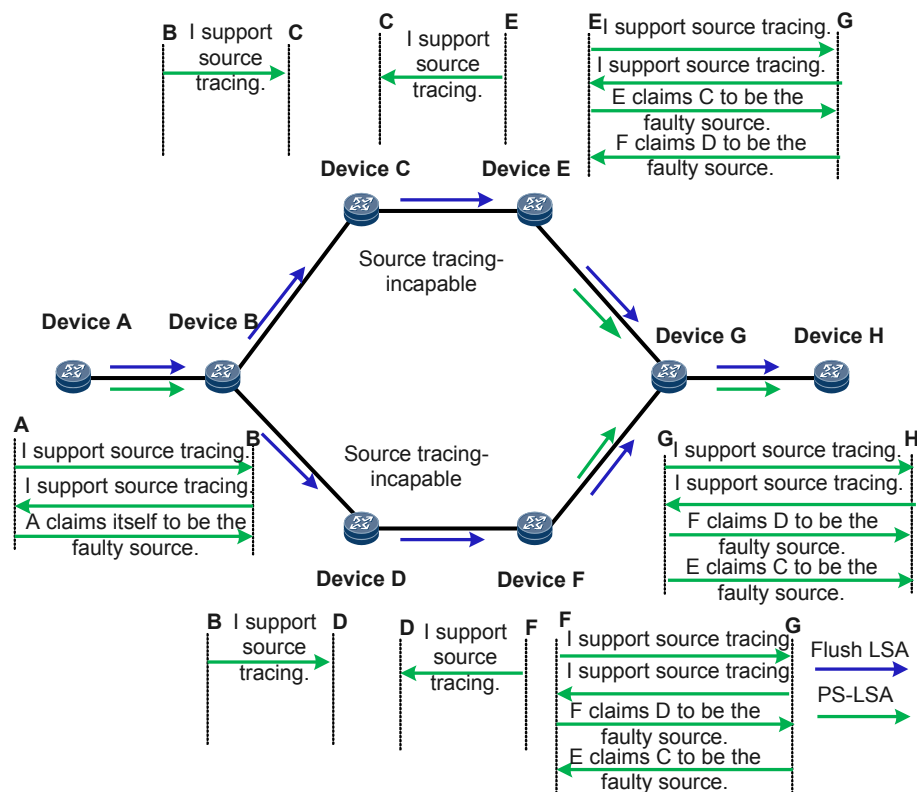
When device A flushes an OSPFv3 LSA, it generates a PS-LSA that carries device A information and brief information about the flush LSA. Then the PS-LSA is flooded on the network hop by hop. When devices B and E negotiate the source tracing capability with device C, they find that device C does not support source tracing. Therefore, after device B receives the PS-LSA from device A, device B sends the PS-LSA to device D, but not to device C. After receiving the flush LSA from device C, device E generates a PS-LSA which carries information about the advertisement source (device E), flush source (device C), and the flush LSA, and floods the PS-LSA on the network.

After the fault occurs, maintenance personnel can log in to any device on the network except device C to locate the faulty node. Two possible faulty nodes can be located in this case: node A and node C, and they both send the same flush LSA. In this case, device A takes precedence over device C when the maintenance personnel determine the most possible faulty source. After device A is isolated, the network recovers.

Scenario where source tracing-incapable nodes are isolated from source tracing-capable nodes

All nodes on the network except devices C and D support source tracing, and node A is the faulty source. In this case, the PS-LSA cannot be flooded on the entire network. **Figure 7-19** shows the networking.

Figure 7-19 Scenario where source tracing-incapable nodes are isolated from source tracing-capable nodes



When device A flushes an OSPFv3 LSA, it generates a PS-LSA that carries device A information and brief information about the flush LSA. However, the PS-LSA can reach only device B because devices C and D do not support source tracing.

During source tracing capability negotiation, device E finds that device C does not support source tracing, and device F finds that device D does not support source tracing. After device E receives the PS-LSA from device C, device E helps device E generate and flood a PS-LSA. Similarly, after device F receives the PS-LSA from device D, device F helps device D generate and flood a PS-LSA.

After the fault occurs:

- If maintenance personnel log in to device A or B, the personnel can locate the faulty source (device A) directly. After device A is isolated, the network recovers.
- If the maintenance personnel log in to device E, F, G, or H, the personnel will find that device E claims device C to be the faulty source and device F claims device D to be the faulty source.
- If the personnel log in to devices C and D and find that the flush LSA was sent by device B, not generated by device C or D.
- If the personnel log in to device B, determine that device A is the faulty device, and isolate device A. After device A is isolated, the network recovers.

7.2.12 OSPFv3 Packet Format

Open Shortest Path First (OSPF) for IPv6 packets are encapsulated into IPv6 packets. The OSPFv3 protocol number is 89. OSPFv3 packets are classified into the following types:

- **Hello packet**
- **Database Description (DD) packet**
- **Link State Request (LSR) packet**
- **Link State Update (LSU) packet**
- **Link State Acknowledgment (LSAck) packet**

Packet Header Format

The five types of OSPFv3 packets have the same packet header format. The length of an OSPFv3 packet header is 24 bytes. **Figure 7-20** shows an OSPFv3 packet header.

Figure 7-20 OSPFv3 packet header

0	7	15	23	31
Version		Type	Packet length	
Router ID				
Area ID				
Checksum		Instance ID	0	

Table 7-11 describes packet header fields.

Table 7-11 Packet header fields

Field	Length	Description
Version	8 bits	OSPF version number. For OSPFv3, the value is 3.
Type	8 bits	OSPFv3 packet type. The values are as follows: ● 1: Hello packet ● 2: DD packet ● 3: LSR packet ● 4: LSU packet ● 5: LSAck packet
Packet length	16 bits	Length of the OSPFv3 packet containing the packet header, in bytes.
Router ID	32 bits	ID of the router that sends the OSPFv3 packet.
Area ID	32 bits	ID of the area to which the router that sends the OSPFv3 packet belongs.
Checksum	16 bits	Checksum of the OSPFv3 packet that does not contain the Authentication field.
Instance ID	8 bits	ID of an OSPFv3 instance.
0	8 bits	Reserved fields.

Hello Packet

Hello packets are commonly used packets, which are periodically sent on OSPFv3 interfaces to establish and maintain neighbor relationships. A Hello packet includes information about the designated router (DR), backup designated router (BDR), timers, and known neighbors.

Figure 7-21 shows the format of a Hello packet.

Figure 7-21 Format of a Hello packet

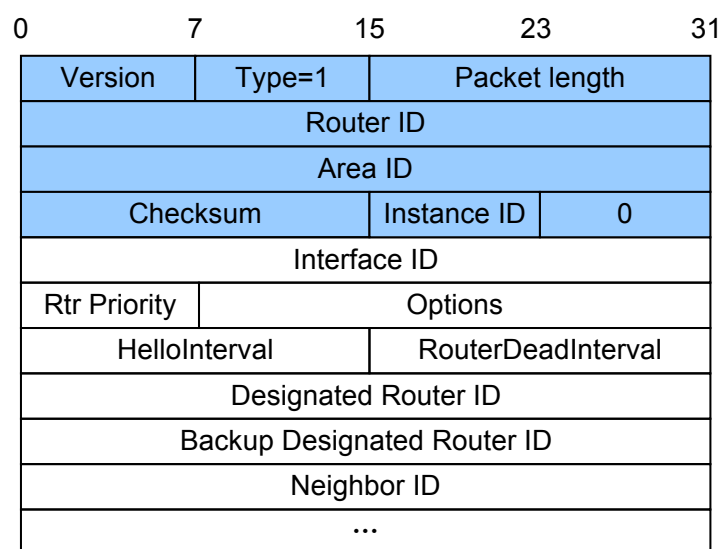


Table 7-12 describes Hello packet fields.

Table 7-12 Hello packet fields

Field	Length	Description
Interface ID	32 bits	ID of the interface that sends the Hello packets.
Rtr Priority	8 bits	DR priority. The default value is 1. NOTE If the DR priority of a router interface is set to 0, the interface cannot participate in a DR or BDR election.
Options	24 bits	The values are as follows: ● E: Type 5 link state advertisements (LSAs) are flooded. ● MC: IP multicast packets are forwarded. ● N/P: Type 7 LSAs are processed. ● DC: On-demand links are processed.
HelloInterval	16 bits	Interval at which Hello packets are sent.
RouterDeadInterval	16 bits	Dead interval. If a router does not receive any Hello packets from its neighbors within a specified dead interval, the neighbors are considered Down.
Designated Router ID	32 bits	Interface address of the DR.

Field	Length	Description
Backup Designated Router ID	32 bits	Interface address of the BDR.
Neighbor ID	32 bits	Router ID of the neighbor.

Table 7-13 lists the address types, interval types, and default intervals used when Hello packets are transmitted on different networks.

Table 7-13 Hello packet characteristics for various network types

Network Type	Address Type	Interval Type	Default Interval
Broadcast	Multicast address	HelloInterval	10 seconds
Non-broadcast multiple access (NBMA)	Unicast address	● HelloInterval for the DR, BDR, and router that can become a DR ● PollInterval for the case when neighbors become Down and HelloInterval for other cases	30 seconds for HelloInterval 120 seconds for PollInterval
Point-to-point (P2P)	Multicast address	HelloInterval	10 seconds
Point-to-multipoint (P2MP)	Multicast address	HelloInterval	30 seconds

NOTE

To establish neighbor relationships between routers on the same network segment, you must set the same HelloInterval, PollInterval, and RouterDeadInterval values for the routers. PollInterval applies only to NBMA networks.

DD Packet

During an adjacency initialization, two routers use DD packets to describe their own link state databases (LSDBs) for LSDB synchronization. A DD packet contains the header of each LSA in an LSDB. An LSA header uniquely identifies an LSA. The LSA header occupies only a small portion of the LSA, which reduces the amount of traffic transmitted between routers. A neighbor can use the LSA header to check whether it already has the LSA. When two routers exchange DD packets, one functions as the master and the other functions as the slave. The master defines a start sequence number. The master increases the sequence number by one

each time it sends a DD packet. After the slave receives a DD packet, it uses the sequence number carried in the DD packet for acknowledgment.

Figure 7-22 shows the format of a DD packet.

Figure 7-22 Format of a DD packet

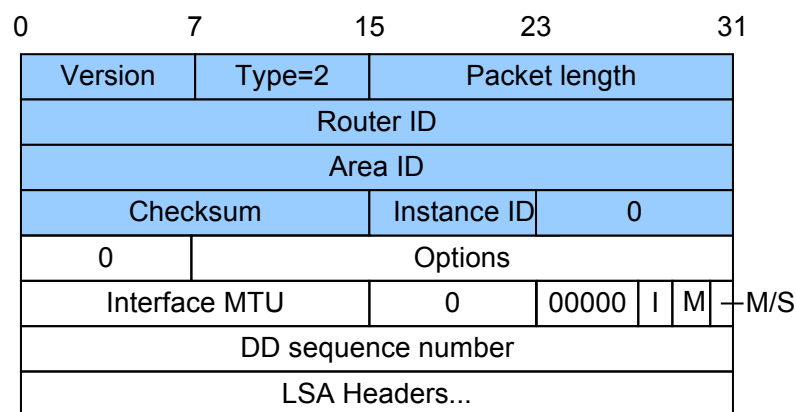


Table 7-14 describes DD packet fields.

Table 7-14 DD packet fields

Field	Length	Description
Options	24 bits	The values are as follows: ● E: Type 5 LSAs are flooded. ● MC: IP multicast packets are forwarded. ● N/P: Type 7 LSAs are processed. ● DC: On-demand links are processed.
Interface MTU	16 bits	Maximum length of the DD packet sent by the interface with packet fragmentation disabled.
I	1 bit	If the DD packet is the first packet among multiple consecutive DD packets sent by a router, this field is set to 1. In other cases, this field is set to 0.
M (More)	1 bit	If the DD packet is the last packet among multiple consecutive DD packets sent by a router, this field is set to 0. In other cases, this field is set to 1.
M/S (Master/Slave)	1 bit	When two routers exchange DD packets, they negotiate a master/slave relationship. The router with a larger router ID becomes the master. If this field is set to 1, the DD packet is sent by the master.
DD sequence number	32 bits	Sequence number of the DD packet. The master and slave use the sequence number to ensure that DD packets are correctly transmitted.

Field	Length	Description
LSA Headers	-	LSA header information included in the DD packet.

LSR Packet

After two routers exchange DD packets, they send LSR packets to request each other's LSAs. The LSR packets contain the summaries of the requested LSAs. **Figure 7-23** shows the format of an LSR packet.

Figure 7-23 Format of an LSR packet

0	7	15	23	31
Version	Type=3	Packet length		
Router ID				
Area ID				
Checksum		Instance ID	0	
0		LS type		
Link State ID				
Advertising Router				
...				

Table 7-15 describes LSR packet fields.

Table 7-15 LSR packet fields

Field	Length	Description
LS type	16 bits	Type of the LSA
Link State ID	32 bits	This field together with the LS type field describes an LSA in an AS.
Advertising Router	32 bits	Router ID of the router that generates the LSA.

NOTE

The LS type, Link State ID, and Advertising Router fields can uniquely identify an LSA. If two LSAs have the same LS type, Link State ID, and Advertising Router fields, a router uses the LS sequence number, LS checksum, and LS age fields to obtain a required LSA.

LSU Packet

A router uses an LSU packet to transmit LSAs requested by its neighbors or to flood its own updated LSAs. The LSU packet contains a set of LSAs. For multicast and broadcast networks, LSU packets are multicast to flood LSAs. To ensure reliable LSA flooding, a router uses an

LSAck packet to acknowledge the LSAs contained in an LSU packet that is received from a neighbor. If an LSA fails to be acknowledged, the router retransmits the LSA to the neighbor. **Figure 7-24** shows the format of an LSU packet.

Figure 7-24 Format of an LSU packet

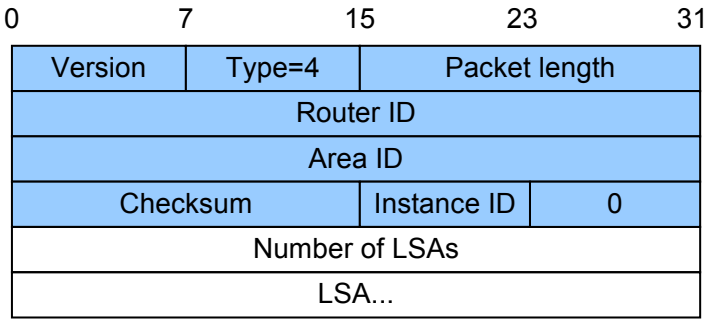


Table 7-16 describes the LSU packet field.

Table 7-16 LSU packet field

Field	Length	Description
Number of LSAs	32 bits	Number of LSAs contained in the LSU packet

LSAck Packet

A router uses an LSAck packet to acknowledge the LSAs contained in a received LSU packet. The LSAs can be acknowledged using LSA headers. LSAck packets can be transmitted over different links in unicast or multicast mode. **Figure 7-25** shows the format of an LSAck packet.

Figure 7-25 Format of an LSAck packet

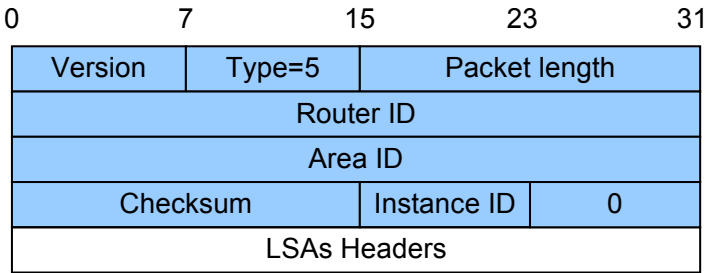


Table 7-17 describes the LSAck packet field.

Table 7-17 LSAck packet field

Field	Length	Description
LSAs Headers	Determined by the header length of the LSA to be acknowledged.	This field is used to acknowledge an LSA.

7.2.13 OSPFv3 LSA Format

Each router in an autonomous system (AS) generates one or more types of link state advertisements (LSAs), depending on the router's type. Multiple LSAs form a link state database (LSDB). Open Shortest Path First (OSPF) for IPv6 encapsulates routing information into LSAs for transmission. Commonly used LSAs include:

- Router-LSA (Type 1)
- Network-LSA (Type 2)
- Inter-Area-Prefix-LSA (Type 3)
- Inter-Area-Router-LSA (Type 4)
- AS-external-LSA (Type 5)
- NSSA LSA (Type 7)
- Link-LSA (Type 8)
- Intra-Area-Prefix-LSA (Type 9)

LSA Header Format

All LSAs have the same header. Figure 7-26 shows an LSA header.

Figure 7-26 LSA header

0	7	15	23	31
LS age		LS type		
Link State ID				
Advertising Router				
LS sequence number				
LS checksum		Length		

Table 7-18 describes LSA header fields.

Table 7-18 LSA header fields

Field	Length	Description
LS age	16 bits	Time that elapses after the LSA is generated, in seconds. The value of this field continually increases regardless of whether the LSA is transmitted over a link or saved in an LSDB.
LS type	16 bits	Type of the LSA. The values are as follows: ● Type 1: Router-LSA. ● Type 2: Network-LSA. ● Type 3: Inter-Area-Prefix-LSA. ● Type 4: Inter-Area-Router-LSA. ● Type 5: AS-external-LSA. ● Type 7: NSSA-LSA. ● Type 8: Link-LSA. ● Type 9: Intra-Area-Prefix-LSA.
Link State ID	32 bits	This field together with the LS type field describes an LSA in an AS.
Advertising Router	32 bits	Router ID of the router that generates the LSA.
LS sequence number	32 bits	Sequence number of the LSA. Neighbors can use this field to identify the latest LSA.
LS checksum	16 bits	Checksum of all fields except the LS age field.
Length	16 bits	Length of the LSA including the LSA header, in bytes.

Router-LSA

A router-LSA describes the link status and cost of a router. Router-LSAs are generated by a router and advertised within the area to which the router belongs. [Figure 7-27](#) shows the format of a router-LSA.

Figure 7-27 Format of a router-LSA

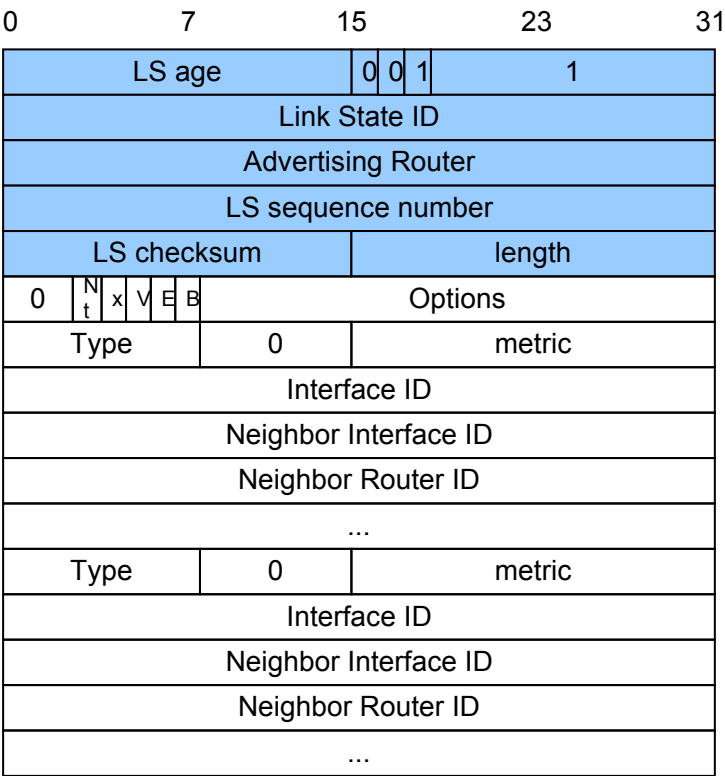


Table 7-19 describes router-LSA fields.

Table 7-19 Router-LSA fields

Field	Length	Description
Nt (NSSA translation)	1 bit	If the router that generates the LSA is an NSSA border router, this field is set to 1. In other cases, this field is set to 0. When this field is set to 1, the router is unconditionally translating NSSA-LSAs into AS-external-LSAs.
x	1 bit	This field is deprecated.
V (Virtual Link)	1 bit	If the router that generates the LSA is located at one end of a virtual link, this field is set to 1. In other cases, this field is set to 0.
E (External)	1 bit	If the router that generates the LSA is an autonomous system boundary router (ASBR), this field is set to 1. In other cases, this field is set to 0.
B (Border)	1 bit	If the router that generates the LSA is an area border router (ABR), this field is set to 1. In other cases, this field is set to 0.
Options	24 bits	The optional capabilities supported by the router.

Field	Length	Description
Type	8 bits	Type of the router link. The values are as follows: ● 1: The router is connected to another router in point-to-point (P2P) mode. ● 2: The router is connected to a transport network. ● 3: Reserved. ● 4: The router is connected to another router over a virtual link.
metric	16 bits	Cost of the link.
Interface ID	32 bits	The Interface ID assigned to the interface.
Neighbor Interface ID	32 bits	The Interface ID the neighbor router has associated with the link. ● For transit links, the value is the interface ID of the DR. ● For links of other types, the value is the interface ID of the neighboring device.
Neighbor Router ID	32 bits	The Router ID of the neighbor router. ● For transit links, the value is the router ID of the DR. ● For links of other types, the value is the router ID of the neighboring device.

Network-LSA

A network-LSA describes the link status of all routers on the local network segment. Network-LSAs are generated by a DR on a broadcast or non-broadcast multiple access (NBMA) network and advertised within the area to which the DR belongs. [Figure 7-28](#) shows the format of a network-LSA.

Figure 7-28 Format of a network-LSA

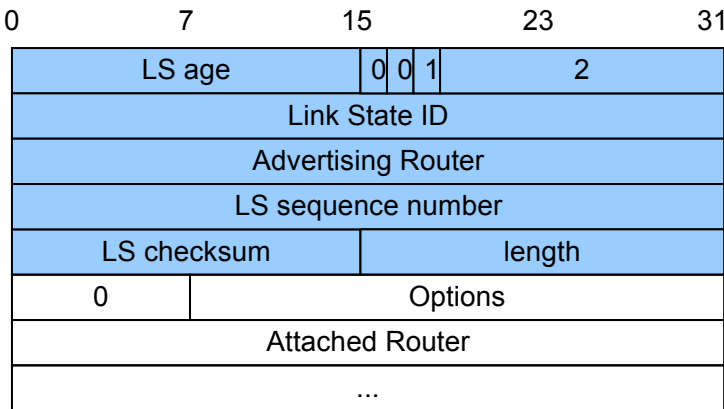


Table 7-20 describes network-LSA fields.

Table 7-20 Network-LSA fields

Field	Length	Description
Options	24 bits	The optional capabilities supported by the router.
Attached Router	32 bits	Router IDs of all routers on the broadcast or NBMA network, including the router ID of the DR

Inter-Area-Prefix-LSA

An inter-area-prefix-LSA describes routes on a network segment in an area. It is generated by the ABR. The routes are advertised to other areas.

Figure 7-29 shows the format of an inter-area-prefix-LSA.

Figure 7-29 Format of An inter-area-prefix-LSA

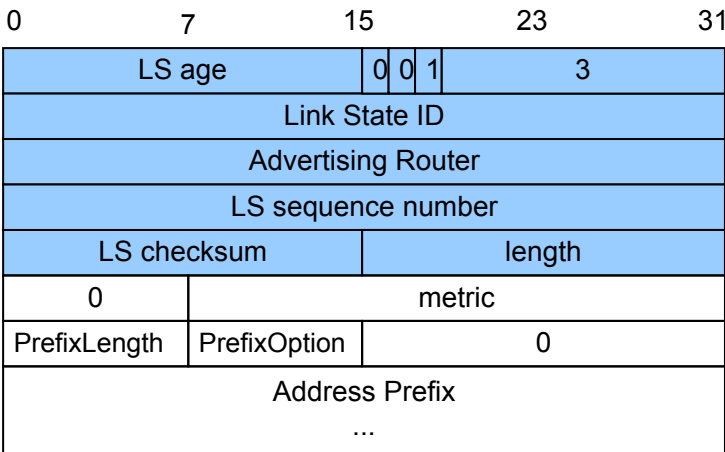


Table 7-21 describes inter-area-prefix-LSA fields.

Table 7-21 Network-summary-LSA fields

Field	Length	Description
PrefixLength	8 bits	Length of the prefix.
PrefixOption	8 bits	The capabilities associated with the prefix.
Address Prefix	32 bits	An encoding of the prefix itself.

Inter-Area-Router-LSA

An inter-area-router-LSA describes routes to the ASBR in other areas. It is generated by the ABR. The routes are advertised to all related areas except the area that the ASBR belongs to.

Figure 7-30 shows the format of an inter-area-prefix-LSA.

Figure 7-30 Format of an inter-area-router-LSA

0	7	15	23	31
LS age		0	0	1
Link State ID				
Advertising Router				
LS sequence number				
LS checksum		length		
0	Options			
0	metric			
Destination Router ID				

Table 7-22 describes inter-area-router-LSA fields.

Table 7-22 Inter-area-router-LSA fields

Field	Length	Description
Destination Router ID	32 bits	The Router ID of the router being described by the LSA.

AS-External-LSAs

An as-external-LSA describes destinations outside the AS, it is originated by ASBR.

Figure 7-31 shows the format of an as-external-LSA.

Figure 7-31 Format of an as-external-LSA

0	7	15	23	31
LS age		0	0	1
Link State ID				
Advertising Router				
LS sequence number				
LS checksum		length		
	E	F	T	metric
PrefixLength		PrefixOption	Referenced LS Type	
Address Prefix				
...				
Forwarding Address (Optional)				
External Route Tag (Optional)				
Referenced Link State ID (Optional)				

Table 7-23 describes as-external-LSA fields.

Table 7-23 As-external-LSA fields

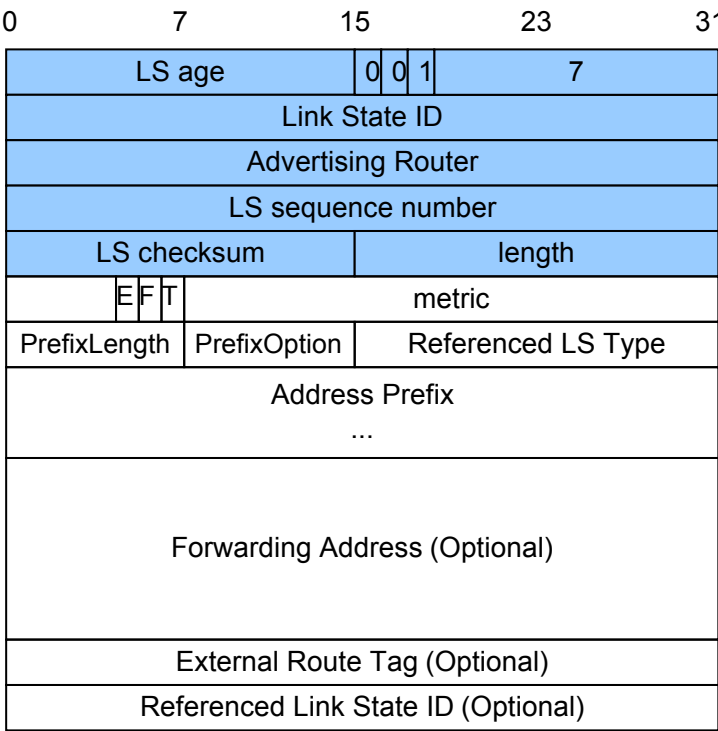
Field	Length	Description
E	1 bit	The type of external metric. ● If this field is 1, the specified metric is a Type 2 external metric. ● If this field is 0, the specified metric is a Type 1 external metric.
F	1 bit	Whether a Forwarding Address has been included in the LSA. ● If this field is 1, a Forwarding Address has been included in the LSA. ● If this field is 0, no Forwarding Address is included in the LSA.
T	1 bit	Whether an External Route Tag has been included in the LSA. ● If this field is 1, an External Route Tag has been included in the LSA. ● If this field is 0, no External Route Tag is included in the LSA.
Referenced LS Type	16 bits	Indicates the referenced LS type. If non-zero, an LSA with this LS type is to be associated with this LSA (see Referenced Link State ID below).
Forwarding Address	128 bits	A fully qualified global IPv6 address.
External Route Tag	32 bits	Indicates the external route tag, which can be used to communicate additional information between ASBRs.
Referenced Link State ID	32 bits	Indicates the referenced link state ID.

NSSA-LSAs

An NSSA-LSA describes destinations outside the AS, it is originated by ASBR.

Figure 7-32 shows the format of an NSSA-LSA.

Figure 7-32 Format of an NSSA-LSA



Link-LSAs

Each router generates a link LSA for each link. A link LSA describes the link-local address and IPv6 address prefix associated with the link and the link option set in the network LSA. It is transmitted only on the link.

Figure 7-33 shows the format of a Link-LSA.

Figure 7-33 Format of a Link-LSA

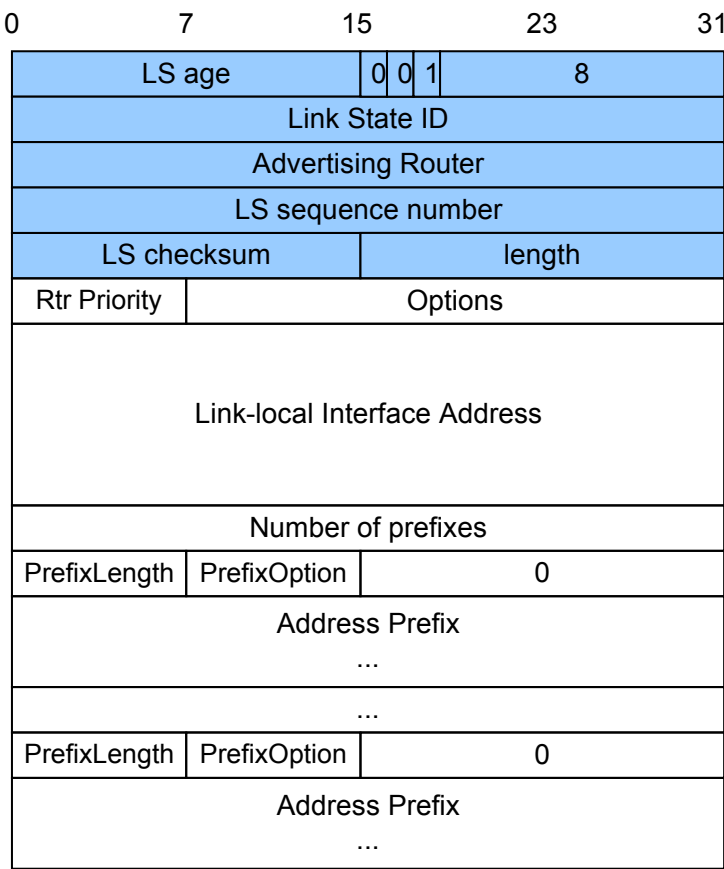


Table 7-24 describes link-LSA fields.

Table 7-24 Link-LSA fields

Field	Length	Description
Rtr Priority	8 bits	The Router Priority of the interface.
Options	24 bits	The set of Options bits that the router would like set in the network-LSA that will be originated by the DR on broadcast or NBMA links.
Link-local Interface Address	128 bits	The originating router's link-local interface address on the link.
Number of prefixes	32 bits	The number of IPv6 address prefixes contained in the LSA.

Intra-Area-Prefix-LSAs

Each router or DR generates one or more intra-area prefix LSAs and transmits it in the local area.

- An LSA generated on a router describes the IPv6 address prefix associated with the router LSA.
- An LSA generated on a DR describes the IPv6 address prefix associated with the network LSA.

Figure 7-34 shows the format of an intra-area-prefix-LSA.

Figure 7-34 Format of an intra-area-prefix-LSA

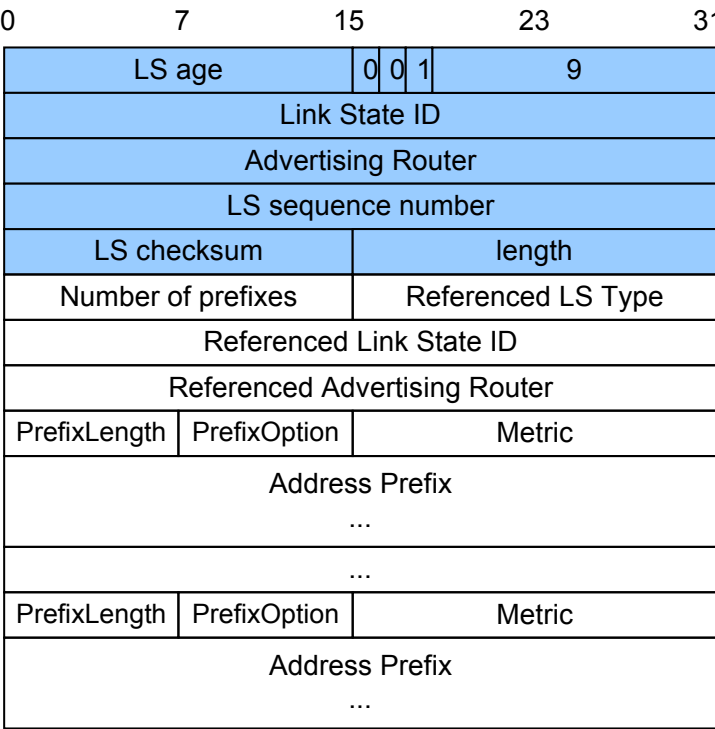


Table 7-25 describes intra-area-prefix-LSA fields.

Table 7-25 Intra-area-prefix-LSA fields

Field	Length	Description
Referenced LS Type	16 bits	This field identifies the router-LSA or network-LSA with which the IPv6 address prefixes should be associated. ● If Referenced LS Type is 0x2001, the prefixes are associated with a router-LSA. ● If Referenced LS Type is 0x2002, the prefixes are associated with a network-LSA.
Referenced Link State ID	32 bits	Indicates the referenced link state ID. ● If Referenced LS Type is 0x2001, Referenced Link State ID should be 0. ● If Referenced LS Type is 0x2002, Referenced Link State ID should be the Interface ID of the link's DR.

Field	Length	Description
Referenced Advertising Router	32 bits	Indicates the ID of the referenced advertising router. ● If Referenced LS Type is 0x2001, Referenced Advertising Router should be the originating router's Router ID. ● If Referenced LS Type is 0x2002, Referenced Advertising Router should be the Designated Router's Router ID.

8 IS-IS

About This Chapter

[8.1 Overview of IS-IS](#)

[8.2 Understanding IS-IS](#)

[8.3 Application Scenarios for IS-IS](#)

8.1 Overview of IS-IS

Definition

Intermediate System to Intermediate System (IS-IS) is a dynamic routing protocol initially designed by the International Organization for Standardization (ISO) for its Connectionless Network Protocol (CLNP).

To support IP routing, the Internet Engineering Task Force (IETF) extends and modifies IS-IS in relevant standards, which enables IS-IS to be applied to both TCP/IP and Open System Interconnection (OSI) environments. This type of IS-IS is called Integrated IS-IS or Dual IS-IS.

In this document, IS-IS refers to Integrated IS-IS, unless otherwise stated.

NOTE

If IS-IS IPv4 and IS-IS IPv6 implement a feature in the same way, details are not provided in this chapter.

Purpose

As an Interior Gateway Protocol (IGP), IS-IS is used in Autonomous Systems (ASs). IS-IS is a link state protocol, and it uses the Shortest Path First (SPF) algorithm to calculate routes.

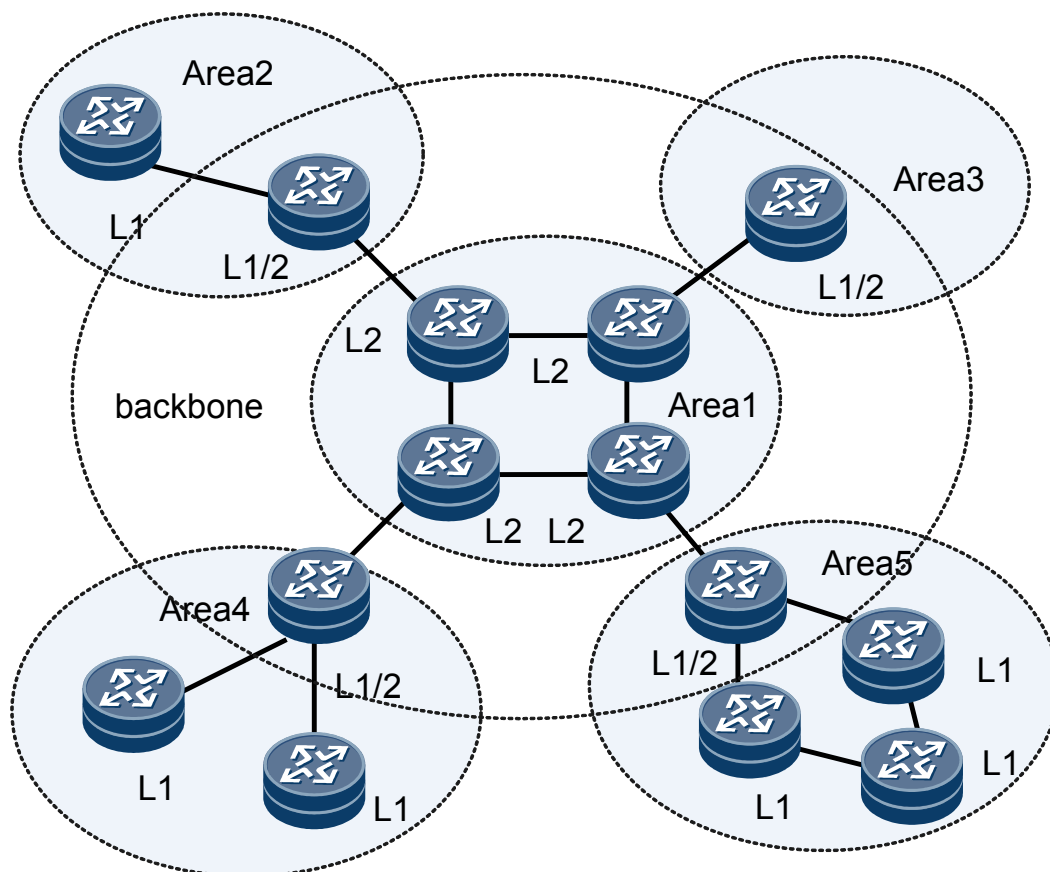
8.2 Understanding IS-IS

8.2.1 Basic Concepts of IS-IS

IS-IS Areas

To support large-scale routing networks, IS-IS adopts a two-level structure in a routing domain. A large domain is divided into areas. **Figure 8-1** shows an IS-IS network. The entire backbone area covers all Level-2 devices in area 1 and Level-1-2 devices in other areas. Three types of devices on the IS-IS network are described as follows:

Figure 8-1 IS-IS topology



- Level-1 device

A Level-1 device manages intra-area routing. It establishes neighbor relationships with only the Level-1 and Level-1-2 devices in the same area and maintains a Level-1 LSDB. The LSDB contains routing information in the local area. A packet to a destination beyond this area is forwarded to the nearest Level-1-2 device.

- Level-2 device

A Level-2 device manages inter-area routing. It can establish neighbor relationships with all Level-2 devices and Level-1-2 devices, and maintains a Level-2 LSDB which contains inter-area routing information.

All Level-2 devices form the backbone network of the routing domain. Level-2 neighbor relationships are set up between them. They are responsible for communications between areas. The Level-2 devices in the routing domain must be in succession to ensure the

continuity of the backbone network. Only Level-2 devices can exchange data packets or routing information with the devices beyond the routing domain.

- Level-1-2 device

A device, which can establish neighbor relationships with both Level-1 devices and Level-2 devices, is called a Level-1-2 device. A Level-1-2 device can establish Level-1 neighbor relationships with Level-1 devices and Level-1-2 devices in the same area. It can also establish Level-2 neighbor relationships with Level-2 devices and Level-1-2 devices in other areas. Level-1 devices can be connected to other areas only through Level-1-2 devices.

A Level-1-2 device maintains two LSDBs: a Level-1 LSDB and a Level-2 LSDB. The Level-1 LSDB is used for intra-area routing, whereas the Level-2 LSDB is used for inter-area routing.

 NOTE

Level-1 devices in different areas cannot establish neighbor relationships. Level-2 devices can establish neighbor relationships with each other, regardless of the areas to which the Level-2 devices belong.

In general, Level-1 devices are located within an area, Level-2 devices are located between areas, and Level-1-2 devices are located between Level-1 devices and Level-2 devices.

Interface level

A Level-1-2 device may need to establish only a Level-1 adjacency with one neighbor and establish only a Level-2 adjacency with another neighbor. In this case, you can set the level of an interface to control the setting of adjacencies on the interface. Specifically, only Level-1 adjacencies can be established on a Level-1 interface, and only Level-2 adjacencies can be established on a Level-2 interface.

Address Structure of IS-IS

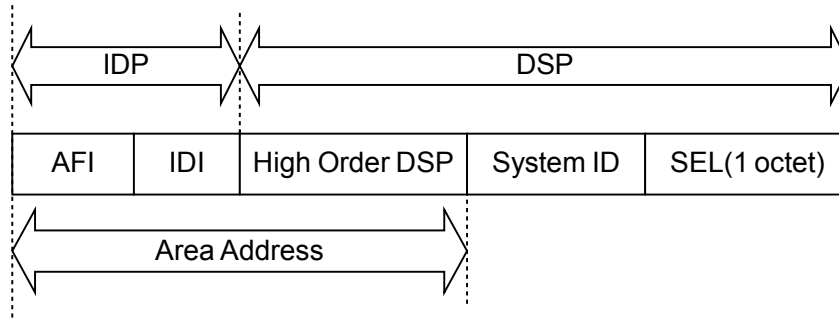
In OSI, the NSAP is used to locate resources. The ISO adopts the address structure shown in [Figure 8-2](#). An NSAP is composed of the Initial Domain Part (IDP) and the Domain Specific Part (DSP). IDP is the counterpart of network ID in an IP address, and DSP is the counterpart of the subnet number and host address in an IP address.

As defined by the ISO, the IDP consists of the Authority and Format Identifier (AFI) and Initial Domain Identifier (IDI). AFI specifies the address assignment mechanism and the address format; the IDI identifies a domain.

The DSP consists of the High Order DSP (HODSP), system ID, and NSAP Selector (SEL). The HODSP is used to divide areas; the system ID identifies a host; the SEL indicates the service type.

The lengths of the IDP and DSP are variable. The length of the NSAP varies from 8 bytes to 20 bytes.

Figure 8-2 IS-IS address structure



- **Area address**

An IDP and HODSP of the DSP can identify a routing domain and the areas in a routing domain; therefore, the combination of the IDP and HODSP is referred to as an area address, equal to an area ID in OSPF. An area address is used to uniquely identify an area in a routing domain. The area addresses of routers in the same Level-1 area must be the same, whereas the area addresses of routers in the Level-2 area can be different.

In general, a router can be configured with only one area address. The area address of all nodes in an area must be the same. In the implementation of a device, an IS-IS process can be configured with a maximum of three area addresses to support seamless combination, division, and transformation of areas.

- **System ID**

A system ID uniquely identifies a host or a router in an area. In the device, the length of the system ID is 48 bits (6 bytes).

A router ID corresponds to a system ID. If a device uses the IP address of Loopback 0 (192.168.1.1) as its router ID, its system ID used in IS-IS can be obtained by performing the following steps:

- Extend each part of the IP address 192.168.1.1 to 3 digits and add 0 or 0s to the front of the part that is shorter than 3 digits.
- Divide the extended address 192.168.001.001 into three parts, with each part consisting of 4 decimal digits.
- The reconstructed 1921.6800.1001 is the system ID.

There are many ways to specify a system ID. Whichever you choose, ensure that the system ID uniquely identifies a host or a device.

NOTE

If the same system ID is configured for more than one device on the same network, network flapping may occur. To address this problem, IS-IS provides the automatic recovery function. With the function, if the system detects an IS-IS system ID conflict, it automatically changes the local system ID to resolve the conflict. The first two bytes of the system ID automatically changed by the system are Fs, and the last four bytes are randomly generated. For example, FFFF:1234:5678 is such a system ID. If the conflict persists after the system automatically changes three system IDs, the system no longer resolves this conflict.

- **SEL**

The role of an SEL (also referred to as NSAP Selector or N-SEL) is similar to that of the "protocol identifier" of IP. A transport protocol matches an SEL. The SEL is "00" in IP.

- **NET**

A Network Entity Title (NET) indicates the network layer information of an IS itself and consists of an area ID and a system ID. It does not contain the transport layer information (SEL = 0). A NET can be regarded as a special NSAP. The length of the NET field is the same as that of an NSAP, varying from 8 bytes to 20 bytes. For example, in NET **ab.cdef.1234.5678.9abc.00**, the area is **ab.cdef**, the system ID is **1234.5678.9abc**, and the SEL is **00**.

In general, an IS-IS process is configured with only one NET. When areas need to be redefined, for example, areas need to be combined or an area needs to be divided into sub-areas, you can configure multiple NETs.

 NOTE

A maximum of three area addresses can be configured in an IS-IS process, and therefore, you can configure only a maximum of three NETs. When you configure multiple NETs, ensure that their system IDs are the same.

The routers in an area must have the same area address.

IS-IS Network Types

IS-IS supports the following types of networks:

- Broadcast network
- Point-to-point (P2P) network

8.2.2 Basic Protocols of IS-IS

Related Concepts

DIS and Pseudo Node

A Designated Intermediate System (DIS) is an intermediate router elected in IS-IS communication. A pseudo node simulates a virtual node on a broadcast network and is not a real router. In IS-IS, a pseudo node is identified by the system ID and 1-byte circuit ID (a non-zero value) of a DIS.

The DIS is used to create and update pseudo nodes and generate the **link state protocol data units (LSPs)** of pseudo nodes. The routers advertise a single link to a pseudo node and obtain routing information about the entire network through the pseudo node. The router does not need to exchange packets with all the other routers on the network. Using the DIS and pseudo nodes simplifies network topology and reduces the length of LSPs generated by routers. When the network changes, fewer LSPs are generated. Therefore, fewer resources are consumed.

SPF Algorithm

The SPF algorithm, also named Dijkstra's algorithm, is used in a link-state routing protocol to calculate the shortest paths to other nodes on a network. In the SPF algorithm, a local router takes itself as the root and generates a shortest path tree (SPT) based on the network topology to calculate the shortest path to every destination node on a network. In IS-IS, the SPF algorithm runs separately in Level-1 and Level-2 databases.

Implementation

All routers on the IS-IS network communicate through the following steps:

- **Establishment of IS-IS Neighbor Relationships**

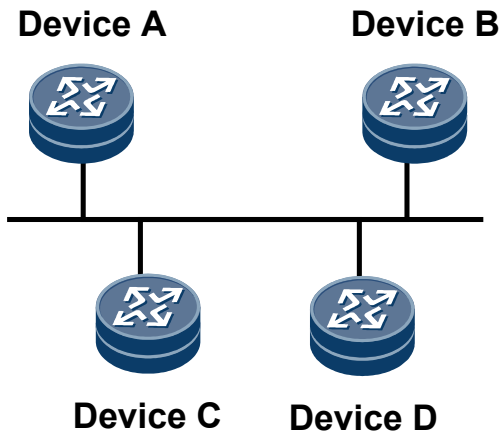
- LSDB Synchronization
- Route Calculation

Establishment of IS-IS Neighbor Relationships

On different types of networks, the modes for establishing IS-IS neighbor relationships are different.

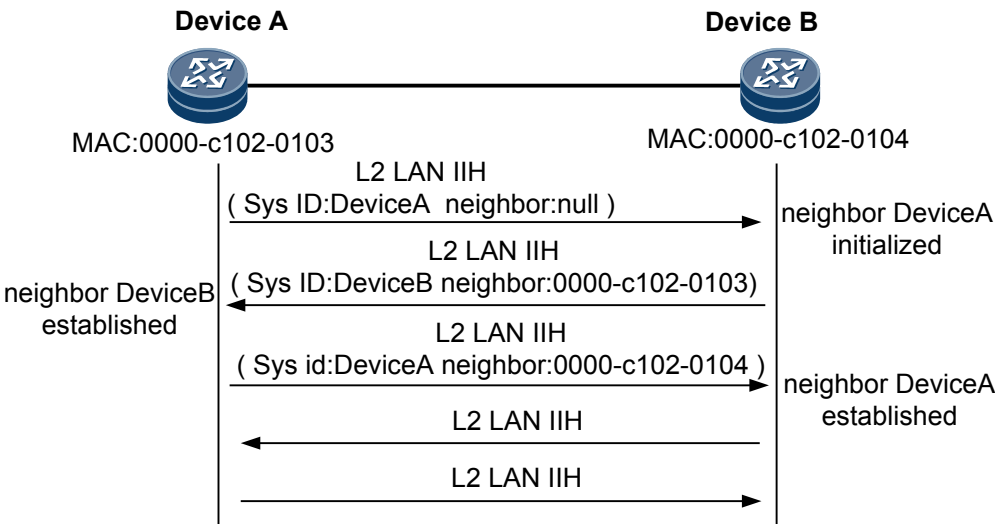
- Establishment of a neighbor relationship on a broadcast link

Figure 8-3 Networking for a broadcast link



Device A, Device B, Device C, and Device D are Level-2 routers. Device A is newly added to the broadcast network. Figure 8-4 demonstrates the process of establishing the neighbor relationship between Device A and Device B, the process of establishing the neighbor relationship between Device A and Device C or Device D is similar to that between Device A and Device B.

Figure 8-4 Establishing a neighbor relationship on a broadcast link



As shown in [Figure 8-4](#), the process for establishing a neighbor relationship on a broadcast link consists of the following phases:

- Device A broadcasts a Level-2 local area network (LAN) **IS-to-IS Hello PDU (IIH)**. After Device B receives the IIH, Device B detects that the neighbor field in the IIH does not contain its media access control (MAC) address, and sets its neighbor status with Device A to **Initial**.
- Device B replies a Level-2 LAN IIH to Device A. After Device A receives the IIH, Device A detects that the neighbor field in the IIH contains its MAC address, and sets its neighbor status with Device B to **Up**.
- Device A sends a Level-2 LAN IIH to Device B. After Device B receives the IIH, Device B detects that the neighbor field in the IIH contains its MAC address, and sets its neighbor status with Device A to **Up**.

DIS Election

On a broadcast network, any two routers exchange information. If n routers are available on the network, $n \times (n - 1)/2$ adjacencies must be established. Each status change of a router is transmitted to other routers, which wastes bandwidth resources. IS-IS resolves this problem by introducing the DIS. All routers send information to the DIS, which then broadcasts the network link status. Using the DIS and pseudo nodes simplifies network topology and reduces the length of LSPs generated by routers. When the network changes, fewer LSPs are generated. Therefore, fewer resources are consumed.

A DIS is elected after a neighbor relationship is established. Level-1 and Level-2 DISs are elected separately. You can configure different priorities for DISs at different levels. In DIS election, a Level-1 priority and a Level-2 priority are specified for every interface on every router. A router uses every interface to send IIHs and advertises its priorities in the IIHs to neighboring routers. The higher the priority, the higher the probability of being elected as the DIS. If there are multiple routers with the same highest priority on a broadcast network, the one with the largest MAC address is elected. The DISs at different levels can be the same router or different routers.

In the DIS election procedure, IS-IS is different from Open Shortest Path First (OSPF). In IS-IS, DIS election rules are as follows:

- The router with the priority of 0 also takes part in the DIS election.
 - When a new router that meets the requirements of being a DIS is added to the broadcast network, the router is selected as the new DIS, which triggers a new round of LSP flooding.
- Establishment of a neighbor relationship on a P2P link
The establishment of a neighbor relationship on a P2P link is different from that on a broadcast link. A neighbor relationship on a P2P link can be established in 2-way or 3-way mode, as shown in [Table 8-1](#). By default, the 3-way handshake mechanism is used to establish a neighbor relationship on a P2P link.

Table 8-1 Comparison between 2-way mode and 3-way mode

Mode	Description	Advantages and Disadvantages	Reliability
2-way mode	When a router receives an IIH, it unidirectionally sets up a neighbor relationship.	Disadvantages: ● The unstable link status causes the loss of complete sequence numbers protocol data units (CSNPs) that are sent once an adjacency is set up. As a result, the link state databases (LSDBs) of two neighboring routers are not synchronized during the LSP update period. ● If two or more links exist between two routers, an adjacency can still be set up when one link is Down and another is Up in the same direction. A router that fails to detect the faulty link may also forward packets over this link.	Low

Mode	Description	Advantages and Disadvantages	Reliability
3-way mode	A neighbor relationship is established after IIHs are sent three times.	Advantages: A neighbor relationship is established only when both ends are Up. This mechanism ensures that packets are transmitted securely.	High

LSDB Synchronization

IS-IS is a link-state protocol. An IS-IS router obtains first-hand information from other routers running link-state protocols. Every router generates information about itself, directly connected networks, and links between itself and directly connected networks. The router then sends the generated information to other routers through adjacent routers. Every router saves link state information without modifying it. Finally, every router has the same network interworking information, and LSDB synchronization is complete. The process of synchronizing LSDBs is called LSP flooding. In LSP flooding, a router sends an LSP to its neighbors and the neighbors send the received LSP to their neighbors except the router that first sends the LSP. The LSP is flooded among the routers at the same level. This implementation allows each router at the same level to have the same LSP information and keep a synchronized LSDB.

All routers in the IS-IS routing domain can generate LSPs. A new LSP is generated in any of the following situations:

- Neighbor goes Up or Down.
- related interface goes Up or Down.
- Imported IP routes change.
- Inter-area IP routes change.
- A new metric value is configured for an interface.
- Periodic updates occur.

A router processes a received LSP as follows:

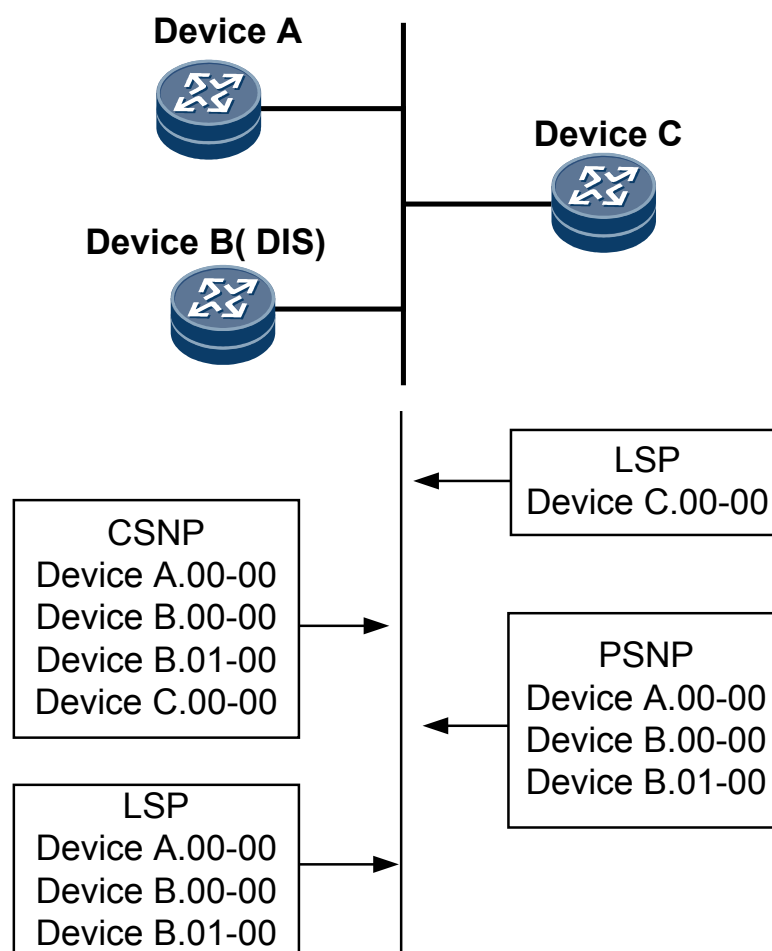
- Updating the LSDB on a broadcast link

The DIS updates the LSDB to synchronize LSDBs on a broadcast network. **Figure 8-5** shows the process of synchronizing LSDBs on a broadcast network.

- a. When the DIS receives an LSP, it searches the LSDB for the related records. If the DIS does not find the LSP in its LSDB, it adds the LSP to its LSDB and broadcasts the new LSDB.
- b. If the sequence number of the received LSP is greater than that of the local LSP, the DIS replaces the local LSP with the received LSP in the LSDB and broadcasts the new LSDB.
- c. If the sequence number of the received LSP is less than that of the local LSP, the DIS sends the local LSP in the LSDB to the inbound interface.

- d. If the sequence number of the received LSP is equal to that of the local LSP, the DIS compares the Remaining Lifetime of the two LSPs. If Remaining Lifetime of the received LSP is 0, the DIS replaces the LSP with the received LSP, and broadcasts the new LSDB. If the Remaining Lifetime of local LSP is 0, the DIS sends the LSP to the inbound interface.
- e. If the sequence number of the received LSP and the local LSP in the LSDB are the same and neither Remaining Lifetime is 0, the DIS compares the checksum of the two LSPs. If the received LSP has a greater checksum than that of the local LSP in the LSDB, the DIS replaces the local LSP in the LSDB with the received LSP and advertises the new LSDB. If the received LSP has a smaller checksum than that of the local LSP in the LSDB, the DIS sends the local LSP in the LSDB to the inbound interface.
- f. If the checksums of the received LSP and the local LSP are the same, the LSP is not forwarded.

Figure 8-5 Process of updating the LSDB on a broadcast link



- Updating the LSDB on a P2P link
 - a. If the sequence number of the received LSP is greater than that of the local LSP in the LSDB, the router adds the received LSP to its LSDB. The router then sends a **PSNP packet** to acknowledge the received LSP and sends the LSP to all its neighbors except the neighbor that sends the LSP.

- b. If the sequence number of the received LSP is less than that of the local LSP, the router directly sends its LSP to the neighbor and waits for a PSNP from the neighbor as an acknowledgement.
- c. If the sequence number of the received LSP is the same as that of the local LSP in the LSDB, the router compares the Remaining Lifetimes of the two LSPs. If Remaining Lifetime of the received LSP is 0, the router adds the LSP to its LSDB. The router then sends a PSNP to acknowledge the received LSP. If Remaining Lifetime of the local LSP is 0, the router directly sends the local LSP to the neighbor and waits for a PSNP from the neighbor.
- d. If the sequence number of the received LSP and the local LSP in the LSDB are the same, and neither Remaining Lifetime is 0, the router compares the checksum of the two LSPs. If the received LSP has a greater checksum than that of the local LSP, the router adds the received LSP to its LSDB. The router then sends a PSNP to acknowledge the received LSP. If the received LSP has a smaller checksum than that of the local LSP, the router directly sends the local LSP to the neighbor and waits for a PSNP from the neighbor. At last, the router sends the LSP to all its neighbors except the neighbor that sends the LSP.
- e. If the checksums of the received LSP and the local LSP are the same, the LSP is not forwarded.

Route Calculation

When LSDB synchronization is complete and network convergence is implemented, IS-IS performs SPF calculation by using LSDB information to obtain the SPT. IS-IS uses the SPT to create a forwarding database (a routing table).

In IS-IS, link costs are used to calculate shortest paths. The default cost for an interface on a Huawei router is 10. The cost is configurable. The cost of a route is the sum of the cost of every outbound interface along the route. There may be multiple routes to a destination, among which the route with the smallest cost is the optimal route.

Level-1 routers can also calculate the shortest path to Level-2 routers to implement inter-area route selection. When a Level-1-2 router is connected to other areas, the router sets the value of the attachment (ATT) bit in its LSP to 1 and sends the LSP to neighboring routers. In the route calculation process, a Level-1 router selects the nearest Level-1-2 router as an intermediate router between the Level-1 and Level-2 areas.

8.2.3 IS-IS Routing Information Control

IS-IS routes calculated using the SPF algorithm may bring about some problems. For example, too many routing entries slow down route lookup, or link usage is unbalanced. As a result, IS-IS routing cannot meet carriers' network planning and traffic management requirements.

IS-IS routing information control can refine control over route selection using the following methods:

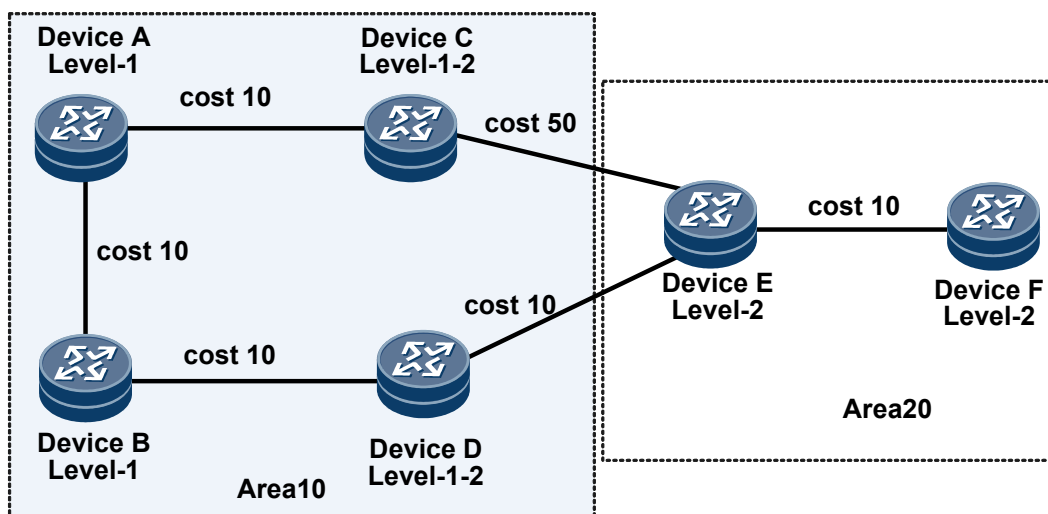
- **Route Leaking**
- **Route Summarization**
- **Load Balancing**
- **Administrative Tag**
- **IS-IS Mesh Group**
- **Link Group**

Route Leaking

When Level-1 and Level-2 areas both exist on an IS-IS network, Level-2 routers do not advertise the learned routing information about a Level-1 area and the backbone area to any other Level-1 area by default. Therefore, Level-1 routers do not know the routing information beyond the local area. As a result, the Level-1 routers cannot select the optimal routes to the destination beyond the local area.

With route leaking, Level-1-2 routers can select routes using routing policies, or tags and advertise the selected routes of other Level-1 areas and the backbone area to the Level-1 area. **Figure 8-6** shows the typical networking for route leaking.

Figure 8-6 Typical networking for route leaking



- Device A, Device B, Device C, and Device D belong to area 10. Device A and Device B are Level-1 routers. Device C and Device D are Level-1-2 routers.
- Device E and Device F belong to area 20 and are Level-2 routers.

If Device A sends a packet to Device F, the selected optimal route should be Device A -> Device B -> Device D -> Device E -> Device F because its cost is 40 ($10 + 10 + 10 + 10 = 40$) which is less than that of Device A -> Device C -> Device E -> Device F ($10 + 50 + 10 = 70$). However, if you check routes on Device A, you can find that the selected route is Device A -> Device C -> Device E -> Device F, which is not the optimal route from Device A to Device F.

This is because Device A does not know the routes beyond the local area, and therefore, the packets sent by Device A to other network segments are sent through the default route generated by the nearest Level-1-2 device.

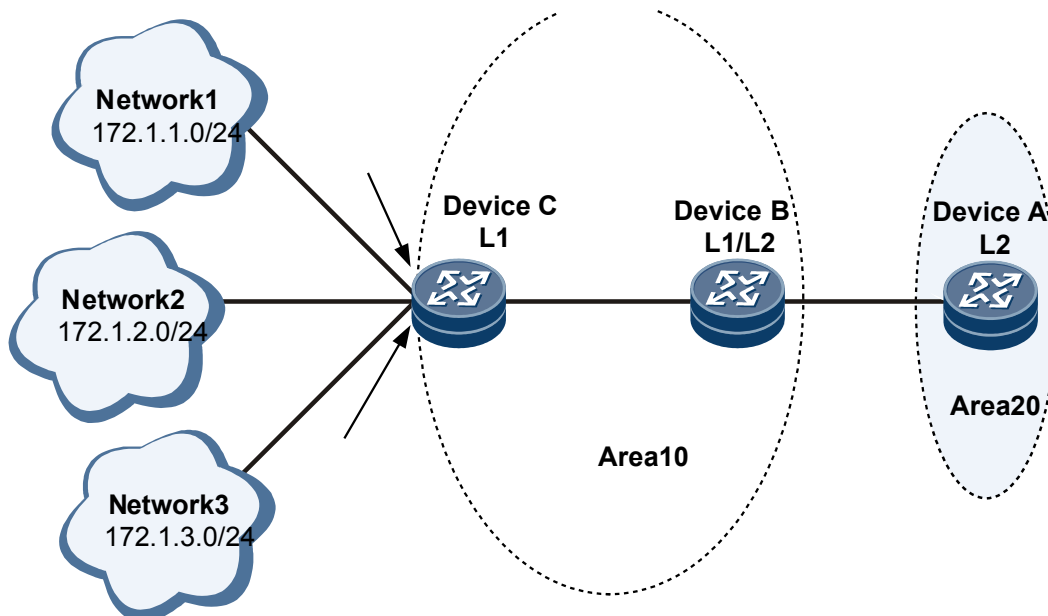
In this case, you can enable route leaking on the Level-1-2 devices (Device C and Device D). Then, check the route and you can find that the selected route is Device A -> Device B -> Device D -> Device E -> Device F.

Route Summarization

On a large-scale IS-IS network, links connected to devices within an IP address range may alternate between Up and Down. With route summarization, multiple routes with the same IP

prefix are summarized into one route, which prevents route flapping, reduces routing entries and system resource consumption, and facilitates route management. **Figure 8-7** shows the typical networking for route summarization.

Figure 8-7 Typical networking for route summarization



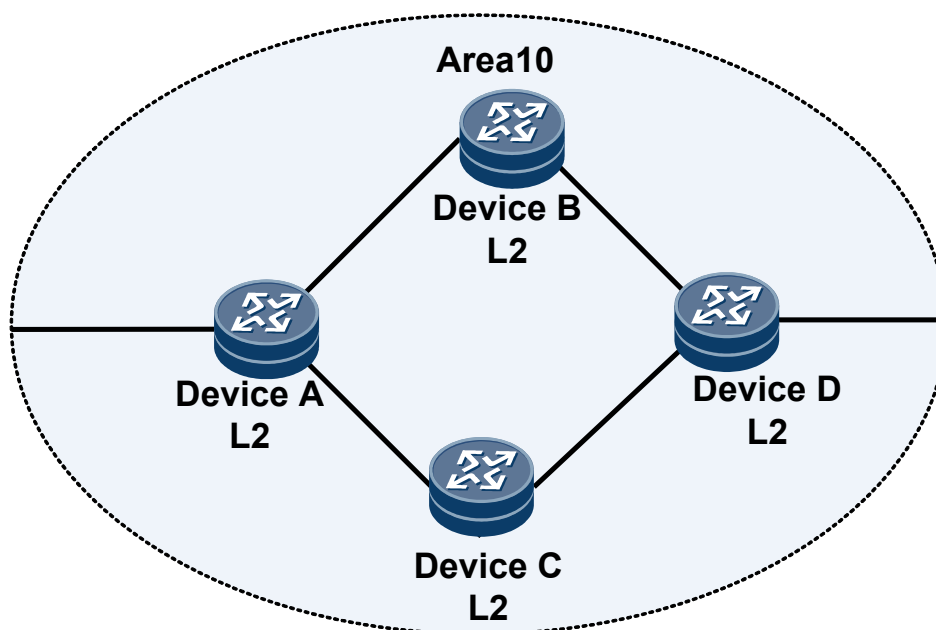
- router A, router B, and router C use IS-IS to communicate with each other.
- Device A belongs to area 20, and Device B and Device C belong to area 10.
- Device A is a Level-2 router. Device B is a Level-1-2 router. Device C is a Level-1 router.
- Device B maintains Level-1 and Level-2 LSDBs and leaks the routes to three network segments (172.16.1.0/24, 172.16.2.0/24, and 172.16.3.0/24) from the Level-1 area to the Level-2 area. If a link fault causes the Device C interface with IP address 172.16.1.1/24 to frequently alternate between Up and Down, the status change is advertised to the Level-2 area, triggering frequent LSP flooding and SPF calculation on Device A. As a result, the CPU usage on Device A increases, and even network flapping occurs.

On Device B, routes to the three network segments in the Level-1 area are summarized to one route to 172.16.0.0/16, which reduces the number of routing entries on Device B and minimizes the impact of route flapping in the Level-1 area on route convergence in the Level-2 area.

Load Balancing

If multiple equal-cost routes are available on a network, they can load-balance traffic, which improves link usage and prevents network congestion caused by link overload. **Figure 8-8** shows the typical networking for load balancing.

Figure 8-8 Typical networking for load balancing



- Device A, Device B, Device C, and Device D communicate with each other on an IP network using IS-IS.
- Device A, Device B, Device C, and Device D belong to area 10 and are Level-2 routers.
- If load balancing is not enabled, traffic on Device A is transmitted along the optimal route obtained using the SPF calculation. Consequently, traffic on different links is unbalanced. Enabling load balancing on Device A sends traffic to router Device D through router Device B and Device C. This transmission mode relieves the load on the optimal route.

Load balancing can work per-destination or per-packet. For details, see "Overview" in *NE40E Feature Description - IP Routing*.

In addition to the support for route balancing within the same process, IS-IS supports ECMP among several processes.

Administrative Tag

Administrative tags carry administrative information about IP address prefixes. When the cost type is wide, wide-compatible, or compatible and the prefix of the reachable IP address to be advertised by IS-IS has this cost type, IS-IS adds the administrative tag to the reachability type-length-value (TLV) in the prefix. In this manner, the administrative tag is advertised throughout the entire IS-IS area so that routes can be imported or filtered based on the administrative tag.

IS-IS Mesh Group

As defined in IS-IS, a router must flood the received LSP to all neighbors. On a network with multiple connections and point-to-point (P2P) links, this flooding method causes repeated LSP flooding and wastes bandwidth resources.

To address this issue, you can add certain interfaces to a mesh group. These interfaces flood the LSPs received from a group only to interfaces of other groups or interfaces no which no

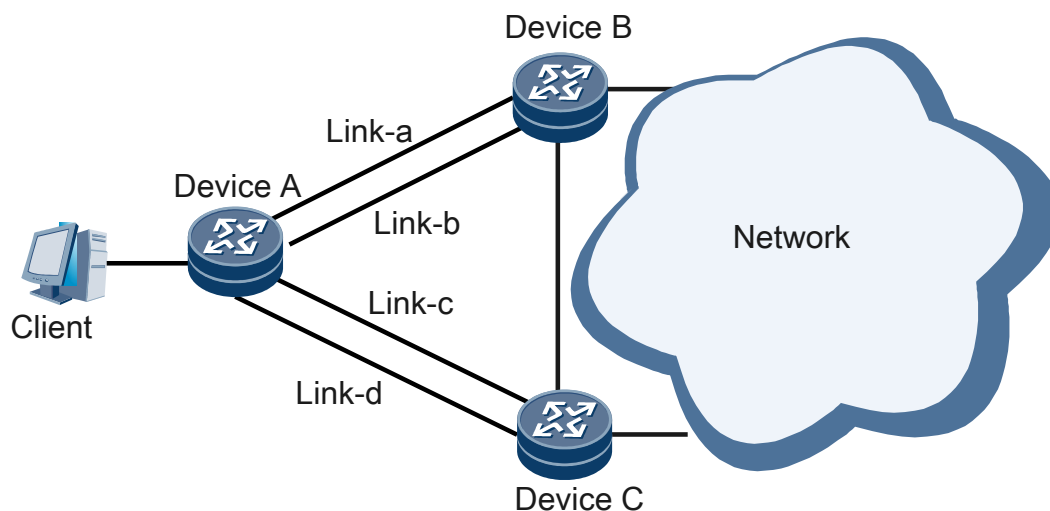
mesh groups are configured. All the interfaces that join a mesh group ensure the synchronization of the LSDBs in the entire network segment using the CSNP and PSNP mechanisms.

Link Group

In [Figure 8-9](#), router A is dual-homed to the IS-IS network through router B and router C. The path router A -> router B is primary and the path router A -> router C is backup. The bandwidth of each link is 100 Gbit/s, and the traffic from Client is transmitted at 150 Gbit/s. In this situation, both links in the path router A -> router B or the path router A -> router C need to carry the traffic. If Link-a fails, Link-b takes over the traffic. However, traffic loss occurs because the bandwidth of Link-b is not sufficient enough to carry the traffic.

To address this problem, configure link groups. You can add multiple links to a link group. If one of the links fails and the bandwidth of all the links in the group is not sufficient enough to carry the traffic, the link group automatically increases the costs of the other links to a configured value so that this link group is not selected. Then, traffic is switched to another link group.

Figure 8-9 IS-IS dual-homing access networking



In [Figure 8-9](#), Link-a and Link-b belong to link group 1, and Link-c and Link-d belong to link group 2.

- If Link-a fails, link group 1 automatically increases the cost of Link-b so that the traffic is switched to link group 2.
- If both Link-a and Link-c fail, the link groups increase the costs of Link-b and Link-d (to the same value) so that Link-b and Link-d load-balance the traffic.

8.2.4 IS-IS Neighbor Relationship Flapping Suppression

IS-IS neighbor relationship flapping suppression works by delaying IS-IS neighbor relationship reestablishment or setting the link cost to the maximum value (16777214 for wide mode and 63 for narrow mode).

Background

If the status of an interface carrying IS-IS services alternates between Up and Down, IS-IS neighbor relationship flapping occurs on the interface. During the flapping, IS-IS frequently

sends Hello packets to reestablish the neighbor relationship, synchronizes LSDBs, and recalculates routes. In this process, a large number of packets are exchanged, adversely affecting neighbor relationship stability, IS-IS services, and other IS-IS-dependent services, such as LDP and BGP. IS-IS neighbor relationship flapping suppression can address this problem by delaying IS-IS neighbor relationship reestablishment or preventing service traffic from passing through flapping links.

Related Concepts

Flapping_event: reported when the status of a neighbor relationship on an interface last changes from **Up** to **Init** or **Down**. The flapping_event triggers flapping detection.

Flapping_count: number of times flapping has occurred.

Detect-interval: interval at which flapping is detected. The interval is used to determine whether to trigger a valid flapping_event.

Threshold: flapping suppression threshold. When the flapping_count exceeds the threshold, flapping suppression takes effect.

Resume-interval: interval used to determine whether flapping suppression exits. If the interval between two valid flapping_events is longer than the resume-interval, flapping suppression exits.

Implementation

Flapping detection

IS-IS interfaces start a flapping counter. If the interval between two flapping_events is shorter than the detect-interval, a valid flapping_event is recorded, and the flapping_count increases by 1. When the flapping_count exceeds the threshold, the system determines that flapping occurs, and therefore triggers flapping suppression, and sets the flapping_count to 0. If the interval between two valid flapping_events is longer than the resume-interval before the flapping_count reaches the threshold again, the system sets the flapping_count to 0 again. Interfaces start the suppression timer when the status of a neighbor relationship last changes to **ExStart** or **Down**.

The detect-interval, threshold, and resume-interval are configurable.

Flapping suppression

Flapping suppression works in either Hold-down or Hold-max-cost mode.

- **Hold-down mode**: In the case of frequent flooding and topology changes during neighbor relationship establishment, interfaces prevent neighbor relationships from being reestablished during the suppression period, which minimizes LSDB synchronization attempts and packet exchanges.
- **Hold-max-cost mode**: If the traffic forwarding path changes frequently, interfaces use the maximum cost of the flapping link during the suppression period, which prevents traffic from passing through the flapping link.

Flapping suppression can also work first in Hold-down mode and then in Hold-max-cost mode.

By default, the Hold-max-cost mode takes effect. The mode and suppression period can be changed manually.

 NOTE

When an interface enters the flapping suppression state, all neighbor relationships on the interface enter the state accordingly.

Exiting from flapping suppression

Interfaces exit from flapping suppression in the following scenarios:

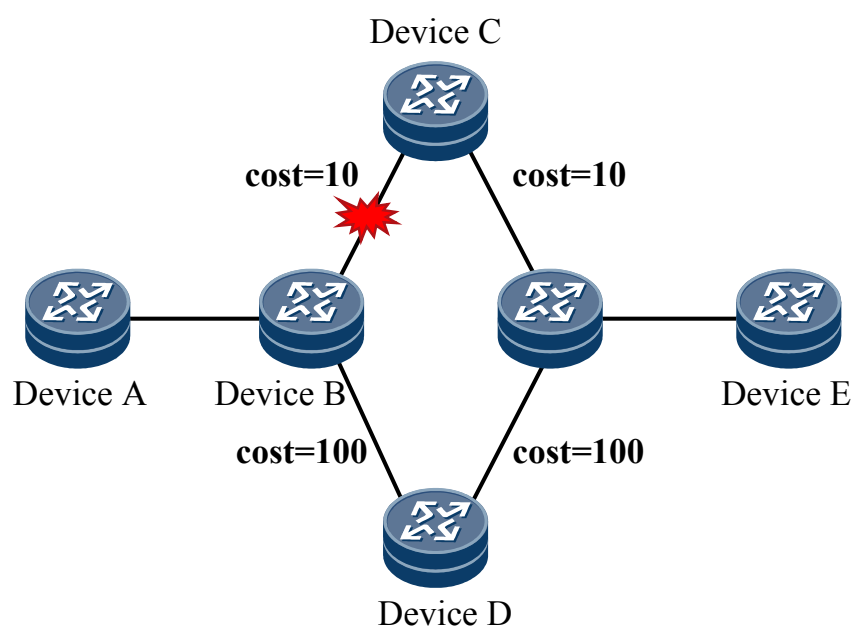
- The suppression timer expires.
- The corresponding IS-IS process is reset.
- A command is run to exit from flapping suppression.
- Three Hello packets in which the padding TLV carries a sub-TLV with the value being 251 are sent consecutively to notify the peer device to forcibly exit flapping suppression.

Basic scenario

In [Figure 8-10](#), the traffic forwarding path is Device A -> Device B -> Device C -> Device E before a link failure occurs. After the link between Device B and Device C fails, the forwarding path switches to Device A -> Device B -> Device D -> Device E. If the neighbor relationship between Device B and Device C frequently flaps at the early stage of the path switchover, the forwarding path will be switched frequently, causing traffic loss and affecting network stability. If the neighbor relationship flapping meets suppression conditions, flapping suppression takes effect.

- If flapping suppression works in Hold-down mode, the neighbor relationship between Device B and Device C is prevented from being reestablished during the suppression period, in which traffic is forwarded along the path Device A -> Device B -> Device D -> Device E.
- If flapping suppression works in Hold-max-cost mode, the maximum cost is used as the cost of the link between Device B and Device C during the suppression period, and traffic is forwarded along the path Device A -> Device B -> Device D -> Device E.

Figure 8-10 Flapping suppression in a basic scenario



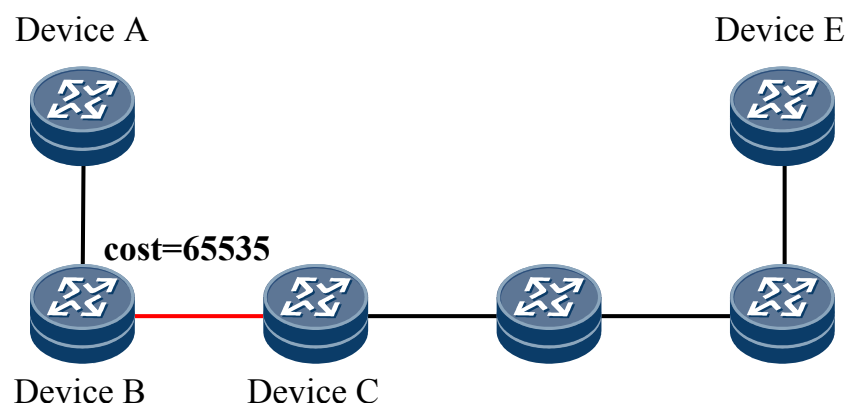
Single-forwarding path scenario

When only one forwarding path exists on the network, the flapping of the neighbor relationship between any two devices on the path will interrupt traffic forwarding. In [Figure 8-11](#), the traffic forwarding path is Device A -> Device B -> Device C -> Device E. If the neighbor relationship between Device B and Device C flaps, and the flapping meets suppression conditions, flapping suppression takes effect. However, if the neighbor relationship between Device B and Device C is prevented from being reestablished, the whole network will be divided. Therefore, Hold-max-cost mode (rather than Hold-down mode) is recommended. If flapping suppression works in Hold-max-cost mode, the maximum cost is used as the cost of the link between Device B and Device C during the suppression period. After the network stabilizes and the suppression timer expires, the link is restored.

NOTE

By default, the Hold-max-cost mode takes effect.

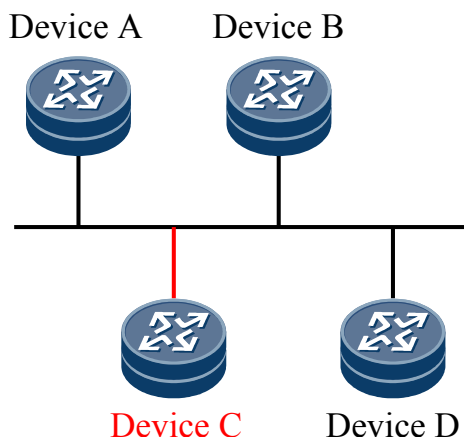
Figure 8-11 Flapping suppression in a single-forwarding path scenario



Broadcast scenario

In [Figure 8-12](#), four devices are deployed on the same broadcast network using switches, and the devices are broadcast network neighbors. If Device C flaps due to a link failure, and Device A and Device B were deployed at different time (Device A was deployed earlier for example) or the flapping suppression parameters on Device A and Device B are different, Device A first detects the flapping and suppresses Device C. Consequently, the Hello packets sent by Device A do not carry Device C's router ID. However, Device B has not detected the flapping yet and still considers Device C a valid node. As a result, the DR candidates identified by Device A are Device B and Device D, whereas the DR candidates identified by Device B are Device A, Device C, and Device D. Different DR candidates result in a different DR election result, which may lead to route calculation errors. To prevent this problem in scenarios where an interface has multiple neighbors, such as on a broadcast, P2MP, or NBMA network, all neighbors on the interface are suppressed when the status of a neighbor relationship last changes to **ExStart** or **Down**. Specifically, if Device C flaps, Device A, Device B, and Device D on the broadcast network are all suppressed. After the network stabilizes and the suppression timer expires, Device A, Device B, and Device D are restored to normal status.

Figure 8-12 Flapping suppression on a broadcast network



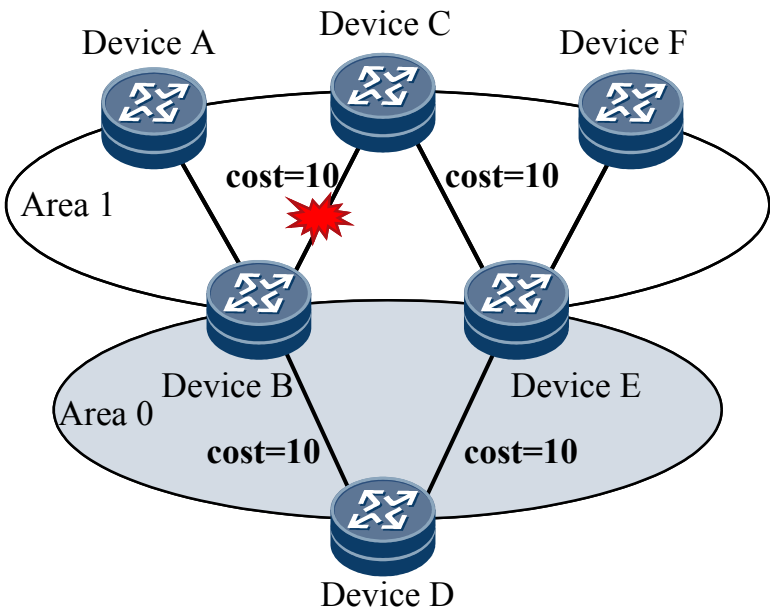
Scenario of multi-level networking

In **Figure 8-13**, Device A, Device B, Device C, Device E, and Device F are connected on Level 1 (Area 1), and Device B, Device D, and Device E are connected on Level 2 (Area 0). Traffic from Device A to Device F is preferentially forwarded along an intra-area route, and the forwarding path is Device A -> Device B -> Device C -> Device E -> Device F. When the neighbor relationship between Device B and Device C flaps and the flapping meets suppression conditions, flapping suppression takes effect in the default mode (Hold-max-cost). Consequently, the maximum cost is used as the cost of the link between Device B and Device C. However, the forwarding path remains unchanged because intra-area routes take precedence over inter-area routes during route selection according to IS-IS route selection rules. To prevent traffic loss in multi-area scenarios, configure Hold-down mode to prevent the neighbor relationship between Device B and Device C from being reestablished during the suppression period. During this period, traffic is forwarded along the path Device A -> Device B -> Device D -> Device E -> Device F.

NOTE

By default, the Hold-max-cost mode takes effect. The mode can be changed to Hold-down manually.

Figure 8-13 Flapping suppression in a multi-area scenario



Scenario with both LDP-IGP synchronization and IS-IS neighbor relationship flapping suppression configured

In [Figure 8-14](#), if the link between PE1 and P1 fails, an LDP LSP switchover is implemented immediately, causing the original LDP LSP to be deleted before a new LDP LSP is established. To prevent traffic loss, LDP-IGP synchronization needs to be configured. With LDP-IGP synchronization, the maximum cost is used as the cost of the new LSP to be established. After the new LSP is established, the original cost takes effect. Consequently, the original LSP is deleted, and LDP traffic is forwarded along the new LSP.

LDP-IGP synchronization and IS-IS neighbor relationship flapping suppression work in either Hold-down or Hold-max-cost mode. If both functions are configured, Hold-down mode takes precedence over Hold-max-cost mode, followed by the configured link cost. [Table 8-2](#) lists the suppression modes that take effect in different situations.

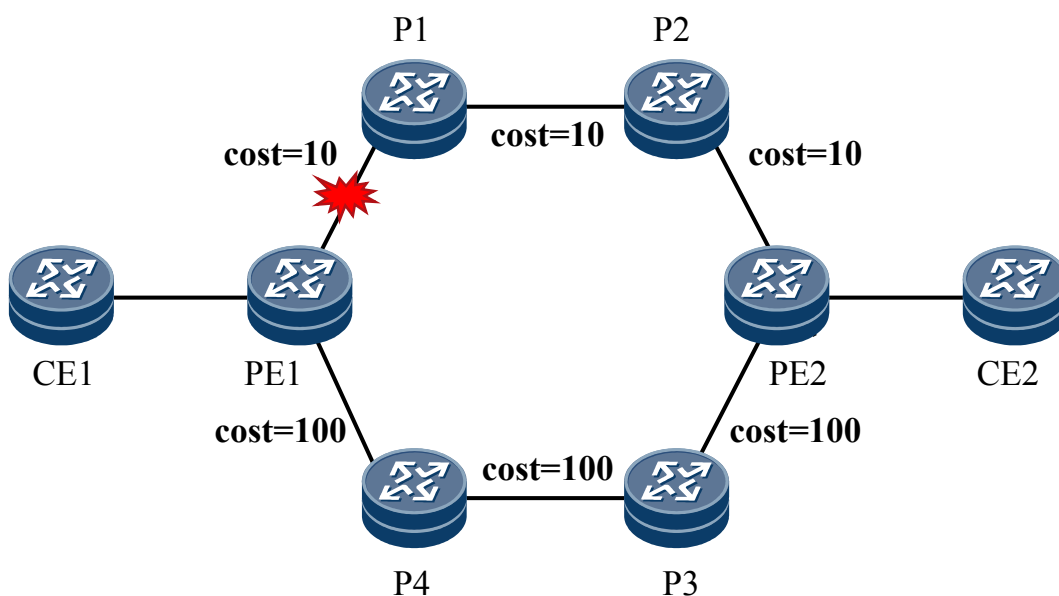
Table 8-2 Principles for selecting the suppression modes that take effect in different situations

LDP-IGP Synchronization/IS-IS Neighbor Relationship Flapping Suppression Mode	LDP-IGP Synchronization Hold-down Mode	LDP-IGP Synchronization Hold-max-cost Mode	Exited from LDP-IGP Synchronization Suppression
IS-IS Neighbor Relationship Flapping Suppression Hold-down Mode	Hold-down	Hold-down	Hold-down

IS-IS Neighbor Relationship Flapping Suppression Hold-max-cost Mode	Hold-down	Hold-max-cost	Hold-max-cost
Exited from IS-IS Neighbor Relationship Flapping Suppression	Hold-down	Hold-max-cost	Exited from LDP-IGP synchronization and IS-IS neighbor relationship flapping suppression

For example, the link between PE1 and P1 frequently flaps in [Figure 8-14](#), and both LDP-IGP synchronization and IS-IS neighbor relationship flapping suppression are configured. In this case, the suppression mode is selected based on the preceding principles. No matter which mode (Hold-down or Hold-max-cost) is selected, the forwarding path is PE1 -> P4 -> P3 -> PE2.

Figure 8-14 Scenario with both LDP-IGP synchronization and IS-IS neighbor relationship flapping suppression configured



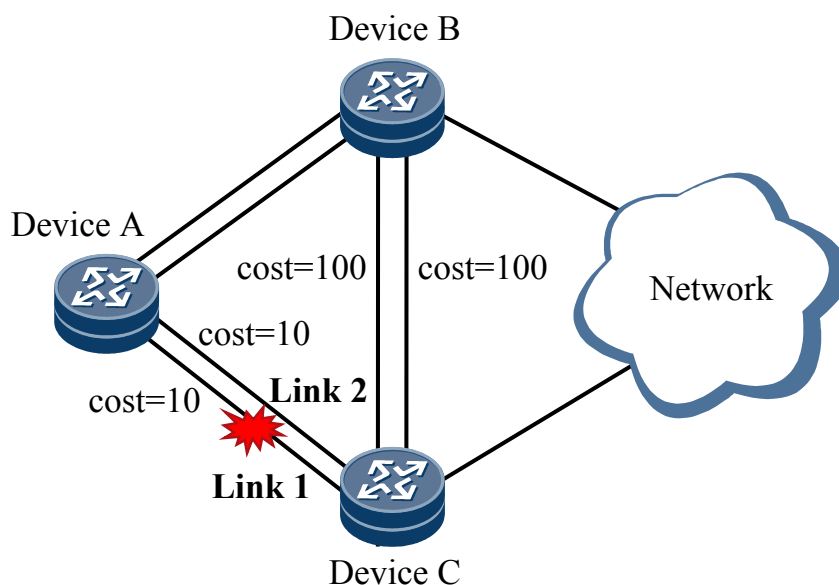
Scenario with both bit-error-triggered protection switching and IS-IS neighbor relationship flapping suppression configured

If a link has poor link quality, services transmitted along it may be adversely affected. If bit-error-triggered protection switching is configured and the bit error rate (BER) along a link exceeds a specified value, a bit error event is reported, and the maximum cost is used as the cost of the link, triggering route reselection. Consequently, service traffic is switched to the backup link. If both bit-error-triggered protection switching and IS-IS neighbor relationship flapping suppression are configured, they both take effect. Hold-down mode takes precedence over Hold-max-cost mode, followed by the configured link cost.

Scenario with both Link-bundle and IS-IS neighbor relationship flapping suppression configured

When the service traffic rate exceeds the capacity of the link, multiple links must be used. If one of the links between two devices is faulty, traffic is switched to another link. Because of limited forwarding capacity on the new link, excessive traffic is discarded. If the number of faulty links reaches the upper threshold, the maximum cost is used as the cost of all links in the link bundle to switch all service traffic to the backup nodes. When both link-bundle and neighbor relationship flapping suppression are configured, if the number of flappy links reaches the upper threshold, the maximum cost must be configured as the cost of all other links in the link bundle to prevent service loss caused by user traffic congestion. As shown in [Figure 8-15](#), two parallel links exist between Device A and Device C. If Link 1 is faulty and Link 2 bears all service traffic, traffic loss occurs. If both link-bundle and neighbor relationship flapping suppression are configured and Link 1 flaps, the maximum cost must be configured for Link 2 to avoid service traffic congestion. Only the Hold-max-cost mode therefore can be configured for neighbor relationship flapping suppression to switch the traffic forwarding path to Device A->Device B->Device C.

Figure 8-15 Scenario with both Link-bundle and IS-IS neighbor relationship flapping suppression configured



8.2.5 IS-IS Overload

The overload (OL) field of LSPs configured on a device prevents other devices from calculating the routes passing through this device.

If a system fails to store new LSPs for LSDB synchronization, the routes calculated by the system are incorrect. In that case, the system enters the Overload state. The user can configure the device to enter the Overload state when the system lacks sufficient memory. At present, users can set the Overload timer when IS-IS is started and configure whether to delete the leaked routes and whether to advertise the imported routes. A device enters the Overload state after an exception occurs on the device or when it is configured to enter the state.

- If IS-IS enters the Overload state after an exception occurs on the device, the system deletes all imported or leaked routes.

- If IS-IS enters the Overload state based on a user configuration, the system only deletes all imported or leaked routes if configured to do so.

Although LSPs with overload fields are flooded throughout the network, they are ignored in the calculation of the routes passing through the device in the Overload state. Specifically, after the overload field of LSPs is configured on a device, other devices do not count the routes that pass through the device when performing SPF calculation, but the direct routes between the device and other devices are still calculated.

If a device in an IS-IS domain is faulty, routes may be incorrectly calculated across the entire domain. The overload field can be configured for the device to isolate it from the IS-IS network temporarily, which facilitates fault isolation.

8.2.6 IS-IS Fast Convergence

IS-IS fast convergence is an extended feature of IS-IS implemented to speed up route convergence. It includes the following concepts:

- I-SPF
Incremental SPF (I-SPF) recalculates only the routes of the changed nodes rather than the routes of all nodes when the network topology changes, which speeds up the calculation of routes.
- PRC
Partial route calculation (PRC) calculates only those routes which have changed when the network topology changes.
- Link State PDUs (LSP) fast flooding
LSP fast flooding speeds up LSP flooding.
- Intelligent timer
The first timeout period of the timer is fixed. If an event that triggers the timer occurs before the set timer expires, the next timeout period of the timer increases.
The intelligent timer applies to LSP generation and SPF calculation.

I-SPF

In ISO 10589, the Dijkstra algorithm was adopted to calculate routes. When a node changes on the network, the algorithm recalculates all routes. The calculation requires a long time to complete and consumes a significant amount of CPU resources, reducing convergence speed.

I-SPF improves the algorithm. Except for the first time the algorithm is run, only the nodes that have changed rather than all nodes in the network are used in the calculation. The SPT generated using I-SPF is the same as that generated using the previous algorithm. This significantly decreases CPU usage and speeds up network convergence.

PRC

Similar to I-SPF, PRC calculates only routes that have changed. PRC, however, does not calculate the shortest path. It updates routes based on the SPT calculated by I-SPF.

In route calculation, a leaf represents a route, and a node represents a device. If the SPT changes after I-SPF calculation, PRC calculates all the leaves only on the changed node. If the SPT remains unchanged, PRC calculates only the changed leaves.

For example, if IS-IS is enabled on an interface of a node, the SPT calculated by I-SPF remains unchanged. In this case, PRC updates only the routes of this interface, which consumes less CPU resources.

PRC working with I-SPF further improves network convergence performance and replaces the original SPF algorithm.

 NOTE

On the NE40E, only I-SPF and PRC are used to calculate IS-IS routes.

LSP Fast Flooding

When an IS-IS device receives new LSPs from other devices, it updates the LSPs in the LSDB and periodically floods the updated LSPs based on a timer. Therefore, the synchronization of all LSDBs is slow.

With LSP fast flooding, when the router receives LSPs that can trigger route calculation or route update, it floods these LSPs before route calculation occurs, which speeds up network convergence and LSDB synchronization throughout the entire network.

Intelligent Timer

Although the route calculation algorithm is improved, the long interval for triggering route calculation also affects the convergence speed. A millisecond-level timer can shorten the interval. Frequent network changes, however, also consume too much CPU resources. The SPF intelligent timer addresses these problems.

In most cases, an IS-IS network running normally is stable. The frequent changes on a network are rather rare, and IS-IS does not calculate routes frequently. Therefore, a short period (within milliseconds) can be configured as the first interval for route calculation. If the network topology changes frequently, the interval set by the intelligent timer increases with the calculation times to reduce CPU consumption.

The LSP generation intelligent timer is similar to the SPF intelligent timer. When the LSP generation intelligent timer expires, the system generates a new LSP based on the current topology. The original mechanism uses a timer with fixed intervals, which results in slow convergence and high CPU consumption. Therefore, the LSP generation timer is designed as an intelligent timer to respond to emergencies (for example, the interface goes Up or Down) quickly and speed up network convergence. In addition, when the network changes frequently, the interval for the intelligent timer becomes longer to reduce CPU consumption.

8.2.7 IS-IS LSP Fragment Extension

If the LSP capacity is insufficient, newly imported routes and new TLVs fail to be added to LSP fragments. In this case, you can use LSP fragment extension to increase the LSP capacity, restoring the LSP space. When the LSP space is restored, the system automatically attempts to re-add these routes and TLVs to LSP fragments.

When the LSPs to be advertised by IS-IS contain a large amount of information, they are advertised in multiple Link State PDUs (LSP) fragments belonging to the same system.

Virtual system IDs can be configured, and virtual LSPs that carry routing information can be generated for IS-IS.

IS-IS LSP fragment extension allows an IS-IS device to generate more LSP fragments and carry more IS-IS information.

Terms

- **Originating system**
The originating system is a device that runs the IS-IS protocol. A single IS-IS process advertises LSPs as virtual devices do, except that the originating system refers to a real IS-IS process.
- **Normal system ID**
The normal system ID is the system ID of the originating system.
- **Additional system ID**
The additional system ID, assigned by the network administrator, is used to generate additional or extended LSP fragments. A maximum of 256 additional or extended LSP fragments can be generated. Like a normal system ID, an additional system ID must be unique in a routing domain.
- **Virtual system**
The virtual system, identified by an additional system ID, is used to generate extended LSP fragments. These fragments carry additional system IDs in their LSP IDs.

Principles

IS-IS LSP fragments are identified by the LSP Number field in their LSP IDs. The LSP Number field is 1 byte. Therefore, an IS-IS process can generate a maximum of 256 fragments. With fragment extension, more information can be carried.

Each system ID represents a virtual system, and each virtual system can generate 256 LSP fragments. In addition, another virtual systems can be configured. Therefore, an IS-IS process can generate more LSP fragments.

After a virtual system and fragment extension are configured, an IS-IS device adds the contents that cannot be contained in its LSPs to the LSPs of the virtual system and notifies other devices of the relationship between the virtual system and itself through a special TLV in the LSPs.

IS Alias ID TLV

Standard protocol defines a special Type-Length-Value (TLV): IS Alias ID TLV.

Table 8-3 IS Alias ID TLV

Field	Length	Description
Type	1 byte	TLV type. If the value is 24, it indicates the IS Alias ID TLV.
Length	1 byte	TLV length.
System ID	6 bytes	System ID.
Pseudonode number	1 byte	Pseudonode number.
Sub-TLVs length	1 byte	Length of sub-TLVs.

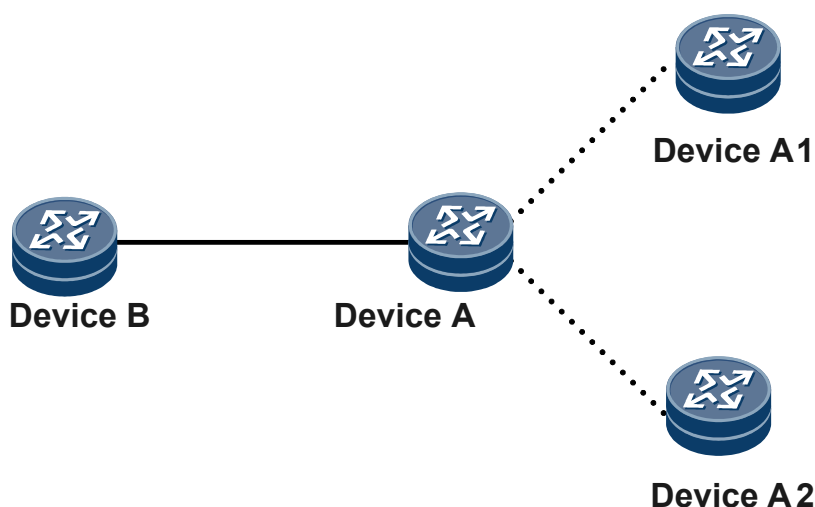
Field	Length	Description
Sub-TLVs	0 to 247 bytes	Sub-TLVs.

LSPs with fragment number 0 sent by the originating system and virtual system carry IS Alias ID TLVs to indicate the originating system.

Operation Modes

IS-IS devices can use the LSP fragment extension feature in the following modes:

Figure 8-16 Networking for IS-IS LSP fragment extension



- Mode-1

Mode-1 is used when some devices on the network do not support LSP fragment extension.

In this mode, virtual systems participate in SPF calculation. The originating system advertises LSPs containing information about links to each virtual system and each virtual system advertises LSPs containing information about links to the originating system. In this manner, the virtual systems function the same as the actual devices connected to the originating system on the network.

Mode-1 is a transitional mode for earlier versions that do not support LSP fragment extension. In the earlier versions, IS-IS cannot identify Alias ID TLVs. Therefore, the LSP sent by a virtual system must look like a common IS-IS LSP.

The LSP sent by a virtual system contains the same area address and overload bit as those in the common LSP. If the LSPs sent by a virtual system contain TLVs specified in other features, the TLVs must be the same as those in common LSPs.

LSPs sent by a virtual system carry information of the neighbor (the originating system), and the carried cost is the maximum value minus 1. LSPs sent by the originating system carry information of the neighbor (the virtual system), and the carried cost is 0. This mechanism ensures that the virtual system is a node downstream of the originating system when other devices calculate routes.

In **Figure 8-16**, Device B does not support LSP fragment extension; Device A supports LSP fragment extension in mode-1; Device A1 and Device A2 are virtual systems of Device A. Device A1 and Device A2 send LSPs carrying partial routing information of Device A. After receiving LSPs from Device A, Device A1, and Device A2, Device B considers there to be three devices at the peer end and calculates routes normally. Because the cost of the route from Device A to Device A1 or Device A2 is 0, the cost of the route from Device B to Device A is equal to that from Device B to Device A1.

- Mode-2

Mode-2 is used when all the devices on the network support LSP fragment extension. In this mode, virtual systems do not participate in SPF calculation. All the devices on the network know that the LSPs generated by the virtual systems actually belong to the originating system.

IS-IS working in mode-2 identifies IS Alias ID TLVs, which are used to calculate the SPT and routes.

In **Figure 8-16**, Device B supports LSP fragment extension, and Device A supports LSP fragment extension in mode-2; Device A1 and Device A2 send LSPs carrying some routing information of Device A. After receiving LSPs from Device A1 and Device A2, Device B obtains IS Alias ID TLV and learns that the originating system of Device A1 and Device A2 is Device A. Device B then considers information advertised by Device A1 and Device A2 to be about Device A.

Whatever the LSP fragment extension mode, LSPs can be resolved. However, if LSP fragment extension is not supported, only LSPs in mode-1 can be resolved.

Table 8-4 Comparison between mode-1 and mode-2

LSP Field	Carried in Mode-1	Carried in Mode-2
IS Alias ID	Yes	Yes
Area	Yes	No
Overload bit	Yes	Yes
IS NBR/IS EXTENDED NBR	Yes	No
Routing	Yes	Yes
ATT bit	Yes, with value 0	Yes, with value 0
P bit	Yes, with value 0	Yes, with value 0

Process

After LSP fragment extension is configured, if information is lost because LSPs overflow, the system restarts the IS-IS process. After being restarted, the originating system loads as much routing information as possible. Any excessive information beyond the forwarding capability of the system is added to the LSPs of the virtual systems for transmission. In addition, if a virtual system with routing information is deleted, the system automatically restarts the IS-IS process.

Usage Scenario

NOTE

If there are non-Huawei devices on the network, LSP fragment extension must be set to mode-1. Otherwise, these devices cannot identify LSPs.

Configuring LSP fragment extension and virtual systems before setting up IS-IS neighbors or importing routes is recommended. If IS-IS neighbors are set up or routes are imported first and the information to be carried exceeds the forwarding capability of 256 fragments before LSP fragment extension and virtual systems are configured, you have to restart the IS-IS process for the configurations to take effect.

8.2.8 IS-IS 3-Way Handshake

IS-IS introduces the 3-way handshake mechanism on P2P links to ensure a reliable data link layer.

Based on ISO 10589, the IS-IS 2-way handshake mechanism uses Hello packets to set up P2P adjacencies between neighboring devices. When a device receives a Hello packet from the other end, it regards the other end as Up and sets up an adjacency with it. However, this mechanism has some serious shortcomings.

When two or more links exist between two devices, an adjacency can still be set up where one link is Down and the other is Up in the same direction. The parameters of the other link are used in SPF calculation. As a result, a device that does not detect any fault along the faulty link will continue trying to forward packets over the link.

The 3-way handshake mechanism resolves these problems on P2P links. In 3-way handshake mode, a device regards a neighbor Up and sets up an adjacency with it only after confirming that the neighbor has received the packet that the device sends.

In addition, the 3-way handshake mechanism uses the 32-bit Extended Local Circuit ID field, which extends the original 8-bit Extended Local Circuit ID field and the limit of only 255 P2P links.

NOTE

By default, the IS-IS 3-way handshake mechanism is implemented on P2P links.

8.2.9 IS-IS NSR

Background

As networks develop, the demand for data, audio, and video services is growing, which imposes increasing requirements on IP network reliability. If an AMB/SMB switchover is performed on a device due to a maintenance operation or a single point of failure, routes may fail to converge, which may result in traffic loss or even a network breakdown. Non-stop routing (NSR) can address this problem and ensure uninterrupted forwarding of key services.

Related Concepts

- High availability (HA): supports data backup between the AMB and SMB.
- Non-stop forwarding (NSF): enables a node to use the GR mechanism to ensure uninterrupted transmission during an AMB/SMB switchover.

- NSR: allows a standby control plane to take over traffic from an active control plane if the active control plane fails, preventing the control planes of neighbors from detecting the fault.
- AMB and SMB: run control plane processes.

Implementation

With IS-IS NSR, IS-IS real-time data is synchronized between the AMB and SMB. After an AMB/SMB switchover is performed on a device, the SMB takes over services from the AMB, and neighbors are unaware of the local fault. After the switchover, the new AMB restores IS-IS immediately based on the synchronized IS-IS real-time data. Therefore, neighbors are unaware of the switchover as well. IS-IS NSR requires synchronization of the following data:

- All configuration data, such as information about neighbors, timer parameters, and process configurations.
- Dynamic data, such as the interface parameters and state, and information about neighbors and the link state database (LSDB).

NOTE

For details about NSR, see "" in *Feature Description - Reliability*.

Usage Scenario

NSR minimizes the impact of control plane faults and prevents route flapping on networks that require high reliability.

Benefits

NSR improves network reliability and ensures uninterrupted traffic forwarding.

8.2.10 IS-IS for IPv6

Standard protocols released by the IETF defines two new TLVs that can support IPv6 routes and a new Network Layer Protocol Identifier (NLPID), which ensures that IS-IS can process and calculate IPv6 routes.

The two new TLVs are as follows:

- IPv6 Reachability
The IPv6 Reachability TLV indicates the reachability of a network by specifying the route prefix and metric. The type value is 236 (0xEC).
- IPv6 Interface Address
The IPv6 Interface Address TLV is similar to the IP interface address TLV of IPv4 in function, except that it changes the original 32-bit IPv4 address to a 128-bit IPv6 address. The type value is 232 (0xE8).

The NLPID is an 8-bit field that identifies network layer protocol packets. The NLPID of IPv6 is 142 (0x8E). If an IS-IS router supports IPv6, it advertises routing information through the NLPID value.

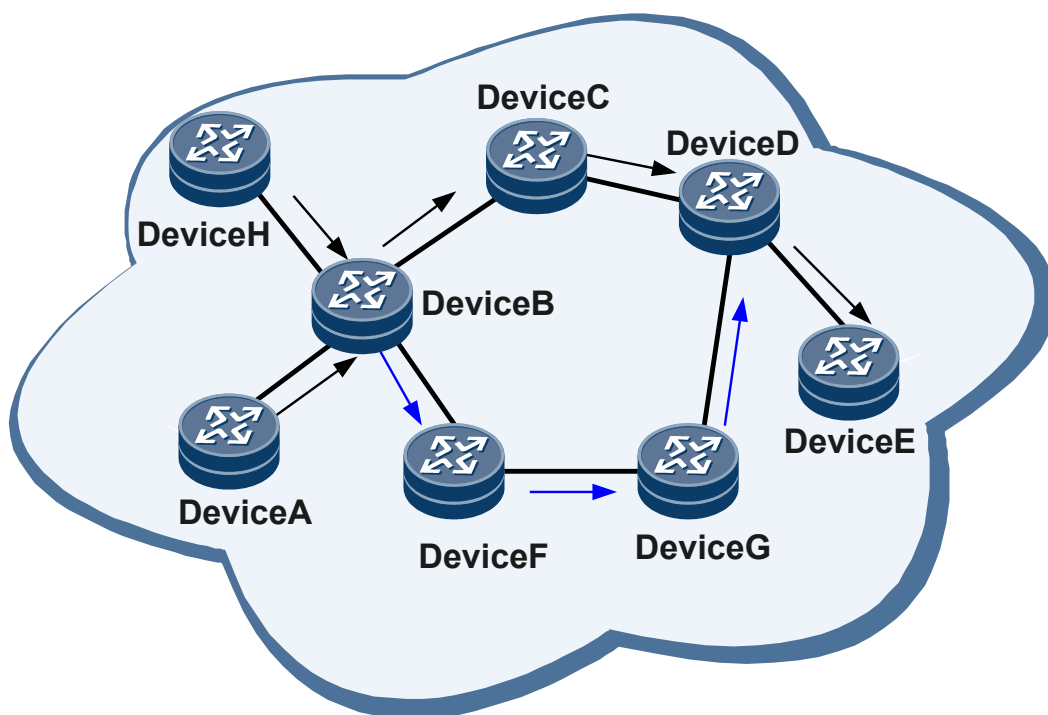
8.2.11 IS-IS TE

IS-IS TE supports MPLS establishment and maintenance of Constraint-based Routed Label Switched Paths (CR-LSPs).

To establish CR-LSPs, MPLS needs to learn the traffic attributes of all the links in the local area. MPLS can acquire the TE information of the links through IS-IS.

Traditional routers select the shortest path as the primary route regardless of other factors, such as bandwidth, even when the path is congested.

Figure 8-17 Networking with IS-IS routing defects



On the network shown in [Figure 8-17](#), all the links have the same cost (10). The shortest path from Device A/Device H to Device E is Device A/Device H → Device B → Device C → Device D → Device E. Data is forwarded along this shortest path. Therefore, the path Device A (Device H) → Device B → Device C → Device D → Device E may be congested whereas the path Device A/Device H → Device B → Device F → Device G → Device D → Device E is idle.

To resolve the preceding problem, the cost of the path from Device B to Device C can be set to 30 so that the traffic is switched to the path Device A/Device H → Device B → Device F → Device G → Device D → Device E.

This method eliminates the congestion on the link Device A/Device H → Device B → Device C → Device D → Device E; however, the other link Device A (Device H) → Device B → Device F → Device G → Device D → Device E may be congested. In addition, on networks with complicated topologies, changing the cost of one link may affect multiple routes.

As an overlay model, MPLS can set up a virtual topology over the physical network topology and map traffic to the virtual topology, effectively combining MPLS and TE technology into MPLS TE.

MPLS TE can resolve network congestion problems by allowing carriers can precisely control the path through which traffic passes and prevent traffic from passing through congested nodes. Meanwhile, MPLS TE can reserve resources during the establishment of LSPs to ensure service quality.

To ensure continuity of services, MPLS TE provides the CR-LSP backup and fast reroute (FRR) mechanisms. If a link fault occurs, traffic can be switched immediately. Through MPLS TE, service providers (SPs) can fully utilize the current network resources to provide diverse services, optimize network resources, and methodically manage the network.

To accomplish the preceding tasks, MPLS TE needs to learn TE information about all devices on the network. However, MPLS TE lacks a mechanism in which each device floods its TE information throughout the entire network for TE information synchronization. However, IS-IS does provide such a mechanism. Therefore, MPLS TE can advertise and synchronize TE information with the help of IS-IS. To support MPLS TE, IS-IS needs to be extended.

In brief, IS-IS TE collects TE information on IS-IS networks and then transmits the TE information to the Constrained Shortest Path First (CSPF) module.

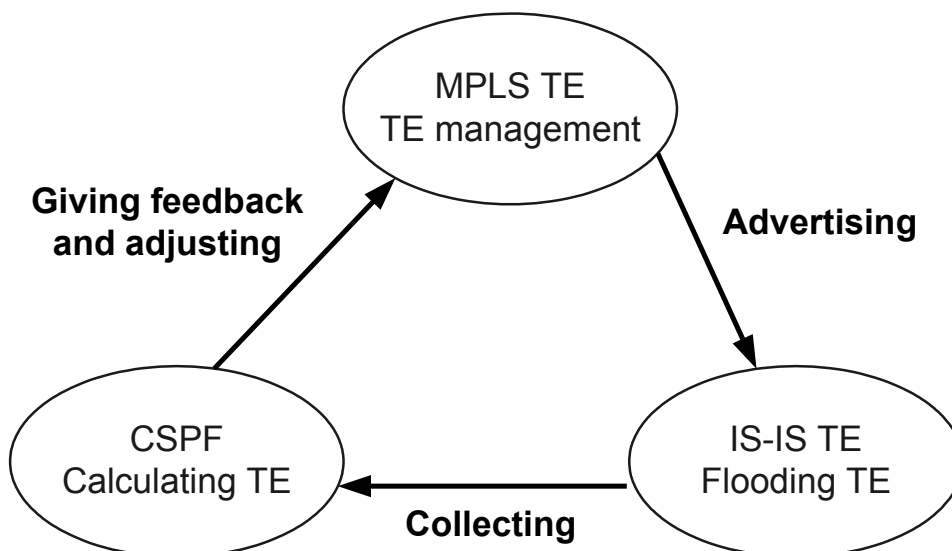
NOTE

Currently, IS-IS TE supports only IPv4.

Basic Principles

IS-IS TE is an extension of IS-IS intended to support MPLS TE. As defined in standard protocols, IS-IS TE defines new TLVs in IS-IS LSPs to carry TE information to help MPLS implement the flooding, synchronization, and resolution of TE information. Then, IS-IS TE transmits the resolved TE information to the CSPF module. In MPLS TE, IS-IS TE plays the role of a porter. **Figure 8-18** illustrates the relationships between IS-IS TE, MPLS TE, and CSPF.

Figure 8-18 Outline of relationships between MPLS TE, CSPF, and IS-IS TE



To carry TE information in LSPs, IS-IS TE defines the following TLVs in standard protocols:

- **Extended IS reachability TLV**
The Extended IS reachability TLV replaces the IS reachability TLV and extends the TLV format using sub-TLVs. The implementation of sub-TLVs in TLVs is the same as that of TLVs in LSPs. Sub-TLVs are used to carry TE information configured on physical interfaces.



All sub-TLVs defined in standard protocols are supported.

Table 8-5 Sub-TLVs defined in the extended IS reachability TLV

Name	Type	Length (Byte)
Administrative Group	3	4
IPv4 Interface Address	6	4
IPv4 Neighbor Address	8	4
Maximum Link Bandwidth	9	4
Maximum Reserved Link Bandwidth	10	4
Unreserved Bandwidth	11	32
Traffic Engineering Default Metric	18	3
Bandwidth Constraints sub-TLV	22	36

- Traffic Engineering router ID TLV
The type of this TLV is 134, and this TLV carries a 4-byte router ID (MPLS LSR-ID). In MPLS TE, each device has a unique router ID.
- Extended IP reachability TLV
The Extended IP reachability TLV replaces the IP reachability TLV and carries routing information. It extends the length of the route cost field to 4 bytes and carries sub-TLVs.

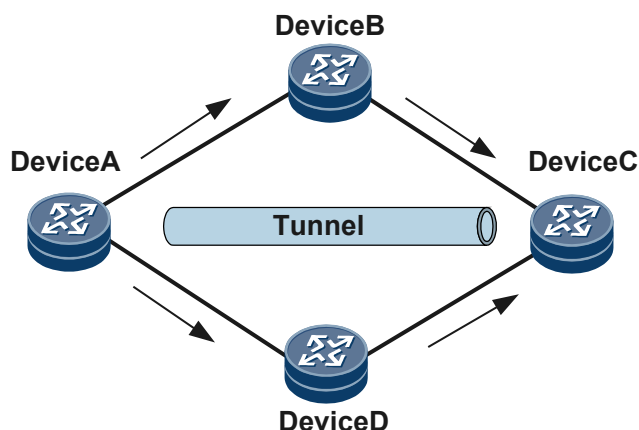
IS-IS TE consists of two procedures:

- Responding to MPLS TE configurations
IS-IS TE functions only after MPLS TE is enabled.
IS-IS TE updates the TE information in IS-IS LSPs based on MPLS TE configurations.
IS-IS TE transmits MPLS TE configurations to the CSPF module.
- Processing TE information in LSPs
IS-IS TE extracts TE information from IS-IS LSPs and transmits the TE information to the CSPF module.

Usage Scenario

IS-IS TE helps MPLS TE set up TE tunnels. In [Figure 8-19](#), a TE tunnel is set up between Device A and Device C.

Figure 8-19 Networking for IS-IS TE



The configuration requirements are as follows:

- MPLS TE and MPLS TE CSPF are enabled on Device A.
- MPLS TE is enabled on Device B, Device C, and Device D.
- IS-IS and IS-IS TE are enabled on Device A, Device B, Device C, and Device D.

After the configurations are complete, IS-IS on Device A, Device B, Device C, and Device D sends LSPs carrying TE information configured on each Device. Device A then obtains the TE information of Device B, Device C, and Device D from the received LSPs. The CSPF module can calculate the path required by the TE tunnel based on the TE information collected across the entire network.

8.2.12 IS-IS Wide Metric

In the earlier ISO 10589, the largest metric of an interface is 63. TLV type 128 and TLV type 130 contain information about routes, and TLV type 2 contains information about IS-IS neighbors. However, on large-scale networks, the metric range cannot meet the requirements. Moreover, IS-IS TE needs to be configured. Therefore, the wide metric was introduced.

As defined in standard protocols, with IS-IS wide metric, the largest metric of an interface is extended to 16777215, and the largest metric of a route is 4261412864.

After IS-IS wide metric is enabled, TLV type 135 contains information about routes; TLV type 22 contains information about IS-IS neighbors.

- The following lists the TLVs used in narrow mode:
 - IP Internal Reachability TLV: carries routes within an area.
 - IP External Reachability TLV: carries routes outside an area.
 - IS Neighbors TLV: carries information about neighbors.
- The following lists the TLVs used in wide mode:
 - Extended IP Reachability TLV: replaces the earlier IP Reachability TLV and carries information about routes. This TLV expands the range of the route cost to 4 bytes and carries sub-TLVs.
 - IS Extended Neighbors TLV: carries information about neighbors.

 NOTE

The metric style can be set to narrow, narrow-compatible, compatible, wide-compatible, or wide mode. [Table 8-6](#) shows which metric styles are carried in received and sent packets. A device can calculate routes only when it can receive, send, and process corresponding TLVs. Therefore, to ensure correct data forwarding on a network, the proper metric style must be configured for each device on the network.

Table 8-6 Metric style carried in received and sent under different metric style configurations

Configured Metric Style	Metric Style Carried in Received Packets	Metric Style Carried in Sent Packets
Narrow	Narrow	Narrow
Narrow-compatible	Narrow and wide	Narrow
Compatible	Narrow and wide	Narrow and wide
Wide-compatible	Narrow and wide	Wide
Wide	Wide	Wide

When the metric style is set to compatible, IS-IS sends the information both in narrow and wide modes.

Process

NOTICE

Once the metric style is changed, the IS-IS process restarts.

- If the metric style carried in sent packets is changed from narrow to wide:
The information previously carried by TLV type 128, TLV type 130, and TLV type 2 is now carried by TLV type 135 and TLV type 22.
- If the metric style carried in sent packets is changed from wide to narrow:
The information previously carried by TLV type 135 and TLV type 22 is now carried by TLV type 128, TLV type 130, and TLV type 2.
- If the metric style carried in sent packets is changed from narrow or wide to narrow and wide:
The information previously carried in narrow or wide mode is now carried by TLV type 128, TLV type 130, TLV type 2, TLV type 135, and TLV type 22.

Usage Scenario

IS-IS wide metric is used to support IS-IS TE, and the metric style needs to be set to wide, compatible or wide compatible.

8.2.13 BFD for IS-IS

In most cases, the interval at which Hello packets are sent is 10s, and the IS-IS neighbor holding time (the timeout period of a neighbor relationship) is three times the interval. If a

device does not receive a Hello packet from its neighbor within the holding time, the device terminates the neighbor relationship.

A device can detect neighbor faults at the second level only. As a result, link faults on a high-speed network may cause a large number of packets to be discarded.

BFD, which can be used to detect link faults on lightly loaded networks at the millisecond level, is introduced to resolve the preceding issue. With BFD, two systems periodically send BFD packets to each other. If a system does not receive BFD packets from the other end within a specified period, the system considers the bidirectional link between them Down.

BFD is classified into the following modes:

- Static BFD

In static BFD mode, BFD session parameters (including local and remote discriminators) are set using commands, and requests must be delivered manually to establish BFD sessions.

- Dynamic BFD

In dynamic BFD mode, the establishment of BFD sessions is triggered by routing protocols.

BFD for IS-IS enables BFD sessions to be dynamically established. After detecting a fault, BFD notifies IS-IS of the fault. IS-IS sets the neighbor status to Down, quickly updates link state protocol data units (LSPs), and performs the partial route calculation (PRC). BFD for IS-IS implements fast IS-IS route convergence.

 NOTE

Instead of replacing the Hello mechanism of IS-IS, BFD works with IS-IS to rapidly detect the faults that occur on neighboring devices or links.

BFD Session Establishment and Deletion

- Conditions for establishing a BFD session

- Global BFD is enabled on each device, and BFD is enabled on a specified interface or process.
- IS-IS is configured on each device and enabled on interfaces.
- Neighbors are Up, and a designated intermediate system (DIS) has been elected on a broadcast network.

- Process of establishing a BFD session

- P2P network

After the conditions for establishing BFD sessions are met, IS-IS instructs the BFD module to establish a BFD session and negotiate BFD parameters between neighbors.

- Broadcast network

After the conditions for establishing BFD sessions are met and the DIS is elected, IS-IS instructs BFD to establish a BFD session and negotiate BFD parameters between the DIS and each device. No BFD sessions are established between non-DISs.

On broadcast networks, devices (including non-DIS devices) of the same level on a network segment can establish adjacencies. In BFD for IS-IS, however, BFD sessions are established only between the DIS and non-DISs. On P2P networks, BFD sessions are directly established between neighbors.

If a Level-1-2 neighbor relationship is set up between the devices on both ends of a link, the following situations occur:

- On a broadcast network, IS-IS sets up a Level-1 BFD session and a Level-2 BFD session.
- On a P2P network, IS-IS sets up only one BFD session.
- Process of tearing down a BFD session
 - P2P network

If the neighbor relationship established between P2P IS-IS interfaces is not Up, IS-IS tears down the BFD session.
 - Broadcast network

If the neighbor relationship established between broadcast IS-IS interfaces is not Up or the DIS is reelected on the broadcast network, IS-IS tears down the BFD session.

If the configurations of dynamic BFD sessions are deleted or BFD for IS-IS is disabled from an interface, all Up BFD sessions established between the interface and its neighbors are deleted. If the interface is a DIS and the DIS is Up, all BFD sessions established between the interface and its neighbors are deleted.

If BFD is disabled from an IS-IS process, BFD sessions are deleted from the process.

 NOTE

BFD detects only the one-hop link between IS-IS neighbors because IS-IS establishes only one-hop neighbor relationships.

- Response to the Down event of a BFD session

When BFD detects a link failure, it generates a Down event and informs IS-IS. IS-IS then suppresses neighbor relationships and recalculates routes. This process speeds up network convergence.

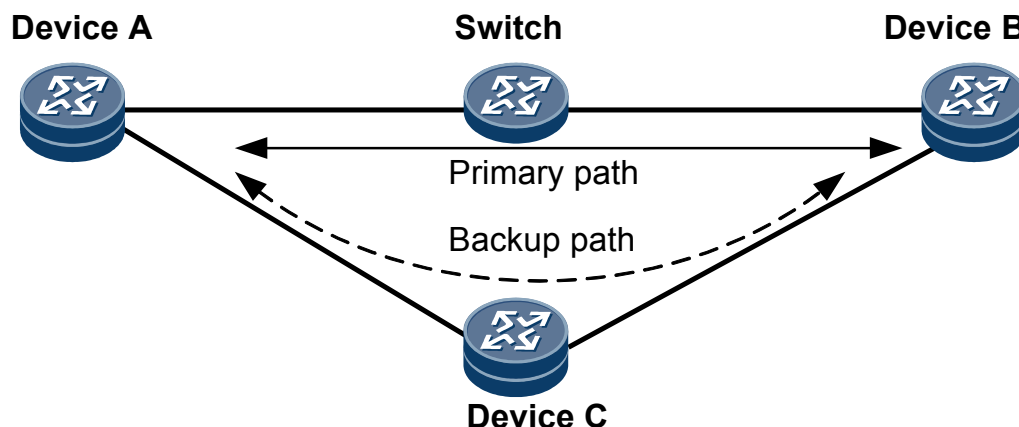
Usage Scenario

NOTICE

Dynamic BFD needs to be configured based on the actual network. If the time parameters are not configured correctly, network flapping may occur.

BFD for IS-IS speeds up route convergence through rapid link failure detection. The following is a networking example for BFD for IS-IS.

Figure 8-20 BFD for IS-IS



The configuration requirements are as follows:

- Basic IS-IS functions are configured on each device shown in [Figure 8-20](#).
- Global BFD is enabled.
- BFD for IS-IS is enabled on Device A and Device B.

If the link between Device A and Device B fails, BFD can rapidly detect the fault and report it to IS-IS. IS-IS sets the neighbor status to Down to trigger an IS-IS topology calculation. IS-IS also updates LSPs so that Device C can promptly receive the updated LSPs from Device B, which accelerates network topology convergence.

8.2.14 IS-IS Auto FRR

Background

As networks develop, voice over IP (VoIP) and online video services pose increasingly higher requirements for real-time transmission. Nevertheless, if an IS-IS fault occurs, multiple operations, including fault detection, LSP updating, LSP flooding, route calculation, and forward information base (FIB) entry delivery, must be performed to switch traffic to a new link. These operations take a long time, much longer than 50 ms, the minimum delay to which users are sensitive. As a result, user experience is affected.

IS-IS Auto fast re-route (FRR) is a dynamic IP FRR technology that minimizes traffic loss by immediately switching traffic to the alternate link pre-computed by an IGP based on the LSDBs on the entire network and stored in the FIB if a link or adjacent node failure is detected. As IP FRR implements route convergence, it is becoming increasingly popular with carriers.

Major Auto FRR techniques include loop-free alternate (LFA), U-turn, Not-Via, TI-LFA, Remote LFA, and MRT, among which IS-IS supports only LFA, TI-LFA, and Remote LFA.

Related Concepts

LFA

LFA is an IP FRR technology that calculates the shortest path from the neighbor that can provide an alternate link to the destination node based on the Shortest Path First (SPF)

algorithm. Then, LFA calculates a loop-free alternate link with the smallest cost based on the inequality: $\text{Distance_opt}(N, D) < \text{Distance_opt}(N, S) + \text{Distance_opt}(S, D)$.

In the preceding inequality, S, D, and N indicate the source node, destination node, and a neighbor of S, respectively, and $\text{Distance_opt}(X, Y)$ indicates the shortest distance from node X to node Y.

Remote LFA

LFA Auto FRR cannot be used to calculate alternate links on large-scale networks, especially on ring networks. Remote LFA Auto FRR addresses this problem by calculating a PQ node and establishing a tunnel between the source node of a primary link and the PQ node. If the primary link fails, traffic can be automatically switched to the tunnel, which improves network reliability.

P space

P space consists of the nodes through which the shortest path trees (SPTs) with the source node of a primary link as the root are reachable without passing through the primary link.

Extended P space

Extended P space consists of the nodes through which the SPTs with neighbors of a primary link's source node as the root are reachable without passing through the primary link.

Q space

Q space consists of the nodes through which the SPTs with the destination node of a primary link as the root are reachable without passing through the primary link.

PQ node

A PQ node exists both in the extended P space and Q space and is used by Remote LFA as the destination of a protection tunnel.

TI-LFA

In some LFA or RLFA scenarios, the P space and Q space do not share nodes or have directly connected neighbors. Consequently, no backup path can be calculated, which does not meet reliability requirements. In this situation, TI-LFA can be used. The TI-LFA algorithm computes the P space and Q space based on a protected path, a shortest path tree (also called a post-convergence tree), and a repair list. The algorithm establishes a segment routing tunnel between the source node and PQ node to provide backup next hop protection. If the protected link fails, traffic automatically switches to the backup path, which improves network reliability. For more information about TI-LFA, see TI-LFA FRR.

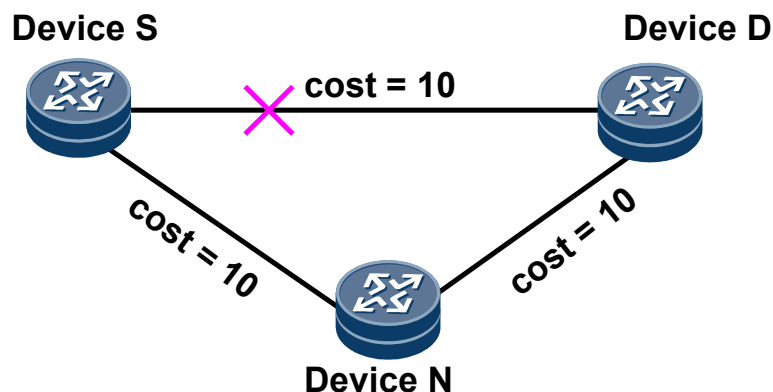
IS-IS LFA Auto FRR

IS-IS LFA Auto FRR protects against both link and node-and-link failures.

- Link protection: Link protection applies to traffic transmitted over specified links.

In the example network shown in [Figure 8-21](#), traffic flows from Device S to Device D, and the link cost meets the preceding link protection inequality. If the primary link (S -> Device D) fails, Device S switches the traffic to the alternate link (S -> Device N -> Device D), minimizing traffic loss.

Figure 8-21 Networking for IS-IS LFA Auto FRR link protection

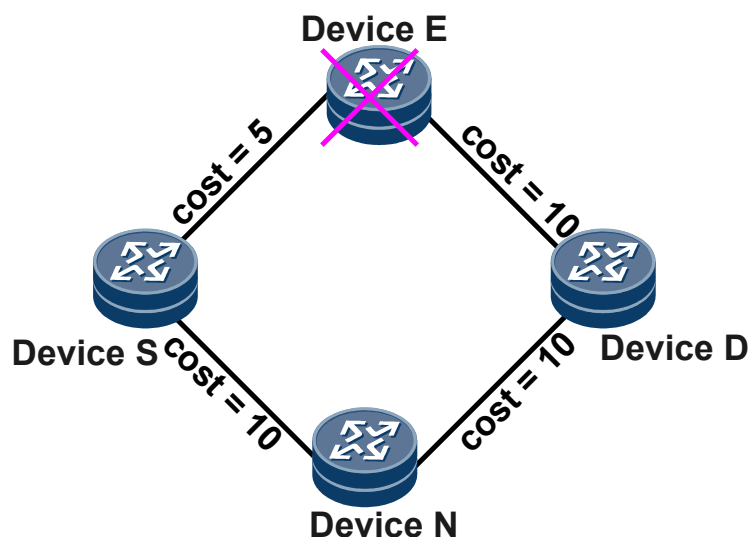


- Node-and-link protection: Node-and-link protection applies to traffic transmitted over specified nodes or links. [Figure 8-22](#) illustrates a network where LFA Auto FRR node-and-link protection is used. Node-and-link protection takes precedence over link protection.

Node-and-link protection takes effect when the following conditions are met:

- The link cost satisfies the inequality: $\text{Distance_opt}(N, D) < \text{Distance_opt}(N, S) + \text{Distance_opt}(S, D)$.
- The interface cost of the device satisfies the inequality: $\text{Distance_opt}(N, D) < \text{Distance_opt}(N, E) + \text{Distance_opt}(E, D)$.

Figure 8-22 Networking for IS-IS LFA Auto FRR node-and-link protection

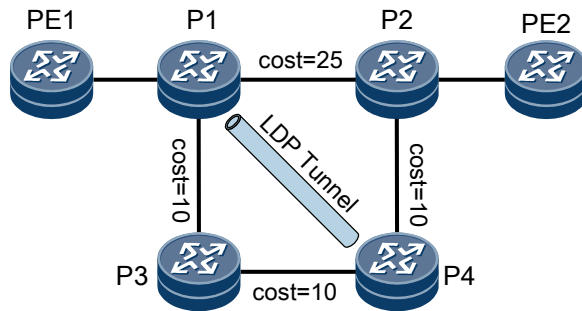


IS-IS Remote LFA Auto FRR

Similar to LFA Auto FRR, Remote LFA protects against both link and node-and-link failures. The following example shows how Remote LFA works to protect against link failures:

In **Figure 8-23**, traffic flows through PE1 -> P1 -> P2 -> PE2, and the primary link is between P1 and P2. Remote LFA calculates a PQ node (P4) and establishes a Label Distribution Protocol (LDP) tunnel between P1 and P4. If P1 detects a failure on the primary link, P1 encapsulates packets into MPLS packets and forwards MPLS packets to P4. After receiving the packets, P4 removes the MPLS label from them and searches the IP routing table for a next hop to forward the packets to PE2. Remote LFA ensures uninterrupted traffic forwarding.

Figure 8-23 Networking for Remote LFA



On the network shown in **Figure 8-23**, Remote LFA calculates the PQ node as follows:

1. Calculates an SPT with P1 at the root. The P space is composed of the root node and its directly connected nodes on the SPT, but excludes the other node on the backup path P1→P2. Therefore, the P space is {PE1, P1, P3}.
2. Calculates each SPT that has a directly connected neighbor of P1 (including PE1 and P3, but excluding P2 on the backup path) as the root. The extended P space is composed of the root nodes and their directly connected nodes on the SPTs, but excludes the node on the backup path P1→P2. Therefore, the extended P space is {PE1, P1, P3, P4}.
3. Calculates a reverse SPT with P2 as the root. The Q space is composed of the root node's directly connected nodes, but excludes the root node. Therefore, the Q space is {PE2, P4}.
4. Determines the PQ node that is in both the extended P space and Q space. Therefore, the PQ node is P4 in this example.

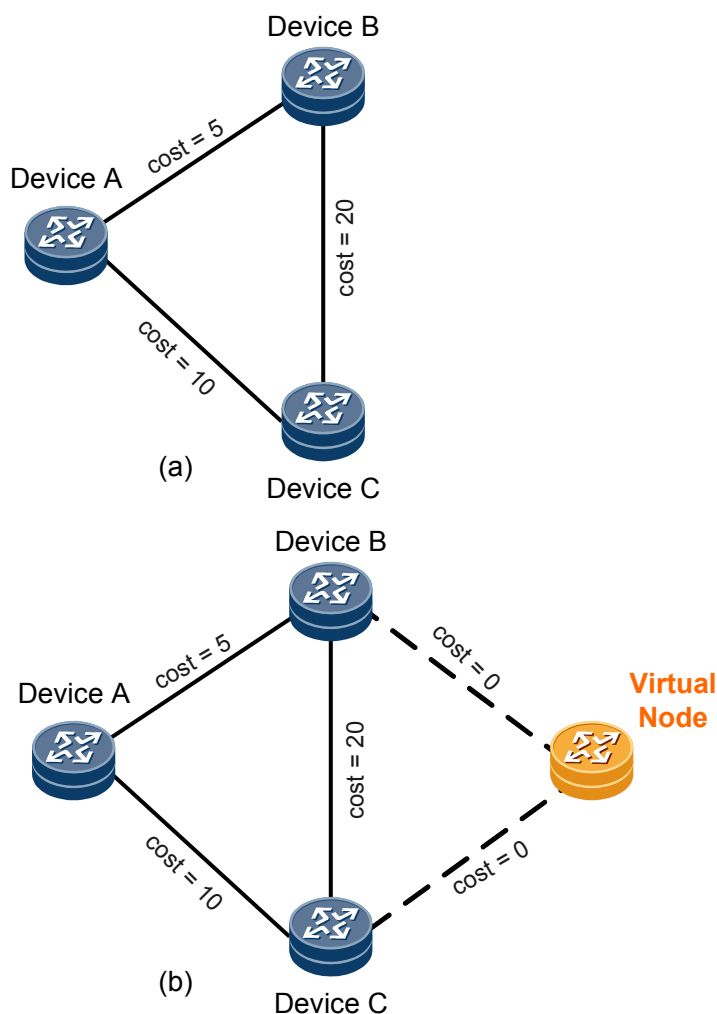
NOTE

IPv6 IS-IS remote LFA auto FRR protects IPv6 traffic and uses IPv4 LDP LSPs. The principle of IPv6 IS-IS remote LFA auto FRR is similar to that of IPv4 IS-IS remote LFA auto FRR.

IS-IS FRR in a Multi-Source Routing Scenario

IS-IS LFA FRR uses the SPF algorithm to calculate the shortest path to the destination node, with each neighbor that provides a backup link as the root node. The backup next hop is node-based, which applies to single-source routing scenarios. With the diversification of networks, multi-source routing scenarios appear, where multiple nodes advertise the same route. Such multi-source routing scenarios do not meet single-source LFA conditions. As a result, the backup next hop cannot be calculated. IS-IS FRR for multi-source routing scenarios can address this problem by using a routing source to protect the primary routing source and improve network reliability.

Figure 8-24 Networking for IS-IS FRR in a multi-source routing scenario



In **Figure 8-24** (a), the cost of the link between Device A and Device B is 5, whereas the cost of the link between Device A and Device C is 10. Both Device B and Device C advertise the route 10.1.1.0/24. IS-IS FRR is enabled on Device A. However, single-source LFA conditions are not met. As a result, Device A fails to calculate the backup next hop of the route 10.1.1.0/24. IS-IS FRR for multi-source routing scenarios can address this problem.

In **Figure 8-24** (b), a virtual node is simulated between Device B and Device C and is connected to Device B and Device C. The cost of the link from Device B or Device C to the virtual node is 0, whereas the cost of the link from the virtual node to Device B or Device C is the maximum value. After the virtual node advertises the route 10.1.1.0/24, Device A uses the LFA algorithm to calculate the backup next hop of the virtual node. Then the route 10.1.1.0/24 inherits the backup next hop from the virtual node. In this example, the primary link to the virtual node is the one from Device A to Device B, and the backup link is the one from Device A to Device C.

IS-IS ECMP FRR

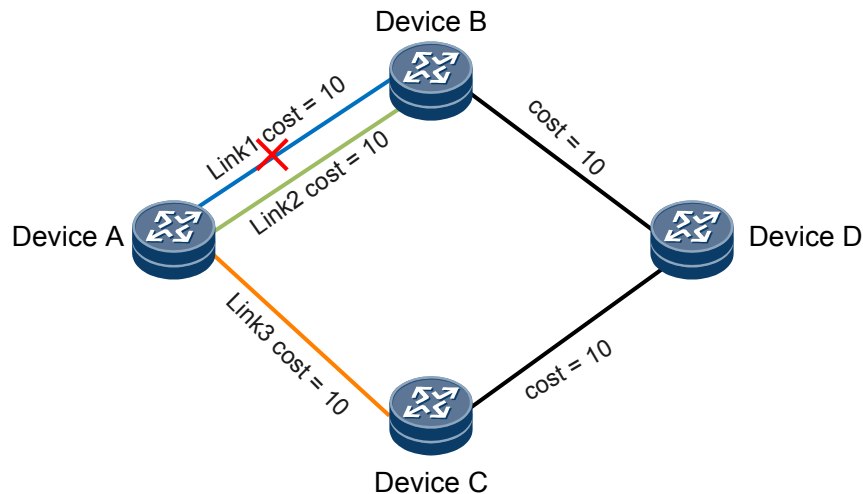
Equal cost multi path (ECMP) evenly balances traffic over multiple equal-cost paths to the same destination.

If the ECMP FRR function is not supported in ECMP scenarios, no backup next hop can be calculated for primary links.

IS-IS ECMP FRR is enabled by default, and a backup next hop is calculated separately for each primary link, which enhances reliability in ECMP scenarios. With ECMP FRR, IS-IS pre-calculates backup paths for load balancing links based on the LSDBs on the entire network. The backup paths are stored in the forwarding table and are used for traffic protection in the case of link failures.

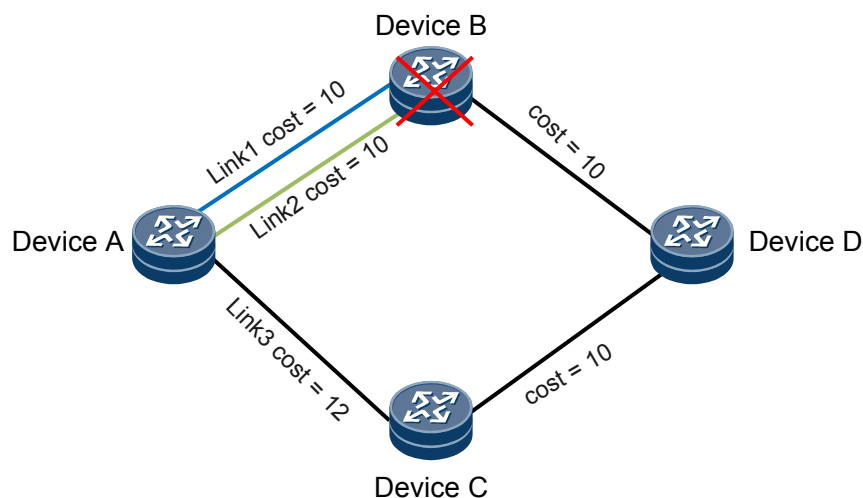
- In **Figure 8-25**, traffic is forwarded from Device A to Device D and is balanced among link 1, link 2, and link 3. Backup paths of the three links are calculated based on ECMP FRR. For example, the backup paths of link 1, link 2, and link 3 are link 3, link 3, and link 2.
 - If the ECMP FRR function is not enabled in the load balancing scenario and link 1 fails, traffic over link 1 is randomly switched to link 2 or link 3, which affects service traffic management.
 - If the ECMP FRR function is enabled in the load balancing scenario and link 1 fails, traffic over link 1 is switched to link 3 according to FRR route selection rules, which enhances service traffic management.

Figure 8-25 Flexible selection of backup paths through IS-IS ECMP FRR



- In **Figure 8-26**, traffic is forwarded from Device A to Device D and is balanced between link 1 and link 2. Backup paths of the two links are calculated based on ECMP FRR. For example, the backup paths of link 1 and link 2 are both link 3.
 - If the ECMP FRR function is not enabled in the load balancing scenario and Device B fails, link 1 and link 2 fail accordingly, leading to a traffic interruption.
 - If the ECMP FRR function is enabled in the load balancing scenario and Device B fails, link 1 and link 2 fail accordingly. However, traffic is switched to link 3, which prevents the traffic interruption.

Figure 8-26 Enhanced traffic protection through IS-IS ECMP FRR



IS-IS SRLG FRR

A shared risk link group (SRLG) is a set of links that share a common physical resource, such as an optical fiber. These links share the same risk level. If one of the links fails, all the other links in the SRLG may also fail.

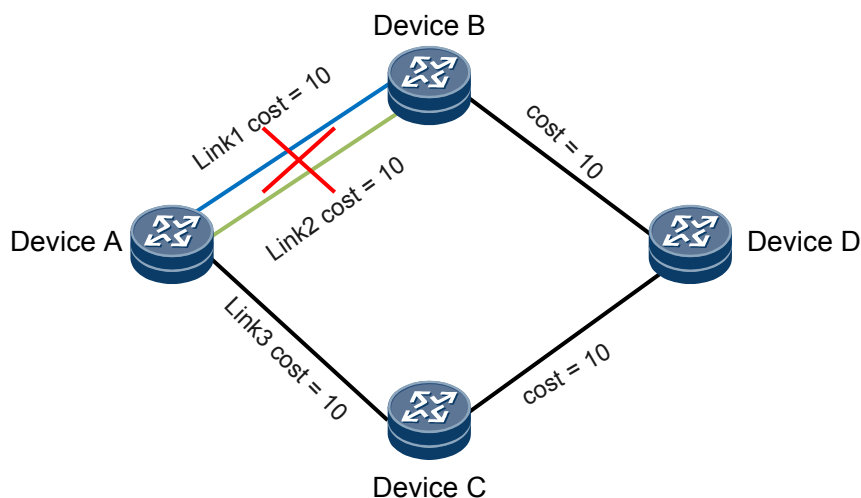
On the network shown in [Figure 8-27](#), traffic between Device A and Device D is balanced by Link 1 and Link 2.

If IS-IS LFA Auto FRR is enabled, it implements protection for the two links by calculating a backup link if either Link 1 or Link 2 fails.

- If Link 1 fails but Link 2 is normal, traffic is not interrupted after being switched to the backup link.
- If both Link 1 and Link 2 fail, traffic is interrupted after being switched to the backup link.

IS-IS SRLG FRR prevents service interruptions in the scenario where links have the same risk of failure. To implement IS-IS SRLG FRR, add the links having the same risk of failure to the same SRLG, and configure IS-IS to preferentially select a backup link not in the same SRLG during route calculation in Auto FRR.

Figure 8-27 Networking for IS-IS SRLG FRR



8.2.15 IS-IS Authentication

Background

As the Internet develops, more data, voice, and video information are exchanged over the Internet. New services, such as e-commerce, online conferencing and auctions, video on demand, and distance learning, emerge gradually. The new services have high requirements for network security. Carriers need to prevent data packets from being intercepted or modified by attackers or unauthorized users. IS-IS authentication applies to the area or interface where packets need to be protected. Using IS-IS authentication enhances system security and helps carriers provide safe network services.

Related Concepts

Authentication Classification

Based on packet types, the authentication is classified as follows:

- Interface authentication: is configured in the interface view to authenticate Level-1 and Level-2 IS-to-IS Hello PDUs (IIHs).
- Area authentication: is configured in the IS-IS process view to authenticate Level-1 CSNPs, PSNPs, and LSPs.
- Routing domain authentication: is configured in the IS-IS process view to authenticate Level-2 CSNPs, PSNPs, and LSPs.

Based on the authentication modes of packets, the authentication is classified into the following types:

- Simple authentication: The authenticated party directly adds the configured password to packets for authentication. This authentication mode provides the lowest password security.
- MD5 authentication: uses the MD5 algorithm to encrypt a password before adding the password to the packet, which improves password security.
- Keychain authentication: further improves network security with a configurable key chain that changes with time.

- HMAC-SHA256 authentication: uses the HMAC-SHA256 algorithm to encrypt a password before adding the password to the packet, which improves password security.

Implementation

IS-IS authentication encrypts IS-IS packets by adding the authentication field to packets to ensure network security. After receiving IS-IS packets from a remote router, a local router discards the packets if the authentication passwords in the packets are different from the locally configured one. This mechanism protects the local router.

IS-IS provides a type-length-value (TLV) to carry authentication information. The TLV components are as follows:

- Type: indicates the type of a packet, which is 1 byte. The value defined by ISO is 10, whereas the value defined by IP is 133.
- Length: indicates the length of the authentication TLV, which is 1 byte.
- Value: indicates the authentication information, including authentication type and authenticated password, which ranges from 1 to 254 bytes. The authentication type is 1 byte:
 - 0: reserved
 - 1: simple authentication
 - 3: general authentication, and only HMAC-SHA256 authentication currently
 - 54: MD5 authentication
 - 255: private authentication

Interface Authentication

Authentication passwords for IIHs are saved on interfaces. The interfaces send authentication packets with the authentication TLV. Interconnected router interfaces must be configured with the same password.

Area Authentication

Every router in an IS-IS area must use the same authentication mode and have the same key chain.

Routing Domain Authentication

Every Level-2 or Level-1-2 router in an IS-IS area must use the same authentication mode and have the same key chain.

For area authentication and routing domain authentication, you can set a router to authenticate SNPs and LSPs separately in the following ways:

- A router sends LSPs and SNPs that carry the authentication TLV and verifies the authentication information of the LSPs and SNPs it receives.
- A router sends LSPs that carry the authentication TLV and verifies the authentication information of the LSPs it receives. The router sends SNPs that carry the authentication TLV and does not verify the authentication information of the SNPs it receives.
- A router sends LSPs that carry the authentication TLV and verifies the authentication information of the LSPs it receives. The router sends SNPs without the authentication TLV and does not verify the authentication information of the SNPs it receives.
- A router sends LSPs and SNPs that carry the authentication TLV but does not verify the authentication information of the LSPs and SNPs it receives.

8.2.16 IS-IS Purge LSP Source Tracing

Background

If network-wide IS-IS LSPs are deleted, purge LSPs are flooded, which adversely affects network stability. In this case, source tracing must be implemented to locate the root cause of the fault immediately to minimize the impact. However, IS-IS itself does not support source tracing. A conventional solution is isolation node by node until the faulty node is located. The solution is complex and time-consuming. Therefore, a fast source tracing method is required.

IS-IS purge LSP source tracing is a Huawei proprietary protocol.

Related Concepts

- PS-PDU: packets that carry information about the node that floods IS-IS purge LSPs.
- CAP-PDU: packets used to negotiate the IS-IS purge LSP source tracing capability between IS-IS neighbors.
- IS-IS purge LSP source tracing port: ID of the UDP port that receives and sends IS-IS purge LSP source tracing packets. The default port ID is 50121, which is configurable.

Implementation

IS-IS purge LSPs do not carry source information. If a device fails on the network, a large number of purge LSPs are flooded. Without a source tracing mechanism, nodes are isolated one by one until the faulty node is located, which is labor-intensive and time-consuming.

IS-IS purge LSPs will trigger route flapping on the network, or even routes become unavailable. In this case, the device that floods the purge LSPs must be located and isolated immediately.

A solution that can address the following issues is required:

- Information about the source that floods IS-IS purge LSPs can be obtained when network routes are unreachable.
- The method used to obtain source information must apply to all devices on the network and support incremental deployment, without compromising routing capabilities.

IS-IS purge LSP source tracing helps locate the device that floods purge LSPs. IS-IS purge LSP source tracing uses a new UDP port. Source tracing packets are carried by UDP packets, and the UDP packets also carry the IS-IS purge LSPs sent by the current device and are flooded hop by hop based on the IS-IS topology.

IS-IS purge LSP source tracing forwards packets along UDP channels which are independent of the channels used to transmit IS-IS packets. Therefore, IS-IS purge LSP source tracing supports incremental deployment. In addition, source tracing does not affect the devices with the related UDP port disabled.

After IS-IS purge LSP source tracing packets are flooded to devices on the network, information about the node that floods purge LSPs can be queried on any of the devices, which speeds up fault locating and faulty node isolation.

Capability Negotiation

IS-IS purge LSP source tracing is Huawei proprietary. It uses UDP to carry packets and listens to the UDP port which is used to receive and send source tracing packets. If a source

tracing-capable Huawei device sends source tracing packets to a source tracing-incapable Huawei device or non-Huawei device, the source tracing-capable Huawei device may be incorrectly identified as an attacker. Therefore, the source tracing capability need to be negotiated between the devices. In addition, the source tracing-capable device needs to help the source tracing-incapable device to send source tracing information, which also requires negotiation.

Source tracing capability negotiation depends on IS-IS neighbor relationships. Specifically, after an IS-IS neighbor relationship is established, the local device initiates source tracing capability negotiation based on the IP address of the neighbor.

PS-PDU Generation

If a device needs to purge an LSP, it generates and floods a PS-PDU to all its source tracing neighbors.

If a device receives a purge LSP from a source tracing-incapable neighbor, the device generates and floods a PS-PDU to all its neighbors. If a device receives the same purge LSP (with the same LSP ID and sequence number) from more than one source tracing-incapable neighbor, the device generates only one PS-PDU.

PS-PDU flooding is similar to IS-IS LSP flooding.

Security Concern

IS-IS purge LSP source tracing uses a UDP port to receive and send source tracing packets. Therefore, the security of the port must be taken into consideration.

IS-IS purge LSP source tracing inevitably increases packet receiving and sending workload and intensifies bandwidth pressure. To minimize the impact on IS-IS, the number of source tracing packets must be controlled.

- Authentication

Source tracing is embedded in IS-IS and uses IS-IS authentication parameters to authenticate packets.

- Generalized TTL security mechanism (GTSM)

GTSM is a security mechanism that checks whether the time to live (TTL) value in each received IP packet header is within a pre-defined range.

IS-IS purge LSP source tracing packets can be flooded as far as one hop. If the TTL of a packet is 255 when it is sent and not 254 when it is received, the packet will be discarded.

- CPU-CAR

The NP chip on interface boards can check the packets to be sent to the CPU for processing and prevent the main control board from being overloaded by a large number of packets that are sent to the CPU.

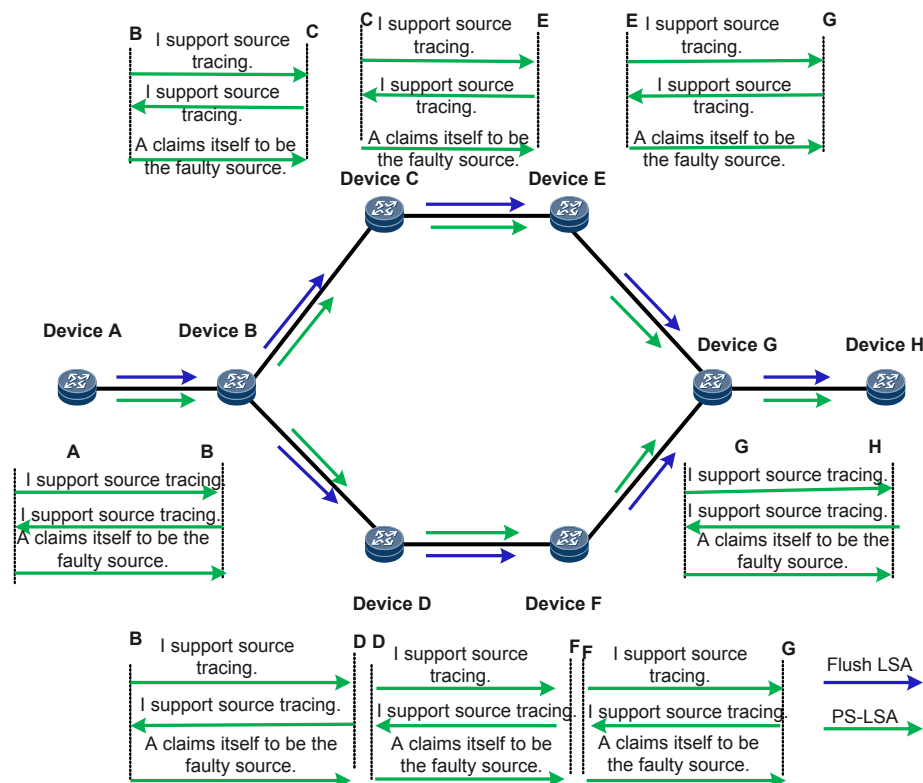
IS-IS purge LSP source tracing needs to apply for an independent CAR channel and has small committed information rate (CIR) and committed burst size (CBS) values configured.

Typical Scenarios

Scenario where all nodes support source tracing

All nodes on the network support source tracing, and Device A is the faulty source. **Figure 8-28** shows the networking.

Figure 8-28 Scenario where all nodes support source tracing



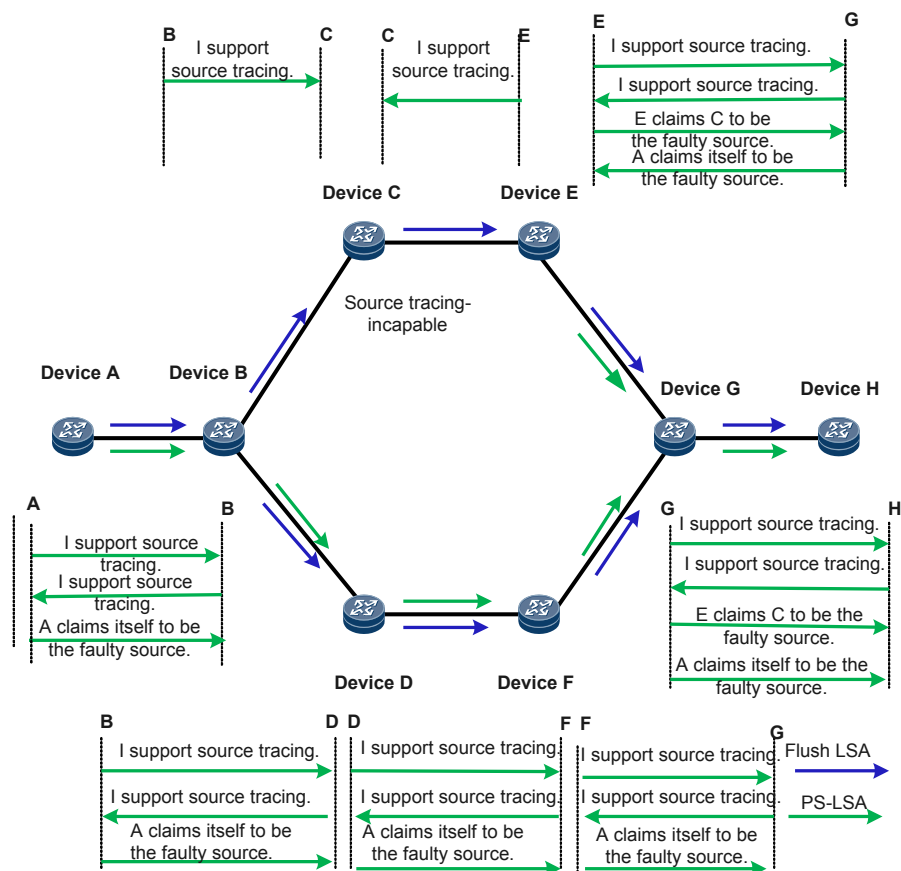
All nodes in the networking support IS-IS purge LSP source tracing, and Device A is the faulty source.

When Device A purges an IS-IS LSP, it floods a source tracing packet that carries Device A information and brief information about the LSP. Then the packet is flooded on the network hop by hop. After the fault occurs, maintenance personnel can log in to any node on the network to locate Device A that keeps sending purge LSPs and isolate Device A from the network.

Scenario where source tracing-incapable nodes are not isolated from source tracing-capable nodes

All nodes on the network except device C support source tracing, and device A is the faulty source. In this case, the PS-LSA can be flooded on the entire network. **Figure 8-29** shows the networking.

Figure 8-29 Scenario where source tracing-incapable nodes are not isolated from source tracing-capable nodes



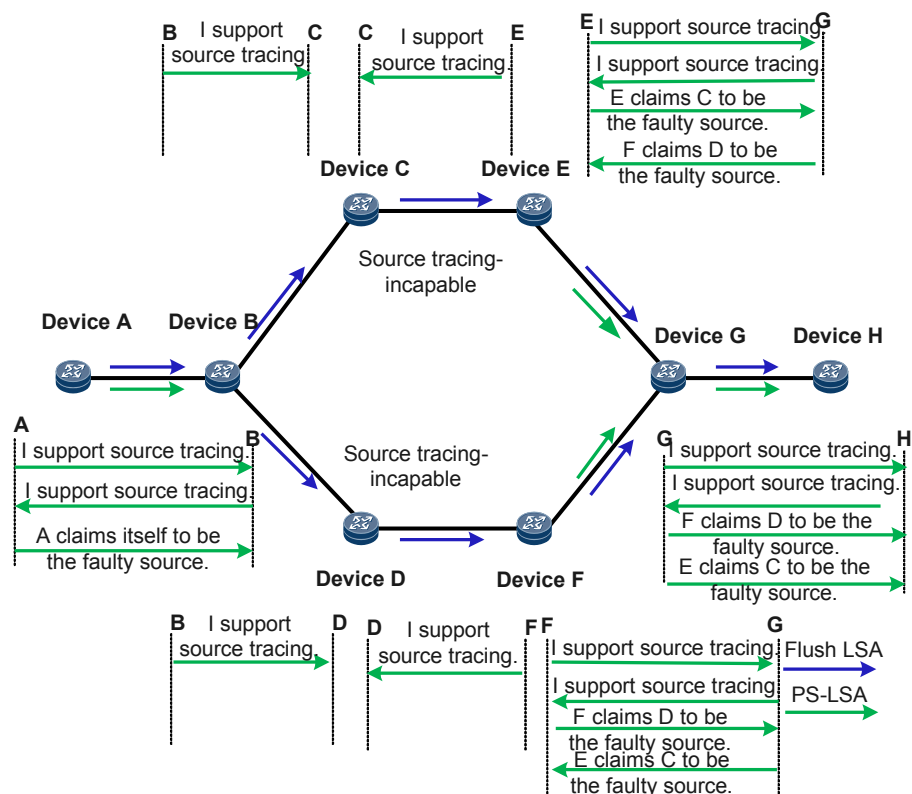
When Device A purges an IS-IS LSP, it floods a source tracing packet that carries Device A information and brief information about the LSP. Then the packet is flooded on the network hop by hop. When Device B and Device E negotiate the source tracing capability with Device C, they find that Device C does not support source tracing. Therefore, after Device B receives the source tracing packet from Device A, Device B sends the packet to Device D, but not to Device C. After receiving the purge LSP from Device C, Device E generates a source tracing packet which carries information about the advertisement source (Device E), purge source (Device C), and the purge LSP, and floods the packet on the network.

After the fault occurs, maintenance personnel can log in to any node on the network except Device C to locate the faulty node. Two possible faulty nodes can be located in this case: Device A and Device C, and they both send the same purge LSP. In this case, Device A takes precedence over Device C when the maintenance personnel determine the most possible faulty source. After Device A is isolated, the network recovers. Then the possibility that Device C is the faulty node is ruled out.

Scenario where source tracing-incapable nodes are isolated from source tracing-capable nodes

All nodes on the network except devices C and D support source tracing, and Device A is the faulty source. In this case, the PS-LSA cannot be flooded on the entire network. [Figure 8-30](#) shows the networking.

Figure 8-30 Scenario where source tracing-incapable nodes are isolated from source tracing-capable nodes



When Device A purges an IS-IS LSP, it floods a source tracing packet that carries Device A information and brief information about the LSP. However, the source tracing packet can reach only Device B because nodes C and Device D do not support IS-IS purge LSP source tracing.

During source tracing capability negotiation, Device E finds that Device C does not support source tracing, and Device F finds that Device D does not support source tracing. After Device E receives the purge LSP from Device C, Device E helps Device E generate and flood a source tracing packet. Similarly, after Device F receives the purge LSP from Device D, Device F helps Device D generate and flood a source tracing packet.

After the fault occurs, if maintenance personnel log in to Device A or Device B, the personnel can locate the faulty source (Device A) directly. After Device A is isolated, the network recovers. If the maintenance personnel log in to Device E, Device F, Device G, or Device H, the personnel will find that Device E claims Device C to be the faulty source and Device F claims Device D to be the faulty source. Then the personnel log in to nodes C and Device D and find that the purge LSP was sent by Device B, not generated by Device C or Device D. Then the personnel log in to Device B, determine that Device A is the faulty node, and isolate Device A. After Device A is isolated, the network recovers.

8.2.17 IS-IS MT

With IS-IS multi-topology (MT), IPv6, multicast, and advanced topologies can have their own routing tables. This feature prevents packet loss if an integrated topology and the IPv4/IPv6 dual stack are deployed, isolates multicast services from unicast routes, improves network resource usage, and reduces network construction cost.

Introduction

On a traditional IP network, IPv4 and IPv6 share the same integrated topology, and only one unicast topology exists, which causes the following problems:

- Packet loss if the IPv4/IPv6 dual stack is deployed: If some routers and links in an IPv4/IPv6 topology do not support IPv4 or IPv6, they cannot receive IPv4 or IPv6 packets sent from the router that supports the IPv4/IPv6 dual stack. As a result, these packets are discarded.
- Multicast services highly depending on unicast routes: Only one unicast forwarding table is available on the forwarding plane because only one unicast topology exists, which forces services transmitted from one router to the same destination address to share the same next hop, and various end-to-end services, such as voice and data services, to share the same physical links. As a result, some links may be heavily congested whereas others remain relatively idle. In addition, the multicast reverse path forwarding (RPF) check depends on the unicast routing table. If the default unicast routing table is used when transmitting multicast services, multicast services depend heavily on unicast routes, a multicast distribution tree cannot be planned independently of unicast routes, and unicast route changes affect multicast distribution tree establishment.

Deploying multiple topologies for different services on a physical network can address these problems. IS-IS MT transmits multicast information by defining new TLVs in IS-IS packets. Users can deploy multiple logical topologies on a physical network based on IP protocols or service types supported by links so that separate SPF calculation operations are performed in different topologies, which improves network usage.

NOTE

If an IPv4 or IPv6 BFD session is Down in a topology on a network enabled with MT, neighbors of the IPv4 or IPv6 address family will be affected.

Related Concepts

IS-IS MT allows multiple route selection subsets to be deployed on a versatile network infrastructure and divides a physical network into multiple logical topologies, where each topology performs its own SPF calculations.

IS-IS MT, an extension of IS-IS, allows multiple topologies to be applied to IS-IS. IS-IS MT complies with standard protocols and transmits multicast information using new TLVs in IS-IS packets. Users can deploy multiple logical topologies on a physical network. Each topology performs its own SPF calculations and maintains its own routing table. Traffic of different services, including the traffic transmitted in different IP topologies, has its own optimal forwarding path.

The MT ID configured on an interface identifies the topology bound to the interface. One or more MT IDs can be configured on a single interface.

With RPF check, upon receiving a packet, a router searches multicast static, unicast, Multiprotocol Border Gateway Protocol (MBGP), and Multicast Interior Gateway Protocol (MIGP) routing tables for an optimal route and sets it as the RPF route to the source IP address of the packet. The packet can be transmitted only when it is destined for the RPF interface.

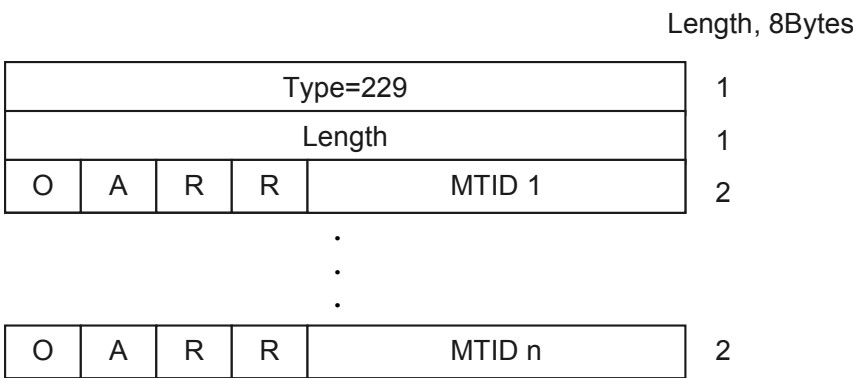
Implementation

In IS-IS MT, the MT ID varies with the topology. Each Hello packet or LSP sent by a router carries one or more MT TLVs of the topologies to which the source interface belongs. If the

router receives from a neighbor a Hello packet or LSP that carries only some of the local MT TLVs, the router assumes that the neighbor belongs to only the default IPv4 topology. On a point-to-point (P2P) link, an adjacency cannot be established between two neighbors that share no common MT ID, while on a broadcast link, the relationship can be established in this case.

Figure 8-31 shows the MT TLV format.

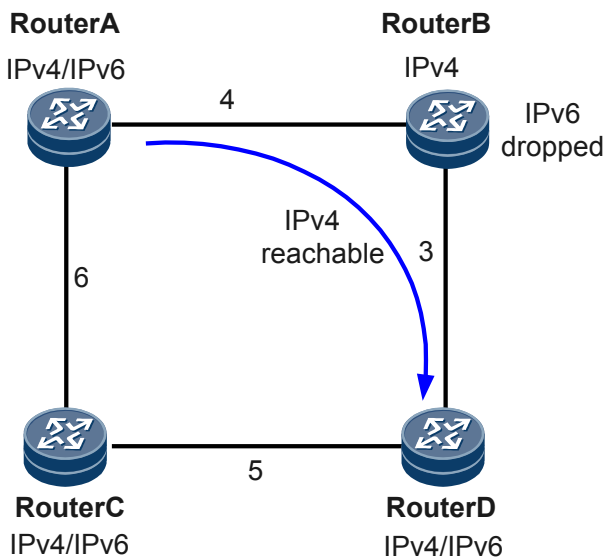
Figure 8-31 MT TLV format



The following section describes separation of the IPv4 topology from the IPv6 topology, and the multicast topology from the unicast topology.

- Figure 8-32 shows the networking for separation of the IPv4 topology from the IPv6 topology. The values in the networking diagram are link costs. Device A, Device C, and Device D support the IPv4/IPv6 dual stack; Device B supports IPv4 only and cannot forward IPv6 packets.

Figure 8-32 Separation of the IPv4 topology from the IPv6 topology



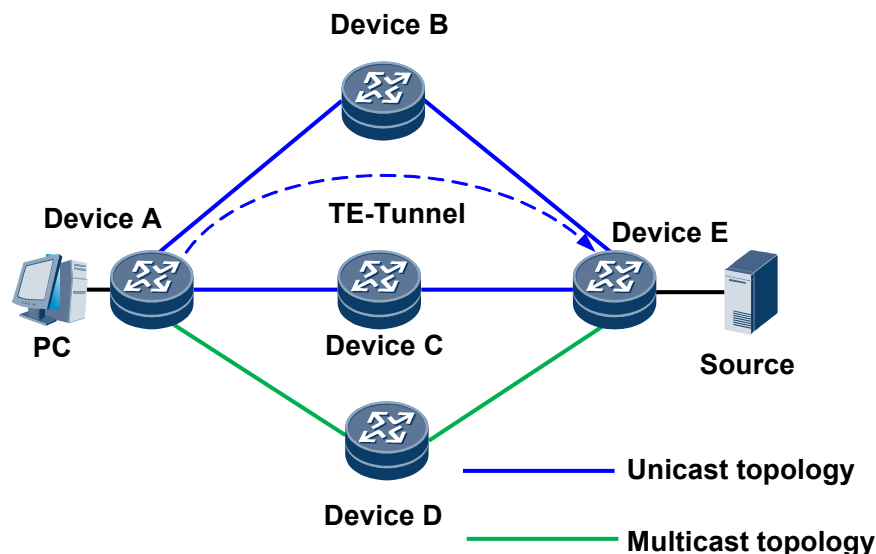
Without IS-IS MT, Device A, Device B, Device C, and Device D use the IPv4/IPv6 topology for the SPF calculation. In this case, the shortest path from Device A to Device

D is Device A -> Device B -> Device D. IPv6 packets cannot reach Device D through Device B because Device B does not support IPv6.

If a separate IPv6 topology is set up using IS-IS MT, Device A chooses only IPv6 links to forward IPv6 packets. In this case, the shortest path from Device A to Device D is Device A -> Device C -> Device D.

- **Figure 8-33** shows the networking for separation between unicast and a multicast topologies using IS-IS MT.

Figure 8-33 Separation of the multicast topology from the unicast topology



On the network shown in **Figure 8-33**, all routers are interconnected using IS-IS. A TE tunnel is set up between Device A (ingress) and Device E (egress). The outbound interface of the route calculated by IS-IS may not be a physical interface but a TE tunnel interface. In this case, router C through which the TE tunnel passes cannot set up multicast forwarding entries. As a result, multicast services cannot be transmitted.

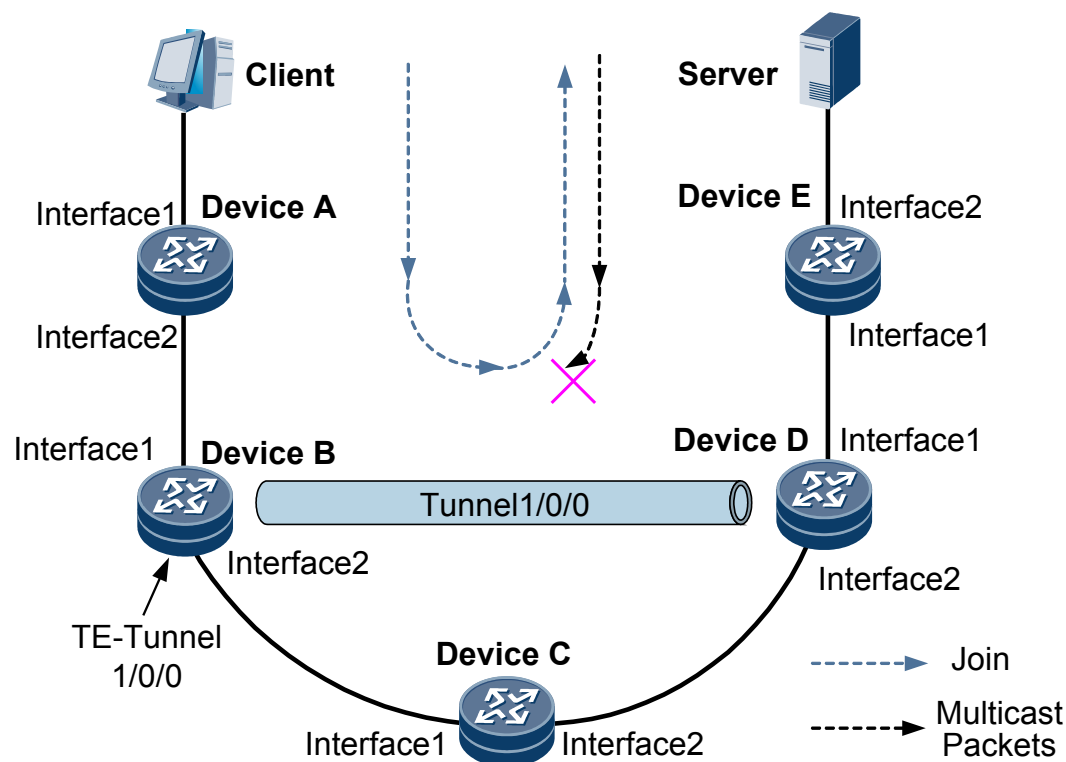
IS-IS MT addresses this problem by establishing separate unicast and multicast topologies. TE tunnels are excluded from a multicast topology. Therefore, multicast services are unaffected by TE tunnels.

8.2.18 IS-IS Local MT

Background

When multicast and an IGP Shortcut-enabled MPLS TE tunnel are configured on a network, the outbound interface of the route calculated by IS-IS may not be a physical interface but a TE tunnel interface. Multicast Join packets are transparent to routers through which the TE tunnel passes. Therefore, these routers cannot generate multicast forwarding entries and discard received multicast packets from the multicast source. **Figure 8-34** shows the conflict between multicast and a TE tunnel.

Figure 8-34 Conflict between multicast and a TE tunnel



Client and Server exchange multicast packets as follows:

1. Client sends a Report message to Device A, requesting to join the multicast group. Upon receipt, Device A sends a Join packet to Device B.
2. After Device B receives the Join packet, it selects TE-Tunnel 1/0/0 as the reverse path forwarding (RPF) interface and adds an MPLS label to the packet before forwarding it from Interface2 to Device C.
3. As the penultimate hop of the TE tunnel, Device C removes the MPLS label from the Join packet before forwarding it from Interface2 to Device D. Because the forwarding is based on MPLS, Device C does not generate any multicast forwarding entries.
4. After Device D receives the Join packet, it generates a multicast forwarding entry in which the upstream and downstream interfaces are Interface1 and Interface2, respectively. Device D then sends the Join packet to Device E, which has already established the shortest path tree.
5. Multicast packets flow from Server to Device C through Device E and Device D. Device C discards these packets because no multicast forwarding entry is available. As a result, multicast services are interrupted.

IS-IS local multicast topology (MT) can address this problem.

Related Concepts

IS-IS local MT is a mechanism that enables the routing management (RM) module to create a separate multicast topology on the local device so that protocol packets exchanged between devices are not erroneously discarded. When the outbound interface of the route calculated by

IS-IS is an IGP Shortcut-enabled TE tunnel interface, IS-IS local MT calculates a physical outbound interface for the route. This mechanism resolves the conflict between multicast and a TE tunnel.

NOTE

The TE tunnel described in this section is IGP Shortcut-enabled.

Implementation

Figure 8-35 shows how multicast packets are processed after IS-IS local MT is enabled.

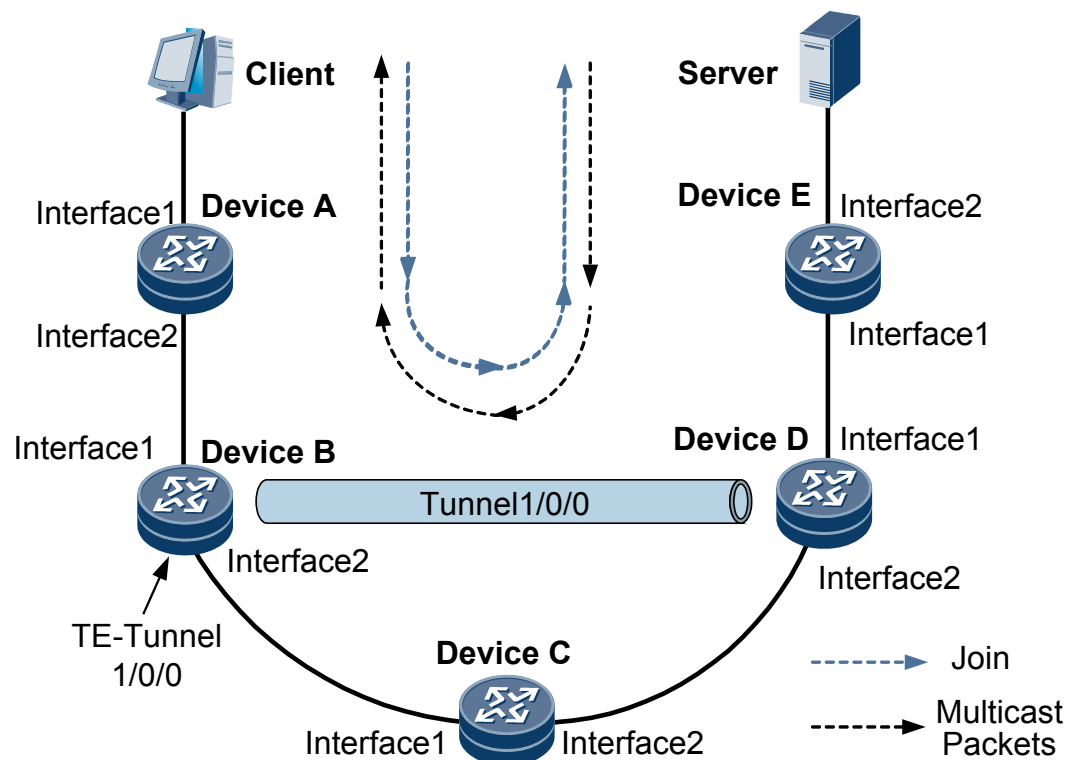
1. Establishment of a multicast IGP (MIGP) routing table

Device B creates an independent MIGP routing table, records the TE tunnel interface, and generates multicast routing entries for multicast packet forwarding. If the outbound interface of a calculated route is a TE tunnel interface, IS-IS calculates a physical outbound interface for the route and adds the route to the MIGP routing table.

2. Multicast packet forwarding

When forwarding multicast packets, a router searches the unicast routing table for a route. If the next hop of the route is a tunnel interface, the router searches the MIGP routing table for the physical outbound interface to forward multicast packets. In this example, the original outbound interface of the route is TE tunnel 1/0/0. IS-IS re-calculates a physical outbound interface (Interface2) for the route and adds the route to the MIGP routing table. Device B then forwards multicast packets through Interface2 based on the MIGP routing table and generates a multicast routing entry in the multicast routing table. Therefore, multicast services are properly forwarded.

Figure 8-35 Local MT networking



Usage Scenario

IS-IS local MT prevents multicast services from being interrupted on networks, which allows multicasting and has an IGP Shortcut-enabled TE tunnel.

Benefits

Local MT resolves the conflict between multicast and a TE tunnel and improves multicast service reliability.

8.2.19 IS-IS Control Messages

IS-IS routers implement routing by exchanging control messages. This section describes IS-IS control messages.

IS-IS PDU Formats

Nine types of IS-IS protocol data units (PDUs) are available for processing control information. Each PDU is identified by a 5-digit type code. IS-IS has three major types of PDUs: Hello PDUs, Link State PDUs (LSPs), and Sequence Number PDUs (SNPs). [Table 8-7](#) shows the mapping between PDUs and type values.

Table 8-7 Mapping between PDUs and type values

PDU Type	Acronym	Type Value
Level-1 LAN IS-IS Hello PDU	L1 LAN IIH	15
Level-2 LAN IS-IS Hello PDU	L2 LAN IIH	16
Point-to-Point IS-IS Hello PDU	P2P IIH	17
Level-1 Link State PDU	L1 LSP	18
Level-2 Link State PDU	L2 LSP	20
Level-1 Complete Sequence Numbers PDU	L1 CSNP	24
Level-2 Complete Sequence Numbers PDU	L2 CSNP	25
Level-1 Partial Sequence Numbers PDU	L1 PSNP	26
Level-2 Partial Sequence Numbers PDU	L2 PSNP	27

The first eight bytes in all IS-IS PDUs are public. [Figure 8-36](#) shows the IS-IS PDU format.

Figure 8-36 IS-IS PDU format

				No. of Octets
Intra-domain Routing Protocol Discriminator				1
Length Indicator				1
Version/Protocol ID Extension				1
ID Length				1
R	R	R	PDU Type	1
Version				1
Reserved				1
Maximum Area Address				1
PDU exclusive				1
TLV				

The main fields are as follows:

- Intradomain Routing Protocol Discriminator: network layer protocol identifier assigned to IS-IS, which is 0x83.
- Length Indicator: length of the fixed header, in bytes.
- ID Length: length of the system ID of network service access point (NSAP) addresses or NETs in this routing domain.
- PDU Type: type of a PDU. For details, see [Table 8-7](#).
- Maximum Area Address: maximum number of area addresses supported by an IS-IS area. The value **0** indicates that a maximum of three area addresses are supported by this IS-IS area.
- Type/Length/Value (TLV): encoding type that features high efficiency and expansibility. Each type of PDU contains a different TLV. [Table 8-8](#) shows the mapping between TLV codes and PDU types.

Table 8-8 Mapping between TLV codes and PDU types

TLV Code	TLV Code Name	PDU Type
1	Area Addresses	IIH, LSP
2	IS Neighbors (LSP)	LSP
4	Partition Designated Level2 IS	L2 LSP
6	IS Neighbors (MAC Address)	LAN IIH
7	IS Neighbors (SNPA Address)	LAN IIH
8	Padding	IIH
9	LSP Entries	SNP

TLV Code	TLV Code Name	PDU Type
10	Authentication Information	IIH, LSP, or SNP
128	IP Internal Reachability Information	LSP
129	Protocols Supported	IIH or LSP
130	IP External Reachability Information	L2 LSP
131	Inter-Domain Routing Protocol Information	L2 LSP
132	IP Interface Address	IIH or LSP

Hello Packet Format

Hello packets, also called the IS-to-IS Hello PDUs (IIHs), are used to set up and maintain neighbor relationships. Level-1 LAN IIHs are applied to the Level-1 routers on broadcast LANs. Level-2 LAN IIHs are applied to the Level-2 routers on broadcast LANs. P2P IIHs are applied to non-broadcast networks. IIHs in different networks have different formats.

- LAN IIHs: [Figure 8-37](#) shows the format of IIHs on a broadcast network.

Figure 8-37 Level-1/Level-2 LAN IIH format

				No. of Octets
Intra-domain Routing Protocol Discriminator				1
Length Indicator				1
Version/Protocol ID Extension				1
ID Length				1
R	R	R	PDU Type	1
Version				1
Reserved				1
Maximum Area Address				1
Reserved/Circuit Type				1
Source ID				ID Length
Holding Time				2
PDU Length				2
R	Priority			1
LAN ID				ID Length+1
Variable Length Fields				

- P2P IIHs: [Figure 8-38](#) shows the format of IIHs on a P2P network.

Figure 8-38 P2P IIH format

				No. of Octets
Intra-domain Routing Protocol Discriminator				1
Length Indicator				1
Version/Protocol ID Extension				1
ID Length				1
R	R	R	PDU Type	1
Version				1
Reserved				1
Maximum Area Address				1
Reserved/Circuit Type				1
Source ID				ID Length
Holding Time				2
PDU Length				2
Local Circuit ID				1
Variable Length Fields				

As shown in [Figure 8-38](#), most fields in a P2P IIH are the same as those in a LAN IIH. The P2P IIH does not have the priority and LAN ID fields but has a local circuit ID field. The local circuit ID indicates the local link ID.

LSP Format

LSPs are used to exchange link-state information. There are two types of LSPs: Level-1 and Level-2. Level-1 IS-IS transmits Level-1 LSPs. Level-2 IS-IS transmits Level-2 LSPs. Level-1-2 IS-IS can transmit both Level-1 and Level-2 LSPs.

Level-1 and Level-2 LSPs have the same format, as shown in [Figure 8-39](#).

Figure 8-39 Level-1 or Level-2 LSP

				No. of Octets
Intra-domain Routing Protocol Discriminator				1
Length Indicator				1
Version/Protocol ID Extension				1
ID Length				1
R	R	R	PDU Type	1
Version				1
Reserved				1
Maximum Area Address				1
PDU Length				2
Remaining Lifetime				2
LSP ID				ID Length+2
Sequency Number				4
Checksum				2
R	ATT	OL	IS Type	1
Variable Length Fields				

The main fields are as follows:

- **ATT: Attached bit**
ATT is generated by a Level-1-2 router to identify whether the originating router is connected to other areas. When a Level-1 router receives a Level-1 LSP with ATT as 1 from a Level-1-2 router, the Level-1 router generates a default route destined for the Level-1-2 router so that data can be transmitted to other areas.
Although ATT is defined in both the Level-1 LSP and Level-2 LSP, it is set only in the Level-1 LSP only by the Level-1-2 router.
- **OL: LSDB overload**
LSPs with the overload bit are still flooded on networks, but the LSPs are not used when routes that pass through a device configured with the overload bit are calculated. That is, after a device is configured with the overload bit, other devices ignore the device when performing the SPF calculation except for the direct routes of the device.
- **IS Type: type of the IS-IS generating the LSP**
IS Type is used to specify whether the IS-IS type is Level-1 or Level-2 IS-IS. The value **01** indicates Level-1; the value **11** indicates Level-2.

SNP Format

SNPs describe the LSPs in all or some of the databases and are used to synchronize and maintain all LSDBs. SNPs consist of complete SNPs (CSNPs) and partial SNPs (PSNPs).

- **CSNPs** carry summaries of all LSPs in LSDBs, which ensures LSDB synchronization between neighboring routers. On a broadcast network, the designated intermediate system (DIS) sends CSNPs at an interval. The default interval is 10 seconds. On a P2P

link, neighboring devices send CSNPs only when a neighbor relationship is established for the first time.

Figure 8-40 shows the CSNP format.

Figure 8-40 Level-1/Level-2 CSNP format

				No. of Octets
Intra-domain Routing Protocol Discriminator				1
Length Indicator				1
Version/Protocol ID Extension				1
ID Length				1
R	R	R	PDU Type	1
Version				1
Reserved				1
Maximum Area Address				1
PDU Length				2
Source ID				ID Length+1
Start LSP ID				ID Length+2
End LSP ID				ID Length+2
Variable Length Fields				

The main fields are as follows:

- Source ID: system ID of the router that sends SNPs
- Start LSP ID: ID of the first LSP in a CSNP
- End LSP ID: ID of the last LSP in a CSNP
- PSNPs list only the sequence numbers of recently received LSPs. A PSNP can acknowledge multiple LSPs at a time. If an LSDB is not updated, PSNPs are also used to request a new LSP from a neighbor.

Figure 8-41 shows the PSNP format.

Figure 8-41 Level-1/Level-2 PSNP format

				No. of Octets
Intradomain Routeing Protocol Discriminator				1
Length Indicator				1
Version/Protocol ID Extension				1
ID Length				1
R	R	R	PDU Type	1
Version				1
Reserved				1
Maximum Area Address				1
PDU Length				2
Source ID				ID Length+1
Variable Length Fields				

8.2.20 IS-IS GR

NOTE

The NE40E can be configured as a GR helper rather than a GR restarter.

Graceful restart (GR) is a technology used to ensure proper traffic forwarding, especially the forwarding of key services, during the restart of routing protocols.

Without GR, the master/slave main control board switchover due to various reasons leads to transient service interruption, and as a result, route flapping occurs on the whole network. Such route flapping and service interruption are unacceptable on large-scale networks, especially carrier networks.

GR is one of the high availability (HA) technologies which comprise a series of comprehensive technologies, such as fault-tolerant redundancy, link protection, faulty node recovery, and traffic engineering technologies. As a fault-tolerant redundancy technology, GR is widely used to ensure non-stop forwarding of key data during the master/slave main control board switchovers and system upgrade.

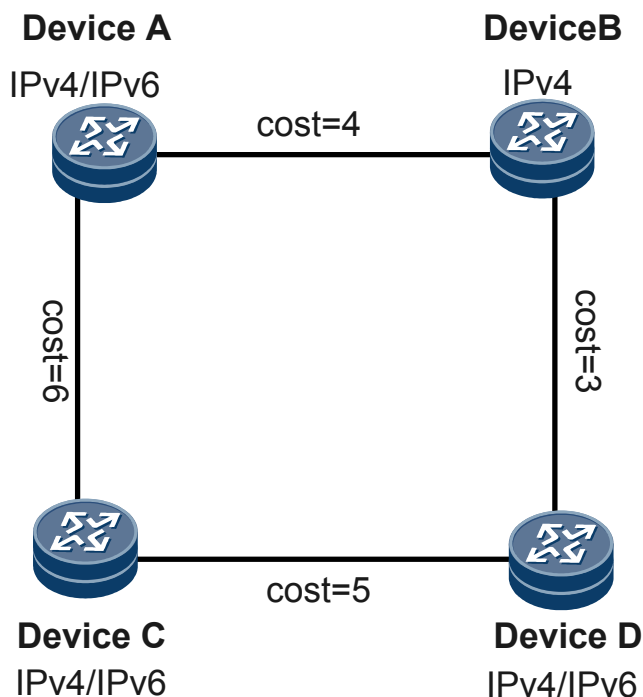
In GR mode, the forwarding plane continues data forwarding during a restart, and operations on the control plane, such as re-establishment of neighbor relationships and route calculation, do not affect the forwarding plane, preventing service interruptions caused by route flapping and improving network reliability.

8.3 Application Scenarios for IS-IS

8.3.1 IS-IS MT

Figure 8-42 shows the use of IS-IS MT to separate an IPv4 topology from an IPv6 topology. Device A, Device C, and Device D support IPv4/IPv6 dual-stack; Device B supports IPv4 only and cannot forward IPv6 packets.

Figure 8-42 Separation of the IPv4 topology from the IPv6 topology

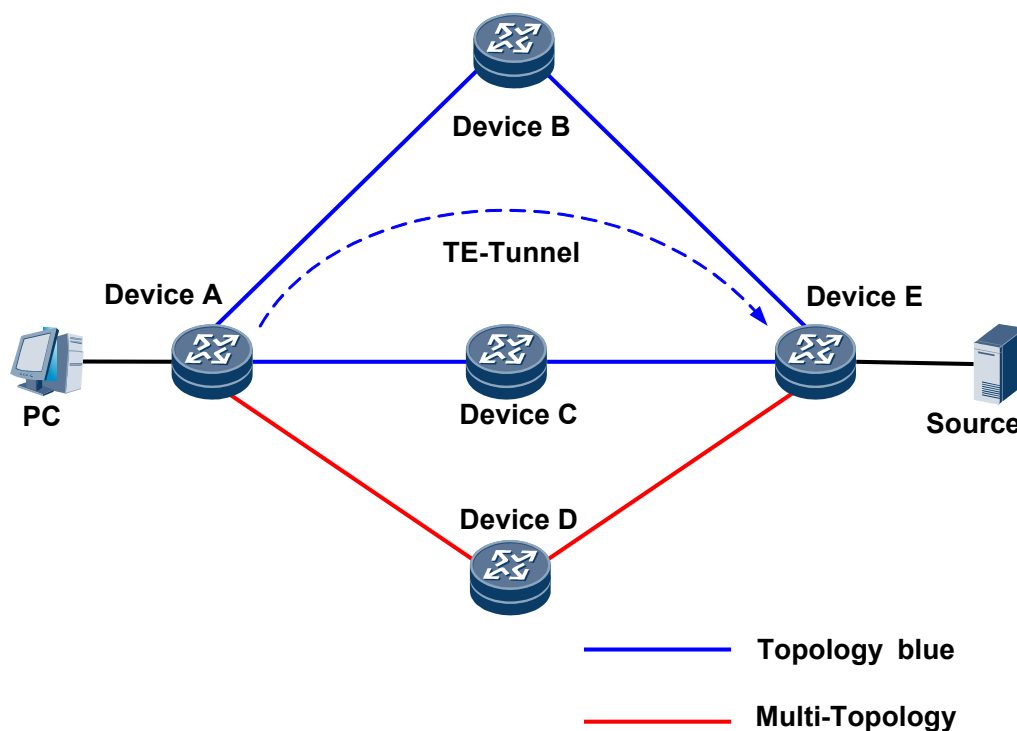


If IS-IS MT is not used, Device A, Device B, Device C, and Device D consider the IPv4 and IPv6 topologies the same when using the SPF algorithm for route calculation. The shortest path from Device A to Device D is Device A -> Device B -> Device D. Device B does not support IPv6 and cannot forward IPv6 packets to Device D.

If IS-IS MT is used to establish a separate IPv6 topology, Device A chooses only IPv6 links to forward IPv6 packets. The shortest path from Device A to Device D changes to Device A -> Device C -> Device D. IPv6 packets are then forwarded.

Figure 8-43 shows the use of IS-IS MT to separate unicast and multicast topologies.

Figure 8-43 Separation of the multicast topology from the unicast topology



All routers in [Figure 8-43](#) are interconnected using IS-IS. A TE tunnel is set up between Device A (ingress) and Device E (egress). The outbound interface of the route calculated by IS-IS may not be a physical interface but a TE tunnel interface. The routers between which the TE tunnel is established cannot set up multicast forwarding entries. As a result, multicast services cannot run properly.

IS-IS MT is configured to solve this problem by establishing separate unicast and multicast topologies. TE tunnels are excluded from a multicast topology; therefore, multicast services can run properly, without being affected by TE tunnels.

9 BGP

About This Chapter

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9.1 Overview of BGP

BGP Definition

Border Gateway Protocol (BGP) is a dynamic routing protocol used between Autonomous Systems (ASs). BGP is widely used by Internet Service Providers (ISPs).

As three earlier-released versions of BGP, BGP-1, BGP-2, and BGP-3 are used to exchange reachable inter-AS routes, establish inter-AS paths, avoid routing loops, and apply routing policies between ASs.

Currently, BGP-4 is used.

BGP has the following characteristics:

- Unlike an Interior Gateway Protocol (IGP), such as Open Shortest Path First (OSPF) and Routing Information Protocol (RIP), BGP is an Exterior Gateway Protocol (EGP) which controls route advertisement and selects optimal routes between ASs rather than discovering or calculating routes.
- BGP uses Transport Control Protocol (TCP) as the transport layer protocol, which enhances BGP reliability.
 - BGP selects inter-AS routes, which poses high requirements on stability. Therefore, using TCP enhances BGP's stability.
 - BGP peers must be logically connected through TCP. The destination port number is 179 and the local port number is a random value.
- BGP supports Classless Inter-Domain Routing (CIDR).
- When routes are updated, BGP transmits only the updated routes, which reduces bandwidth consumption during BGP route distribution. Therefore, BGP is applicable to the Internet where a large number of routes are transmitted.

- BGP is a distance-vector routing protocol.
- BGP is designed to prevent loops.
 - Between ASs: BGP routes carry information about the ASs along the path. The routes that carry the local AS number are discarded to prevent inter-AS loops.
 - Within an AS: BGP does not advertise routes learned in an AS to BGP peers in the AS to prevent intra-AS loops.
- BGP provides many routing policies to flexibly select and filter routes.
- BGP provides a mechanism that prevents route flapping, which effectively enhances Internet stability.
- BGP can be easily extended.

BGP4+ Definition

As a dynamic routing protocol used between ASs, BGP4+ is an extension of BGP.

Traditional BGP4 manages IPv4 routing information but does not support the inter-AS transmission of packets encapsulated by other network layer protocols (such as IPv6).

To support IPv6, BGP4 must have the additional ability to associate an IPv6 protocol with the next hop information and network layer reachable information (NLRI).

Two NLRI attributes that were introduced to BGP4+ are as follows:

- Multiprotocol Reachable NLRI (MP_REACH_NLRI): carries the set of reachable destinations and the next hop information used for packet forwarding.
- Multiprotocol Unreachable NLRI (MP_UNREACH_NLRI): carries the set of unreachable destinations.

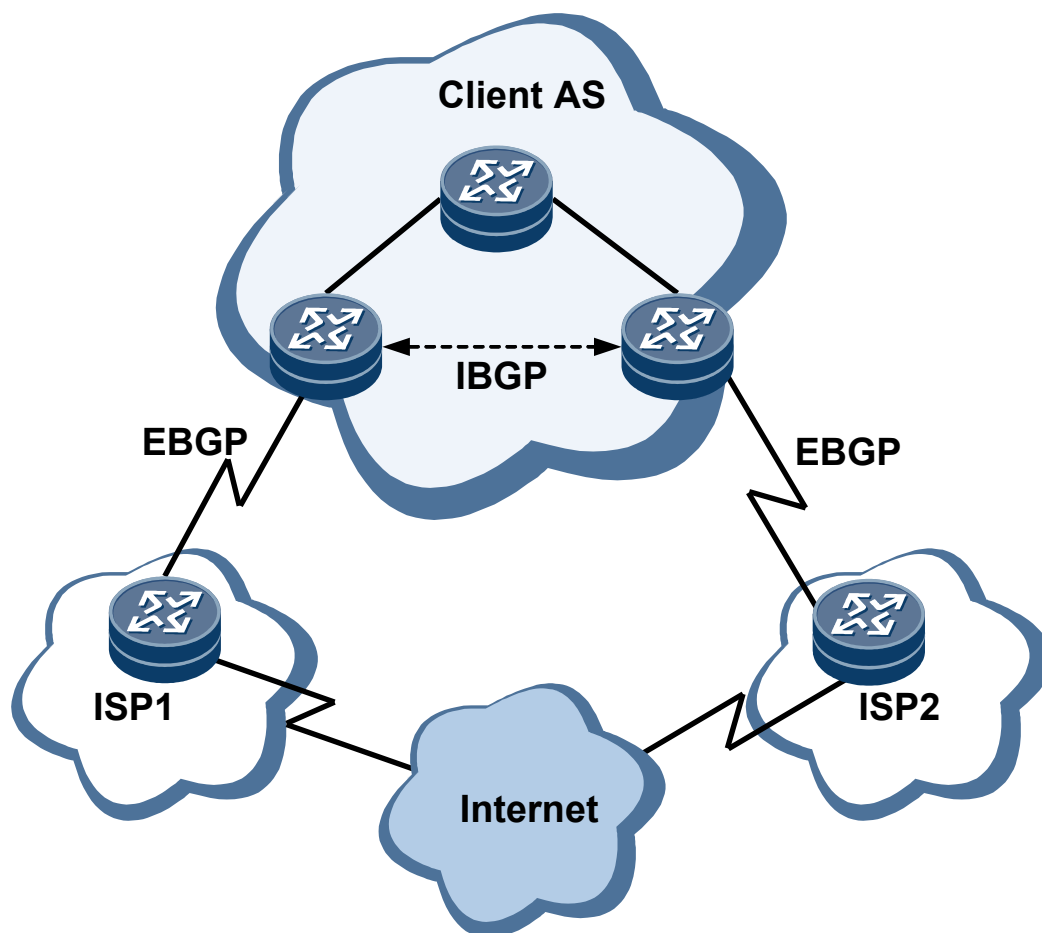
The Next_Hop attribute in BGP4+ is in the format of an IPv6 address, which can be either a globally unique IPv6 address or a next hop link-local address.

Using multiple protocol extensions of BGP4, BGP4+ is applicable to IPv6 networks without changing the messaging and routing mechanisms of BGP4.

Purpose

BGP transmits route information between ASs. It, however, is not required in all scenarios.

Figure 9-1 BGP networking



BGP is required in the following scenarios:

- On the network shown in [Figure 9-1](#), users need to be connected to two or more ISPs. The ISPs need to provide all or part of the Internet routes for the users. Routers, therefore, need to select the optimal route through the AS of an ISP to the destination based on the attributes carried in BGP routes.
- The AS_Path attribute needs to be transmitted between users in different organizations.
- Users need to transmit VPN routes through a Layer 3 VPN. For details, see the *HUAWEI NetEngine40E Feature Description - VPN*.
- Users need to transmit multicast routes and construct a multicast topology. For details, see the *HUAWEI NetEngine40E Feature Description - IP Multicast*.

BGP is not required in the following scenarios:

- Users are connected to only one ISP.
- The ISP does not need to provide Internet routes for users.
- ASs are connected through default routes.

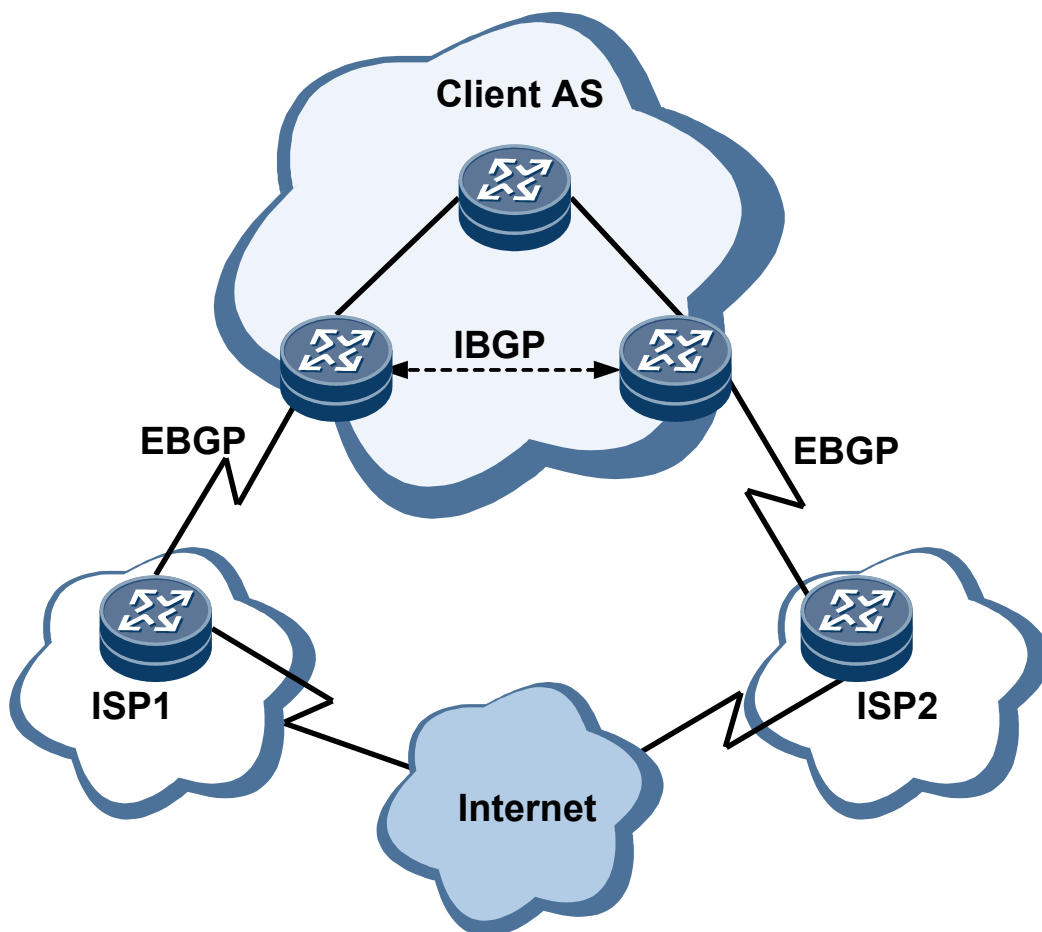
9.2 Understanding BGP

9.2.1 BGP Fundamentals

BGP Operating Modes

BGP is called Internal BGP (IBGP) when it runs within an AS; it is called External BGP (EBGP) when it runs between ASs, as shown in [Figure 9-2](#).

Figure 9-2 BGP operating modes



Roles in Transmitting BGP Messages

- Speaker: Any router that sends BGP messages is called a BGP speaker. The speaker receives or generates new routing information and then advertises the routing information to other BGP speakers. After receiving a route from another AS, a BGP speaker compares the route with its local routes. If the route is better than its local routes, or the route is new, the speaker advertises this route to all other BGP speakers.
- Peer: BGP speakers that exchange messages with each other are called peers.

BGP Messages

BGP runs by sending five types of messages: Open, Update, Notification, Keepalive, and Route-refresh.

- **Open:** The first message sent after a TCP connection is set up is an Open message, which is used to set up BGP peer relationships. After a peer receives an Open message and the peer negotiation is successful, the peer sends a Keepalive message to confirm and maintain the peer relationship. Then, peers can exchange Update, Notification, Keepalive, and Route-refresh messages.
- **Update:** This type of message is used to exchange routes between BGP peers.
 - An Update message can advertise multiple reachable routes with the same attributes. These route attributes are applicable to all destination addresses (expressed by IP prefixes) in the Network Layer Reachability Information (NLRI) field of the Update message.
 - An Update message can be used to delete multiple unreachable routes. Each route is identified by its destination address (using the IP prefix), which identifies the routes previously advertised between BGP speakers.
 - An Update message can be used only to delete routes. In this case, it does not need to carry the route attributes or NLRI. In addition, an Update message can be used only to advertise reachable routes. In this case, it does not need to carry information about the deleted routes.
- **Notification:** When BGP detects an error, it sends a Notification message to its peer. The BGP connection is then torn down immediately.
- **Keepalive:** BGP periodically sends Keepalive messages to peers to maintain peer relationships.
- **Route-refresh:** This type of message is used to request that the peer resend all reachable routes.

If all BGP routers are enabled with the Route-refresh capability and the import policy of BGP changes, the local BGP router sends a Route-refresh message to its peers. After receiving the Route-refresh message, the peers resend their routing information to the local BGP router. In this manner, BGP routing tables are dynamically refreshed and new routing policies are used without tearing down BGP connections.

BGP Finite State Machine

The BGP Finite State Machine (FSM) has six states: Idle, Connect, Active, OpenSent, OpenConfirm, and Established.

Three common states during the establishment of BGP peer relationships are Idle, Active, and Established.

- In the Idle state, BGP denies all connection requests. This is the initial status of BGP.
- In the Connect state, BGP decides subsequent operations after a TCP connection is established.
- In the Active state, BGP attempts to establish a TCP connection.
- In the OpenSent state, BGP is waiting for an Open message from the peer.
- In the OpenConfirm state, BGP is waiting for a Notification or Keepalive message.
- In the Established state, BGP peers can exchange Update, Route-refresh, Keepalive, and Notification messages.

The BGP peer relationship can be established only when both BGP peers are in the Established state. Both peers send Update messages to exchange routes.

BGP Processing

- BGP adopts TCP as its transport layer protocol. Therefore, a TCP connection must be available between the peers. BGP peers negotiate parameters by exchanging Open messages to establish a BGP peer relationship.
- After the peer relationship is established, BGP peers exchange BGP routing tables. BGP does not periodically update a routing table. When BGP routes change, BGP updates the changed BGP routes in the BGP routing table by sending Update messages.
- BGP sends Keepalive messages to maintain the BGP connection between peers.
- After detecting an error on a network, BGP sends a Notification message to report the error and the BGP connection is torn down.

BGP Attributes

BGP route attributes are a set of parameters that describe specific BGP routes. With BGP route attributes, BGP can filter and select routes. BGP route attributes are classified into the following types:

- Well-known mandatory: This type of attribute can be identified by all BGP routers and must be carried in Update messages. Without this attribute, errors occur in the routing information.
- Well-known discretionary: This type of attribute can be identified by all BGP routers. This type of attribute is optional and, therefore, is not necessarily carried in Update messages.
- Optional transitive: This indicates the transitive attribute between ASs. A BGP router may not recognize this attribute, but the router still receives it and advertises it to other peers.
- Optional non-transitive: If a BGP router does not recognize this type of attribute, the router does not advertise it to other peers.

The most common BGP route attributes are as follows:

- Origin

The Origin attribute defines the origin of a route. The Origin attribute is classified into the following types:

 - Interior Gateway Protocol (IGP): This attribute type has the highest priority. IGP is the Origin attribute for routes obtained through an IGP in the AS from which the routes originate. For example, the Origin attribute of the routes imported to the BGP routing table using the **network** command is IGP.
 - Exterior Gateway Protocol (EGP): This attribute type has the second highest priority. The Origin attribute of the routes obtained through EGP is EGP.
 - Incomplete: This attribute type has the lowest priority. Incomplete is the Origin attribute type of all routes that do not have the IGP or EGP Origin attribute. For example, the Origin attribute of the routes imported using the **import-route** command is Incomplete.
- AS_Path

The AS-Path attribute records all ASs through which a route passes from the local end to the destination in distance-vector (DV) order.

When a BGP speaker advertises a local route:

- When advertising the route beyond the local AS, the BGP speaker adds the local AS number to the AS_Path list and then advertises it to the neighboring routers through Update messages.
- When advertising the route within the local AS, the BGP speaker creates an empty AS_Path list in an Update message.

When a BGP speaker advertises a route learned from the Update messages of another BGP speaker:

- When advertising the route beyond the local AS, the BGP speaker adds the local AS number to the left of the AS_Path list. From the AS_Path attribute, the BGP router that receives the route learns the ASs through which the route passes to the destination. The number of the AS that is nearest to the local AS is placed on the left of the list, whereas other AS numbers are listed in sequence.
- When advertising the route within the local AS, the BGP speaker does not change the AS_Path attribute.

The AS_Path attribute has four types:

- AS_Sequence: records in reverse order all the ASs through which a route passes from the local device to the destination.
- AS_Set: records without an order all the ASs through which a route passes through from the local device to the destination. The AS_Set attribute is used in route summarization scenarios. After route summarization, the device records the unsequenced AS numbers because it cannot sequence the numbers of ASs through which specific routes pass. No matter how many AS numbers an AS_Set contains, BGP regards the AS_Set as one AS number when calculating routes.
- AS_Confed_Sequence: records in reverse order all the sub-ASs within a BGP confederation through which a route passes from the local device to the destination.
- AS_Confed_Set: records without an order all the sub-ASs within a BGP confederation through which a route passes from the local device to the destination. The AS_Confed_Set attribute is used in route summarization scenarios in a confederation.

The AS_Confed_Sequence and AS_Confed_Set attributes are used to prevent routing loops and to select routes among the various sub-ASs in a confederation.

- Next_Hop

Different from the Next_Hop attribute in an IGP, the Next_Hop attribute in BGP is not necessarily the IP address of a neighboring router. In most cases, the Next_Hop attribute in BGP complies with the following rules:

- When advertising a route to an EBGP peer, a BGP speaker sets the Next_Hop of the route to the address of the local interface through which the BGP peer relationship is established.
- When advertising a locally generated route to an IBGP peer, a BGP speaker sets the Next_Hop of the route to the address of the local interface through which the BGP peer relationship is established.
- When advertising a route learned from an EBGP peer to an IBGP peer, the BGP speaker does not change the Next_Hop of the route.

- MED

The Multi-Exit-Discriminator (MED) is transmitted only between two neighboring ASs. The AS that receives the MED does not advertise it to a third AS.

Similar to the cost used by an IGP, the MED is used to determine the optimal route when traffic enters an AS. When a BGP peer learns multiple routes that have the same

destination address but different next hops from EBGp peers, the route with the smallest MED value is selected as the optimal route if all other attributes are the same.

- Local_Pref

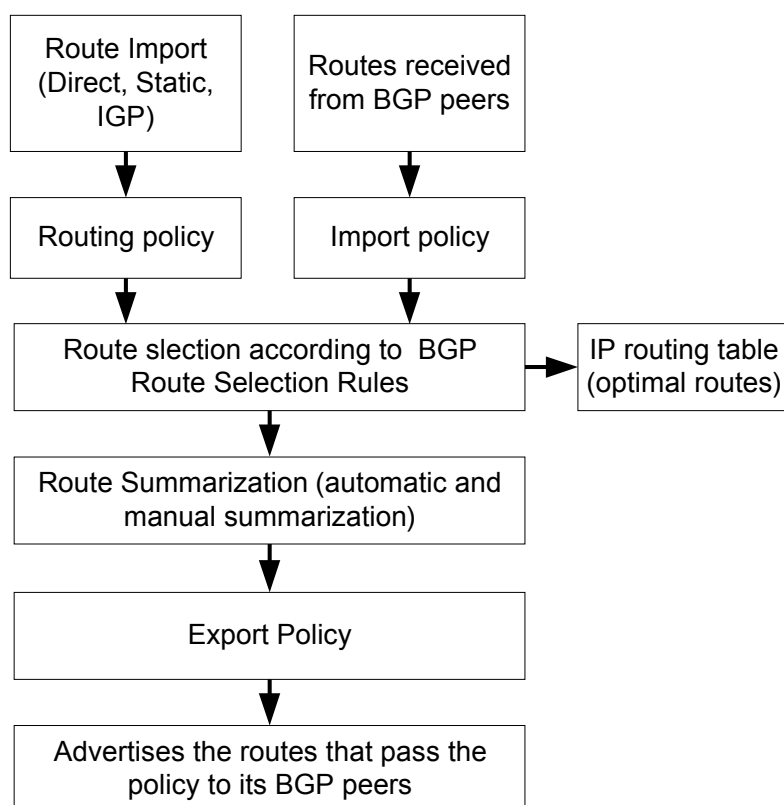
The Local_Pref attribute indicates the BGP priority of a route. It is available only to IBGP peers and is not advertised to other ASs.

The Local_Pref attribute is used to determine the optimal route when traffic leaves an AS. When a BGP router obtains multiple routes to the same destination address but with different next hops through IBGP peers, the route with the largest Local_Pref value is selected.

9.2.2 BGP Route Processing

Figure 9-3 shows how BGP processes routes. BGP routes can be imported from other protocols or learned from peers. To reduce the routing size, you can configure route summarization after BGP selects routes. In addition, you can configure route-policies and apply them to route import, receipt, or advertisement to filter routes or modify route attributes.

Figure 9-3 BGP route processing



For details about route import, see [Route Import](#); for details about BGP route selection rules, see [BGP Route Selection](#); for details about route summarization, see [Route Summarization](#); for details about advertising routes to BGP peers, see [BGP Route Advertisement](#).

For details about import or export policies, see "Routing Policies" in *NE40E Feature Description — IP Routing*.

Route Import

BGP itself cannot discover routes. Therefore, it needs to import other protocol routes, such as IGP routes or static routes, to the BGP routing table. Imported routes can be transmitted within an AS or between ASs.

BGP routes are imported in either of the following modes:

- The **import** command imports routes based on protocol types, such as RIP routes, OSPF routes, Intermediate System to Intermediate System (IS-IS) routes, static routes, or direct routes.
- The **network** command imports a route with the specified prefix and mask to the BGP routing table, which is more precise than the previous mode.

BGP Route Selection

On the NE40E, when multiple routes to the same destination are available, BGP selects routes based on the following rules:

1. Prefers routes in descending order of Valid, Not Found, and Invalid after BGP origin AS validation states are applied to route selection in a scenario where the device is connected to an RPKI server.
2. Prefers routes without bit errors.
If the **bestroute bit-error-detection** command is run, BGP preferentially selects routes without bit error events.
3. Prefers the route with the largest PreVal value.
PrefVal is Huawei-specific. It is valid only on the device where it is configured.
4. Prefers the route with the largest Local_Pref value.
If a route does not carry Local_Pref, the default value 100 takes effect. To change the value, run the **default local-preference** command.
5. Prefers a locally originated route to a route learned from a peer.
Locally originated routes include routes imported using the **network** or **import-route** command, as well as manually and automatically summary routes.
 - a. Prefers a summary route over a non-summary route.
 - b. Prefers a route obtained using the **aggregate** command over a route obtained using the **summary automatic** command.
 - c. Prefers a route imported using the **network** command over a route imported using the **import-route** command.
6. Prefers a route that carries the Accumulated Interior Gateway Protocol Metric (AIGP) attribute.
 - The priority of a route that carries the AIGP attribute is higher than the priority of a route that does not carry the AIGP attribute.
 - If two routes both carry the AIGP attribute, the route with a smaller AIGP attribute value plus IGP metric of the recursive next hop is preferred over the other route.
7. Prefers the route with the shortest AS_Path.
 - The AS_CONFED_SEQUENCE and AS_CONFED_SET are not included in the AS_Path length.
 - During route selection, a router assumes that an AS_SET carries only one AS number regardless of the actual number of ASs it carries.

- If the **bestroute as-path-ignore** command is run, BGP no longer compares the AS_Path attribute.

 NOTE

After the **load-balancing as-path-ignore** command is run, the routes with different As_Path values can load-balance traffic.

8. Prefers the route with the Origin type as IGP, EGP, and Incomplete in descending order.
9. Prefers the route with the smallest MED value.

 NOTE

If the **bestroute med-plus-igp** command is run, BGP preferentially selects the route with the smallest sum of MED multiplied by a MED multiplier and IGP cost multiplied by an IGP cost multiplier.

- BGP compares the MEDs of only routes from the same AS (excluding confederation sub-ASs). MEDs of two routes are compared only when the first AS number in the AS_Sequence (excluding AS_Confed_Sequence) of one route is the same as its counterpart in the other route.
 - If a route does not carry MED, BGP considers its MED as the default value (0) during route selection. If the **bestroute med-none-as-maximum** command is run, BGP considers its MED as the largest MED value (4294967295).
 - If the **compare-different-as-med** command is run, BGP compares MEDs of routes even when the routes are received from peers in different ASs. Do not run this command unless the ASs use the same IGP and route selection mode. Otherwise, a loop may occur.
 - If the **deterministic-med** command is run, routes are no longer selected in the sequence in which they are received.
10. Prefers local VPN routes, LocalCross routes, and RemoteCross routes in descending order.

 NOTE

LocalCross routes indicate local VPN cross routes or routes imported between public network and VPN instances.

If the ERT of a VPNv4 route in the routing table of a VPN instance on a PE matches the IRT of another VPN instance on the PE, the VPNv4 route is added to the routing table of the second VPN instance. This route is called a LocalCross route. If the ERT of a VPNv4 route learned from a remote PE matches the IRT of a VPN instance on the local PE, the VPNv4 route is added to the routing table of that VPN instance. This route is called a RemoteCross route.

11. Prefers EBGP routes to IBGP routes.
12. Prefers the route that recurses to an IGP route with the smallest cost.

If the **bestroute igp-metric-ignore** command is run, BGP no longer compares the IGP cost.

13. Prefers the route with the shortest Cluster_List.

 NOTE

By default, Cluster_List takes precedence over Router ID during BGP route selection. To enable Router ID to take precedence over Cluster_List during BGP route selection, run the **bestroute routerid-prior-clusterlist** command.

14. Prefers the route advertised by the router with the smallest router ID.

If the **bestroute router-id-ignore** command is run, router IDs do not determine which route is selected for BGP.

 NOTE

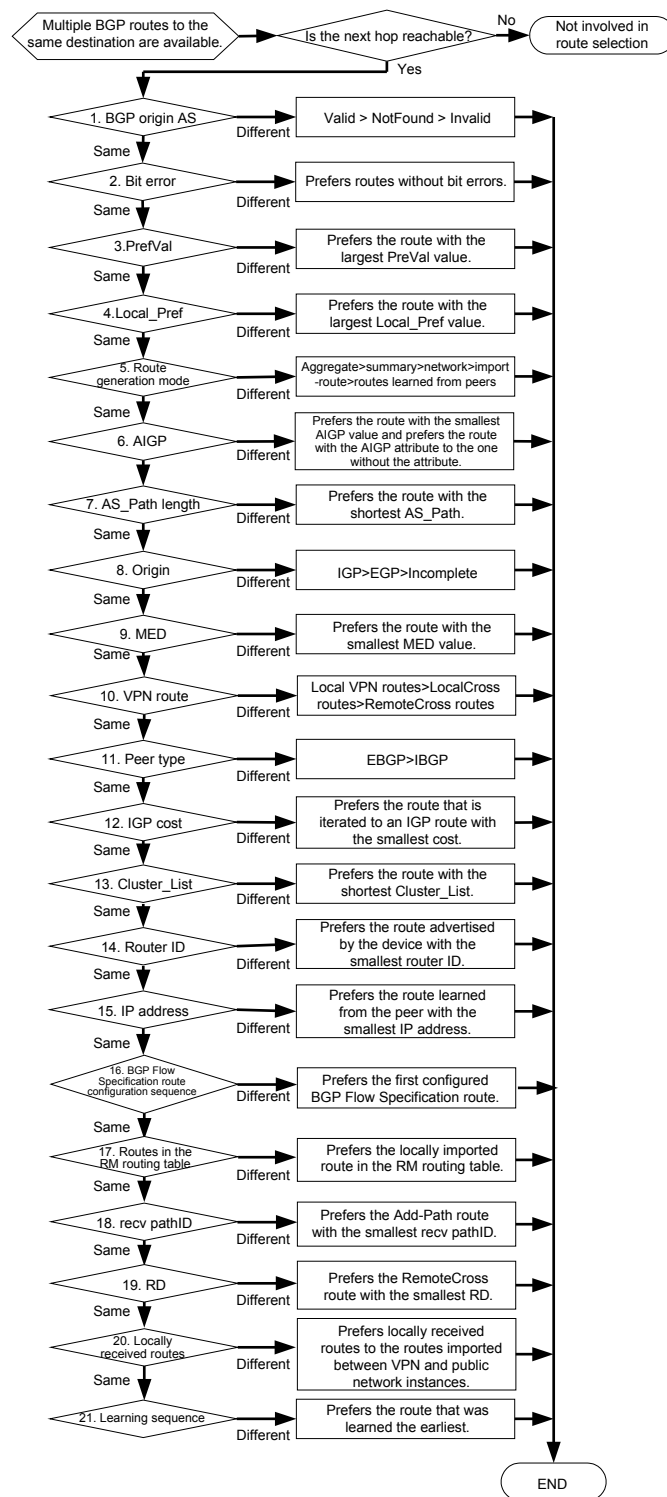
If each route carries an Originator_ID, the originator IDs rather than router IDs are compared during route selection. The route with the smallest Originator_ID is preferred.

15. Prefers the route learned from the peer with the smallest IP address.
16. If BGP Flow Specification routes are configured locally, the first configured BGP Flow Specification route is preferentially selected.
17. Prefers the locally imported route in the RM routing table.
If a direct route, static route, and IGP route are imported, BGP preferentially selects the direct route, static route, and IGP route in descending order.
18. Prefers the Add-Path route with the smallest recv pathID.
19. Prefers the RemoteCross route with the smallest RD.
20. Prefers locally received routes to the routes imported between VPN and public network instances.
21. Prefers the route that was learned the earliest.

For details about BGP route attributes, see [BGP Attributes](#).

For details about the BGP route selection process, see [Figure 9-4](#).

Figure 9-4 BGP route selection process



Route Summarization

On a large-scale network, the BGP routing table can be very large. Route summarization can reduce the size of the routing table.

Route summarization is the process of summarizing specific routes with the same IP prefix into a summary route. After route summarization, BGP advertises only the summary route rather than all specific routes to BGP peers.

BGP supports automatic and manual route summarization.

- Automatic route summarization: takes effect on the routes imported by BGP. With automatic route summarization, the specific routes for the summarization are suppressed, and BGP summarizes routes based on the natural network segment and sends only the summary route to BGP peers. For example, 10.1.1.1/24 and 10.2.1.1/24 are summarized into 10.0.0.0/8, which is a Class A address.
- Manual route summarization: takes effect on routes in the local BGP routing table. With manual route summarization, users can control the attributes of the summary route and determine whether to advertise the specific routes.

IPv4 supports both automatic and manual route summarization, whereas IPv6 supports only manual route summarization.

BGP Route Advertisement

BGP adopts the following policies to advertise routes:

- When there are multiple valid routes, a BGP speaker advertises only the optimal route to its peers.
- A BGP speaker advertises the routes learned from EBGP peers to all BGP peers, including EBGP peers and IBGP peers.
- A BGP speaker does not advertise the routes learned from an IBGP peer to other IBGP peers.
- A BGP speaker advertises the routes learned from IBGP peers to its EBGP peers.
- A BGP speaker advertises all BGP optimal routes to new peers after peer relationships are established.

9.2.3 AIGP

Background

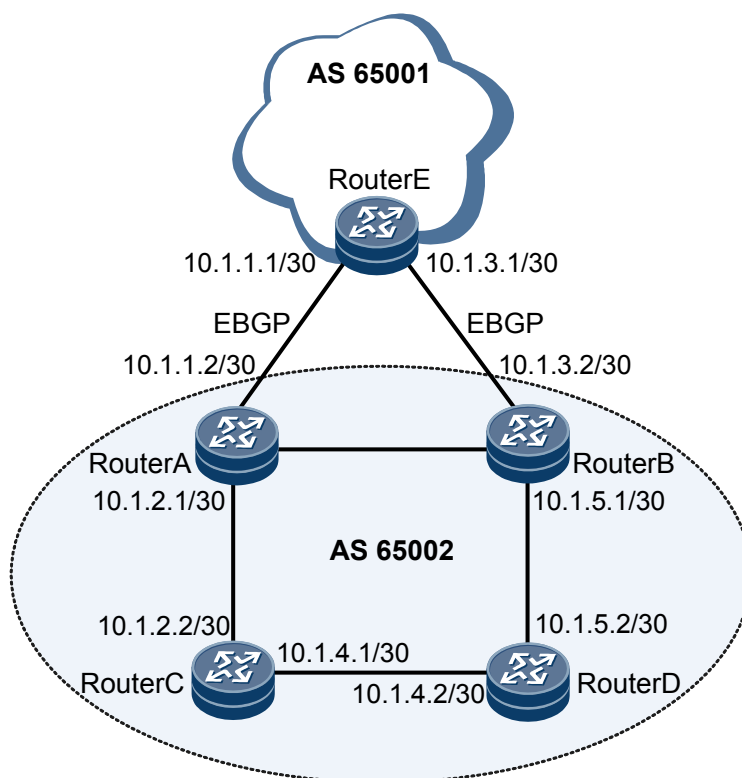
The Accumulated Interior Gateway Protocol Metric (AIGP) attribute is an optional non-transitive Border Gateway Protocol (BGP) path attribute. The attribute type code assigned by the Internet Assigned Numbers Authority (IANA) for the AIGP attribute is 26.

Routing protocols, such as IGP, that have been designed to run within a single administrative domain, generally assign a metric to each link, and then choose the path with the smallest metric as the optimal path between two nodes. BGP, designed to provide routing over a large number of independent administrative domains, does not select paths based on metrics. If a single administrative domain runs several contiguous BGP networks, it is desirable for BGP to select paths based on metrics, just as an IGP does. The AIGP attribute enables BGP to select paths based on metrics.

Related Concepts

An AIGP administrative domain is a set of autonomous systems (ASs) in a common administrative domain. The AIGP attribute takes effect only in an AIGP administrative domain. [Figure 9-5](#) shows the networking diagram of AIGP application.

Figure 9-5 AIGP application networking



Implementation

AIGP Attribute Origination

The AIGP attribute can be added to a route only through a route-policy. You can configure a BGP route to add an AIGP value when routes are imported, received, or sent. If no AIGP value is configured, BGP routes do not contain AIGP attributes.

AIGP Attribute Delivery

BGP does not allow the AIGP attribute to leak out of an AIGP administrative domain boundary onto the Internet. If the AIGP attribute of a route changes, BGP sends Update packets for BGP peers to update information about this route. In a scenario in which A, a BGP speaker, sends a route that carries the AIGP attribute to B, its BGP peer:

- If B does not support the AIGP attribute or does not have the AIGP capability enabled for a peer, B ignores the AIGP attribute and does not transmit the AIGP attribute to other BGP peers.
- If B supports the AIGP attribute and has the AIGP capability enabled for a peer, B can modify the AIGP attribute of the route only after B has set itself to be the next hop of the route. To modify the AIGP attribute of the route, B complies with the following rules:
 - If the BGP peer relationship between A and B is established over an IGP route, or a static route that does not require recursive next hop resolution, B uses the IGP or static route metric value plus the received AIGP attribute value as the new AIGP attribute value of the received route and sends the new AIGP attribute along with the route to other BGP peers.
 - If the BGP peer relationship between A and B is established over a BGP route, or a static route that requires recursive next hop resolution, route recursion occurs when

B sends data to A. Each route recursion requires a pre-existing route. B uses the sum of metric values for recursive routes along the path from B to A plus the received AIGP attribute value as the new AIGP attribute value of the received route and sends the new AIGP attribute along with the route to other BGP peers.

Role of the AIGP Attribute in BGP Route Selection

If multiple active routes exist between two nodes, BGP will make a route selection decision. If BGP cannot determine the optimal route based on PrefVal, Local_Pref, and Route-type, BGP compares the AIGP attributes of these routes. BGP route selection rules are as follows:

- If BGP cannot determine the optimal route based on Route-type, BGP compares the AIGP attributes. If this method still cannot determine the optimal route, BGP proceeds to compare the AS_Path attributes.
- The priority of a route that carries the AIGP attribute is higher than the priority of a route that does not carry the AIGP attribute.
- If all routes carry the AIGP attribute, the route with the smallest AIGP attribute value plus the IGP metric value of the recursive next hop is preferred over the other routes.

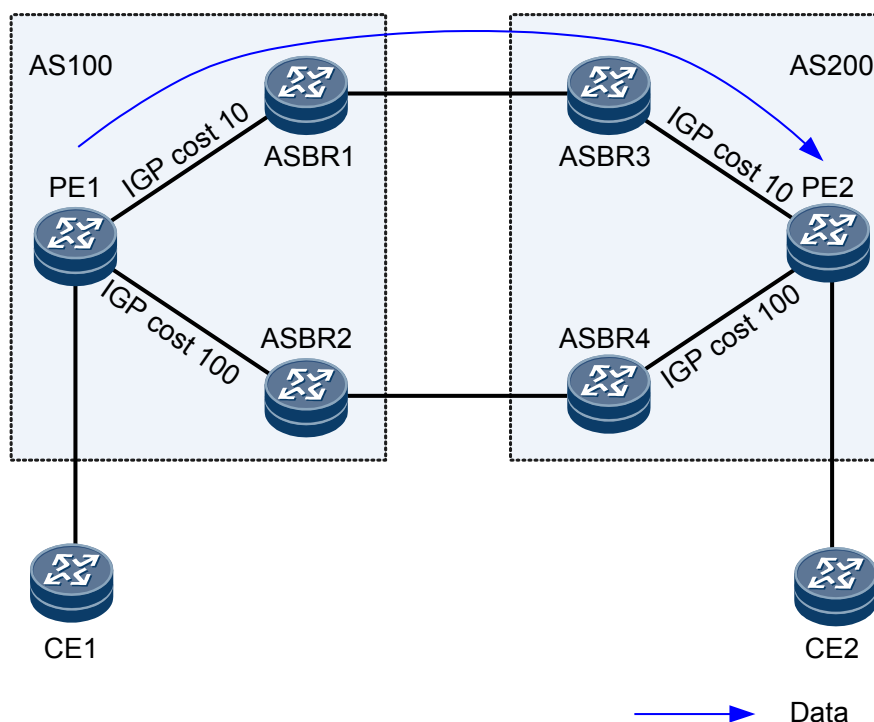
Usage Scenario

The AIGP attribute is used to select the optimal route in an AIGP administrative domain.

The AIGP attribute can be transmitted between BGP unicast peers as well as between BGP VPNv4/VPNv6 peers. Transmitting the AIGP attribute between BGP VPNv4/VPNv6 peers allows L3VPN traffic to be transmitted along the path with the smallest AIGP attribute value.

On the inter-AS VPN Option B network shown in [Figure 9-6](#), BGP VPNv4 peer relationships are established between PE1 and ASBRs 1 and 2; BGP VPNv4 peer relationships are also established between PE2 and ASBRs 3 and 4. Two paths with different IGP costs exist between PE1 and PE2. If you want the PEs to select a path with a lower IGP cost to carry traffic, you can enable the AIGP capability in the BGP VPNv4 address family view and configure a route policy to add the same AIGP initial value to BGP VPNv4 routes. Take PE1 as an example. After this configuration, PE1 receives two BGP VPNv4 routes destined for CE2 from ASBR1 and ASBR2, and the BGP VPNv4 route sent by ASBR1 has a lower AIGP value. If higher-priority route selection conditions of the routes are the same, PE1 preferentially selects the BGP VPNv4 route with a lower AIGP value so that traffic can be transmitted over the PE1 -> ASBR1 -> ASBR3 -> PE2 path.

Figure 9-6 Applying AIGP in inter-AS VPN Option B scenarios



Benefits

After the AIGP attribute is configured in an AIGP administrative domain, BGP selects paths based on metrics, just as an IGP. Consequently, all devices in the AIGP administrative domain use the optimal routes to forward data.

9.2.4 Peer Group

A peer group is a set of peers with the same policies. When a peer joins a peer group, it inherits the configurations of the peer group. If the configurations of the peer group change, the configurations of all the peers in the group changes accordingly.

On a large-scale BGP network, there are many peers and most of them need the same policies. Therefore, some commands need to be run repeatedly on each peer. Configuring a peer group can simplify the configuration.

Each peer in a peer group can be configured with unique policies to advertise and receive routes.

9.2.5 Route Dampening

Route instability is reflected by route flapping. When a route flaps, it repeatedly disappears from the routing table and then reappears.

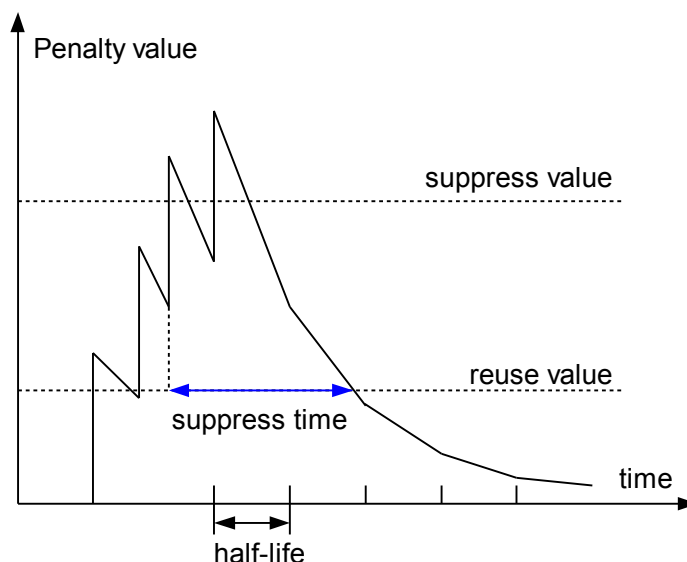
If route flapping occurs, a router sends an Update packet to its peers. After the peers receive the Update packet, they recalculate routes and update their routing tables. Frequent route flapping consumes lots of bandwidth and CPU resources and can even affect network operations.

Route dampening can address this problem. In most cases, BGP is deployed on complex networks where routes change frequently. To reduce the impact of frequent route flapping, BGP adopts route dampening to suppress unstable routes.

BGP dampening measures route stability using a penalty value. The greater the penalty value, the less stable a route. Each time route flapping occurs (a device receives a Withdraw or an Update packet), BGP adds a penalty value to the route carried in the packet. If a route changes from active to inactive, the penalty value increases by 1000. If a route is updated when it is active, the penalty value increases by 500. When the penalty value of a route exceeds the Suppress value, the route is suppressed. As a result, BGP does not add the route to the routing table or advertise any Update message to BGP peers.

The penalty value of a suppressed route reduces by half after a half-life period. When the penalty value decreases to the Reuse value, the route becomes reusable, and BGP adds the route to the IP routing table and advertises an Update packet carrying the route to BGP peers. The penalty value, suppression threshold, and half-life are configurable. **Figure 9-7** shows the process of BGP route dampening.

Figure 9-7 BGP route dampening



Route dampening applies only to EBGp routes and VPNv4 IBGP routes. IBGP routes (except VPNv4 IBGP routes) cannot be dampened because IBGP routing tables contain the routes from the local AS, which require that the forwarding entries be the same on IBGP peers in the AS. If IBGP routes are dampened, the forwarding entries may be inconsistent because dampening parameters may vary among these IBGP peers.

9.2.6 Community Attribute

The community attribute is a group of destination addresses with the same characteristics and consists of a set of 4-byte values that specify a community. On the NE40E, the community attribute is expressed in the format of aa:nn or as a community number.

- aa:nn: aa indicates an AS number and nn indicates the community identifier defined by an administrator. The value of aa or nn ranges from 0 to 65535, which is configurable.

For example, if a route is from AS 100 and the community identifier defined by the administrator is 1, the community is 100:1.

- **Community number:** It is an integer ranging from 0 to 4294967295. As defined in standard protocols, numbers from 0 (0x00000000) to 65535 (0x0000FFFF) and from 4294901760 (0xFFFF0000) to 4294967295 (0xFFFFFFFF) are reserved.

The community attribute is used to simplify the application, maintenance, and management of routing policies. With the community attribute, a group of BGP peers in multiple ASs can share the same routing policy. The community attribute is a route attribute. It is transmitted between BGP peers and is not restricted by the AS. Before advertising a route with the community attribute to peers, a BGP peer can change the original community attribute of this route.

The peers in a peer group share the same policy, while the routes with the same community attribute share the same policy.

The well-known communities are described in the next section. Users can also create their own communities to filter routes.

Well-known Community

Table 9-1 lists well-known communities of BGP routes.

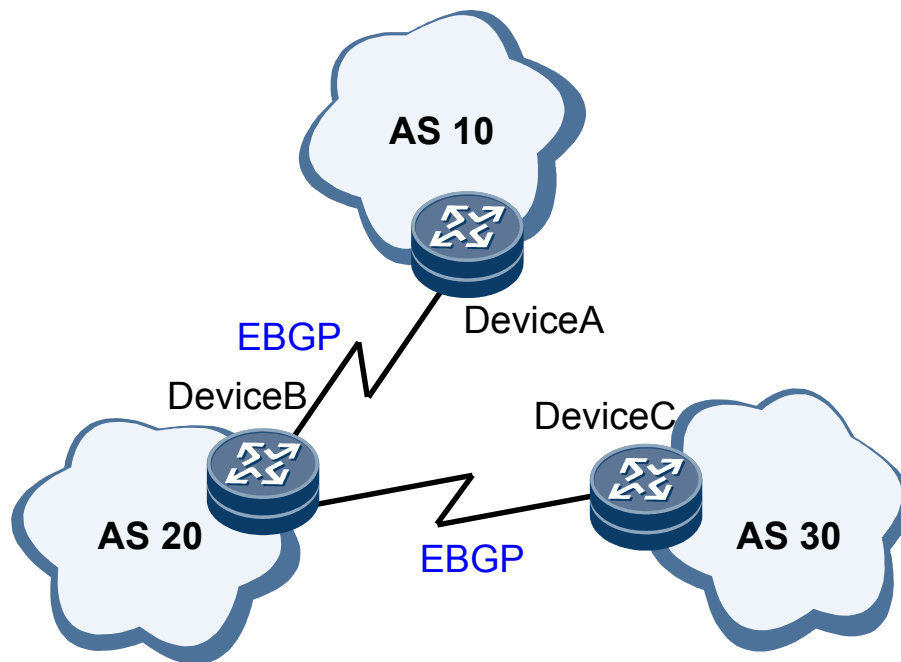
Table 9-1 Well-known communities of BGP routes

Community Name	Community Identifier	Description
Internet	0 (0x00000000)	By default, all routes belong to the Internet community. A route with this attribute can be advertised to all BGP peers.
No_Export	4294967041 (0xFFFFFFFF01)	A route with this attribute cannot be advertised beyond the local AS.
No_Advertise	4294967042 (0xFFFFFFFF02)	A route with this attribute cannot be advertised to any other BGP peers.
No_Export_Subconf ed	4294967043 (0xFFFFFFFF03)	A route with this attribute cannot be advertised beyond the local AS or to other sub-ASs.

Usage Scenario

On the network shown in **Figure 9-8**, EBGP connections are established between Device A and Device B, and between Device B and Device C. If the community attribute of No_Export is configured on Device A and Device A sends a route with the community attribute to Device B, Device B does not advertise the route to other ASs after receiving it.

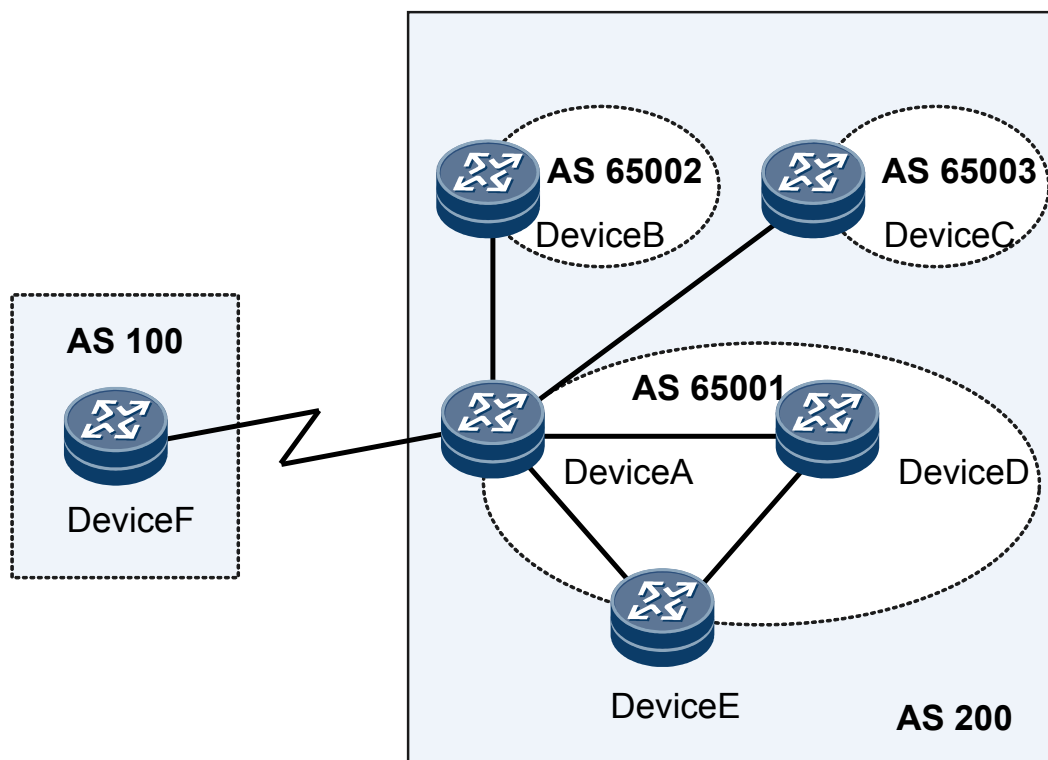
Figure 9-8 Networking for BGP communities



9.2.7 BGP Confederation

Besides RR, BGP confederation can also reduce IBGP connections in an AS. It divides an AS into several sub-ASs. Fully meshed IBGP connections are established in each sub-AS, and fully meshed EBGP connections are established between sub-ASs.

Figure 9-9 BGP confederation



In **Figure 9-9**, there are multiple BGP routers in AS 200. To reduce the number of IBGP connections, AS 200 is divided into three sub-ASs: AS 65001, AS 65002, and AS 65003. In AS 65001, fully meshed IBGP connections are established between the three routers.

BGP speakers outside a confederation such as Router F in AS 100, do not know the existence of the sub-ASs (AS 65001, AS 65002, and AS 65003) in the confederation. The confederation ID is the AS number that is used to identify the entire confederation. For example, AS 200 in **Figure 9-9** is the confederation ID.

Applications and Limitations

The confederation needs to be configured on each router, and the router that joins the confederation must support the confederation function.

BGP speakers need to be reconfigured when a network in non-confederation mode switches to confederation mode. As a result, the logical topology changes accordingly.

On large-scale BGP networks, the RR and confederation can both be used.

9.2.8 Route Reflector

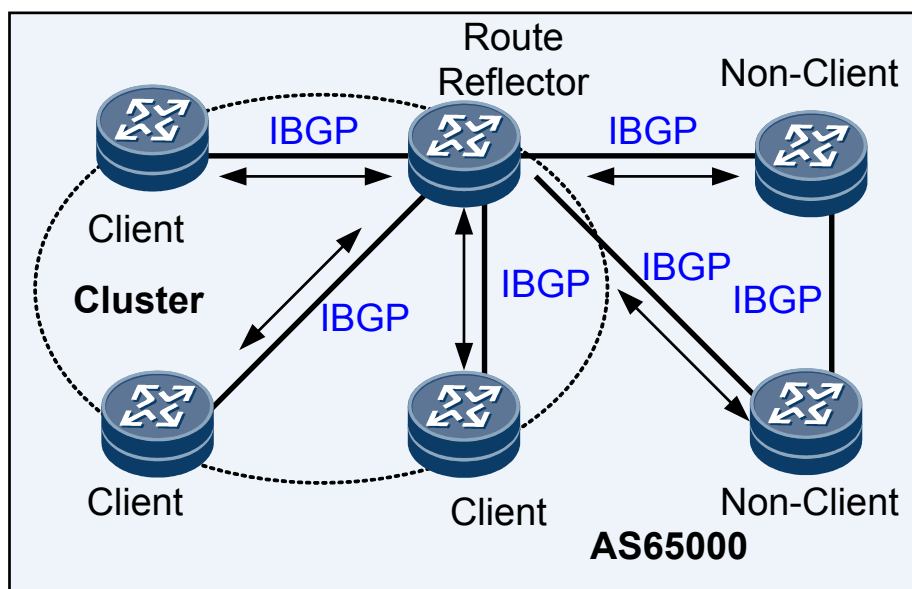
Fully meshed connections need to be established between IBGP peers to ensure the connectivity between IBGP peers. If there are n routers in an AS, $n \times (n-1)/2$ IBGP connections need to be established. When there are a lot of IBGP peers, network resources and CPU resources are greatly consumed. Route reflection can solve the problem.

In an AS, one router functions as a Route Reflector (RR) and the other routers as clients. The clients establish IBGP connections with the RR. The RR and its clients form a cluster. The RR

reflects routes among clients, and BGP connections do not need to be established between the clients.

A BGP peer that functions as neither an RR nor a client is called a non-client. A non-client must establish full meshed connections with the RR and all the other non-clients, as shown in [Figure 9-10](#).

Figure 9-10 Networking with an RR



Applications

After an RR receives routes from its peers, it selects the optimal route based on BGP route selection policies and performs one of the following operations:

- If the optimal route is from a non-client IBGP peer, the RR advertises the route to all clients.
- If the optimal route is from a client, the RR advertises the route to all non-clients and clients.
- If the optimal route is from an EBGP peer, the RR advertises the route to all clients and non-clients.

An RR is easy to configure because it only needs to be configured on the RR itself, and clients do not need to know whether they are clients.

On some networks, if fully meshed connections have already been established among clients of an RR, they can exchange routing information directly. In this case, route reflection among the clients through the RR is unnecessary and occupies bandwidth. For example, on the NE40E, route reflection can be disabled, but the routes between clients and non-clients are still exchanged. By default, route reflection between clients through the RR is enabled.

On the NE40E, an RR can change various attributes of BGP routes, such as the AS_Path, MED, Local_Pref, and community attributes.

Originator_ID

Originator_ID and Cluster_List are used to detect and prevent routing loops.

The Originator_ID attribute is four bytes long and is generated by an RR. It carries the router ID of the route originator in the local AS.

- When a route is reflected by an RR for the first time, the RR adds the Originator_ID to this route. If a route already carries the Originator_ID attribute, the RR does not create a new one.
- After receiving the route, a BGP speaker checks whether the Originator_ID is the same as its router ID. If Originator_ID is the same as its router ID, the BGP speaker discards this route.

Cluster_List

To prevent routing loops between ASs, a BGP router uses the AS_Path attribute to record the ASs through which a route passes. Routes with the local AS number are discarded by the router. To prevent routing loops within an AS, IBGP peers do not advertise routes learned from the local AS.

With RR, IBGP peers can advertise routes learned from the local AS to each other. However, the Cluster_List attribute must be deployed to prevent routing loops within the AS.

An RR and its clients form a cluster. In an AS, each RR is uniquely identified by a Cluster_ID.

Similar to an AS_Path, a Cluster_List is composed of a series of Cluster_IDs and is generated by an RR. The Cluster_List records all the RRs through which a route passes.

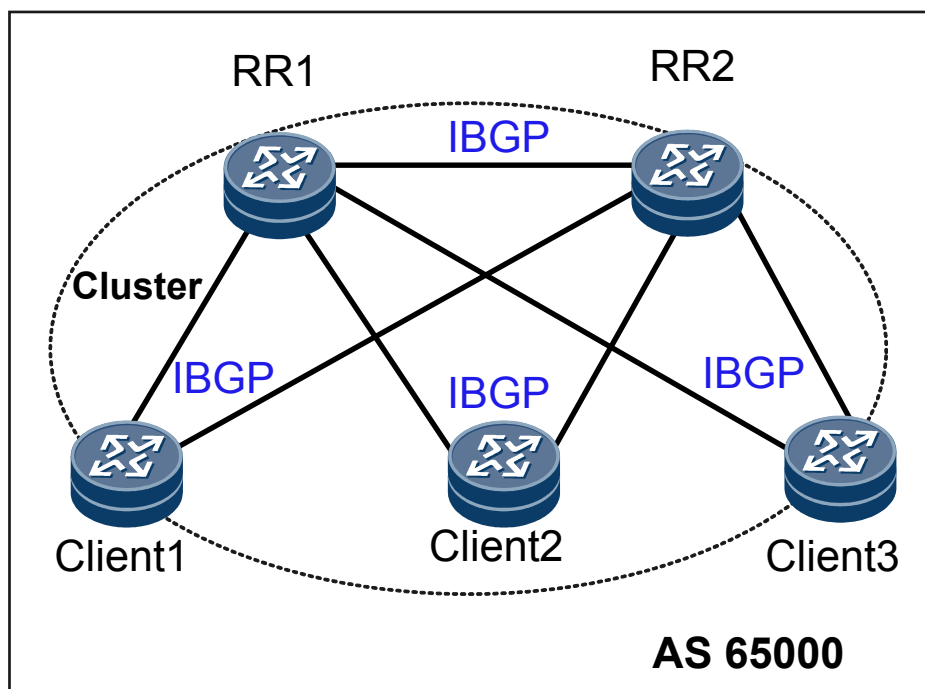
- Before an RR reflects a route between its clients or between its clients and non-clients, the RR adds the local Cluster_ID to the head of the Cluster_List. If a route does not carry any Cluster_List, the RR creates one for the route.
- After the RR receives an updated route, it checks the Cluster_List of the route. If the RR finds that its cluster ID is included in the Cluster_List, the RR discards the route. If its cluster ID is not included in the Cluster_List, the RR adds its cluster ID to the Cluster_List and then reflects the route.

Backup RR

To enhance network reliability and prevent single points of failure, more than one route reflector needs to be configured in a cluster. The route reflectors in the same cluster must share the same Cluster_ID to prevent routing loops.

With backup RRs, clients can receive multiple routes to the same destination from different RRs. The clients then apply route selection policies to choose the optimal route.

Figure 9-11 Backup RR



On the network shown in [Figure 9-11](#), RR1 and RR2 are in the same cluster. RR1 and RR2 establish an IBGP connection so that each RR is a non-client of the other RR.

- If Client 1 receives an updated route from an external peer, Client 1 advertises the route to RR1 and RR2 through IBGP.
- After receiving the updated route, RR1 reflects the route to other clients (Client 2 and Client 3) and the non-client (RR2) and adds the local Cluster_ID to the head of the Cluster_List.
- After receiving the reflected route, RR2 checks the Cluster_List. RR2 finds that its Cluster_ID is contained in the Cluster_List; therefore, it discards the updated route.

If RR1 and RR2 are configured with different Cluster_IDs, each RR receives both the route from Client 1 and the updated route reflected from the other RR. Therefore, configuring the same Cluster_ID for RR1 and RR2 reduces the number of routes that each RR receives and memory consumption.

NOTE

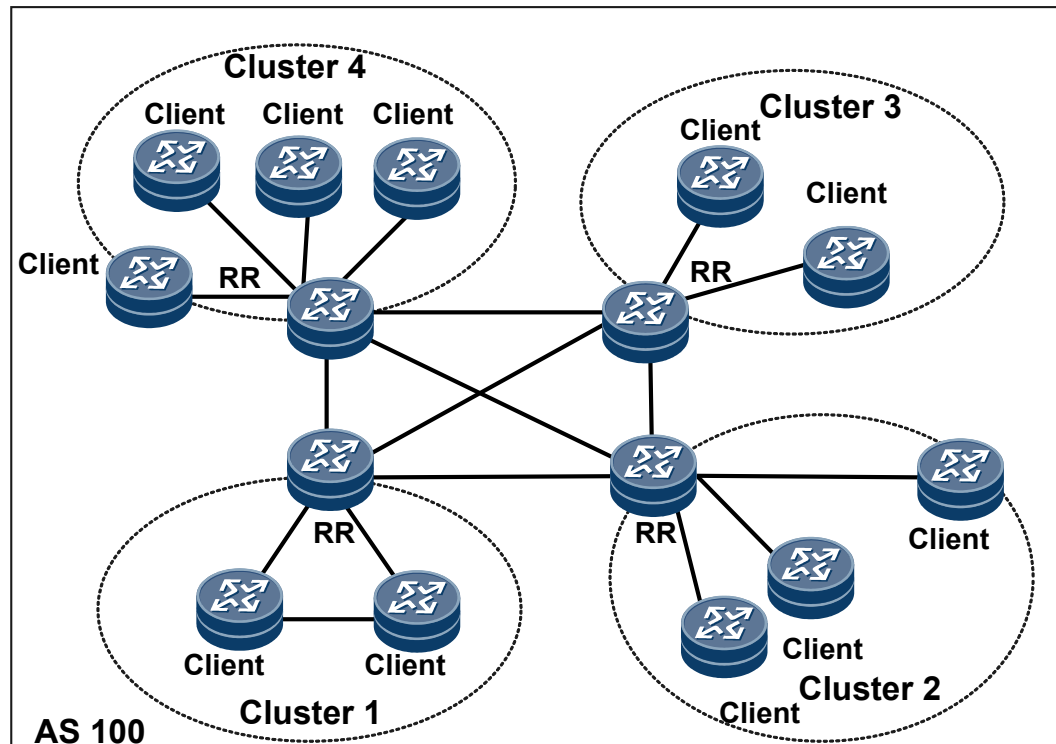
The application of Cluster_List prevents routing loops among RRs in the same AS.

Multiple Clusters in an AS

Multiple clusters may exist in an AS. RRs are IBGP peers of each other. An RR can be configured as a client or non-client of another RR.

For example, the backbone network shown in [Figure 9-12](#) is divided into multiple clusters. Each RR is configured as a non-client of the other RRs, and these RRs are fully meshed. Each client establishes IBGP connections with only the RR in the same cluster. In this manner, all BGP peers in the AS can receive reflected routes.

Figure 9-12 Multiple clusters in an AS

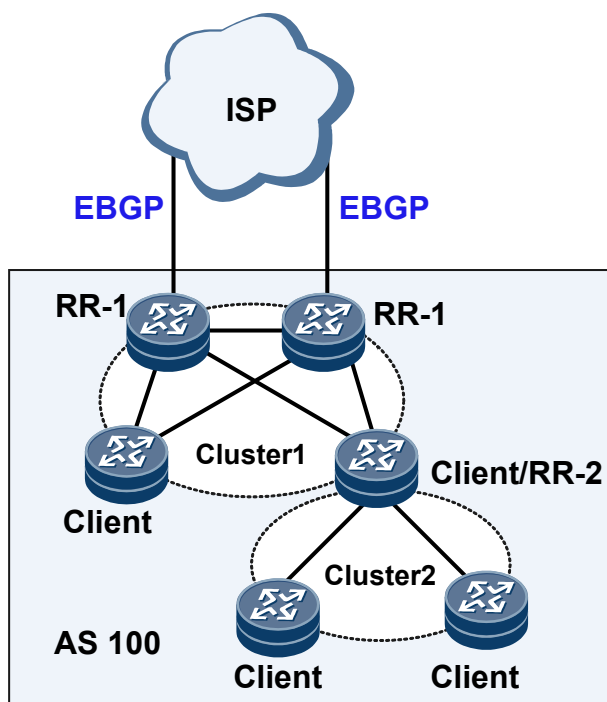


Hierarchical Reflector

Hierarchical reflectors are deployed on live networks. On the network shown in [Figure 9-13](#), the ISP provides Internet routes for AS 100. Two EBGP connections are established between the ISP and AS 100. AS 100 is divided into two clusters. The four routers in Cluster 1 are core routers.

- Two Level-1 RRs (RR-1s) are deployed in Cluster 1, which ensures the reliability of the core layer of AS 100. The other two routers in the core layer are clients of RR-1s.
- One Level-2 RR (RR-2) is deployed in Cluster 2. RR-2 is a client of RR-1.

Figure 9-13 Hierarchical reflector



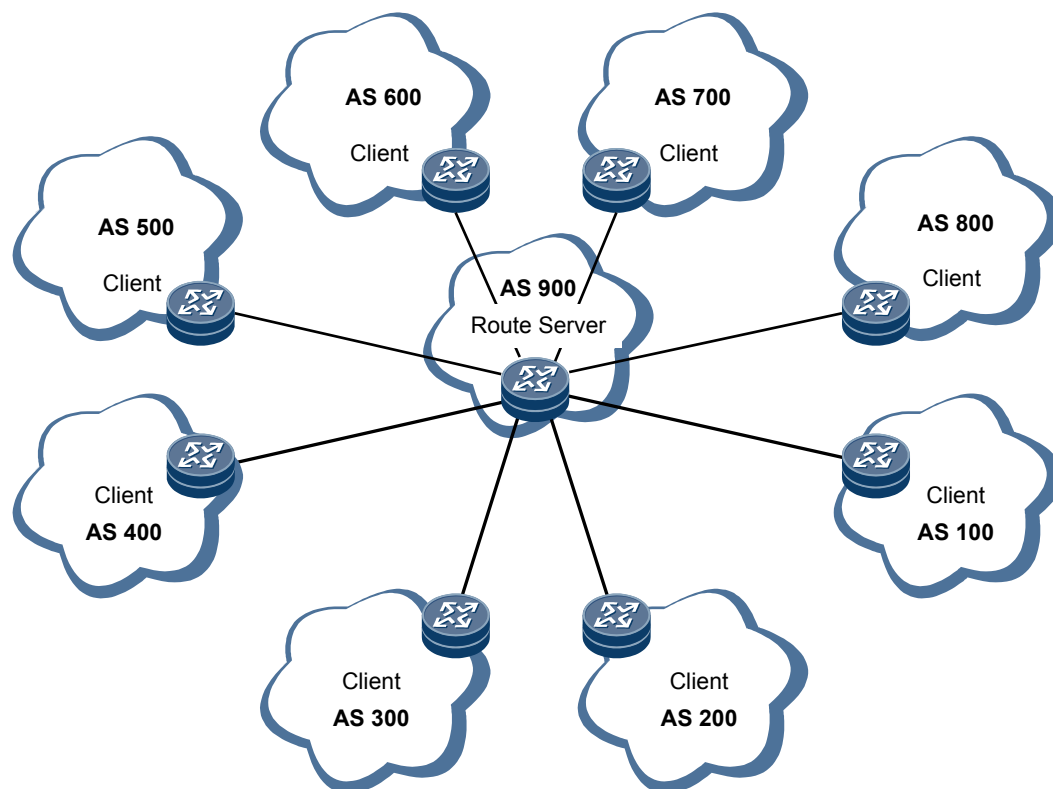
9.2.9 Route Server

The route server function is similar to the RR function in IBGP scenarios and allows devices to advertise routes to their clients (ASBR devices) without changing route attributes, such as AS_Path, Nexthop, and MED. With the route server function, EBGP full-mesh connections are not required among the ASBR devices, which reduces network resource consumption.

Application

In some scenarios on the live network, to achieve network traffic interworking, EBGP full-mesh connections may be required. However, establishing full-mesh connections among devices that function as ASBRs is costly and places high requirements on the performance of the devices, which adversely affects the network topology and device expansion. In [Figure 9-14](#), the route server can advertise routes to all its EBGP peers, without requiring EBGP full-mesh connections among ASBRs. Therefore, the route server function reduces network resource consumption.

Figure 9-14 Route server networking



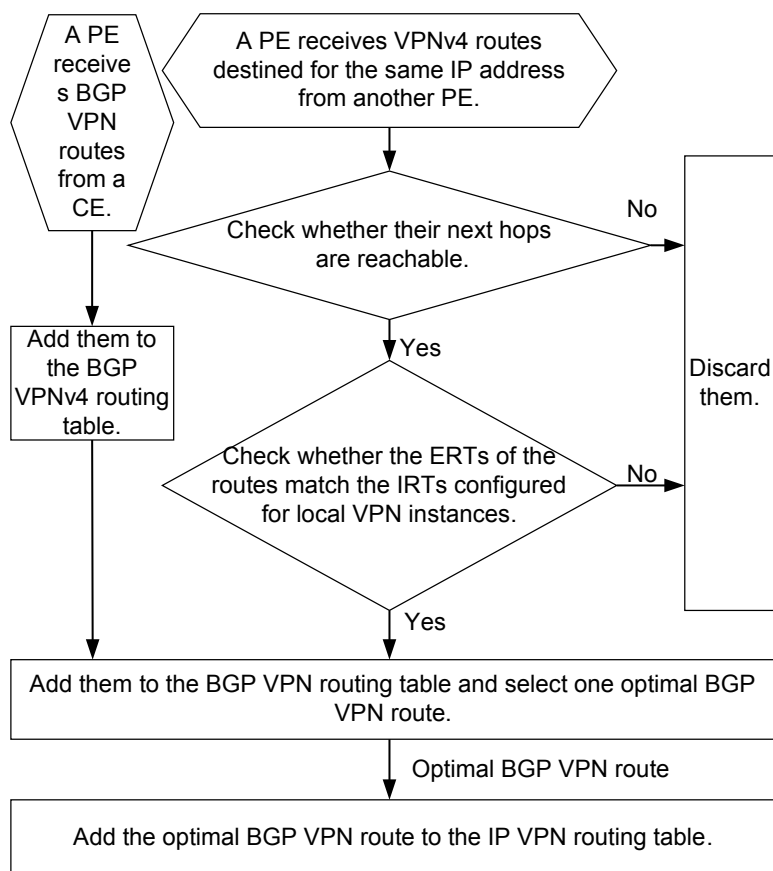
9.2.10 BGP VPN Route Crossing

Route crossing refers to the process of adding a BGP VPN route to the routing table of the local or remote VPN instance in a BGP/MPLS IP VPN scenario. Route crossing can be classified as local route crossing and remote route crossing based on the source of the BGP VPN route.

- Remote route crossing: After a PE receives a BGP VPNv4 route from a remote PE, the local PE matches the export target (ERT) of the route against the import targets (IRTs) configured for local VPN instances. If the ERT matches the IRT of a local VPN instance, the PE converts the BGP VPNv4 route to a BGP VPN route and adds the BGP VPN route to the routing table of this local VPN instance.
- Local route crossing: A PE matches the ERT of a BGP VPN route in a local VPN instance against the IRTs configured for other local VPN instances. If the ERT matches the IRT of a local VPN instance, the PE adds the BGP VPN route to the routing table of this local VPN instance. Locally crossed routes include both locally imported routes and routes learned from VPN peers.

After a PE receives VPNv4 routes destined for the same IP address from another PE or VPN routes from a CE, the local PE implements route crossing by following the steps in [Figure 9-15](#).

Figure 9-15 Flowchart for BGP VPN route crossing



In [Figure 9-16](#), PEs have the same VPN instance (**vpna**) and RTs (including the ERT and IRT). The RD configured for PE2 and PE3 is 2:2, and the RD configured for PE4 is 3:3. Site 2 has a route destined for 10.1.1.0/24. The route is sent to PE2, PE3, and PE4, who convert this route to multiple BGP VPNv4 routes and send them to PE1. On receipt of the BGP VPNv4 routes, PE1 implements route crossing as shown in [Figure 9-17](#). The detailed process is as follows:

1. After receiving the BGP VPNv4 routes from PE2, PE3, and PE4, PE1 adds them to the BGP VPNv4 routing table.
2. PE1 converts the BGP VPNv4 routes to BGP VPN routes by removing their RDs, adds the BGP VPN routes to the routing table of the VPN instance, selects an optimal route from the BGP VPN routes based on BGP route selection policies, and adds the optimal BGP VPN route to the IP VPN instance routing table.

Figure 9-16 Route crossing networking

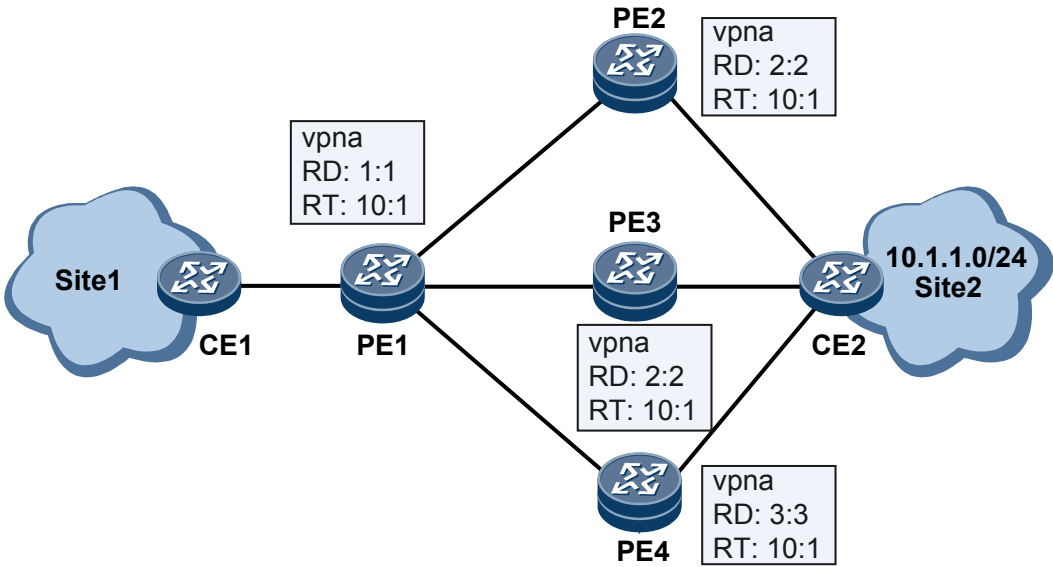
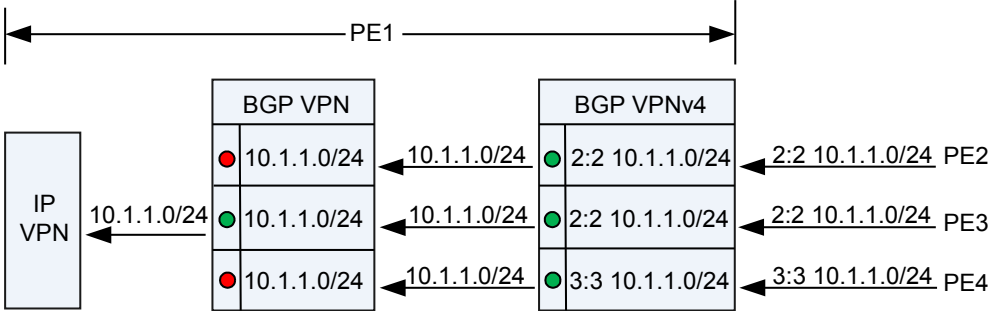


Figure 9-17 BGP VPN route crossing process



9.2.11 MP-BGP

Conventional BGP-4 manages only IPv4 unicast routing information, and inter-AS transmission of packets of other network layer protocols, such as multicast, is limited.

To support multiple network layer protocols, the Internet Engineering Task Force (IETF) extends BGP-4 to Multiprotocol Extensions for BGP-4 (MP-BGP). MP-BGP is forward compatible. Specifically, routers supporting MP-BGP can communicate with the routers that do not support MP-BGP.

As an enhancement of BGP-4, MP-BGP provides routing information for various routing protocols, including IPv6 (BGP-4) and multicast.

- MP-BGP maintains both unicast and multicast routes. It stores them in different routing tables to separate unicast routing information from multicast routing information.
- MP-BGP supports both unicast and multicast address families and can build both the unicast routing topology and multicast routing topology.

- Most unicast routing policies and configurations supported by BGP-4 can be applied to multicast, and BGP-4 can maintain unicast and multicast routes according to these routing policies.

Extended Attributes

BGP-4 Update packets carry three IPv4-related attributes: NLRI (Network Layer Reachable Information), Next_Hop, and Aggregator. Aggregator contains the IP address of the BGP speaker that performs route aggregation.

To carry information about multiple network layer protocols in NLRI and Next_Hop, MP-BGP introduces the following route attributes:

- MP_REACH_NLRI: indicates the multiprotocol reachable NLRI. It is used to advertise a reachable route and its next hop.
- MP_UNREACH_NLRI: indicates the multiprotocol unreachable NLRI. It is used to delete an unreachable route.

The preceding two attributes are optional non-transitive. Therefore, the BGP speakers that do not support MP-BGP will ignore the information carried in the two attributes and do not advertise the information to other peers.

Address Family

The Address Family Information field consists of a 2-byte Address Family Identifier (AFI) and a 1-byte Subsequent Address Family Identifier (SAFI).

BGP uses address families to distinguish different network layer protocols. For the values of address families, see relevant standards. The NE40E supports multiple MP-BGP extension applications, such as VPN extension and IPv6 extension, which are configured in their respective address family views.

For details about the BGP VPNv4 address family and BGP VPN instance address family, see the *HUAWEI NetEngine40E Universal Service Router Feature Description - VPN*.

9.2.12 BGP Security

BGP Authentication

BGP can work properly only after BGP peer relationships are established. Authenticating BGP peers can improve BGP security. BGP supports the following authentication modes:

- MD5 authentication
BGP uses TCP as the transport layer protocol. Message Digest 5 (MD5) authentication can be used when establishing TCP connections to improve BGP security. MD5 authentication sets the MD5 authentication password for the TCP connection, and TCP performs the authentication. If the authentication fails, the TCP connection cannot be established.
- Keychain authentication
Keychain authentication is performed on the application layer. It ensures smooth service transmission and improves security by periodically changing the password and encryption algorithms. When keychain authentication is configured for BGP peer relationships over TCP connections, BGP packets as well as the establishment process of

a TCP connection can be authenticated. For details about keychain, see "Keychain" in *HUAWEI NetEngine40E Feature Description - Security*.

GTSM

During network attacks, attackers may simulate BGP packets and continuously send them to the router. If the packets are destined for the router, it directly forwards them to the control plane for processing without validating them. As a result, the increased processing workload on the control plane results in high CPU usage.

The Generalized TTL Security Mechanism (GTSM) defends against attacks by checking whether the time to live (TTL) value in each IP packet header is within a pre-defined range. TTL refers to the maximum number of routers through which a packet can pass.

In actual networking, packets whose TTL values are not within the specified range are either allowed to pass or discarded by the GTSM. To configure the GTSM to discard packets, you need to set an appropriate TTL value range according to the network topology. Then, packets whose TTL values are not within the specified range are discarded, which prevents the local device from potential attacks.

You can also enable the log function to record discarded packets for further fault location.

RPKI

Resource Public Key Infrastructure (RPKI) improves BGP security by validating the origin ASs of BGP routes.

Attackers can steal user data by advertising routes that are more specific than those advertised by carriers. For example, if a carrier has advertised a route destined for 10.10.0.0/16, an attacker can advertise a route destined for 10.10.153.0/16, which is more specific than 10.10.0.0/16. According to the longest match rule, 10.10.153.0/16 is preferentially selected for traffic forwarding. As a result, the attacker succeeds in intercepting user data.

To address this issue, establish an RPKI session between a router and an RPKI server. The router will then query Route Origin Authorizations (ROAs) from the RPKI server through the RPKI session and match the origin AS of each received BGP route against the ROAs. This mechanism ensures that only the routes that originate from the trusted ASs are accepted. The validation result can also be applied to BGP route selection to ensure that hosts in the local AS can communicate with hosts in other ASs.

SSL/TLS authentication

Secure Sockets Layer (SSL) is a security protocol that protects data privacy on the Internet. Transport Layer Security (TLS) is a successor of SSL. TLS protects data integrity and privacy by preventing attackers from eavesdropping the data exchanged between a client and server. To ensure data transmission security on a network, SSL/TLS authentication can be enabled for BGP message encryption.

9.2.13 BGP GR

Graceful restart (GR) is one of the high availability (HA) technologies, which comprise a series of comprehensive technologies such as fault-tolerant redundancy, link protection, faulty node recovery, and traffic engineering. As a fault-tolerant redundancy technology, GR ensures normal forwarding of data during the restart of routing protocols to prevent interruption of key services. Currently, GR has been widely applied to the master/slave switchover and system upgrade.

GR is usually used when the active route processor (RP) fails because of a software or hardware error, or used by an administrator to perform the master/slave switchover.

Prerequisite for Implementation

On a traditional routing device, a processor implements both control and forwarding. The processor finds routes based on routing protocols, and maintains the routing table and forwarding table of the device. Mid-range and high-end devices generally adopt the multi-RP structure to improve forwarding performance and reliability. The processor in charge of routing protocols is located on the main control board, whereas the processor responsible for data forwarding is located on the interface board. The design helps to ensure the continuity of packet forwarding on the interface board during the restart of the main processor. The technology that separates control from forwarding satisfies the prerequisite for GR implementation.

At present, a GR-capable device must have two main control boards. In addition, the interface board must have an independent processor and memory.

Related Concepts

The concepts related to GR are as follows:

- GR Restarter: indicates a device that performs master/slave switchover triggered by the administrator or a failure. A GR Restarter must support GR.
- GR Helper: indicates the neighbor of a GR Restarter. A GR Helper must support GR.
- GR session: indicates a session, through which a GR Restarter and a GR Helper can negotiate GR capabilities.
- GR time: indicates the time when the GR Helper finds that the GR Restarter is Down but keeps the topology information or routes obtained from the GR Restarter.
- End-of-RIB (EOR): indicates BGP information, notifying a peer BGP that the first route upgrade is finished after the negotiation.
- EOR timer: indicates a maximum time of a local device waiting for the EOR information sent from the peer. If the local device does not receive the EOR information from the peer within the EOR timer, the local device will select an optimal route from the current routes.

Principles

Principles of BGP GR are as follows:

1. During BGP peer relationship establishment, devices negotiate GR capabilities by sending supported GR capabilities to each other.
2. When detecting the master/slave switchover of the GR Restarter, a GR Helper does not delete the routing information and forwarding entries related to the GR Restarter within the GR time, but waits to re-establish a BGP connection with the GR Restarter.
3. After the master/slave switchover, the GR Restarter receives routes from all the negotiated peers with GR capabilities before the switchover, and starts the EOR timer. The GR Restarter selects a route when either of the following conditions is met:
 - The GR Restarter receives the EOR information of all peers and the EOR timer is deleted.
 - The EOR timer times out but the GR Restarter receives no EOR information from all peers.

4. The GR Restarter sends the optimal route to the GR Helper and the GR Helper starts the EOR timer. The GR Helper quits GR when either of the following conditions is met:
 - The GR Helper receives the EOR information from the GR Restarter and the EOR timer is deleted.
 - The EOR timer times out and the GR Helper receives no EOR information from the GR Restarter.

GR Reset

Currently, BGP does not support dynamic capability negotiation. Therefore, each time a new BGP capability (such as the IPv4, IPv6, VPNv4, and VPNv6 capabilities) is enabled on a BGP speaker, the BGP speaker tears down existing sessions with its peer and renegotiates BGP capabilities. This process will interrupt ongoing services.

To prevent the service interruptions, the NE40E provides the GR reset function that enables the NE40E to reset a BGP session in GR mode. With the GR reset function configured, when you enable a new BGP capability on the BGP speaker, the BGP speaker enters the GR state, resets the BGP session, and renegotiates BGP capabilities with the peer. In the whole process, the BGP speaker re-establishes the existing sessions but does not delete the routing entries for the existing sessions, so that the existing services are not interrupted.

Benefits

Through BGP GR, the forwarding is not interrupted. In addition, the flapping of BGP occurs only on the neighbors of the GR Restarter, and does not occur in the entire routing domain. This is important for BGP that needs to process a large number of routes.

9.2.14 BFD for BGP

The Border Gateway Protocol (BGP) periodically sends Keepalive packets to a peer to monitor the peer's status. However, BGP takes more than 1 second for fault detection. When traffic is transmitted at gigabit rates, lengthy fault detection causes packet loss, which does not meet carrier-class network requirements for high reliability.

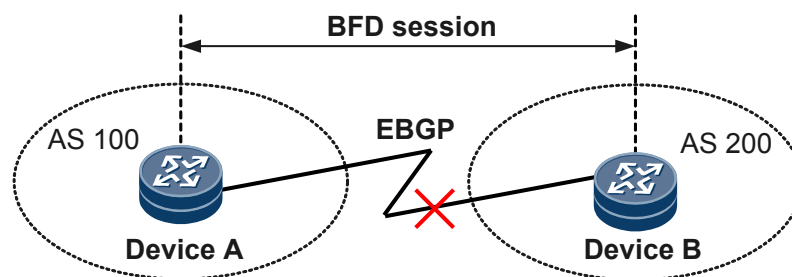
Bidirectional Forwarding Detection (BFD) for BGP can quickly detect faults on the link between BGP peers and notify BGP of the faults, which implements fast BGP route convergence.

Networking

As shown in [Figure 9-18](#), Device A and Device B belong to ASs 100 and 200, respectively. The two routers are directly connected and establish an External Border Gateway Protocol (EBGP) peer relationship.

BFD is enabled to detect the EBGP peer relationship between Device A and Device B. If the link between Device A and Device B fails, BFD can quickly detect the fault and notify BGP.

Figure 9-18 BFD for BGP



9.2.15 BGP 6PE

Background

As IPv6 technology becomes more popular, an increasing number of separate IPv6 networks take shape. IPv6 provider edge (6PE), a technology designed to provide IPv6 services over IPv4 networks, allows service providers to provide IPv6 services without constructing IPv6 backbone networks. The 6PE solution connects separate IPv6 networks using multiprotocol label switching (MPLS) tunnels. The 6PE solution implements IPv4/IPv6 dual stack on the provider edge devices (PEs) of Internet service providers and uses the Multi-protocol Extensions for Border Gateway Protocol (MP-BGP) to assign labels to IPv6 routes. In this manner, the 6PE solution connects separate IPv6 networks over IPv4 tunnels between PEs.

Related Concepts

In practical application, different metropolitan area networks (MANs) of a service provider or collaborative backbone networks of different service providers often span multiple autonomous systems (ASs). The 6PE solution can be intra-AS 6PE or inter-AS 6PE, depending on whether separate IPv6 networks connect to the same AS. Standard protocol provides three inter-AS 6PE modes: inter-AS 6PE OptionB, inter-AS 6PE OptionB with autonomous system boundary routers (ASBRs) as PEs, and inter-AS OptionC. This section describes the following 6PE modes:

- Intra-AS 6PE: Separate IPv6 networks are connected by the same AS. PEs in the AS exchange IPv6 routes by establishing MP-IBGP peer relationships.
- Inter-AS 6PE OptionB: ASBRs in different ASs exchange labeled IPv6 routes by establishing MP-EBGP peer relationships.
- Inter-AS 6PE OptionB (with ASBRs as PEs): ASBRs in different ASs exchange IPv6 routes using MP-EBGP.
- Inter-AS 6PE OptionC: PEs in different ASs exchange labeled IPv6 routes over multi-hop MP-EBGP peer sessions.

Intra-AS 6PE

Figure 9-19 shows intra-AS 6PE networking. 6PE runs on the edge of a service provider network. PEs that connect to IPv6 networks are IPv4/IPv6 dual-stack devices. PEs and customer edge devices (CEs) exchange IPv6 routes using the IPv6 Interior Gateway Protocol (IGP), or IPv6 External Border Gateway Protocol (EBGP). PEs exchange IPv4 routes with each other or with provider devices (Ps) using an IPv4 routing protocol. PEs must establish

tunnels to transparently transmit IPv6 packets. PEs often use MPLS label switched paths (LSPs) and MPLS Local IFNET tunnels. By default, a PE uses an MPLS LSP to transmit IPv6 packets. If no MPLS LSP is available, a PE uses an MPLS Local IFNET tunnel to transmit IPv6 packets.

Figure 9-19 Intra-AS 6PE networking diagram

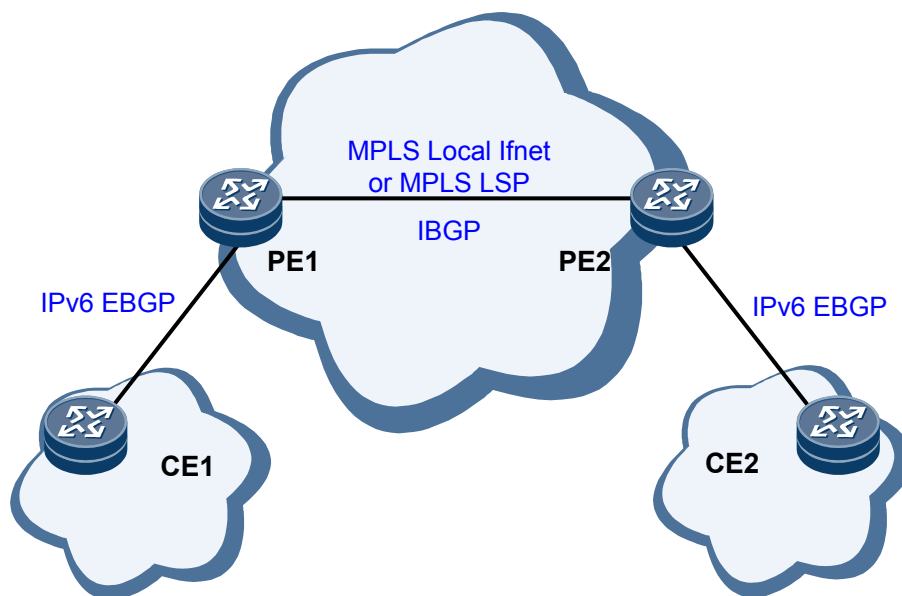


Figure 9-20 shows route and packet transmission in an intra-AS 6PE scenario. I-L indicates an inner label, and O-L indicates an outer label. The outer label directs the packet to the BGP next hop, and the inner label identifies the outbound interface or CE to which the packet should be forwarded.

The route transmission process is as follows:

1. CE2 sends an IPv6 route to PE2, its EBGP peer.
2. Upon receipt, PE2 changes the next hop of the IPv6 route to itself and assigns a label to the IPv6 route. Then, PE2 sends the labeled IPv6 route to PE1, its IBGP peer.
3. Upon receipt, PE1 relays the labeled IPv6 route to a tunnel and adds information about the route to the local forwarding table. Then, PE1 changes the next hop of the route to itself, removes the label of the route, and sends the route to CE1, its EBGP peer.

The IPv6 route transmission from CE2 to CE1 is complete.

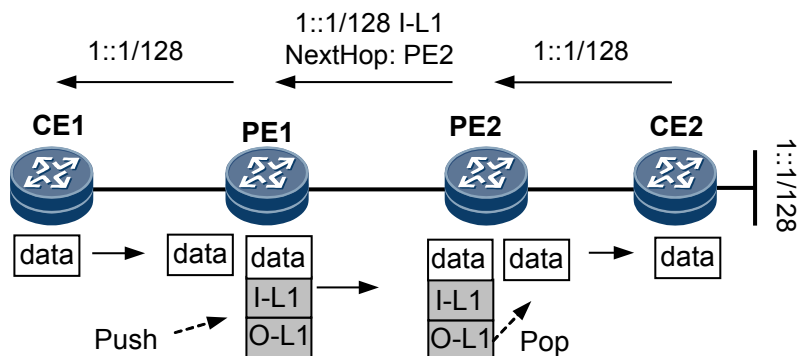
The packet transmission process is as follows:

1. CE1 sends an ordinary IPv6 packet to PE1 over an IPv6 link on the public network.
2. Upon receipt, PE1 searches its local forwarding table for the forwarding entry based on the destination address of the packet and encapsulates the packet with inner and outer labels. Then, PE1 sends the IPv6 packet to PE2 over a public network tunnel.
3. Upon receipt, PE2 removes the inner and outer labels and forwards the IPv6 packet to CE2 over an IPv6 link.

As a result, the IPv6 packet is transmitted from CE1 to CE2.

The route and packet transmission processes show that whether the public network is an IPv4 or IPv6 network does not matter to the CEs.

Figure 9-20 Route and packet transmission in an intra-AS 6PE scenario

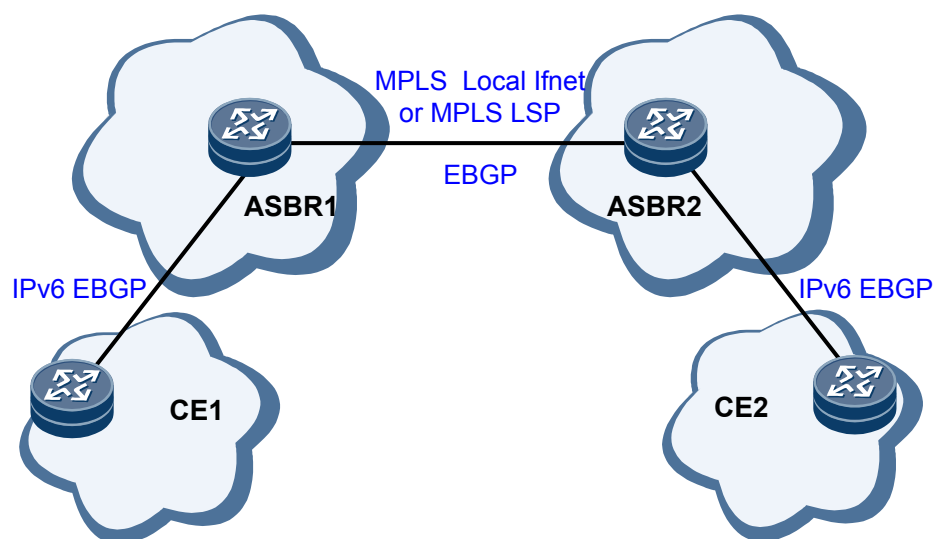


Inter-AS 6PE

- **Inter-AS 6PE OptionB (with ASBRs as PEs)**

Figure 9-21 shows inter-AS 6PE OptionB (with ASBRs as PEs) networking. Inter-AS 6PE OptionB (with ASBRs as PEs) is similar to intra-AS 6PE. The only difference is that in an inter-AS 6PE OptionB scenario in which ASBRs also function as PEs, ASBRs establish EBGP peer relationships between each other. The route and packet transmission processes in an inter-AS 6PE OptionB scenario in which ASBRs also function as PEs are similar to those in an intra-AS 6PE scenario.

Figure 9-21 Networking diagram for inter-AS 6PE OptionB (with ASBRs as PEs)



- **Inter-AS 6PE OptionB**

Figure 9-22 shows inter-AS 6PE OptionB networking. ASBRs exchange labeled IPv6 routes with each other or with PEs using an IPv4 routing protocol. Tunnels must be established between ASBRs and between PEs and ASBRs to transparently transmit IPv6 packets. MPLS LSPs, MPLS Local IFNET tunnels, GRE tunnels, and MPLS TE tunnels are often used between ASBRs to transmit IPv6 packets. By default, an ASBR uses an MPLS LSP to transmit IPv6 packets. If no MPLS LSP is available, an ASBR uses an MPLS Local IFNET tunnel to transmit IPv6 packets. If you want an ASBR to transmit IPv6 packets over an MPLS TE or a Generic Routing Encapsulation (GRE) tunnel, configure a tunnel policy on the ASBR.

Figure 9-22 Networking diagram for inter-AS 6PE OptionB

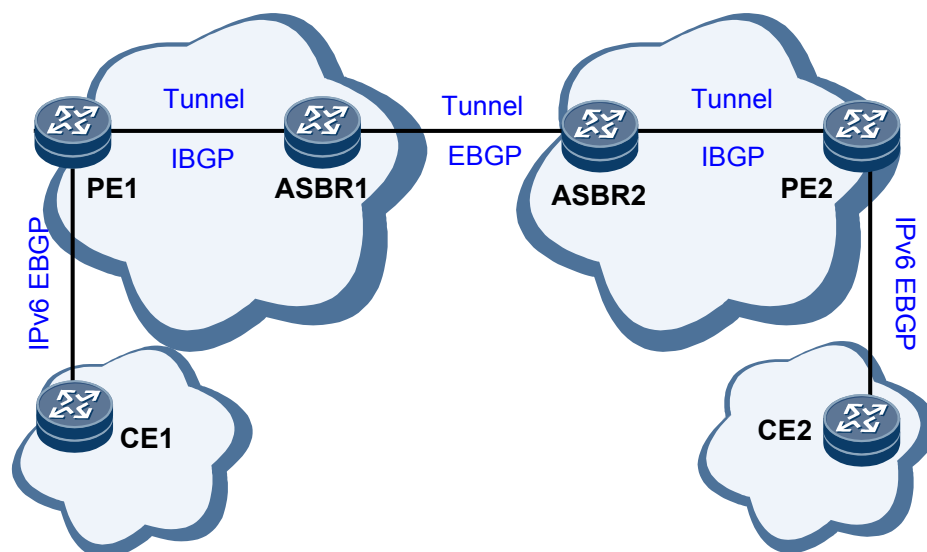
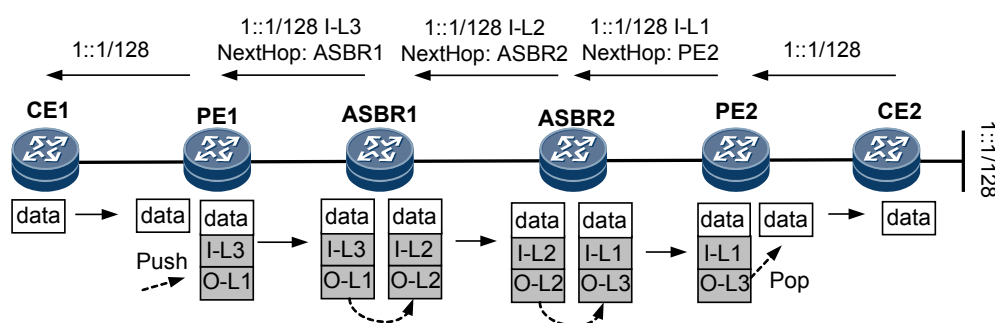


Figure 9-23 shows route and packet transmission in an inter-AS 6PE OptionB scenario. I-L indicates an inner label, and O-L indicates an outer label.

Figure 9-23 Route and packet transmission in an inter-AS 6PE OptionB scenario



The route transmission process is as follows:

- CE2 sends an IPv6 route to PE2, its EBGP peer.
- Upon receipt, PE2 changes the next hop of the IPv6 route to itself and assigns a label to the IPv6 route. Then, PE2 sends the labeled IPv6 route to ASBR2 over an IBGP peer session.
- Upon receipt, ASBR2 relays the route to a tunnel and adds information about the route to the local forwarding table. Then, ASBR2 changes the next hop of the route to itself, replaces the label of the route, and sends the route to ASBR1, its EBGP peer.
- Upon receipt, ASBR1 relays the route to a tunnel and adds information about the route to the local forwarding table. Then, ASBR1 changes the next hop of the route to itself, replaces the label of the route, and sends the route to PE1, its IBGP peer.
- Upon receipt, PE1 relays the route to a tunnel and adds information about the route to the local forwarding table. Then, PE1 changes the next hop of the route to itself, removes the label of the route, and sends the route to CE1, its EBGP peer.

As a result, the IPv6 route is transmitted from CE2 to CE1.

The packet transmission process is as follows:

- a. CE1 sends an ordinary IPv6 packet to PE1 over an IPv6 link on the public network.
- b. Upon receipt, PE1 looks up its local forwarding table based on the destination address of the packet and encapsulates the packet with inner and outer labels. Then, PE1 sends the IPv6 packet to ASBR1 over a public network tunnel.
- c. Upon receipt, ASBR1 removes the inner and outer labels of the packet, looks up the local forwarding table based on the destination address of the packet, and encapsulates the packet with new inner and outer labels. Then, ASBR1 sends the IPv6 packet to ASBR2 over a public network tunnel.
- d. Upon receipt, ASBR2 removes the inner and outer labels of the packet, looks up the local forwarding table based on the destination address of the packet, and encapsulates the packet with new inner and outer labels. Then, ASBR2 sends the IPv6 packet to PE2 over a public network LSP.
- e. Upon receipt, PE2 removes the inner and outer labels and forwards the IPv6 packet to CE2 over an IPv6 link.

As a result, the IPv6 packet is transmitted from CE1 to CE2.

- **Inter-AS 6PE OptionC**

Figure 9-24 shows inter-AS 6PE OptionC networking. In an inter-AS 6PE OptionC scenario, PEs establish multi-hop MP-EBGP peer relationships between each other and exchange labeled IPv6 routes using an IPv4 routing protocol. PEs exchange IPv6 packets over end-to-end BGP LSPs.

 NOTE

Two inter-AS 6PE OptionC solutions are available, depending on the establishment methods of end-to-end LSPs. In an inter-AS 6PE OptionC scenario, PEs establish multi-hop MP-EBGP peer relationships to exchange labeled IPv6 routes and establish end-to-end BGP LSPs to transmit IPv6 packets. The way in which an end-to-end BGP LSP is established does not matter much to inter-AS 6PE OptionC and therefore is not described here.

Figure 9-24 Networking diagram for inter-AS 6PE OptionC

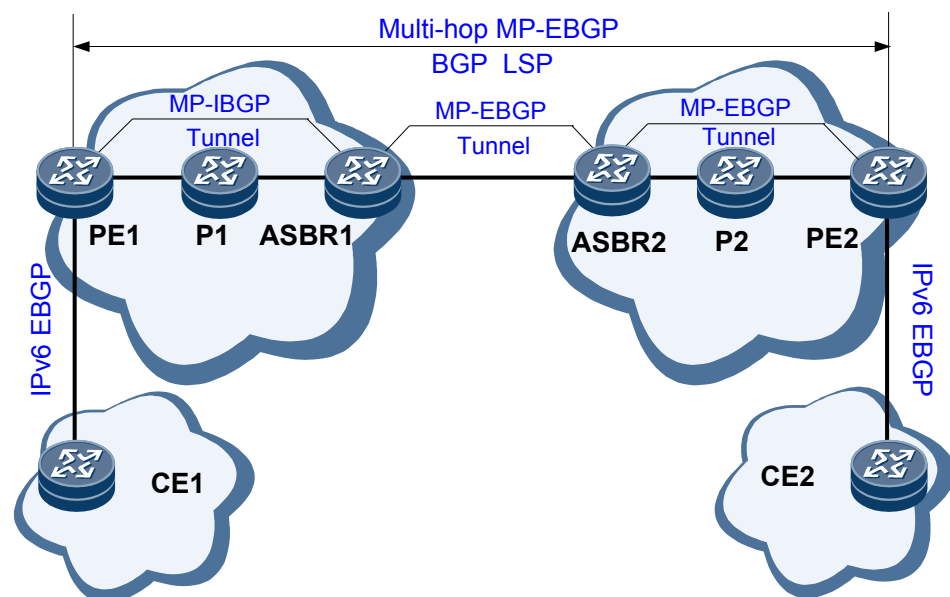


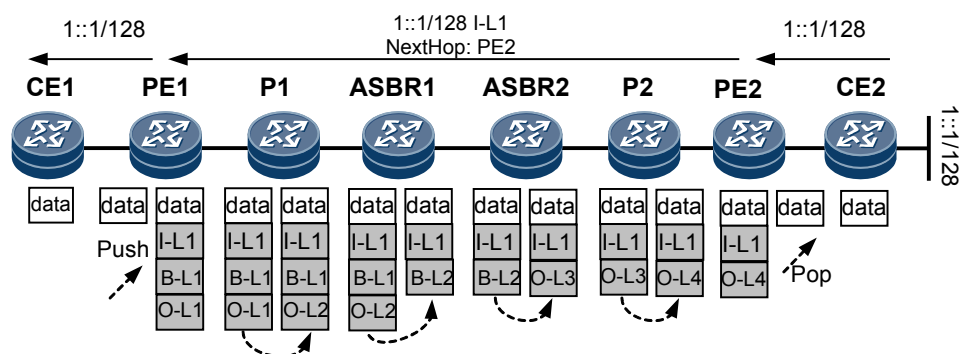
Figure 9-25 shows route and packet transmission in an inter-AS 6PE OptionC scenario. I-L indicates an inner label, B-L indicates a BGP LSP label, and O-L indicates an outer label.

NOTE

To simplify the description of the figure, it is assumed that:

- The two ASBRs are connected by an MPLS Local IFNET tunnel.
- MPLS does not use the penultimate hop popping (PHP) function.

Figure 9-25 Route and packet transmission in an inter-AS 6PE OptionC scenario



The route transmission process is as follows:

- CE2 sends an IPv6 route to PE2, its EBGP peer.
- Upon receipt, PE2 changes the next hop of the IPv6 route to itself and assigns a label to the IPv6 route. Then, PE2 sends the labeled IPv6 route to PE1, its MP-EBGP peer.
- Upon receipt, PE1 relays the route to a tunnel and adds information about the route to the local forwarding table. Then, PE1 changes the next hop of the route to itself, removes the label of the route, and sends the route to CE1, its EBGP peer.

The IPv6 route transmission from CE2 to CE1 is complete. During this process, ASBRs transparently transmit information about the labeled IPv6 route.

The packet transmission process is as follows:

- CE1 sends an ordinary IPv6 packet to PE1 over an IPv6 link on the public network.
- Upon receipt, PE1 searches its local forwarding table for the forwarding entry based on the destination address of the packet, changes the next hop of the packet based on the search result, and encapsulates the packet with an inner label, a BGP LSP label, and an outer label. Then, PE1 sends the IPv6 packet to P1 over a public network tunnel.
- Upon receipt, P1 replaces the outer label of the packet and forwards the packet to ASBR1 over a public network tunnel.
- Upon receipt, ASBR1 removes the outer and BGP LSP labels and encapsulates the packet with a new BGP LSP label. Then, ASBR1 sends the IPv6 packet to ASBR2 over a public network tunnel.
- Upon receipt, ASBR2 removes the BGP LSP label and encapsulates the packet with an outer label. Then, ASBR2 sends the IPv6 packet to PE2 over a public network tunnel.
- Upon receipt, P2 replaces the outer label of the packet and forwards the packet to PE2 over a public network tunnel.

- g. Upon receipt, PE2 removes the inner and outer labels and forwards the IPv6 packet to CE2 over an IPv6 link.

As a result, the IPv6 packet is transmitted from CE1 to CE2.

Usage Scenarios

Each 6PE mode has its advantages and usage scenarios. The intra-AS 6PE mode is best suited for scenarios in which separate IPv6 networks connect to the same AS. Inter-AS 6PE modes are best suited for scenarios in which separate IPv6 networks connect to different ASs. [Table 9-2](#) lists the usage scenarios for inter-AS 6PE modes.

Table 9-2 Usage scenarios for inter-AS 6PE modes

Mode	Characteristic	Usage Scenario
Inter-AS 6PE OptionB (with ASBRs as PEs)	Advantage: Configuration is similar to that for intra-AS 6PE and additional inter-AS configuration is not required. Disadvantage: Network expansibility is poor. ASBRs must have high performance to manage information about all labeled IPv6 routes.	A small network with ASBRs in different ASs to which separate IPv6 networks are connected. The smaller the number of ASs spanned, the more obvious the advantage of this solution is.
Inter-AS 6PE OptionB	Advantage: MPLS tunnels are established segment by segment and easy to manage. Disadvantage: Information about labeled IPv6 routes is stored on and advertised by ASBRs. If a large number of VPN routes exist, the overburdened ASBRs are likely to encounter bottlenecks.	An inter-AS OptionB public network with tunnels established between the PEs in different ASs and with separate IPv6 networks connected to the ASs.
Inter-AS 6PE OptionC	Advantage: Labeled IPv6 routes are directly exchanged between the ingress and egress PEs. Information about labeled IPv6 routes is managed by PEs only and ASBRs are no longer the bottlenecks. Disadvantage: Management costs of end-to-end BGP LSP connections are high.	An inter-AS OptionC public network with end-to-end tunnels established between the PEs in different ASs and with separate IPv6 networks connected to the ASs. The greater the number of ASs spanned, the more obvious the advantage of this solution is.

Benefits

6PE offers the following benefits:

- Easy maintenance: All configurations are performed on PEs and network maintenance is simple. IPv6 services are carried over IPv4 networks, but the users on IPv6 networks are unaware of IPv4 networks.
- Low network construction costs: Service providers can provide IPv6 services over existing MPLS networks without upgrading the networks. 6PE devices can provide multiple types of services, such as IPv6 VPN and IPv4 VPN.

9.2.16 BGP ORF

Outbound Route Filtering (ORF) is used to enable a BGP device to send the local routing policy to its BGP peer. The peer can use the local routing policy to filter out unwanted routes before route advertisement.

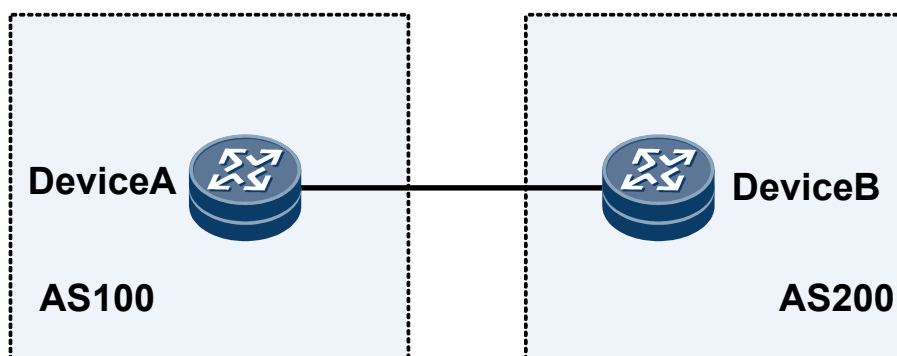
In most cases, users expect the carrier to send them only the routes they require. Therefore, the carrier needs to maintain a separate outbound policy for each user. ORF allows carriers to send only required routes to each user without maintaining a separate outbound policy for each user. ORF supports on-demand route advertisement, which greatly reduces bandwidth consumption and the manual configuration workload.

Prefix-based ORF, defined in standard protocols, can be used to send prefix-based inbound policies configured by users to a carrier through Route-Refresh packets. The carrier then filters out unwanted routes before route advertisement based on the received inbound policies, which prevents users from receiving a large number of unwanted routes and saves resources.

Applications

On the network shown in [Figure 9-26](#), Device A and Device B are directly connected, and prefix-based ORF is enabled on them; after negotiating the prefix-based ORF capability with Device B, Device A adds the local prefix-based inbound policy to a Route-Refresh packet and then sends the Route-Refresh packet to Device B. Device B uses the information in the packet to work out an outbound policy to advertise routes to Device A.

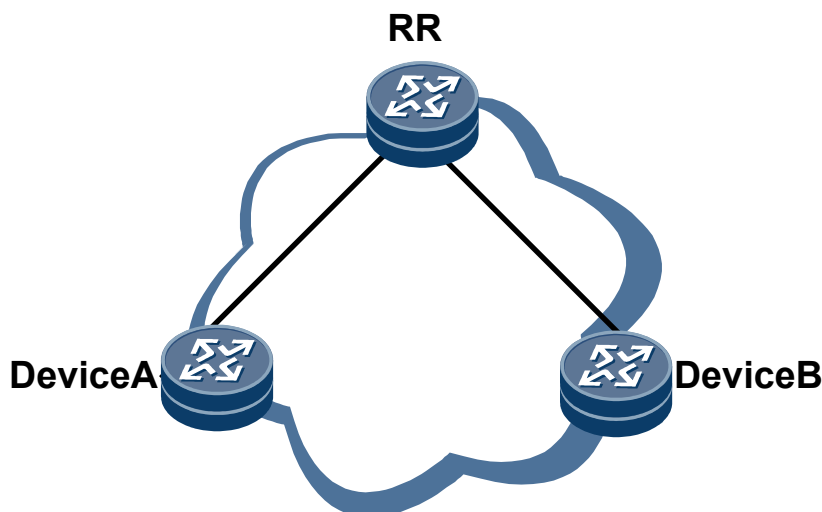
Figure 9-26 Applying ORF to directly connected BGP peers



As shown in [Figure 9-27](#), Device A and Device B are clients of the RR in the domain. Prefix-based ORF is enabled on all three NEs. After negotiating prefix-based ORF with the RR, Device A and Device B add the local prefix-based inbound policies Route-Refresh packets

and then send the packets to the RR. Based on the Route-Refresh packets, the RR uses the information in the Route-Refresh packets to work out the outbound policies to reflect routes to Device A and Device B.

Figure 9-27 Applying ORF to a domain with an RR



9.2.17 VPN ORF

VPN outbound route filtering (ORF) enables VPNv4/VPNv6/NGMVPN/L2VPN-AD peers to filter route target-MEM-network layer reachability information (RT-MEM-NLRI) on the BGP multi-service bearer framework.

ORF applies a local routing policy to the outbound interface of a peer so that the peer advertises only desired routes to the local device.

VPN ORF enables a PE to receive only desired routes, reducing pressure on the routing table capacity of the route reflector (RR) and autonomous system boundary router (ASBR).

Background

As networks develop, users keep increasing. The broadcast outbound policies used by carriers no longer meet user requirements because the routes that users desire vary. If the carrier maintains an export policy for each user, the O&M costs are extremely high. ORF allows users to receive only desired routes, without requiring the carrier to maintain an export policy for each user.

Related Concepts

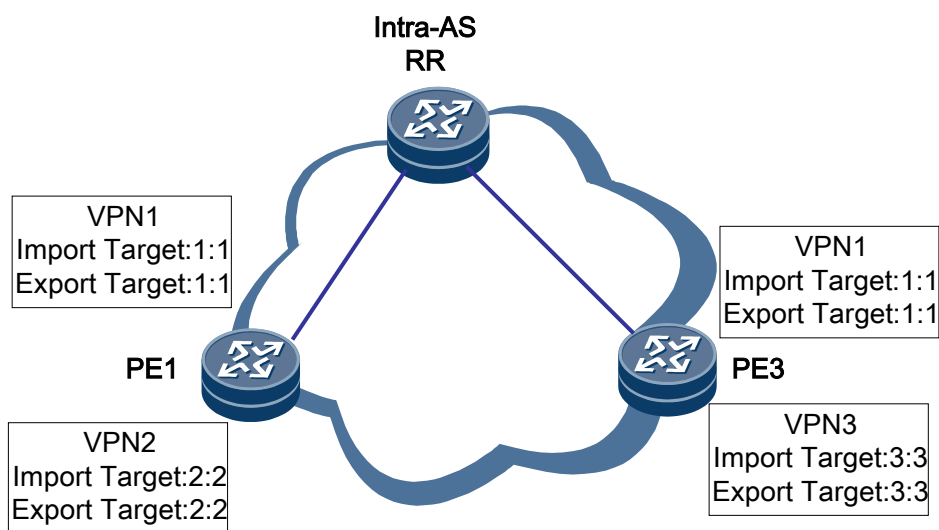
- RT-MEM-NLRI: VPN ORF routes
- PE: provider edge
- RR: route reflector
- ASBR: autonomous system boundary router

Implementation

Each PE with VPN instances bound sends to BGP peers VPN ORF routes carrying the import route targets (IRTs) and the original AS number of desired routes, based on which the peers generate an export policy for the PE so that the PE receives only desired routes.

In **Figure 9-28**, if VPN ORF is disabled, the RR sends to PE3 all routes of VPN instances received from PE1. However, among these routes, PE3 only desires the routes with ERT 1:1. In addition, the RR sends to PE1 all routes of VPN instances received from PE3. However, among these routes, PE1 only desires the routes with ERT 1:1.

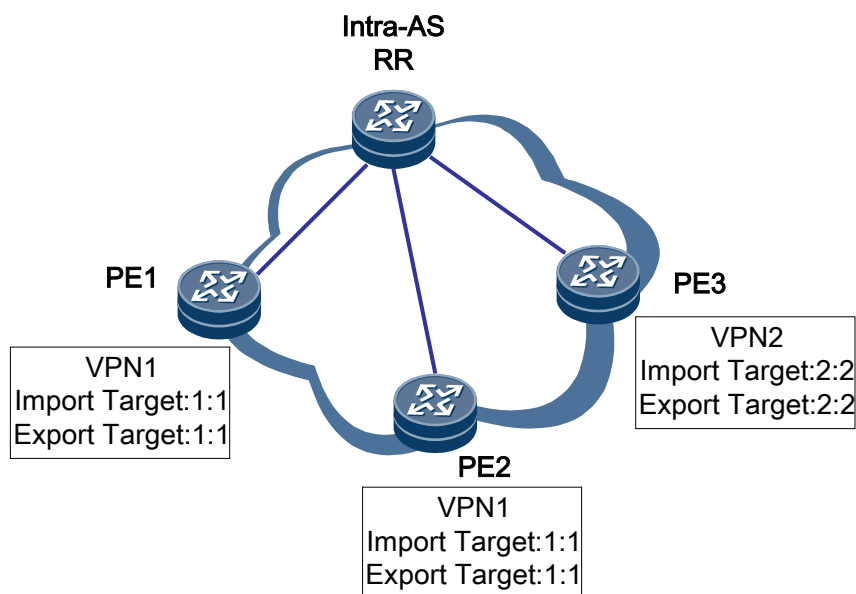
Figure 9-28 Basic usage scenario of VPN ORF



After VPN ORF is enabled, a BGP peer relationship is established in the VPN target address family view. As shown in **Figure 9-28**, a BGP peer relationship is established between the RR and PE1, and another BGP peer relationship is established between the RR and PE3. The BGP peers exchange VPN ORF routes. PE1 and PE3 each send VPN ORF routes carrying the IRTs of desired routes to the RR. Based on the received VPN ORF routes, the RR generates an export policy. In other words, before advertising BGP VPN routes, PE1 filters these routes based on the VPN ORF routes received from its peer.

In **Figure 9-29**, PE1, PE2, and PE3 are clients of the RR, VPN1 is bound to PE1 and PE2, and VPN2 is bound to PE3. If VPN ORF is disabled, PE1, PE2, and PE3 send routes with ERTs 1:1 and 2:2 to the RR. The RR reflects all routes to the PEs, and among the routes, PE1 and PE2 only accept the routes with ERT 1:1, and PE3 only accepts the routes with ERT 2:2.

Figure 9-29 Networking where a VPN RR has three clients



If VPN ORF is enabled, PE1, PE2, and PE3 send VPN ORF routes to the RR. The VPN ORF routes sent by PE1 and PE2 are the same. If the RR prefers the route sent by PE1 to the one sent by PE2, the RR reflects the route sent by PE1 and the route sent by PE3 to all clients. Then PE1 and PE2 learn the route with ERT 1:1, and PE3 learns the route with ERT 2:2.

Usage Scenario

VPN ORF applies to the following scenarios:

- Intra-AS scenario where a VPN RR has clients
- Inter-AS VPN scenario
- Scenario where some routers do not support VPN ORF
- Intra-AS scenario where an RR has clients and non-clients

Benefits

VPN ORF offers the following benefits:

- Reduced bandwidth consumption
- Reduced configuration workload

9.2.18 BGP Auto FRR

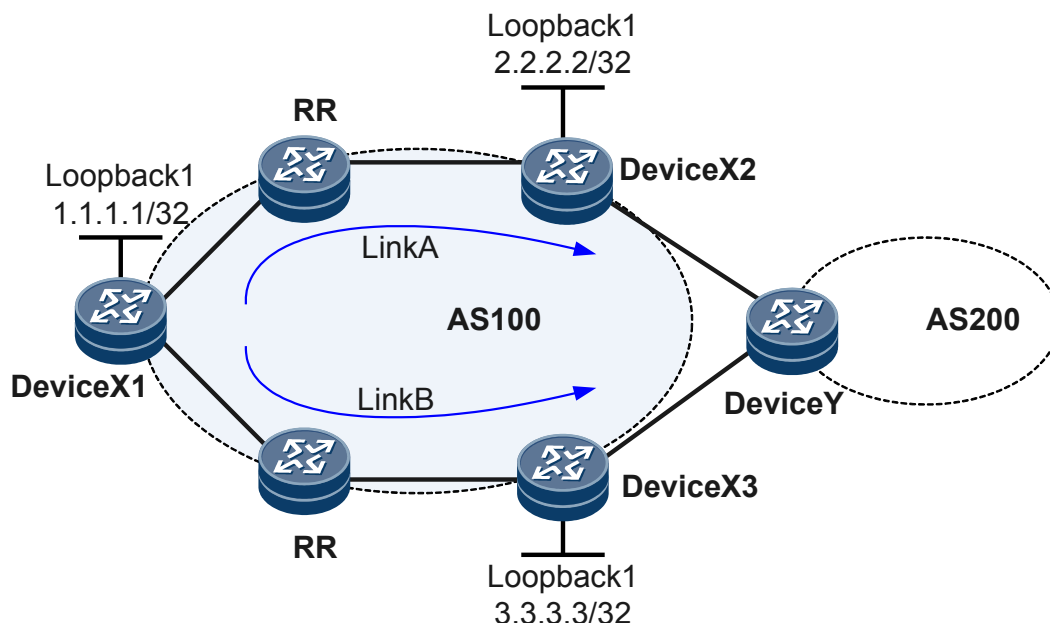
As a protection measure against faults over links, BGP Auto Fast Reroute (FRR) is applicable to networks with primary and backup links. With BGP Auto FRR, traffic can be switched between two BGP peers or next hops within sub-seconds.

With BGP Auto FRR, if a peer has multiple routes with the same prefix that are learned from different peers, the peer uses the optimal route as the primary link to forward packets and the less optimal route as a backup link. If the primary link fails, the peer rapidly notifies other peers that the BGP route has become unreachable and then switches traffic from the primary link to the backup link.

Usage Scenario

On the network shown in [Figure 9-30](#), Device Y advertises a learned BGP route to Device X2 and Device X3 in AS 100; Device X2 and Device X3 then advertise the BGP route to Device X1 through RR. Therefore, Device X1 receives two routes whose next hops are Device X2 and Device X3 respectively. Then, Device X1 selects a route based on a configured routing policy. Assume that the route sent by Device X2 (Link A) is preferred. The route sent by Device X3 (Link B) then functions as a backup link.

Figure 9-30 Networking for BGP Auto FRR



If a node along Link A fails or faults occur on Link A, the next hop of the route from Device X1 to Device X2 becomes unavailable. If BGP Auto FRR is enabled on Device X1, the forwarding plane then quickly switches to Link B the traffic from Device X1 to Device Y, which ensures uninterrupted traffic transmission. In addition, Device X1 reselects the route sent by Device X3 based on the forwarding prefixes and then updates the FIB table.

9.2.19 BGP Dynamic Update Peer-Groups

As the routing table increases in size and the network topology increases in complexity, BGP needs to be able to support more peers. When the router needs to send a large number of routes to many BGP peers and most of the peers share the same configuration, if the router groups each route and then send the route to each peer, the efficiency is low.

The dynamic update peer-groups feature can address this problem. It considers all the BGP peers with the same configuration as an update-group. With the dynamic update peer-groups feature, each route to be sent is grouped once for all and then sent to all peers in the update-group, improving grouping efficiency and forwarding performance exponentially.

Usage Scenario

The BGP dynamic update peer-groups feature is applicable to the following scenarios:

- Scenario with an international gateway
- Scenario with an RR
- Scenario where routes received from EBGPeers need to be sent to all IBGP peers

Figure 9-31 Networking for the international gateway

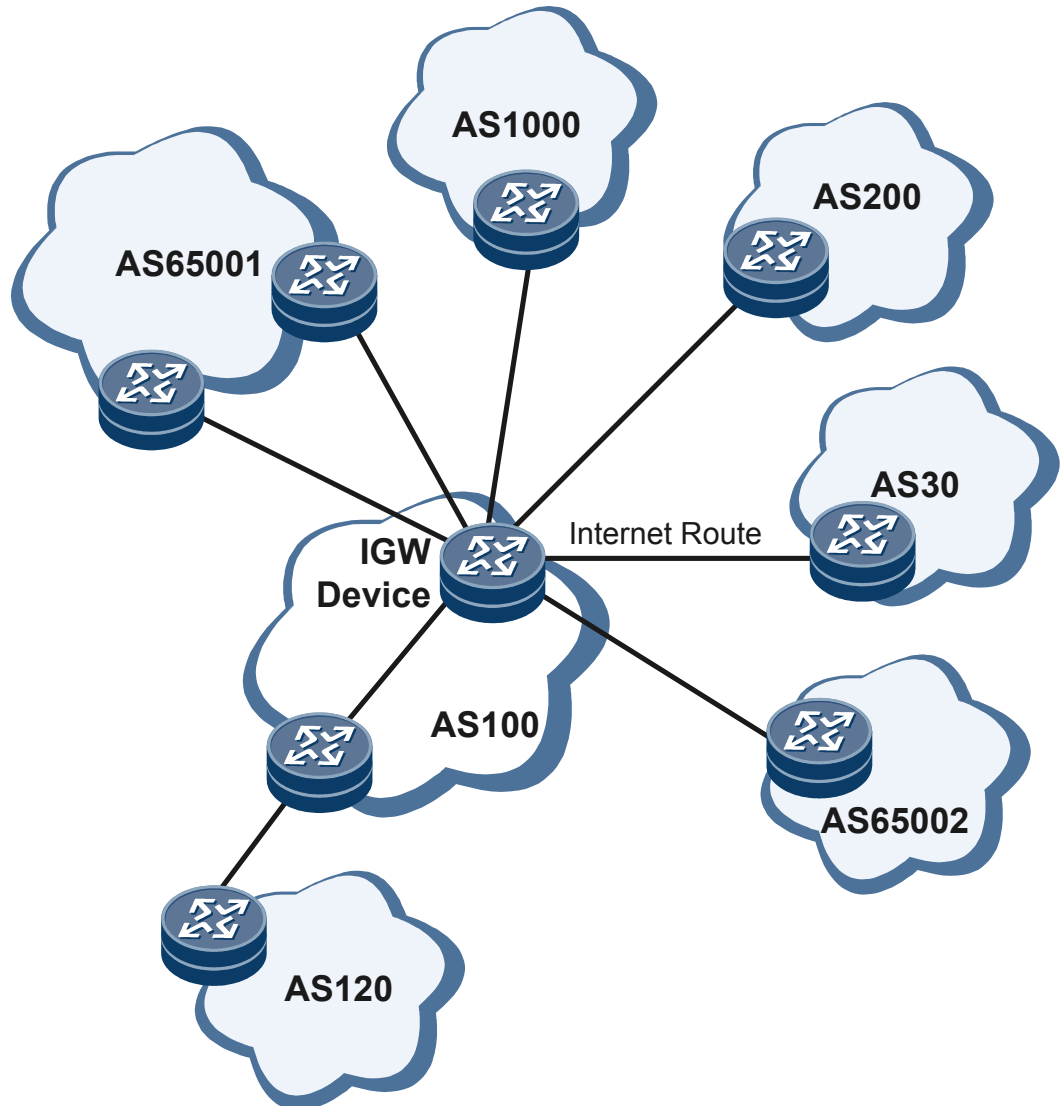


Figure 9-32 Networking for RRs with many clients

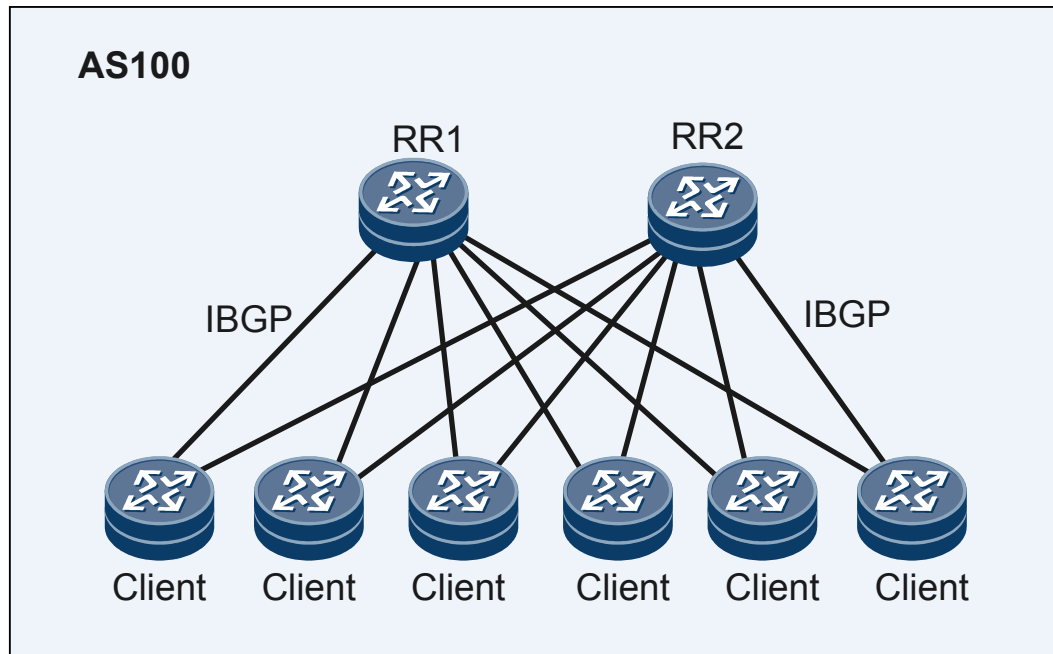
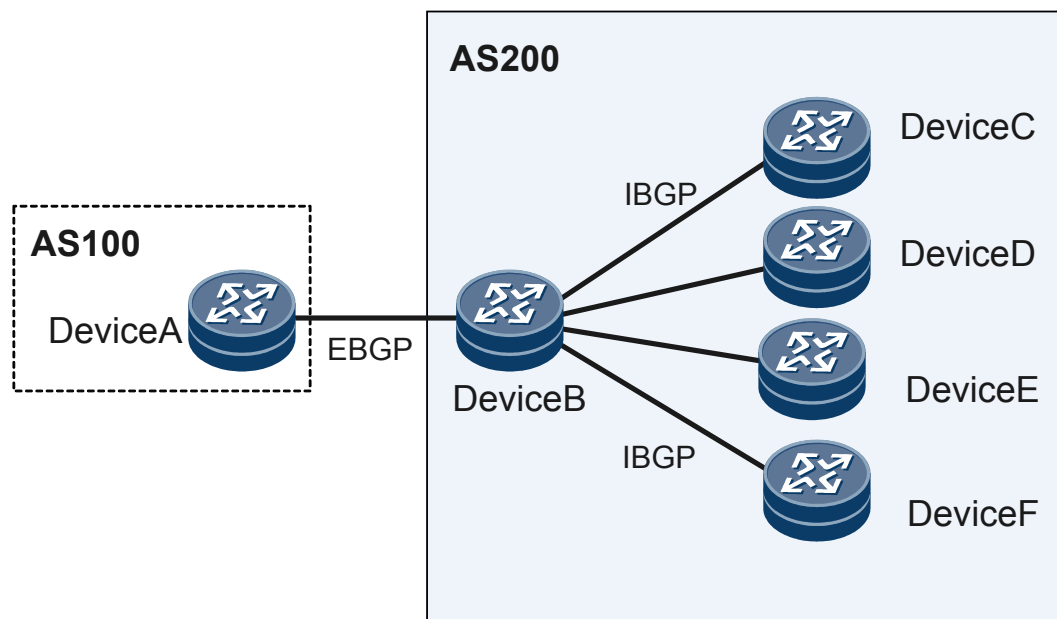


Figure 9-33 Networking for a PE connected to multiple IBGP peers



The preceding scenarios have in common that a router needs to send routes to a large number of BGP peers, most of which share the same configuration. This situation is most evident in the networking shown in [Figure 9-32](#).

For example, an RR has 100 clients and needs to reflect 100,000 routes to them. If the RR groups the routes for each peer before sending the routes to 100 clients, the total number of

times that all routes are grouped is 100,000 x 100. After the dynamic update peer-groups feature is applied, the total number of times that all routes are grouped changes to 100,000 x 1. The efficiency is 100 times higher than before.

9.2.20 Active-Route-Advertise

Background

Active-route-advertise allows a device to advertise only optimal routes in an IP routing table.

Active-route-advertise prevents changes to data forwarding paths that may occur after independent BGP route selection is deployed. Independent BGP route selection enables a device to advertise an optimal BGP route in a BGP routing table to BGP peers, regardless of whether this optimal BGP route is optimal in an IP routing table. Therefore, if a device is upgraded to support independent BGP route selection, previously selected data forwarding paths may change. If you do not expect the changes, configure active-route-advertise on the device.

Related Concepts

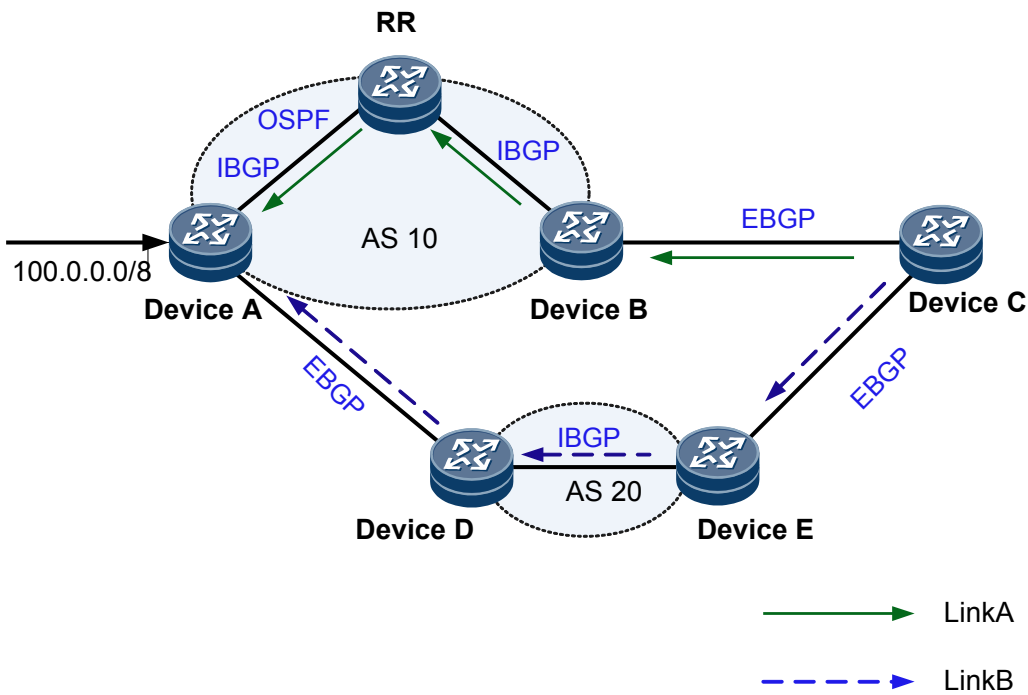
IP routing table: stores routes that are optimal in each available protocol routing table, selects optimal routes from the stored routes, and delivers the selected optimal routes to the forwarding information base (FIB) table.

Independent BGP route selection: enables a device to advertise an optimal BGP route in a BGP routing table to BGP peers, regardless of whether this optimal BGP route is optimal in an IP routing table.

Implementation

As shown in [Figure 9-34](#), an RR is deployed between Device A and Device B, an Open Shortest Path First (OSPF) neighbor relationship is established between Device A and the RR, an External BGP (EBGP) peer relationship is established between Device B and Device C, and Device A and Device C are not directly connected.

Figure 9-34 Active-route-advertise networking diagram



The route 100.0.0.0/8 is imported to the BGP and OSPF routing tables on Device A, and the RR learns both the BGP route 100.0.0.0/8 and OSPF route 100.0.0.0/8. By default, an OSPF route has a higher priority over a BGP route. Therefore, the OSPF route 100.0.0.0/8 is an optimal route in the IP routing table, and the BGP route 100.0.0.0/8 is inactive in the IP routing table. [Table 9-3](#) describes the changes after the RR is upgraded to support independent BGP route selection, including the changes in the optimal route in the BGP routing table, route advertisement, and data forwarding path.

Table 9-3 Change description

Item	In the Source Version	In the Target Version Without Active-Route-Advertise	In the Target Version with Active-Route-Advertise
Optimal route in the BGP routing table	The BGP route 100.0.0.0/8 is an optimal route in the BGP routing table, but not optimal in the IP routing table.		

Item	In the Source Version	In the Target Version Without Active-Route-Advertise	In the Target Version with Active-Route-Advertise
Route advertisement	Independent BGP route selection is not supported. The RR does not advertise the BGP route 100.0.0.0/8 to Device B. Device C learns routes advertised by Device E but not by Device B.	Independent BGP route selection is supported. The RR advertises the BGP route 100.0.0.0/8 to Device B. Device C learns routes advertised by both Device B and Device E.	The BGP route 100.0.0.0/8 is not an optimal route in the IP routing table on the RR. The RR does not advertise the BGP route 100.0.0.0/8 to Device B. Device C learns routes advertised by Device E but not by Device B.
Available data forwarding path	Link B	Link A and Link B	Link B
Actually selected data forwarding path	Device C selects Link B to send data to 100.0.0.0/8.	Device C selects Link A to send data to 100.0.0.0/8.	Device C selects Link B to send data to 100.0.0.0/8.

As described in [Table 9-3](#), active-route-advertise ensures that the data forwarding path Link B is still used after independent BGP route selection is deployed.

NOTE

Before you configure active-route-advertise, check whether BGP routes are optimal routes in the IP routing table. If all BGP routes are optimal routes in the IP routing table, data forwarding paths do not change after active-route-advertise is configured. If only some BGP routes are optimal routes in the IP routing table, analyze the impacts of active-route-advertise and determine whether to configure this feature.

Advantages

Active-route-advertise prevents changes to data forwarding paths that may occur after independent BGP route selection is deployed.

9.2.21 4-Byte AS Number

Purpose

2-byte autonomous system (AS) numbers used on networks range from 1 to 65535, and the available AS numbers are close to exhaustion as networks expand. Therefore, the AS number range needs to be extended. 4-byte AS numbers ranging from 1 to 4294967295 can address this problem. New speakers that support 4-byte AS numbers can co-exist with old speakers that support only 2-byte AS numbers.

Definition

4-byte AS numbers are extended from 2-byte AS numbers. Border Gateway Protocol (BGP) peers use a new capability code and optional transitive attributes to negotiate the 4-byte AS number capability and transmit 4-byte AS numbers. This mechanism enables communication between new speakers and between old speakers and new speakers.

Open capability code (0x41), defined by standard protocols, indicates that the local end supports 4-byte capability extension.

The following new optional transitive attributes are defined by standard protocols and used to transmit 4-byte AS numbers in old sessions:

- **AS4_Path** coded 0x11
- **AS4_Aggregator** coded 0x12

If a new speaker with an AS number greater than 65535 communicates with an old speaker, the old speaker needs to set the peer AS number to **AS_TRANS**. The value of **AS_TRANS** is 23456 and reserved.

Related Concepts

- New speaker: a peer that supports 4-byte AS numbers
- Old speaker: a peer that does not support 4-byte AS numbers
- New session: a BGP connection established between new speakers
- Old session: a BGP connection established between a new speaker and an old speaker, or between old speakers

Principles

BGP speakers negotiate capabilities by exchanging Open messages. **Figure 9-35** shows the format of Open messages exchanged between new speakers. The header of a BGP Open message is fixed, in which **My AS Number** is supposed to be the local AS number. However, **My AS Number** can carry only 2-byte AS numbers. Therefore, a new speaker adds 23456 to **My AS Number** and its local AS number to **Optional parameters** before it sends an Open message to a peer. After the peer receives the message, it can determine whether the new speaker supports 4-byte AS numbers by checking **Optional parameters** in the message.

Figure 9-35 Format of Open messages sent by new speakers

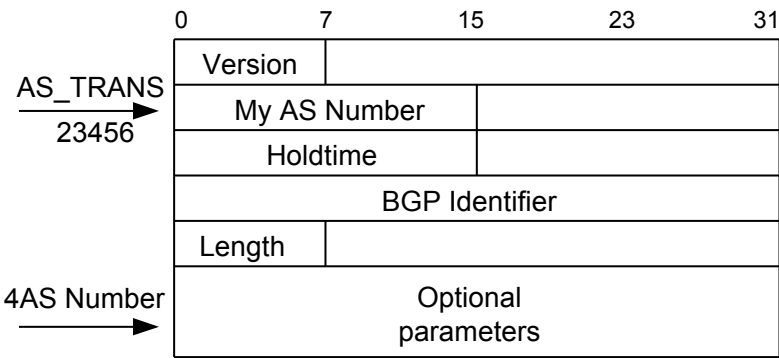
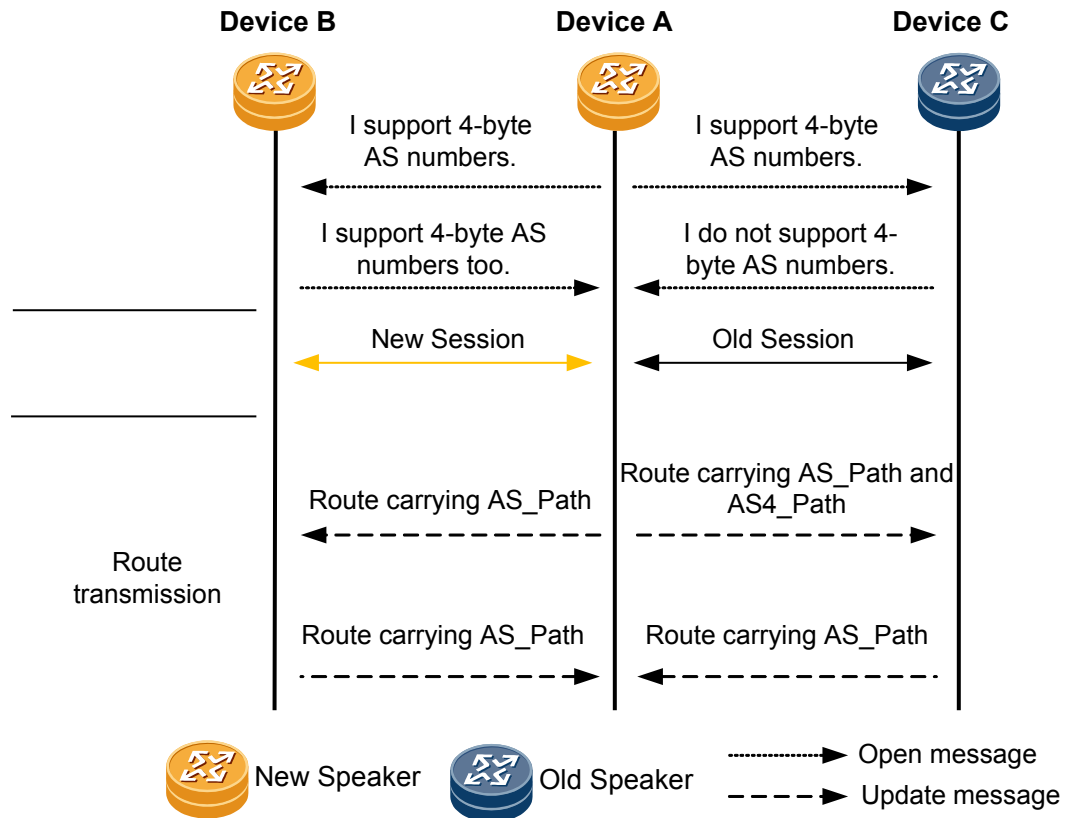


Figure 9-36 shows how peer relationships are established between new speakers, and between an old speaker and a new speaker. BGP speakers notify each other of whether they

support 4-byte AS numbers by exchanging Open messages. After the capability negotiation, new sessions are established between new speakers, and old sessions are established between a new speaker and an old speaker.

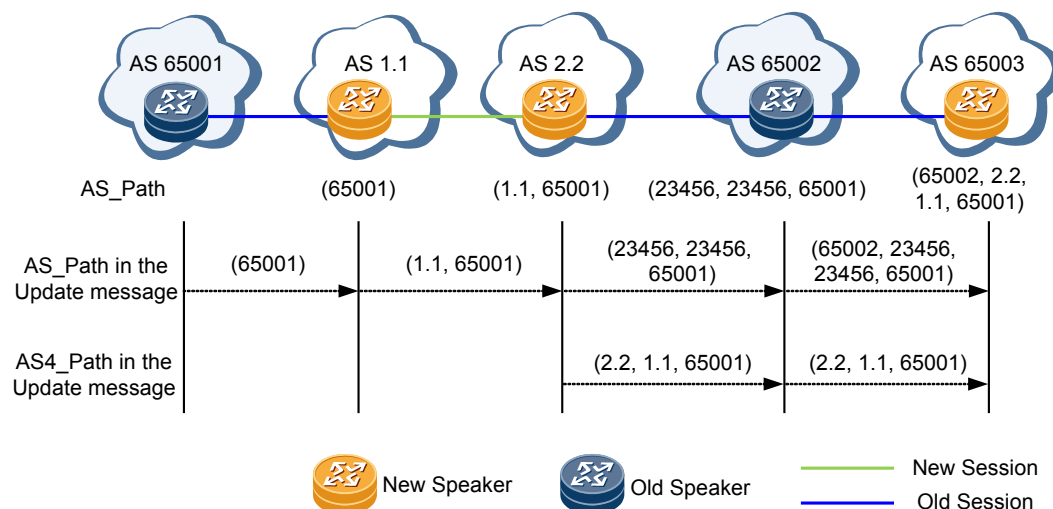
Figure 9-36 Process of establishing a BGP peer relationship



AS_Path and **Aggregator** in Update messages exchanged between new speakers carry 4-byte AS numbers, whereas **AS_Path** and **Aggregator** in Update messages sent by an old speaker carry 2-byte AS numbers.

- When a new speaker sends an Update message carrying an AS number greater than 65535 to an old speaker, the new speaker uses **AS4_Path** and **AS4_Aggregator** to assist **AS_Path** and **AS_Aggregator** in transferring 4-byte AS numbers. **AS4_Path** and **AS4_Aggregator** are transparent to the old speaker. In the networking shown in [Figure 9-37](#), before the new speaker in AS 2.2 sends an Update message to the old speaker in AS 65002, the new speaker replaces each 4-byte AS number (2.2, 1.1) with 23456 in **AS_Path**. Therefore, the **AS_Path** carried in the Update message is (23456, 23456, 65001), and the carried **AS4_Path** is (2.2, 1.1). After the old speaker in AS 65002 receives the Update message, it transparently transmits the message to other ASs.
- When the new speaker receives an Update message carrying **AS_Path**, **AS4_Path**, **AS_Aggregator**, and **AS4_Aggregator** from the old speaker, the new speaker uses the reconstruction algorithm to reconstruct the actual **AS_Path** and **AS_Aggregator**. In the networking shown in [Figure 9-37](#), after the new speaker in AS 65003 receives an Update message carrying **AS_Path** (65002, 23456, 23456, 65001) and **AS4_Path** (2.2, 1.1) from the old speaker in AS 65002, the new speaker reconstructs the actual **AS_Path** (65002, 2.2, 1.1, 65001).

Figure 9-37 Process of transmitting a BGP Update message



Format of 4-byte AS numbers

A 4-byte AS number can be an integer or in dotted notation. The system stores 4-byte AS numbers as unsigned integers, regardless of their formats. 4-byte AS numbers in dotted notation are in the format of A.B. The formula of the conversion between 4-byte AS numbers for the two formats is as follows: Integer 4-byte AS number = $A \times 65536 + B$. For example, the 4-byte AS number in dotted notation **2.3** can be converted to the integer 4-byte AS number 131075 ($2 \times 65536 + 3$).

The NE40E supports 4-byte AS numbers of both formats. The 4-byte AS numbers displayed in the configuration files are in the format configured by users.

By default, the 4-byte AS numbers displayed in the **display** and **debugging** command outputs are in dotted notation, regardless of the configured format. If users change the default display format of 4-byte AS numbers from dotted notation to an integer, the displayed 4-byte AS numbers are integers.

NOTICE

Adjusting the display format of 4-byte AS numbers affects the matching results of **AS_Path** regular expressions and **extended community** filters. If you adjust the display format of 4-byte AS numbers on a system that uses an **AS_Path** regular expression or **extended community** filter as the export or import policy, reconfigure the **AS_Path** regular expression or **extended community** filter. If you do not reconfigure the **AS_Path** regular expression or **extended community** filter, routes cannot match the export or import policy, and a network error may occur.

Benefits

4-byte AS numbers alleviate AS number exhaustion and therefore are beneficial to carriers who need to expand the network scale.

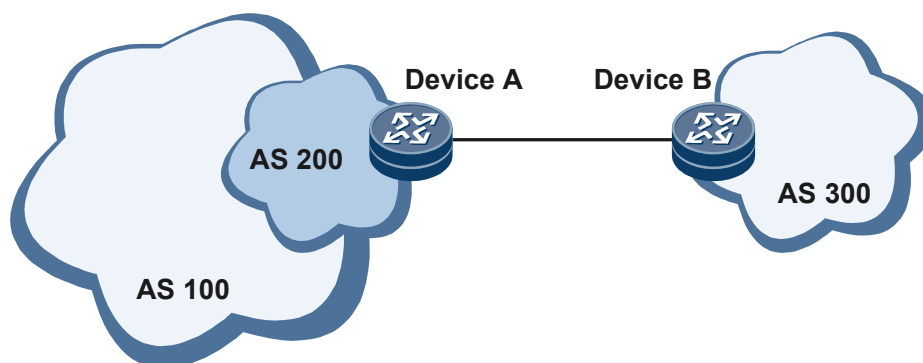
9.2.22 Fake AS Number

In the acquisition and merger scenarios between carriers, if the acquiree and the acquirer are located in different ASs, the BGP peers of the acquiree need to be migrated to the AS of the

acquirer. However, during network migration, the customer of the acquirer may not want to have local BGP configurations modified right away or at all. As a result, the BGP peer relationships may be interrupted for a long time.

In [Figure 9-38](#), the AS number of carrier A is 100, whereas the AS number of carrier B is 200. Device A belongs to carrier B. Then carrier A acquires carrier B. In this case, the AS number of device A needs to be changed from 200 to 100. Because device A already has a BGP peer relationship established with device B in AS 300 using AS 200, device A's AS number used to establish the BGP peer relationship needs to be changed to 100. The carrier of AS 100 and the carrier of AS 300 then need to communicate about the change. In addition, the AS number configured on device A and peer AS number configured on device B may not be changed at the same time, which will lead to a lengthy interruption of the BGP peer relationship between the two devices. To ensure a smooth merger, you can run the **peer fake-as** command on device A to set AS 200 of carrier B as a fake AS number so that device A's AS number used to establish the BGP peer relationship between devices A and B does not need to be changed.

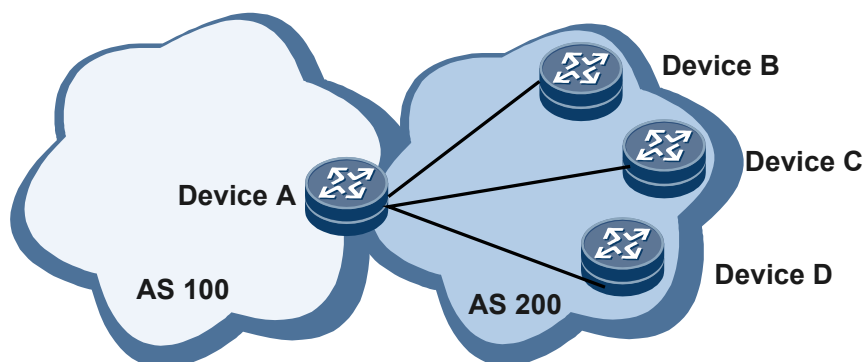
Figure 9-38 Network 1 with a fake AS number



In addition, the AS number of the original BGP speakers of carrier B may be changed to the actual AS number at any time when BGP peer relationships are established with devices of carrier A after the merger. If carrier B has a large number of BGP speakers and some of the speakers use the actual AS number whereas other speakers use the fake AS number during BGP peer relationship establishment with devices of carrier A, the local configuration on BGP speakers of carrier B needs to be changed based on the configuration of the peer AS number, which increases the workload of maintenance. To address this problem, you can run the **peer fake-as** command with **dual-as** specified to allow the local end to use the actual or fake AS number to establish a BGP peer relationship with the specified peer.

In [Figure 9-39](#), the AS number of carrier A is 100, whereas the AS number of carrier B is 200; devices A, B, C, and D belong to carrier B, and device A establishes an IBGP peer relationship with device B, device C, and device D each. Then carrier A acquires carrier B. In this case, the AS number of device A needs to be changed from 200 to 100. Because the AS number used by device A to establish the IBGP peer relationship with devices B, C, and D is 200, the AS number needs to be changed to 100. In this case, carrier A and carrier B need to communicate about the change. In addition, the AS number configured on device A and peer AS number configured on devices B, C, and D may not be changed at the same time, which will lead to a lengthy interruption of the IBGP peer relationships. To ensure a smooth merger, you can run the **peer fake-as** command on device A to set AS 200 of carrier B as a fake AS number so that device A's AS number used to establish the IBGP peer relationships with devices B, C, and D does not need to be changed.

Figure 9-39 Network 2 with a fake AS number



9.2.23 BMP

Background

The BGP Monitoring Protocol (BMP) is designed to monitor BGP running status, such as BGP peer relationship establishment and termination and route updates.

Without BMP, manual query is required if you want to know about BGP running status.

With BMP, a router can be connected to a monitoring server and configured to report BGP running statistics to the server for monitoring, which improves the network monitoring efficiency.

BMP Messages

Routers send BMP packets carrying Initiation, Peer Up Notification (PU), Route Monitoring (RM), Peer Down Notification (PD), Status Report (SR), or Termination messages to the monitoring server to report BGP running statistics. The functions of these messages are listed as follows:

- Initiation message: Reports to the monitoring server such information as the router vendor and its software version.
- PU message: Notifies the monitoring server that a BGP peer relationship has been established.
- RM message: Sends to the monitoring server all routes received from BGP peers and notifies the server of route addition or deletion in real time.
- PD message: Notifies the monitoring server that a BGP peer has been disconnected.
- SR message: Reports router running statistics to the monitoring server.
- Termination message: Reports to the monitoring server the cause of BMP session termination.

NOTE

BMP sessions are unidirectional. Routers send messages to the monitoring server but ignore messages replied by the server.

Implementation

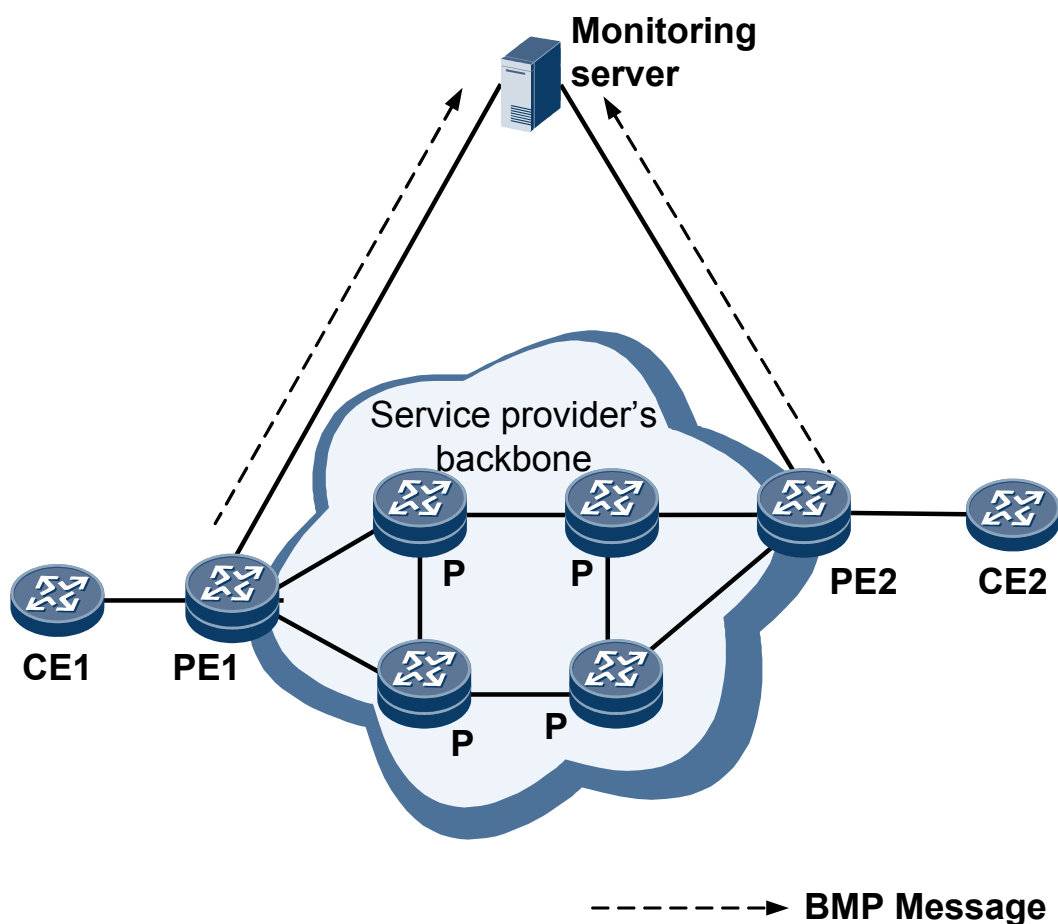
In [Figure 9-40](#), a TCP connection is established between the monitoring server and PE1 and between the monitoring server and PE2. PE1 and PE2 send unsolicited BMP packets to the

monitoring server to report BGP running statistics. After receiving these BMP packets, the monitoring server parses them and displays the BGP running status in the monitoring view. The BMP packets carry headers. By analyzing the headers, the monitoring server can decide which BGP peers have advertised the routes carried in these packets.

When establishing a connection between a router and a monitoring server, note the following rules:

- You can specify a port for the TCP connection between the router and the monitoring server.
- One router can connect to multiple monitoring servers, and one monitoring server can also connect to multiple routers.
- In each BMP instance, one router can connect to multiple monitoring servers so that:
 - Multiple monitoring servers back up each other, which improves reliability.
 - The load of a single server in monitoring BGP peers is reduced.
 - Each server can monitor BGP peers in a unique address family, allowing different BGP services to be controlled.
- A monitoring server can monitor BGP peers in all address families or in a specified address family.

Figure 9-40 Networking with BMP



Benefits

BMP facilitates the monitoring of BGP running status and reports security threats in real time so that preventive measures can be taken promptly.

9.2.24 BGP Best External

Background

If multiple routes to the same destination are available, a BGP device selects one optimal route based on BGP route selection policies and advertises the route to its BGP peers.

NOTE

For details about BGP route selection policies, see *BGP Principles*.

However, in scenarios with master and backup provider edges (PEs) or route reflectors (RRs), if routes are selected based on the preceding policies and the primary link fails, the BGP route convergence takes a long time because no backup route is available. To address this problem, the BGP Best External feature was introduced.

Related Concepts

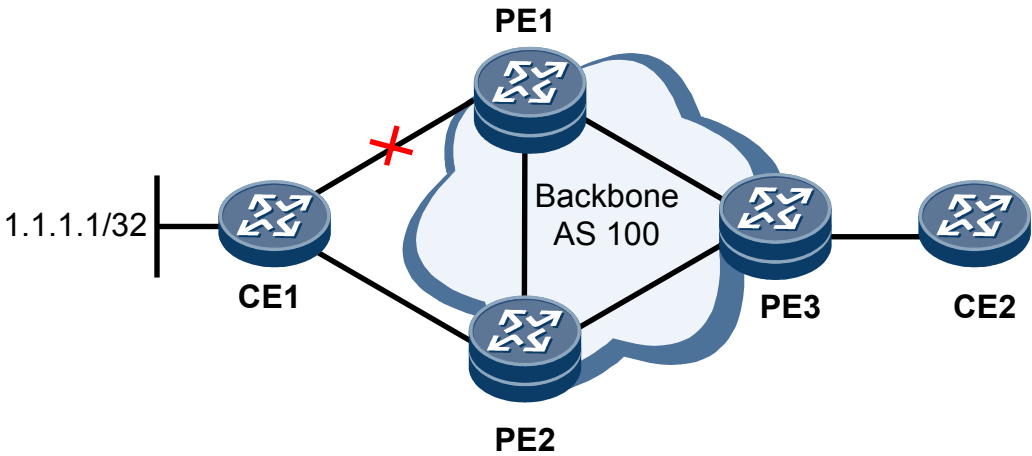
BGP Best External: A mechanism that enables a backup device to select a sub-optimal route and send the route to its BGP peers if the route preferentially selected based on BGP route selection policies is an Internal Border Gateway Protocol (IBGP) route advertised by the master device. Therefore, BGP Best External speeds up BGP route convergence if the primary link fails.

Best External route: The sub-optimal route selected after BGP Best External is enabled.

Networking with Master and Backup PEs

In the networking shown in [Figure 9-41](#), CE1 is dual-homed to PE1 and PE2. PE1 has a greater Local_Pref value than PE2 and CE1. EBGP peer relationships are established between CE1 and PE1, and between CE1 and PE2. In addition, IBGP peer relationships are established among PE1, PE2, and PE3. PE1 and PE2 receive the same route to 1.1.1.1/32 from CE1. After receiving this route, PE1 advertises it to PE2 and PE3. Therefore, PE2 has two routes to 1.1.1.1/32. Of the two routes, PE2 preferentially selects the route from PE1 because PE1 has a larger Local_Pref value than CE1. PE2 does not advertise the selected route to PE3. Therefore, PE3 has only one route to 1.1.1.1/32, which is advertised by PE1. If the link between CE1 and PE1 fails, a new route must be selected to take over traffic after routes are converged, which takes a long time.

Figure 9-41 Networking with master and backup PEs



BGP Best External can be enabled on PE2 to address this problem. With BGP Best External, PE2 selects the EBGP route from CE1 and advertises it to PE3. Therefore, PE3 has two routes to 1.1.1.1/32, in which the route CE1 -> PE2 -> PE3 backs up CE1 -> PE1 -> PE3. [Table 9-4](#) lists the differences with and without BGP Best External.

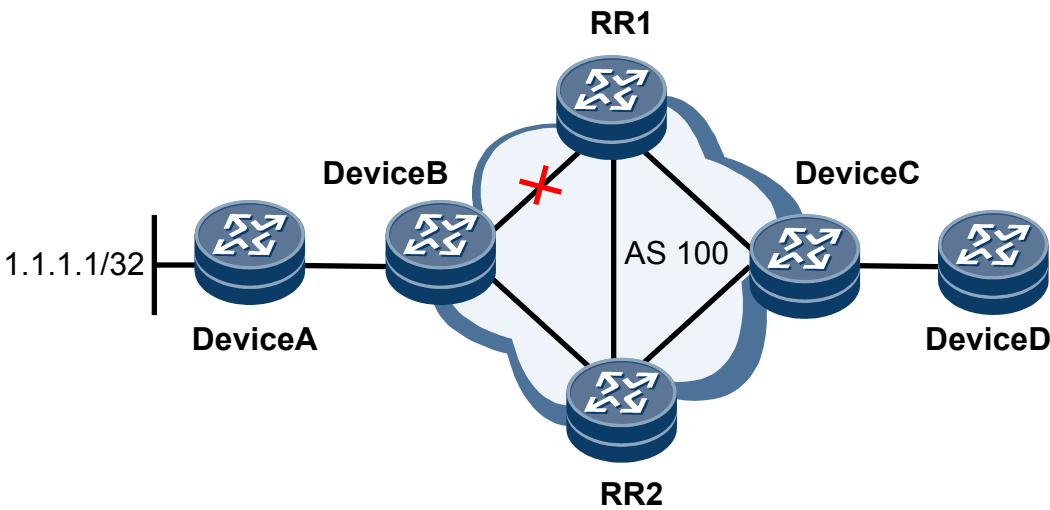
Table 9-4 Differences with and without BGP Best External

BGP Best External	Route Available on PE3	Optimal Route	If the Link Between CE1 and PE1 Fails
Not enabled	CE1 -> PE1 -> PE3	CE1 -> PE1 -> PE3	A new route must be selected to take over traffic after routes are converged.
Enabled	CE1 -> PE1 -> PE3 CE1 -> PE2 -> PE3	CE1 -> PE1 -> PE3	Traffic is switched to CE1 -> PE2 -> PE3 immediately.

Networking with Master and Backup RRs

In the networking shown in [Figure 9-42](#), an EBGP peer relationship is established between Device A and Device B. An IBGP peer relationship is established between each two devices among RR1, RR2, Device B, and Device C except between Device B and Device C. Device B is a client of RR1 and RR2. RR1 has a greater Local_Pref value than RR2, and therefore RR1 is the master device whereas RR2 is the backup device. RR1 and RR2 receive the same route to 1.1.1.1/32 from Device B. After receiving this route, RR1 advertises it to RR2 and Device C. Therefore, RR2 has two routes to 1.1.1.1/32. Of the two routes, RR2 preferentially selects the route from RR1 because RR1 has a greater Local_Pref value. RR2 does not advertise the selected route to Device C. Therefore, Device C has only one route to 1.1.1.1/32, which is advertised by RR1. If the link between Device B and RR1 fails, a new route must be selected to take over traffic after routes are converged, which takes a long time.

Figure 9-42 Networking with master and backup RRs



BGP Best External can be enabled on RR2 to address this problem. With BGP Best External, RR2 selects the EBGp route from Device B and advertises it to Device C. Therefore, Device C has two routes to 1.1.1.1/32, in which the route Device A -> Device B -> RR2 -> Device C backs up Device A -> Device B -> RR1 -> Device C. [Table 9-5](#) lists the differences with and without BGP Best External.

Table 9-5 Differences with and without BGP Best External

BGP Best External	Route Available on Device C	Optimal Route	If the Link Between Device B and RR1 Fails
Not enabled	Device A -> Device B -> RR1 -> Device C	Device A -> Device B -> RR1 -> Device C	A new route must be selected to take over traffic after routes are converged.
Enabled	Device A -> Device B -> RR1 -> Device C Device A -> Device B -> RR2 -> Device C	Device A -> Device B -> RR1 -> Device C	Traffic is switched to Device A -> Device B -> RR2 -> Device C immediately.

Usage Scenario

The BGP Best External feature applies to scenarios in which master and backup PE or RRs are deployed and the backup PE or RR needs to advertise the sub-optimal route (Best External route) to its BGP peers to speed up BGP route convergence.

Advantages

As networks develop, services, such as Voice over Internet Protocol (VoIP), online video, and financial services, pose higher requirements for real-time transmission. With BGP Best External, the backup device selects the sub-optimal route and advertises the route to its BGP peers, which speeds up BGP route convergence and minimizes service interruptions.

9.2.25 BGP Add-Path

Background

In a scenario with a route reflector (RR) and clients, if the RR has multiple routes to the same destination (with the same prefix), the RR selects an optimal route from these routes and then sends only the optimal route to its clients. Therefore, the clients have only one route to the destination. If a link along this route fails, route convergence takes a long time, which cannot meet the requirements on high reliability.

To address this issue, deploy the BGP Add-Path feature on the RR. With BGP Add-Path, the RR can send two or more routes with the same prefix to its clients. After reaching the clients, these routes can back up each other or load-balance traffic, which ensures high reliability in data transmission.

NOTE

- For details about route selection and advertisement policies, see [9.2.1 BGP Fundamentals](#).
- BGP Add-Path is deployed on RRs in most cases although it can be configured on any router.
- With BGP Add-Path, you can configure the maximum number of routes with the same prefix that an RR can send to its clients. The actual number of routes with the same prefix that an RR can send to its clients is the smaller value between the configured maximum number and the number of available routes with the same prefix.

Related Concepts

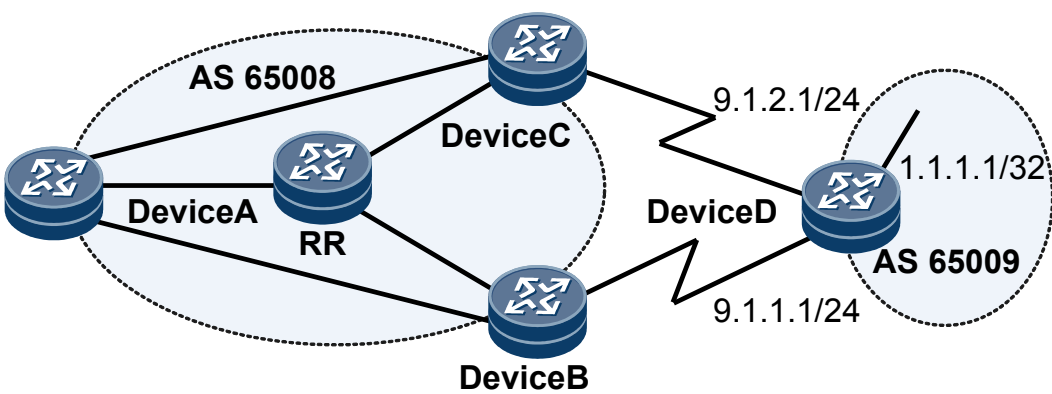
Add-Path route: The routes selected by BGP after BGP Add-Path is configured.

Typical Networking

On the network shown in [Figure 9-43](#), Device A, Device B, and Device C are clients of the RR, and Device D is an EBGP peer of Device B and Device C.

Each of Device B and Device C receives a route to 1.1.1.1/32 from Device D, with 9.1.1.1/24 and 9.1.2.1/24 as the next hops, respectively. Then, each of Device B and Device C sends the received route to the RR. After receiving the two routes, the RR selects an optimal route from them and sends it to Device A. Therefore, Device A has only one route to 1.1.1.1/32.

Figure 9-43 Networking with BGP Add-Path



BGP Add-Path can be configured to allow the RR to send more than one route with the same prefix to Device A. Suppose that the configured maximum number of routes with the same prefix that the RR can send to Device A is 2 and that the optimal route selected by the RR is the one from Device B. Table 9-6 lists the differences with and without BGP Add-Path.

Table 9-6 Differences with and without BGP Add-Path

Scenario	Route Available on Device A	If the Link Between Device D and Device B Fails
Not enabled	Only one route to 1.1.1.1/32, with 9.1.1.1/24 as the next hop	A new route must be selected to take over traffic after route convergence.
Enabled	Two routes to 1.1.1.1/32, with 9.1.1.1/24 and 9.1.2.1/24 as the next hops	If the two routes back up each other, the link between Device D and Device C takes over traffic. If the two routes load-balance traffic, the link between Device D and Device C takes over all the traffic that ran along the two routes.

Usage Scenario

The BGP Add-Path feature applies to scenarios in which an RR and clients are deployed and the RR needs to send more than one route with the same prefix to its clients to ensure high reliability in data transmission.

BGP Add-Path is used in traffic optimization scenarios and allows multiple routes to be sent to the controller.

Benefits

Deploying BGP Add-Path can improve network reliability.

9.2.26 BGP Recursion Suppression in Case of Next Hop Flapping

Background

In some scenarios, if a large number of routes recurse to the same next hop that flaps frequently, the system will be busy processing reselection and re-advertisement of these routes, which consumes excessive resources and leads to high CPU usage. BGP recursion suppression in case of next hop flapping can address this problem.

Principles

After this function is enabled, BGP calculates the penalty value that starts from 0 by comparing the flapping interval with configured intervals if next hop flapping occurs. When the penalty value exceeds 10, BGP suppresses route recursion to the corresponding next hop. For example, if the intervals for increasing, retaining, and clearing the penalty value are T1, T2, and T3, respectively, BGP calculates the penalty value as follows:

- Increases the penalty value by 1 if the flapping interval is less than T1.
- Retains the penalty value if the flapping interval is greater than or equal to T1, but less than T2.
- Reduces the penalty value by 1 if the flapping interval is greater than or equal to T2, but less than T3.
- Clears the penalty value if the flapping interval is greater than or equal to T3.

When the penalty value exceeds 10, the system processes reselection and re-advertisement of the routes that recurse to a flapping next hop much slower.

Benefits

BGP recursion suppression in case of next hop flapping prevents the system from frequently processing reselection and re-advertisement of a large number of routes that recurse to a flapping next hop, which reduces system resource consumption and CPU usage.

9.2.27 BGP-LS

BGP-link state (LS) enables BGP to report topology information collected by IGP to the controller.

Background

BGP-LS is a new method of collecting topology information.

Without BGP-LS, the router uses an IGP (OSPF or IS-IS) to collect topology information of each AS, and the IGP reports the information to the controller. This topology information collection method has the following disadvantages:

- The controller must have high computing capabilities and support the IGP and its algorithm.
- The controller cannot gain the complete inter-AS topology information and therefore is unable to calculate optimal E2E paths.

- Different IGP report topology information separately to the controller, which complicates the controller's analysis and processing.

For details on how OSPF collects topology information, see *NE40E Feature Description - OSPF*.

For details on how IS-IS collects topology information, see *NE40E Feature Description - IS-IS*.

With powerful routing capabilities of BGP, BGP-LS has the following advantages:

- Reduces computing capability requirements and spares the necessity of IGPs on the controller.
- Facilitates route selection and calculation on the controller by using BGP to summarize process or AS topology information and report the complete information to the controller.
- Requires only one routing protocol (BGP) to report topology information to the controller.

Related Concepts

BGP-LS provides a simple and efficient method of collecting topology information.

BGP-LS routes carry topology information and are classified into three types of routes that carry node, link, and route prefix information, respectively. These routes collaborate in carrying topology information.

BGP-LS Route Formats

Format of node routes

For example, a node route is in the format of [NODE][ISIS-LEVEL-1][IDENTIFIER0][LOCAL[as100][bgp-ls-identifier11.1.1.2][ospf-area-id0.0.0.0][igp-router-id0000.0000.0001.00]].

Node routes carry node information.

[Table 9-7](#) describes the fields in node routes.

Table 9-7 Description of the fields in node routes

Item	Description
NODE	Field indicating that the BGP-LS route is a node route.
ISIS-LEVEL-1	Protocol that collects topology information. The protocol is IS-IS in this example.
IDENTIFIER0	Identifier of the protocol that collects topology information.
LOCAL	Field indicating information of a local node.
as	BGP-LS domain AS number.
bgp-ls-identifier	BGP-LS domain ID.

Item	Description
ospf-area-id	OSPF area ID.
igp-router-id	IGP router ID, generated by the IGP that collects topology information. The router ID is obtained from the NET of an IS-IS process in this example.

Format of link routes

For example, a link route is in the format of [LINK][ISIS-LEVEL-1][IDENTIFIER0][LOCAL[as255.255][bgp-ls-identifier192.168.102.4][ospf-area-id0.0.0.0][igp-router-id0000.0000.0002.01]][REMOTE[as255.255][bgp-ls-identifier192.168.102.4][ospf-area-id0.0.0.0][igp-router-id0000.0000.0002.00]][LINK[if-address0.0.0.0][peer-address0.0.0.0][if-address::][peer-address::][mt-id0]].

Link routes carry information about links between devices.

[Table 9-8](#) describes the fields in link routes.

Table 9-8 Description of the fields in link routes

Item	Description
LINK	Field indicating that the BGP-LS route is a link route.
ISIS-LEVEL-1	Protocol that collects topology information. The protocol is IS-IS in this example.
IDENTIFIER0	Identifier of the protocol that collects topology information.
LOCAL	Field indicating information of a local node.
as	BGP-LS domain AS number.
bgp-ls-identifier	BGP-LS domain ID.
ospf-area-id	OSPF area ID.
igp-router-id	IGP router ID, generated by the IGP that collects topology information. The router ID is obtained from the NET of an IS-IS process in this example.
REMOTE	Field indicating information of a remote node.
if-address	IP address of the local interface.
peer-address	IP address of the remote interface.
mt-id	ID of the topology.

Format of prefix routes

For example, a prefix route is in the format of [IPV4-PREFIX][ISIS-LEVEL-1][IDENTIFIER0][LOCAL[as100][bgp-ls-identifier192.168.102.3][ospf-area-id0.0.0.0][igp-router-id0000.0000.0001.00]][PREFIX[mt-id0][ospf-route-type0][prefix192.168.102.0/24]].

Prefix routes carry information about reachable network segments.

Table 9-9 describes the fields in link routes.

Table 9-9 Description of the fields in prefix routes

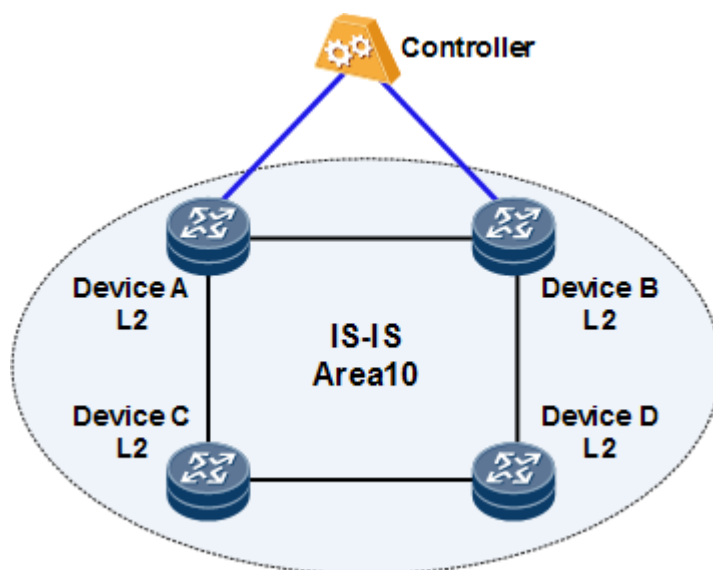
Item	Description
IPV4-PREFIX	Field indicating an IPv4 or IPv6 prefix route. The router cannot generate IPv6 prefix routes, but it can process the IPv6 prefix routes received from non-Huawei devices.
ISIS-LEVEL-1	Protocol that collects topology information. The protocol is IS-IS in this example.
IDENTIFIER0	Identifier of the protocol that collects topology information.
LOCAL	Field indicating information of a local node.
as	BGP-LS domain AS number.
bgp-ls-identifier	BGP-LS domain ID.
ospf-area-id	OSPF area ID.
igp-router-id	IGP router ID, generated by the IGP that collects topology information. The router ID is obtained from the NET of an IS-IS process in this example.
PREFIX	Field indicating an IGP route.
mt-id	ID of the topology.
ospf-route-type	OSPF route type: ● 1: Intra-Area ● 2: Inter-Area ● 3: External 1 ● 4: External 2 ● 5: NSSA 1 ● 6: NSSA 2
prefix	Prefix of an IGP route.

Typical Networking

Networking in which topology information is collected within an IGP area

In [Figure 9-44](#), Device A, Device B, Device C, and Device D use IS-IS to communicate with each other at the network layer. They are all Level-2 devices in the same area (area 10). Only one of the four devices needs to have BGP-LS deployed and establish a BGP-LS peer relationship with the controller so that BGP-LS can collect and report topology information to the controller. To improve reliability, deploying BGP-LS on two or more devices and establishing a BGP-LS peer relationship between each BGP-LS device and the controller are recommended. The BGP-LS devices collect the same topology information, and they back up each other in case one of them fails.

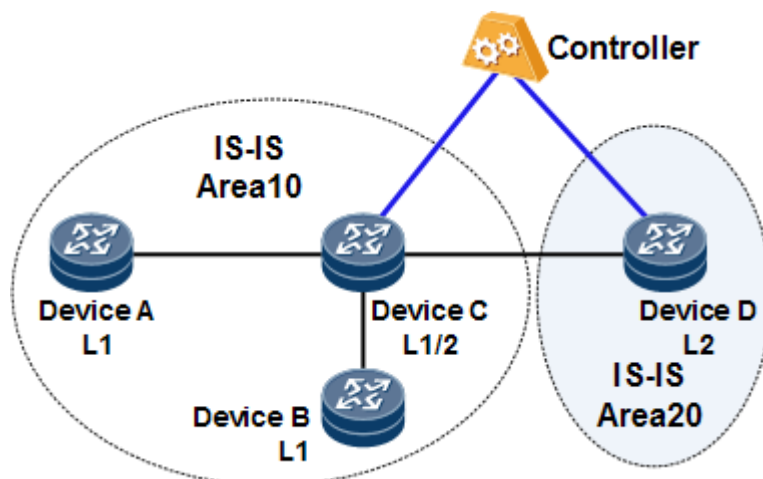
Figure 9-44 Networking in which topology information is collected within an IGP area



Networking in which topology information is collected between IGP areas

In [Figure 9-45](#), Device A, Device B, Device C, and Device D use IS-IS to communicate with each other at the network layer. Device A, Device B, and Device C reside in area 10, whereas Device D resides in area 20. Device A and Device B are Level-1 devices, Device C is a Level-1-2 device, and Device D is a Level-2 device. Only one of the four devices needs to have BGP-LS deployed and establish a BGP-LS peer relationship with the controller so that BGP-LS can collect and report topology information to the controller. To improve reliability, deploying BGP-LS on two or more devices and establishing a BGP-LS peer relationship between each BGP-LS device and the controller are recommended. The BGP-LS devices collect the same topology information, and they back up each other in case one of them fails.

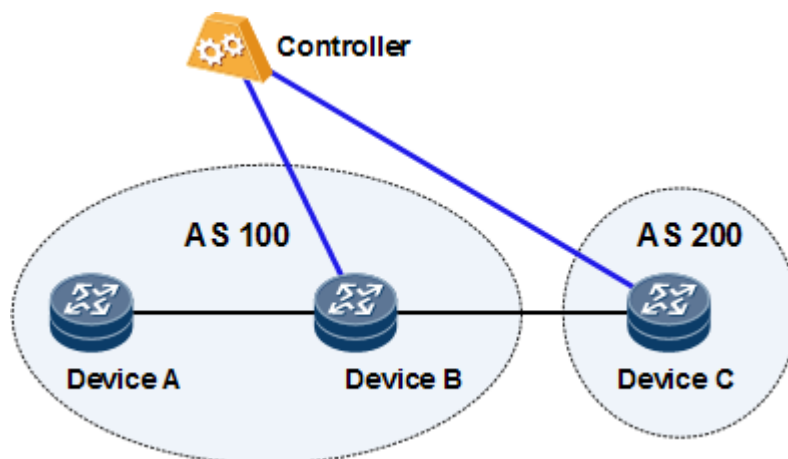
Figure 9-45 Networking in which topology information is collected between IGP areas



Networking in which topology information is collected between BGP ASs

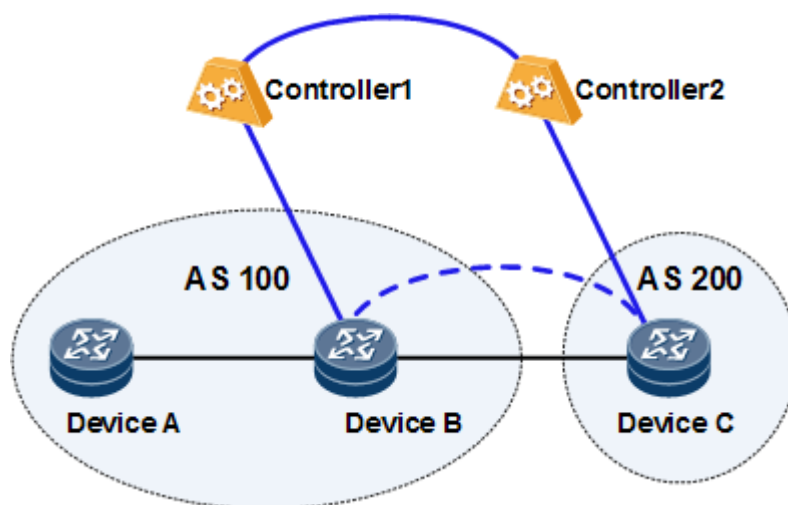
In [Figure 9-46](#), Device A and Device B belong to the same AS, and an IS-IS neighbor relationship is established between them. BGP is not enabled on Device A. An EBGP peer relationship is established between Device B and Device C. If BGP-LS is not enabled, BGP cannot exchange topology information between AS 100 and AS 200. As a result, the topology information collected in AS 100 is different from that collected in AS 200. To address this problem, enable BGP-LS on at least one device (two or more is recommended for higher reliability) in each AS and establish a BGP-LS peer relationship between each of the devices and the controller.

Figure 9-46 Networking 1 in which topology information is collected between BGP ASs



If two controllers are deployed and are connected to different ASs, for example in [Figure 9-47](#), a BGP-LS peer relationship must be established between the two controllers or between Device B and Device C so that both controllers can obtain topology information on the whole network.

Figure 9-47 Networking 2 in which topology information is collected between BGP ASs



NOTE

To reduce the number of connections to the controller, deploy one or more BGP-LS RRs and establish BGP-LS peer relationships between each RR and the devices that require BGP-LS peer relationships with the controller.

Usage Scenario

The router functions as a forwarder and reports topology information to the controller for topology monitoring and traffic control.

Benefits

BGP-LS offers the following benefits:

- Reduces computing capability requirements of the controller.
- Allows the controller to gain the complete inter-AS topology information.
- Requires only one routing protocol (BGP) to report topology information to the controller.

9.2.28 RPD

Background

Route policy distribution (RPD) is used to distribute route policies dynamically.

Without RPD, route-policies can be generated only through manual configuration, and then the route-policies are applied to peers. Such a generation mode is not applicable when the route-policies need to be adjusted dynamically and frequently. For example, in the inbound traffic optimization scenario with an NCE, the NCE monitors the traffic bandwidth usage on the network in real time, and users perform traffic optimization based on the analysis result. Specifically, for traffic optimization purposes, route-policies need to be used to modify route attributes to control the route selection on the peer end. However, the traffic bandwidth usage constantly changes, leading to the constant changes of traffic optimization policies. In this case, route-policies configured manually are not suitable. RPD provides a dynamic route-policy distribution mode for the NCE. With RPD, route-policy information is transmitted through the BGP RPD address family.

After RPD is configured, you can use the NCE to monitor and control traffic in real time. The traffic optimization policy configurations are performed on the NCE, not on forwarders. Forwarders receive RPD routes from the NCE, generate route-policies based on the routes, and implement the route-policies.

Related Concepts

RPD route: Carries route-policy information and distributes the information to peers in the BGP RPD address family. After learning the RPD route, the receiver converts it into a route-policy and applies the policy.

RPD Route Format

RPD routes are in the format of policy type (export policy)/peer address/policy ID, for example, 1/1.1.1.1/1.

Table 9-10 describes the fields in RPD routes.

Table 9-10 RPD route description

Field	Description
Policy type	Specifies the policy type. Currently, only export policies are supported.
Peer address	Specifies the peer address used by the policy.
Policy ID	Specifies the ID of a policy.

Route-policies carried in RPD routes are encapsulated through the WideCommunity attribute. Figure 9-48 shows the WideCommunity format.

Figure 9-48 WideCommunity format

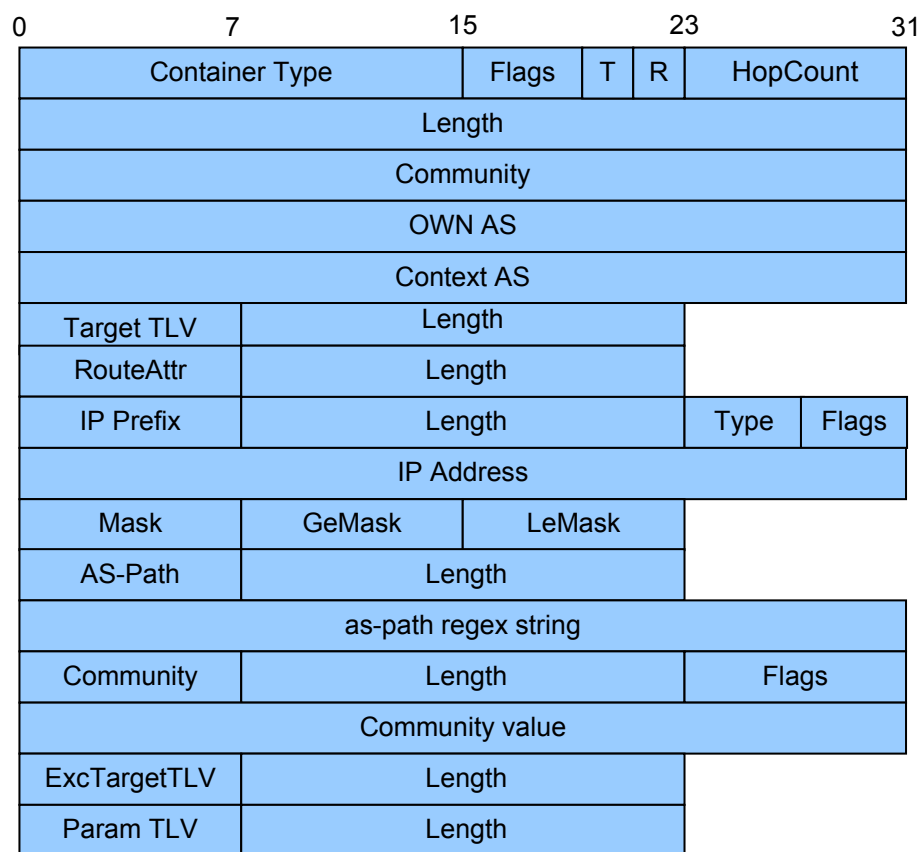


Table 9-11 Description of fields in the WideCommunity format

Field Name	Length (in bits)	Description
Container Type	16	The value is 1, indicating the WideCommunity attribute.
Flags	8	The R bit is 0, indicating a private attribute. The T bit is 0, which is not used currently.
HopCount	8	This field is not used currently. The value is 1.
Length	32	Total packet length.
Community	32	Value of WideCommunity. Each value identifies a specific function of WideCommunity. If the value is MATCH AND SET ATTR , the attributes of the routes that match specific conditions are modified. If the value is MATCH AND NOT ADVERTISE , the routes that match specific conditions are not advertised.
OWN AS	32	AS number of the NCE.
Context AS	32	AS number of the local device (forwarder).

Field Name	Length (in bits)	Description
Target TLV	8	Target TLV.
Length	16	Target TLV length.
RouteAttr	8	Route attribute type. The TLV has three sub-TLVs: IP Prefix, AS-Path, and Community.
Length	16	Length of the route attribute type.
IP Prefix	8	IP address prefix.
Length	16	Length of IP address prefix.
Type	4	Matching type of the IP address prefix. ● 0: exact matching ● 1: matches the routes that carry the prefix and a mask whose length is greater than or equal to the specified mask length. ● 2: matches the routes that carry the prefix and a mask whose length is less than or equal to the specified mask length. ● 3: matches the routes that carry the prefix and a mask whose length is within the specified range.
Flags	4	This field is not used currently.
IP Address	32	IP address.
Mask	8	Mask of the IP address.
GeMask	8	Used to specify a range. The value of this field must be less than or equal to the length of the Mask or be 0.
LeMask	8	Used to specify a range. The value of this field must be greater than the length of the Mask or be 0.
AS-Path	8	AS_Path.
Length	16	AS_Path length.
as-path regex string	32	Content of the AS_Path, which is presented using a regular expression. The maximum length of the AS_Path is 1024 bytes.
Community	8	Community attribute.
Length	16	Length of the community attribute.
Flags	8	This field is not used currently.

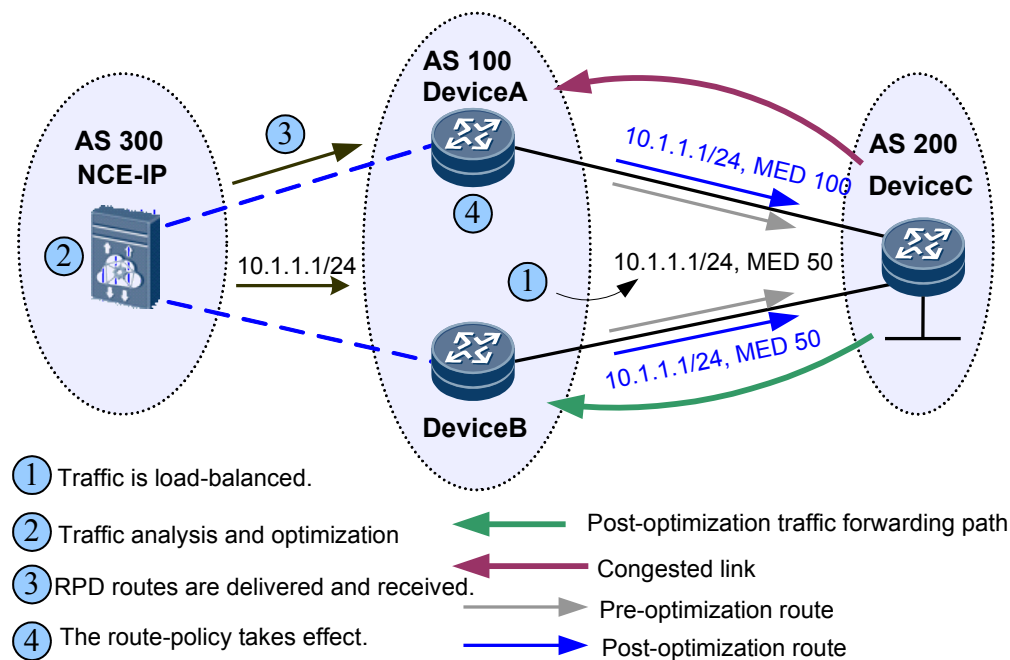
Field Name	Length (in bits)	Description
Community value	32	Community attribute value.
ExcTargetTLV	8	This field is not used currently. Even if the field has a value, it is also ignored.
Length	16	ExcTargetTLV length.
Param TLV	8	Content of the action to be performed. The format depends on the community value, and the TLV content also varies according to the community value. If the community value is MATCH AND SET ATTR , the MED or AS_Path attribute is modified. If the community value is MATCH AND NOT ADVERTISE , the TLV is empty, without any sub-TLV.
Length	16	Param TLV length.

Implementation

In the following example, the typical networking of inbound traffic optimization is used to describe how RPD works to implement traffic optimization:

Figure 9-49 shows an inbound traffic optimization scenario. The NCE collects traffic information from devices and performs analysis and calculation to identify the routes to be adjusted. After a traffic optimization policy is configured on the NCE, the NCE converts the policy to an RPD route, and delivers the route to forwarders.

Figure 9-49 Typical networking of inbound traffic optimization



In [Figure 9-49](#), the implementation of inbound traffic optimization is as follows:

1. Before traffic optimization is implemented, the MED values of the BGP routes advertised from Device A to Device C and from Device B to Device C are 50, and traffic is load balanced.
2. The NCE collects traffic information from devices and performs calculation and finds that the path from Device C to Device A is congested. In this case, it is expected that the traffic that is from AS 200 and destined for 10.1.1.1/24 enters AS 100 through Device B rather than Device A. The NCE configures a traffic optimization policy, converts it to an RPD route, and delivers the route to Device A to instruct Device A to increase the MED value of the route advertised to AS 200 to 100. The MED value of the route advertised by Device B to AS 200 remains unchanged.
3. After receiving the RPD route delivered by the NCE, Device A generates a route-policy based on the RPD route and executes the policy.
4. The RPD route-policy takes effect. After receiving the routes destined for 10.1.1.1/24 from A and B, Device C selects the route received from Device B because its MED value is smaller than the MED value of the route received from Device A. In this case, traffic that is from AS 200 and destined for 10.1.1.1/24 enters AS 100 through Device B rather than Device A.

 NOTE

In this scenario, forwarders receive RPD policies delivered by the NCE and adjust route attributes (MED or AS_Path) based on the policies. The forwarders follow the policies strictly when advertising routes, but are not responsible for the traffic optimization results. You can check the traffic optimization results in real time through the NCE.

Usage Scenario

The IP network optimization solution provides users with a method of on-demand traffic scheduling to make full use of network bandwidth. In the IP network optimization solution, RPD ensures inbound traffic optimization in MAN ingress or IGW scenarios. In this solution, the router functions as a forwarder, receives RPD routes delivered by the NCE, generates route-policies based on the RPD routes, and executes the policies to implement dynamic traffic optimization.

Benefits

In traffic optimization scenarios, RPD spares manual BGP route-policy maintenance which is complex, time-consuming, and error-prone. Therefore, RPD reduces maintenance workload and improves maintenance quality.

10 Routing Policy

About This Chapter

[10.1 Overview of Routing Policies](#)

[10.2 Understanding Routing Policies](#)

[10.3 Application Scenarios for Routing Policies](#)

10.1 Overview of Routing Policies

Definition

Routing policies are used to filter routes and control how routes are received and advertised. If route attributes, such as reachability, are changed, the path along which network traffic passes changes accordingly.

Purpose

When advertising, receiving, and importing routes, the router implements certain routing policies based on actual networking requirements to filter routes and change the route attributes. Routing policies serve the following purposes:

- Control route advertising
Only routes that match the rules specified in a policy are advertised.
- Control route receiving
Only the required and valid routes are received, which reduces the routing table size and improves network security.
- Filter and control imported routes
A routing protocol may import routes discovered by other routing protocols. Only routes that satisfy certain conditions are imported to meet the requirements of the protocol.
- Modify attributes of specified routes
To enrich routing information, a routing protocol may import routing information discovered by other routing protocols. Only the routing information that satisfies the

conditions is imported. Some attributes of the imported routing information are changed to meet the requirements of the routing protocol.

Benefits

Routing policies have the following benefits:

- Control the routing table size, saving system resources.
- Control route receiving and advertising, improving network security.
- Modify attributes of routes for proper traffic planning, improving network performance.

Differences Between the Routing Policy and Policy-based Routing

Unlike the routing mechanism that searches the forwarding table for matching routes based on the destination addresses of IP packets, policy-based routing (PBR) is based on the user-defined routing policies. PBR selects routes based on the user-defined routing policies, with reference to the source IP addresses and lengths of incoming packets. PBR can be used to improve security and implement load balancing.

A routing policy and PBR have different mechanisms. [Table 10-1](#) shows the differences between them.

Table 10-1 Differences between the routing policy and PBR

Routing Policy	Policy-based Routing
Forwards packets based on destination addresses in the routing table.	Forwards packets based on the policy. The device searches the routing table for packet forwarding only after packets fail to be forwarded based on the policy.
Based on the control plane, serves routing protocols and routing tables.	Based on the forwarding plane, serves forwarding.
Combines with a routing protocol to form a policy.	Needs to be configured hop by hop to ensure that packets are forwarded based on the policies.
Is configured using the route-policy command.	Is configured using the policy-based-route command.

10.2 Understanding Routing Policies

Implementation

Routing policies are implemented in the following steps:

1. Define rules. Characteristics of routing information to which routing policies are applied need to be defined. Specifically, you need to define a set of matching rules regarding different attributes of routing information, such as the destination address and AS number.
2. Apply rules. Matching rules are applied to advertise, receive, or import routes.

Filter

A filter is the core of a routing policy and is defined using a set of matching rules. The NE40E provides the filters listed in [Table 10-2](#).

Table 10-2 Comparisons between filters

Filter	Application Scope	Matching Rules
Access control list (ACL)	Dynamic routing protocols	Inbound interface, source or destination IP address, protocol type, and source or destination port number
IP prefix list	Dynamic routing protocols	Source and destination IP addresses and next hop address
AS_Path	BGP	AS_Path attribute
Community	BGP	Community attribute
Extended community	VPN	Extended community attribute
Route distinguisher (RD)	VPN	RD attribute
Route-policy	Dynamic routing protocols	Destination IP address, next hop address, cost, interface information, route type, ACL, IP prefix list, AS_Path, community, extended community, Layer 2 VNI list, Layer 3 VNI list, MAC address list, Ethernet Tag list, and RD.

The ACL, IP prefix list, AS_Path, community, Extended community, and RD filters can be used to filter routes but cannot modify route attributes. A route-policy is a comprehensive filter and can use the matching rules of the ACL, IP prefix list, AS_Path, community, Extended community, and RD filters to filter routes and change route attributes.

ACL

An ACL is a set of sequential filtering rules. Users can define rules based on packet information, such as inbound interfaces, source or destination IP addresses, protocol types, or source or destination port numbers and specify an action to deny or permit packets. After an ACL is configured, the system classifies received packets based on the rules defined in the ACL and denies or permits the packets accordingly.

An ACL only classifies packets based on defined rules and filters packets only after it is applied to a routing policy.

ACLs can be configured for both IPv4 packets and IPv6 packets. Based on the usage, ACLs are classified as interface-based ACLs, basic ACLs, or advanced ACLs. Users can specify the IP address and subnet address range in an ACL to match the source IP address, destination network segment address, or the next hop address of a route.

ACLs can be configured on access or core devices to:

- Protect the devices against IP, TCP, and Internet Control Message Protocol (ICMP) packet attacks.
- Control network access. For example, ACLs can be used to control the access of enterprise network users to external networks, the specific network resources that users can access, and the period for which users can access networks.
- Limit network traffic and improve network performance. For example, ACLs can be used to limit bandwidth for upstream and downstream traffic, charge for the bandwidth that users have applied for, and fully use high-bandwidth network resources.

For details about ACL features, see ACL.

IP Prefix List

An IP prefix list contains a group of route filtering rules. Users can specify the prefix and mask length range to match the destination network segment address or the next hop address of a route. An IP prefix list is used to filter routes that are advertised and received by various dynamic routing protocols.

An IP prefix list is easier and more flexible than an ACL. However, if a large number of routes with different prefixes need to be filtered, configuring an IP prefix list to filter the routes is complex.

IP prefix lists can be configured for both IPv4 routes and IPv6 routes, and they share the same implementation process. An IP prefix list filters routes based on the mask length or mask length range.

- Mask length: An IP prefix list filters routes based on IP address prefixes. An IP address prefix is defined by an IP address and the mask length. For example, for route 10.1.1.1/16, the mask length is 16 bits, and the valid prefix is 16 bits (10.1.0.0).
- Mask length range: Routes with the IP address prefix and mask length within the range defined in the IP prefix list meet the matching rules.

NOTE

0.0.0.0 is a wildcard address. If the IP prefix is 0.0.0.0, specify either a mask or a mask length range, with the following results:

- If a mask is specified, all routes with the mask are permitted or denied.
- If a mask length range is specified, all routes with the mask length in the range are permitted or denied.

AS_Path

An AS_Path filter is used to filter BGP routes based on AS_Path attributes contained in BGP routes. The AS_Path attribute is used to record in distance-vector (DV) order the numbers of all ASs through which a BGP route passes from the local end to the destination. Therefore, AS_Path attributes can be used to filter BGP routes.

The matching condition of an AS_Path is specified using a regular expression. For example, ^30 indicates that only the AS_Path attribute starting with 30 is matched. Using a regular expression can simplify configurations. For details about regular expressions, see Configuration Guide - Basic Configurations.

 NOTE

The AS_Path attribute is a private attribute of BGP and is therefore used to filter BGP routes only. For details about the AS_Path attribute, see [9.2.1 BGP Fundamentals](#).

Community

A community filter is used to filter BGP routes based on the community attributes contained in BGP routes. The community attribute is a group of destination addresses with the same characteristics. Therefore, community attributes can be used to filter BGP routes.

In addition to the well-known community attributes, users can define community attributes using digits. The matching condition of a community filter can be specified using a community ID or a regular expression.

 NOTE

Like AS_Path filters, community filters are used to filter only BGP routes because the community attribute is also a private attribute of BGP. For details about the community attribute, see [9.2.6 Community Attribute](#).

Extended Community

An extended community is used to filter BGP routes based on extended community attributes. BGP extended community attributes are classified as follows:

- **VPN target:** A VPN target controls route learning between VPN instances, isolating routes of VPN instances from each other. A VPN target may be either an import or export VPN target. Before advertising a VPNv4 or VPNv6 route to a remote MP-BGP peer, a PE adds an export VPN target to the route. After receiving a VPNv4 or VPNv6 route, the remote MP-BGP peer compares the received export VPN target with the local import VPN target. If they are the same, the remote MP-BGP peer adds the route to the routing table of the local VPN instance.
- **Source of Origin (SoO):** Several CEs at a VPN site may be connected to different PEs. The VPN routes advertised from the CEs to the PEs may be re-advertised to the VPN site where the CEs reside after the routes have traversed the backbone network, causing routing loops at the VPN site. In this situation, configure an SoO attribute for VPN routes. With the SoO attribute, routes advertised from different VPN sites can be distinguished and will not be advertised to the source VPN site, preventing routing loops.

The formats of VPN target and SoO extended community attributes are the same. The matching condition of an extended community can be specified using an extended community ID or a regular expression.

 NOTE

An extended community is used to filter only BGP routes because the extended community attribute is also a private attribute of BGP. For details about the extended community attribute, see BGP/MPLS IP VPN Fundamentals.

RD

An RD is used to filter BGP routes based on RDs in VPN routes. RDs are used to distinguish IPv4 and IPv6 prefixes in the same address segment in VPN instances. An RD filters specify matching rules regarding RD attributes.

For details about how to configure an RD, see *HUAWEI NetEngine40E Universal Service Router Configuration Guide - VPN*.

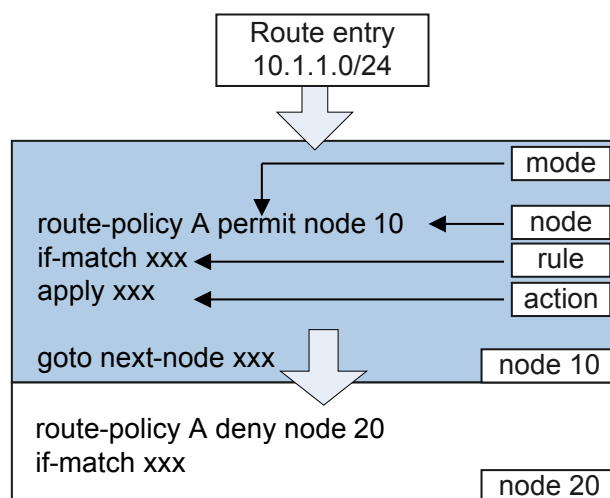
Route-Policy

A route-policy is a comprehensive filter. It is used to match attributes of specified routes and change route attributes when specific conditions are met. A route-policy can use the preceding six filters to define its matching rules.

- **Composition of a Route-Policy**

As shown in [Figure 10-1](#), a route-policy consists of node IDs, matching mode, **if-match** clauses, **goto next-node** clauses and **apply** clauses.

Figure 10-1 Composition of a route-policy



- Node ID

A route-policy consists of one or more nodes. Node IDs are specified as indexes in the IP prefix list. In a route-policy, routes are filtered based on the following rules:

- **Sequential matching:** The system checks entries based on node IDs in ascending order. Therefore, specifying the node IDs in the required sequence is recommended.
- **One-time matching:** The relationship between the nodes of a route-policy is "OR". If a route matches one node, the route matches the route-policy and will not be matched against the next node.

- Matching mode

Either of the following matching modes can be used:

- **permit:** specifies the **permit** mode of a node. If a route matches the **if-match** clauses of a node, all the actions defined by **apply** clauses are performed, and the matching is complete. If a route does not match the **if-match** clauses of the node, the route continues to match against the next node.
- **deny:** specifies the **deny** mode of a node. In **deny** mode, the **apply** clauses are not used. If a route matches all the **if-match** clauses of the node, the route is denied by the node and no longer matches against the next node. If the route does not match any of the **if-match** clauses, the route continues to match against the next node.

 NOTE

To allow other routes to pass through, a route-policy that contains no **if-match** or **apply** clause in **permit** mode needs to be configured for a node next to multiple nodes in **deny** mode.

– **if-match** clause

The **if-match** clause defines the matching rules.

Each node of a route-policy can comprise multiple **if-match** clauses or no **if-match** clause at all. If no **if-match** clause is configured for a node in **permit** mode, all IPv4 and IPv6 routes match the node. If an **if-match** clause is configured to match only IPv4 routes for a node in **permit** mode, matching IPv4 routes and all IPv6 routes match the node. If an **if-match** clause is configured to match only IPv6 routes for a node in **permit** mode, matching IPv6 routes and all IPv4 routes match the node.

– **apply** clause

The **apply** clauses specify actions. When a route matches a route-policy, the system sets some attributes for the route based on the **apply** clause.

Each node of a route-policy can comprise multiple **apply** clauses or no **apply** clause at all. The **apply** clause is not used when routes need to be filtered but attributes of the routes do not need to be changed.

– **goto next-node** clause

goto next-node clauses further match routes against a specified node after the routes match the current node.

● **Matching results of a route-policy**

The matching results of a route-policy are obtained based on the following aspects:

- Matching mode of the node, either **permit** or **deny**
- Matching rules (either **permit** or **deny**) contained in the **if-match** clause (such as ACLs or IP prefix lists)

The matching results are listed in [Table 10-3](#).

Table 10-3 Matching results of a Route-Policy

Rule (Matching Rule Contained in if-match Clauses)	Mode (Matching Mode of a Node)	Matching Result
permit	permit	● Routes matching the if-match clauses of the node match the route-policy, and the matching is complete. ● Routes not matching the if-match clauses of the node continue to match against the next node of the route-policy.
	deny	● Routes matching the if-match clauses of the node are denied by the route-policy, and the matching is complete. ● Routes not matching the if-match clauses of the node continue to match against the next node of the route-policy.

Rule (Matching Rule Contained in if-match Clauses)	Mode (Matching Mode of a Node)	Matching Result
deny	permit	- Routes matching the if-match clauses of the node are denied by the route-policy and continue to match against the next node. - Routes not matching the if-match clauses of the node continue to match against the next node of the route-policy.
	deny	- Routes matching the if-match clauses of the node are denied by the route-policy and continue to match against the next node. - Routes not matching the if-match clauses of the node continue to match against the next node of the route-policy. NOTE If all if-match clauses and nodes of the route-policy are in deny mode, all the routes to be filtered are denied by the route-policy.

 NOTE

On the HUAWEI NetEngine40E, all routes that fail to match a route-policy are denied by the route-policy by default. If more than one node is defined in a route-policy, at least one of them must be in **permit** mode. The reason is as follows:

- If a route fails to match any of the nodes, the route is denied by the route-policy.
- If all the nodes in the route-policy are set in **deny** mode, all the routes to be filtered are denied by the route-policy.

Other Functions

In addition to the preceding functions, routing policies have an enhanced feature: BGP to IGP.

In some scenarios, when an IGP uses a routing policy to import BGP routes, route attributes, the cost for example, can be set based on private attributes, such as the community in BGP routes. However, without the BGP to IGP feature, BGP routes are denied because the IGP fails to identify private attributes, such as community attributes in these routes. As a result, **apply** clauses used to set route attributes do not take effect.

With the BGP to IGP feature, route attributes can be set based on private attributes, such as the community, extended community, and AS_Path attributes in BGP routes. The BGP to IGP implementation process is as follows:

- When an IGP imports BGP routes through a route-policy, route attributes can be set based on private attributes, such as the community attribute in BGP routes.
- If BGP routes carry private attributes, such as community attributes, the system filters the BGP routes based on the private attributes. If the BGP routes meet the matching rules, the routes match the route-policy, and **apply** clauses take effect.

- If BGP routes do not carry private attributes, such as community attributes, the BGP routes fail to match the route-policy and are denied, and **apply** clauses do not take effect.

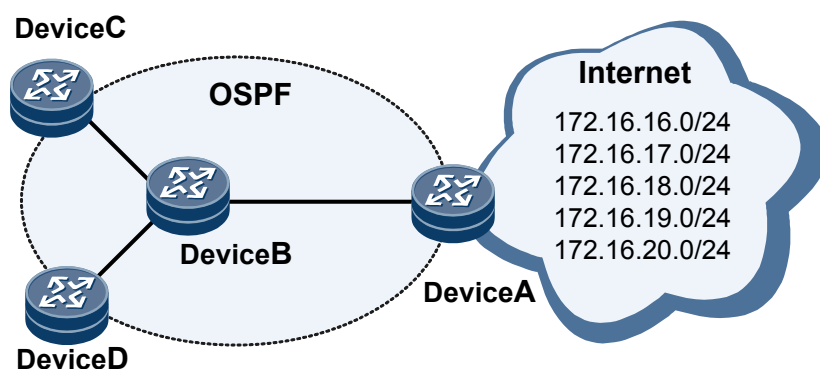
10.3 Application Scenarios for Routing Policies

Specific Route Filtering

On the OSPF-enabled network shown in [Figure 10-2](#), Device A receives routes from the Internet and advertises some of the routes to Device B.

- Device A advertises only routes 172.16.17.0/24, 172.16.18.0/24, and 172.16.19.0/24 to Device B.
- Device C accepts only the route 172.16.18.0/24.
- Device D accepts all the routes advertised by Device B.

Figure 10-2 Filtering received and advertised routes



There are multiple approaches to meet the preceding requirements, and the following two approaches are used in this example:

- Use IP prefix lists
 - Configure an IP prefix list for Device A and configure the IP prefix list as an export policy on Device A for OSPF.
 - Configure another IP prefix list for Device C and configure the IP prefix list as an import policy on Device C for OSPF.
- Use route-policies
 - Configure a route-policy (the matching rules can be the IP prefix list, cost, or route tag) for Device A and configure the route-policy as an export policy on Device A for OSPF.
 - Configure another route-policy on Device C and configure the route-policy as an import policy on Device C for OSPF.

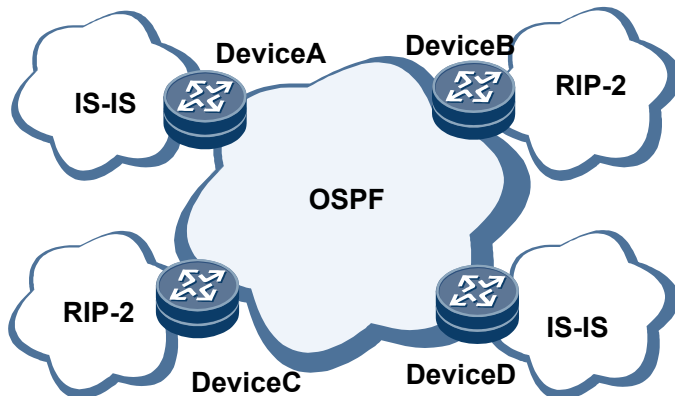
Compared with an IP prefix list, a route-policy can change route attributes and control routes more flexibly, but it is more complex to configure.

Transparent Transmission of Routes of Other Protocols Through an OSPF AS

On the network shown in [Figure 10-3](#), an AS runs OSPF and functions as a transit AS for other areas. Routes from the IS-IS area connected to Device A need to be transparently

transmitted through the OSPF AS to the IS-IS area connected to Device D. Routes from the RIP-2 area connected to Device B need to be transparently transmitted through the OSPF AS to the RIP-2 area connected to Device C.

Figure 10-3 Transparently transmitting routes of other protocols through an OSPF AS

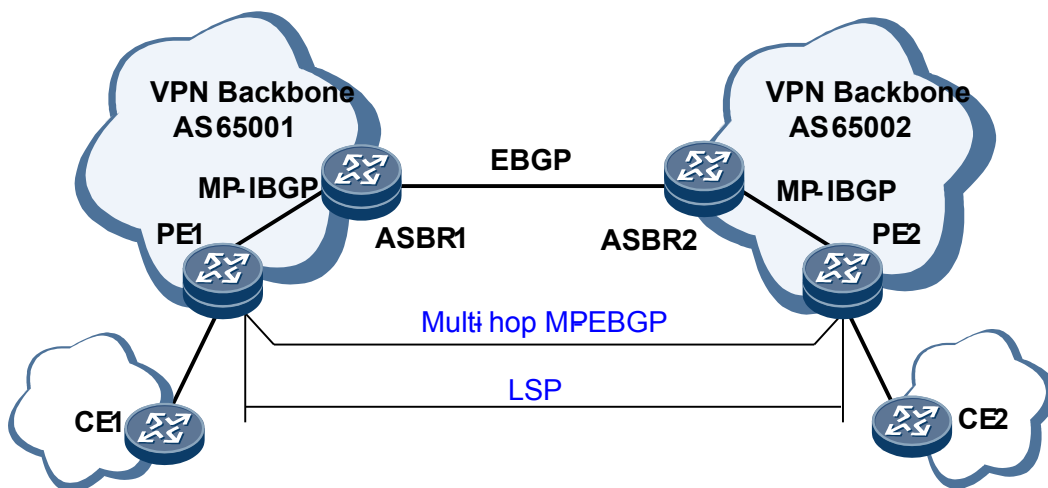


To meet the preceding requirements, configure a route-policy for Device A to set a tag for the imported IS-IS routes. Device D identifies the IS-IS routes from OSPF routes based on the tag.

Routing Policy Application in Inter-AS VPN Option C

On the network shown in [Figure 10-4](#), CE1 and CE2 communicate with each other through inter-AS VPN Option C.

Figure 10-4 Implementing route-policies in the inter-AS VPN Option C scenario



To establish an inter-AS label switched path (LSP) between PE1 and PE2, route-policies need to be configured for autonomous system boundary routers (ASBRs).

- When an ASBR advertises the routes received from a PE in the same AS to the peer ASBR, the ASBR allocates MPLS labels to the routes using a route-policy.

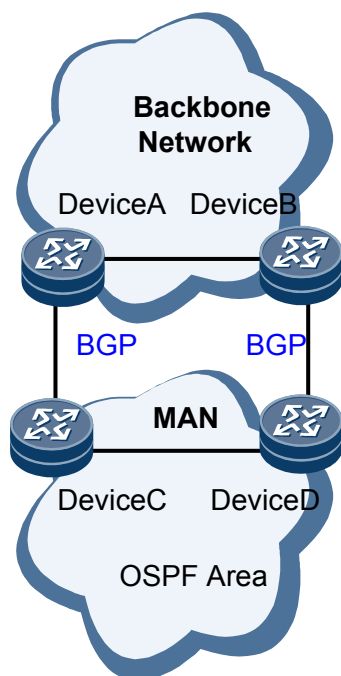
- When an ASBR advertises labeled IPv4 routes to a PE in the same AS, the ASBR reallocates MPLS labels to the routes using another route-policy.

In addition, to control route transmission between different VPN instances on a PE, configure a route-policy for the PE and configure the route-policy as an import or export policy for the VPN instances.

Application of BGP to IGP

On the network shown in **Figure 10-5**, Device A and Device B are aggregation devices on a backbone network, and Device C and Device D are egress devices of a metropolitan area network (MAN). BGP peer relationships are established between Device A and Device C as well as between Device B and Device D. External routes are advertised to the MAN using BGP. The MAN runs OSPF to implement interworking.

Figure 10-5 BGP to IGP



To enable devices on the MAN to access the backbone network, Device C and Device D need to import routes. When OSPF imports BGP routes, a routing policy can be configured to control the number of imported routes based on private attributes (such as the community) of the imported BGP routes or modify the cost of the imported routes to control the MAN egress traffic.

11 XPL

About This Chapter

- 11.1 Overview of XPL
- 11.2 Understanding XPL
- 11.3 Application Scenarios for XPL

11.1 Overview of XPL

Definition

Extended routing-policy language (XPL) is a language used to filter routes and modify route attributes. By modifying route attributes (including reachability), XPL changes the path through which network traffic passes. XPL provides the same functions as routing policies do, but it uses different editing and filtering methods from routing policies. Therefore, XPL can meet different customer requirements.

Table 11-1 compares XPL and routing policies.

Table 11-1 Comparison between XPL and routing policies

Item	Key Functions	Editing Method	Filtering Method	User Experience
XPL	Filters routes and modifies route attributes.	Line-by-line or paragraph-by-paragraph editing	Uses sets or single elements to filter routes.	Users can configure or modify policies as required in a text editor.
Routing policies	Filter routes and modify route attributes.	Line-by-line editing	Use filters or single elements to filter routes.	Users must follow strict command configuration rules.

 NOTE

For details about routing policies, see "Routing Policies" in *HUAWEI NetEngine40E Universal Service Router Feature Description — IP Routing*.

Line-by-Line and Paragraph-by-Paragraph Editing

XPL supports line-by-line editing and paragraph-by-paragraph editing, whereas routing policies support line-by-line editing only. Line-by-line editing is a traditional configuration method, whereas paragraph-by-paragraph editing is an innovative configuration method.

Table 11-2 compares the two methods.

Table 11-2 Line-by-line and paragraph-by-paragraph editing comparison

Item	Applicable to	Differences	Help and Error Correction Mechanisms
Line-by-line editing	Users who are used to the traditional configuration method or unfamiliar with XPL	Each command is run in a command view, and one command is presented in one line, which is considered a configuration unit.	The desired command can be suggested using the command association function. If any configuration error occurs, it is reported after the command is configured.
Paragraph-by-paragraph editing	Users who are familiar with XPL clause configuration and want to simplify the configuration process	The paragraph editing UI functions as a text editor, in which users edit XPL clauses. The XPL clauses are committed after a paragraph of them are configured, and each paragraph is considered a configuration unit.	The command association function is not supported, and complete clauses must be entered in the paragraph editing UI. If any configuration error occurs, it is reported after the configurations of the whole paragraph are committed.

Purpose

When advertising, receiving, or importing routes, the router can use XPL based on actual networking requirements to filter routes and modify route attributes. XPL serves the following purposes:

- Controls route advertisement.
Only routes that match the rules specified in the XPL are advertised.
- Controls route acceptance.
Only necessary and valid routes are accepted, which reduces the routing table size and improves network security.
- Filters and controls imported routes.
A routing protocol may import routes discovered by other routing protocols. XPL ensures that only the routes that meet certain conditions are imported and route attributes of the imported routes are modified to meet the requirements of the protocol.

- Modifies route attributes.
Attributes of the routes that match the specified route-filter can be modified as required.

Benefits

XPL offers the following benefits:

- Saves system resources by controlling the routing table size.
- Improves network security by controlling route advertisement and acceptance.
- Improves network performance by modifying route attributes for effective traffic planning.
- Simplifies routing policy configurations.

11.2 Understanding XPL

XPL Implementation

XPL implementation involves the following two steps:

1. Define rules: Define route characteristics for route matching. Specifically, you need to define a set of matching rules based on route attributes, such as the destination address and the address of the router that advertises the routes. For details, see [Route-Filters](#).
2. Apply rules: Apply the matching rules to route advertisement, acceptance, and import.

Sets

A set is a group of data that XPL uses as matching rules. Sets are classified as global variable sets and route attribute sets.

- Global variable set: A global variable set is a group of frequently used values that are defined as global variables. Global variables are variables that can be referenced by all route-filters on a device. To enable a route-filter to reference a global variable, enter \$+variable name, for example, **\$glovar1**. The global variables on a device must be unique. A new global variable will override an existing global variable with the same name.
- Route attribute set: A route attribute set is a group of data concerning a route attribute. If the routes to be filtered have the same or similar route attribute, for example, they are destined for the same network segment or originate from the same AS, you can configure a route attribute set for the routes as matching rules. The application scopes and matching items vary with the route attribute set. [Table 11-3](#) shows the application scopes and matching items of different route attribute sets.

NOTE

Sets do not have the permit or deny function as routing policies do. Instead, sets are only groups of data used as matching rules, and the actions to be applied are specified in route-filters.

Table 11-3 Route attribute sets

Name	Application Scope	Matching Attribute
IPv4 prefix set	Routes of all dynamic routing protocols	Source, destination, and next hop IP addresses
IPv6 prefix set	Routes of all dynamic routing protocols	Source, destination, and next hop IP addresses
AS_Path set	BGP routes	AS_Path
Community set	BGP routes	Community
Extended community set	VPN routes	Route target and Site-of-Origin
RD set	VPN routes	RD

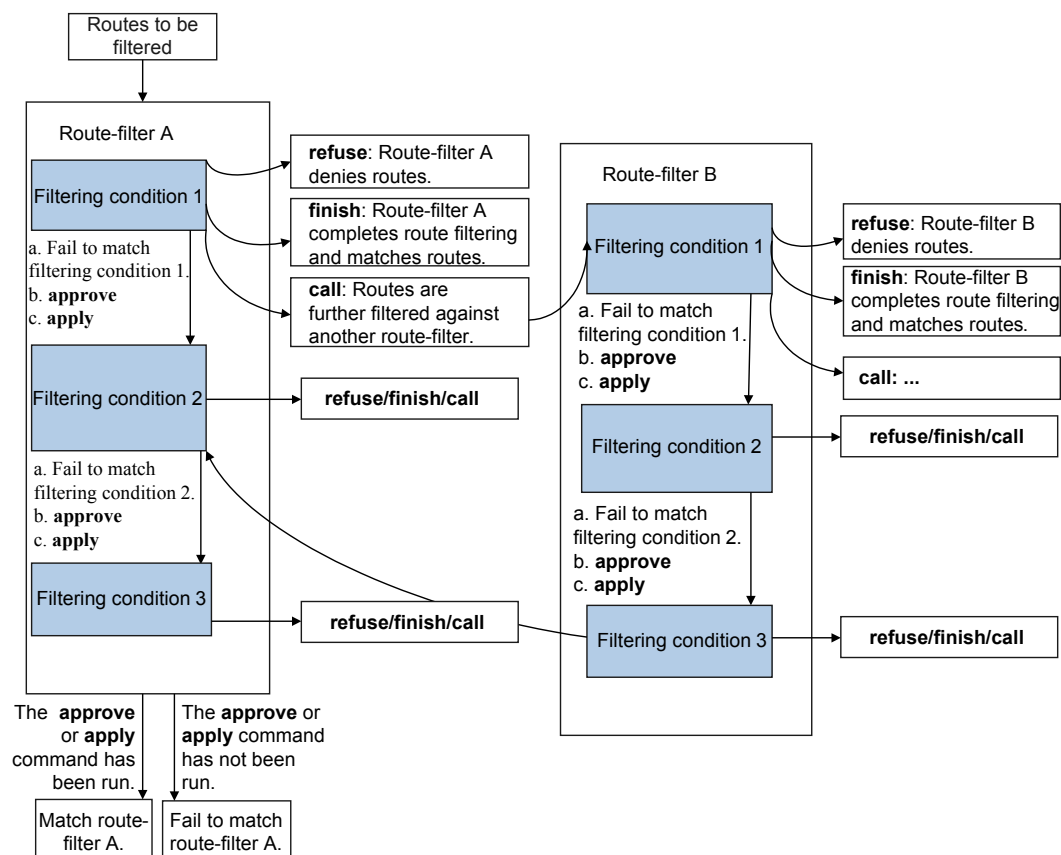
Route-Filters

Route-filters are used to filter routes based on sets or a single element and modify route attributes of the routes that match specified rules. Route-filters consist of condition and action clauses.

- Condition clause: A condition clause is defined based on a set or single element. The action specified in the action clause is applied only to the routes that match the conditions specified in the condition clause.
- Action clause: An action clause specifies an action to be applied to the routes that match the conditions specified in the condition clause. An action clause determines whether the routes match the route-filter or modifies their route attributes.

Figure 11-1 shows how a route-filter is used to filter routes. For details about condition and action clauses, see [XPL Statements](#).

Figure 11-1 Filtering process



XPL Statements

XPL statements are used to convert matching rules to sets and route-filters. XPL statements include the remark, set definition, set element, condition clause, action clause, and route-filter with pre-defined variables.

Remark

A remark is an explanation attached to an XPL policy configuration line, beginning with an exclamatory mark (!).

NOTE

If the list is not empty, no remarks can be configured in the last line (the line above the end-list).

```

xpl ip-prefix-list prefix-list1
! prefix-list1 is the name of a ip-prefix-list.
 10.0.1.0 24,
 10.0.3.0 24 eq 26,
 10.0.2.0 24 le 28,
 10.0.4.0 24 ge 26 le 30
end-list
  
```

Set definition

A set definition specifies matching rules and begins and ends with apparent clauses.

For example, an IPv4 prefix set begins with **xpl ip-prefix-list** and ends with **end-list**, with a group of IPv4 prefixes in between.

```
xpl ip-prefix-list prefix-list2
  10.0.1.0 24,
  10.0.3.0 24 eq 26,
  10.0.2.0 24 le 28,
  10.0.4.0 24 ge 26 le 30
end-list
```

Set element

Set elements include elements such as IP prefixes, AS_Path values, and communities. The elements are separated with commas. Elements in a route-filter must have the same type as the route-filter.

```
xpl ip-prefix-list prefix-list3
  10.0.1.0 24,
! element
  10.0.3.0 24 eq 26,
! element
  10.0.2.0 24 le 28,
! element
  10.0.4.0 24 ge 26 le 30
end-list
```

Condition clause

Condition clauses are used in route-filters. Condition clauses can be used with sets to define matching rules. Condition clauses can be **if**, **elseif**, or **else** clauses. Condition clauses may include **eq** (equal to), **ge** (greater than or equal to), **le** (less than or equal to), and **in** (included in) expressions, which can be used in conjunction with the Boolean condition **not**, **and**, or **or**.

in can be followed by a set so that the elements in the set are used as matching rules.

```
xpl route-filter route-filter1
! route-filter1 is the name of Route-Filter
  if med eq 20 then
    apply community { 100:1 } additive
  endif
end-filter
```

Action clause

Action clauses specify the actions to be applied to given routes and include the following clauses:

- **approve** clause: permits routes.
- **refuse** clause: denies routes.
- **finish** clause: completes route filtering and indicates that the route matches the route-filter.
- **abort** clause: aborts the route-filter or set modification.
- **apply** clause: modifies route attributes.
- **call** clause: references other route-filters.
- **break** clause: enables the device to exit from the current route-filter. If the current route-filter is referenced by a parent route-filter, the device keeps implementing remaining condition and action clauses of the parent route-filter.

```
xpl route-filter route-filter2
! Name of Route-Filter
if med eq 10 then
    approve
endif
end-filter
```

Route-filter with pre-defined variables

XPL supports route-filters with pre-defined variables. Route-filters with pre-defined variables can be referenced during route-filter configuration through **call** clauses.

```
xpl route-filter param-route-filter ($mytag)
! Configure a route-filter with a pre-defined variable.
    apply community { 1234:$mytag } additive
end-filter
xpl route-filter origin-10
! Reference the route-filter with the pre-defined variable.
if med eq 20 then
    call route-filter param-route-filter (10)
else
    permit
endif
end-filter
```

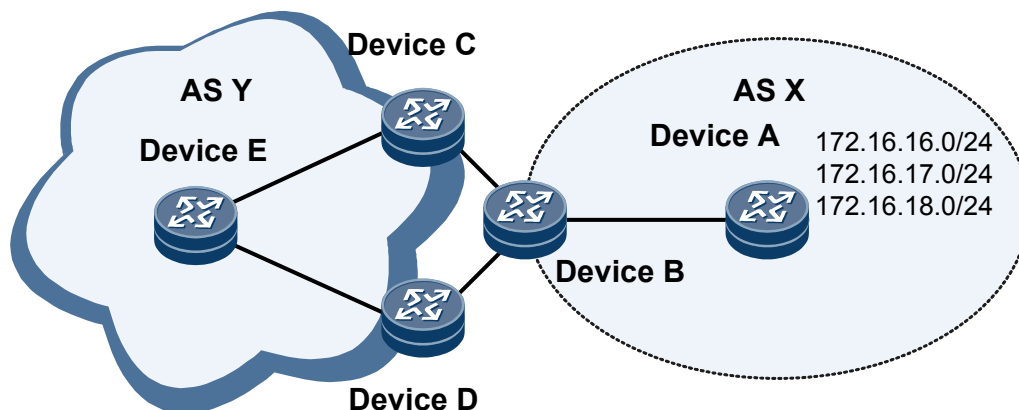
11.3 Application Scenarios for XPL

Using XPL to Filter Routes

On the network shown in [Figure 11-2](#), Device A has routes to 172.16.16.0/24, 172.16.17.0/24, and 172.16.18.0/24. The networking requirements are as follows:

- Device A advertises only route 172.16.17.0/24 to Device B.
- After receiving the route from Device B, Device C forwards it directly to Device E, and Device D increases the MED attribute of the route before forwarding it to Device E to ensure that Device C functions as the egress for the traffic from Device E to 172.16.17.0/24.

Figure 11-2 Route filtering scenario



The preceding requirements can be met using an IPv4 prefix set:

1. Configure an IPv4 prefix set named **ip-prefix1**, which includes only the 172.16.17.0 24 element, on Device A.
2. Configure a route-filter named **route-filter1** on Device A, which permits the route carrying the element in **ip-prefix1** and denies other routes.
3. Configure **route-filter1** as an export policy on Device A so that Device A advertises only route 172.16.17.0/24 to Device B.
4. Configure an IPv4 prefix set named **ip-prefix2**, which includes only 172.16.17.0 24, on Device D.
5. Configure a route-filter named **route-filter2**, which increases the MED value of the route carrying the element in **ip-prefix2**, on Device D.
6. Configure **route-filter2** as an export policy on Device D so that the MED value of the route advertised by Device D is greater than that of the route advertised by Device C, making Device C the egress for the traffic from Device E to 172.16.17.0/24.

12 Route Monitoring Group

About This Chapter

[12.1 Overview of Route Monitoring Groups](#)

[12.2 Understanding Route Monitoring Groups](#)

[12.3 Application Scenarios for Route Monitoring Groups](#)

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12.1 Overview of Route Monitoring Groups

Definition

All monitored network-side routes of the same type can be added to a group called a route monitoring group. Each route monitoring group is identified by a unique name.

A route monitoring group monitors the status of its member routes, each of which has a down-weight. The down-weight indicates the link quality. The higher the value, the more important the route. The down-weight can be set based on parameters, such as the link bandwidth, rate, and cost.

- If a route in the route monitoring group goes Down, its down-weight is added to the down-weight sum of the route monitoring group.
- If a route in the route monitoring group goes Up again, its down-weight is subtracted from the down-weight sum of the route monitoring group.

Purpose

Service modules can be associated with the route monitoring group, with a threshold configured for each service module for triggering a primary/backup access-side link switchover. If the down-weight sum of the route monitoring group reaches the threshold of a service module, the routing management (RM) module notifies the service module to switch services from the primary link to the backup link. If the down-weight sum of the route monitoring group falls below the threshold, the RM module notifies the service module to switch services back.

Benefits

If a service module is associated with a route monitoring group in a dual-system backup scenario, services can be switched to the backup link if the primary link fails, therefore preventing traffic overload and forwarding failures.

12.2 Understanding Route Monitoring Groups

12.2.1 Route Monitoring Group Fundamentals

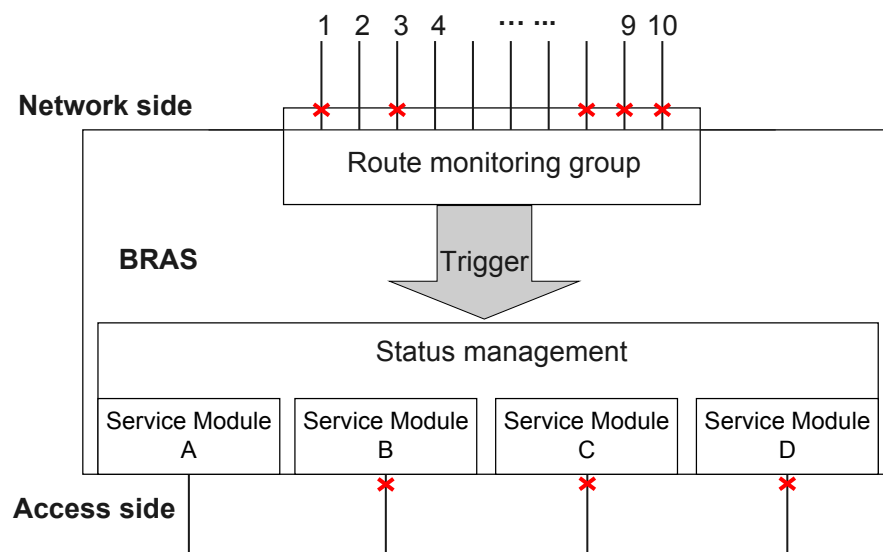
All monitored network-side routes of the same type can be added to a route monitoring group. A down-weight can be set for each route in the route monitoring group based on link attributes, such as the bandwidth, and a threshold can be set for each service module that is associated with the route monitoring group to trigger a primary/backup access-side link switchover.

- A route monitoring group monitors the status of its member routes. If a route in the group goes Down, its down-weight is added to the down-weight sum of the route monitoring group. If the down-weight sum of the route monitoring group reaches the threshold of a service module, the RM module notifies the service module to switch services from the primary link to the backup link.
- If a route in the route monitoring group goes Up again, its down-weight is subtracted from the down-weight sum of the route monitoring group. If the down-weight sum of the route monitoring group falls below the threshold of a service module, the RM module notifies the service module to switch services back. You can specify the switchback delay time based on your actual network requirements.

As shown in [Figure 12-1](#), the route monitoring group contains 10 routes, each having a down-weight of 10. The route monitoring group is associated with service modules A, B, C, and D whose thresholds for a primary/backup switchover are 80, 50, 30, and 20, respectively.

- If two routes in the route monitoring group go Down, the RM module notifies service module D to switch services from the primary link to the backup link. If one more route goes Down, the RM module notifies service module C to perform a primary/backup link switchover. If five routes go Down, the RM module notifies service module B to perform a primary/backup link switchover. If the number of routes that go Down reaches eight, the RM module notifies service module A to perform a primary/backup link switchover.
- If the number of routes that are Down in the route monitoring group falls below eight, the RM module notifies service module A to switch services back. If the number of routes that are Down in the route monitoring group falls below five, the RM module notifies service module B to switch services back. If the number of routes that are Down in the route monitoring group falls below three, the RM module notifies service module C to switch services back. If the number of routes that are Down in the route monitoring group falls below two, the RM module notifies service module D to switch services back.

Figure 12-1 Route monitoring group



12.3 Application Scenarios for Route Monitoring Groups

12.3.1 Applications of route monitoring groups

Service Overview

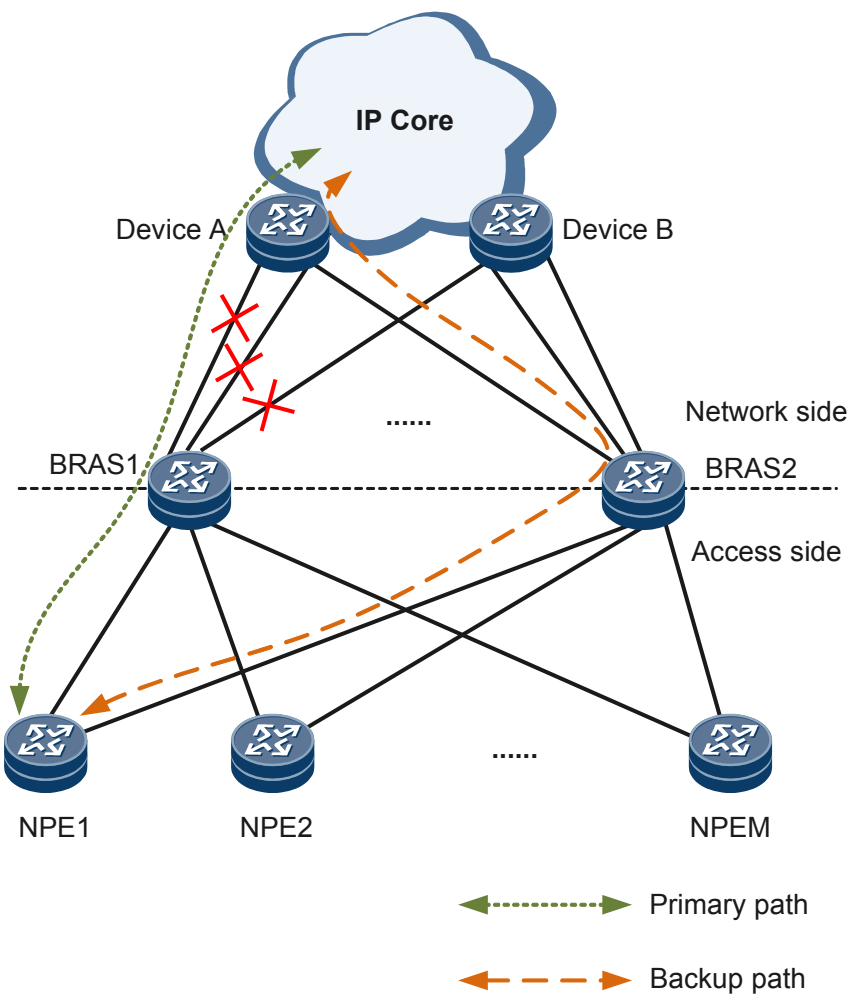
To improve network reliability, most carriers implement device-level redundancy by deploying two devices. The two devices back up each other or share traffic load. If one of the devices fails, the other device takes over services. Despite the benefit of enhanced network reliability, you must dual-home other devices to the two devices, which may introduce link reliability and load balancing issues.

Networking Description

As shown in [Figure 12-2](#), BRAS 2 backs up BRAS 1. NPEs on the user side are dual-homed to the two BRASs to load-balance traffic, and the two BRASs are connected to routers on the network side.

- If the link between BRAS 1 and Device A or between BRAS 1 and Device B fails, the link bandwidth between BRAS 1 and the IP core network decreases. The NPEs, however, cannot detect the link failure and keep sending packets to the IP core network through BRAS 1. As a result, the other link between BRAS 1 and the IP core network may be overloaded.
- If the links between BRAS 1 and Device A and between BRAS 1 and Device B both fail, only the links between BRAS 2 and the IP core network are available. The NPEs, however, cannot detect the link failure and keep sending packets to the IP core network through BRAS 1. As a result, the packets are discarded.

Figure 12-2 Route monitoring group



Feature Deployment

To address the packet drop issue, deploy a route monitoring group on each BRAS, and add network-side routes of the BRAS to the route monitoring group. If the down-weight sum of the route monitoring group reaches the threshold of a service module that is associated with the group, the RM module will notify the service module to trigger a primary/backup access-side link switchover. This mechanism prevents traffic overload and service interruptions.

12.4 Terminology for Route Monitoring Group

Terms

Term	Definition
Route monitoring group	A group that consists of monitored network-side routes of the same type.